

Protocol

This trial protocol has been provided by the authors to give readers additional information about their work.

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This supplement contains the following items:

1. Original protocol, final protocol, summary of changes
2. First **and** final statistical analysis plan



Developed on
behalf of the NCRI
Prostate Clinical
Studies Group

Part the National
Cancer Research
Network Portfolio

STAMPEDE

Systemic **T**herapy in **A**dvancing or
Metastatic **P**rostate Cancer: **E**valuation of
Drug **E**fficacy

A 5-stage multi-arm randomised controlled trial

MRC PR08

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PROTOCOL VERSION 1.0

20 May 2004

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GENERAL INFORMATION

This document describes a trial coordinated by the Medical Research Council (MRC) Clinical Trials Unit (CTU) and provides information about procedures for entering patients into it. The protocol should not be used as an aide-memoire or guide for the treatment of other patients; every care was taken in its drafting, but corrections or amendments may be necessary. These will be circulated to the known investigators in the trial, but centres entering patients for the first time are advised to contact the Cancer Division, MRC CTU, London to confirm they have the most up to date version. Clinical problems relating to this study should be referred to the Chief Investigator.

Sponsor

Medical Research Council, 20 Park Crescent, London, W1B 4AL

Funding

Clinical Trials Advisory Awards Committee (on behalf of Cancer Research UK, Medical Research Council, and other charities) together with educational grants from Novartis and Aventis.

Compliance

This trial will adhere to the principles outlined in the International Conference on Harmonization (ICH) Good Clinical Practice (GCP) guidelines. It will be conducted in compliance with the protocol, MRC GCP, Data Protection Act (DPA number: G0027154) and other regulatory requirements, as appropriate.

Authorisation

The following persons are authorised to sign the final protocol and protocol amendments for the sponsor: Professor N James (Chief Investigator) and Professor M Parmar (Head, Cancer Division, MRC Clinical Trials Unit).

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ABBREVIATIONS AND GLOSSARY

ACE	Angiotensin-Converting Enzyme
AS	Androgen suppression
<i>bid</i>	Twice a day (<i>bis in die</i>)
BP	Blood pressure
BSA	Body surface area
CERES	Consumers for Ethics in Research
CF	Consent Form
CI	Confidence interval
COSTART	Coding Symbols for a Thesaurus of Adverse Reaction Terms
Cox-2	Cyclooxygenase-2
CRF	Case Report Form
CRUK	Cancer Research UK
CT	Computerised tomography
CTA	Clinical Trials Authorisation
CTAAC	Clinical Trials Advisory and Awards Committee
CTC	Common Toxicity Criteria
CTU	Clinical Trials Unit
CXR	Chest X-ray
DDX	Doctors and Dentists Exemption
DNA	Deoxyribonucleic Acid
DPA	Data Protection Act
ERC	Endpoint Review Committee
ICH	International Conference on Harmonization
ECG	Electro cardiogram
FBC	Full Blood Count
FFS	Failure Free Survival
GCP	Good Clinical Practice
GP	General Practitioner
GRO	General Register Office
HE	Health Economics
hr	Hour
HR	Hazard Ratio
HRPC	Hormone Refractory Prostate Cancer
IDMC	Independent Data Monitoring Committee
IM	Intramuscular
ISRCTN	International Standard Randomised Controlled Trial Number
IU	International Units
IV	Intravenous
LD	Longest diameter
LFT's	Liver Function Tests
LHRH	Luteinising Hormone Releasing Hormone
LREC	Local Research Ethics Committee
MHRA	Medicine and Healthcare Products Regulatory Agency
min	Minutes
MRC	Medical Research Council
MREC	Multi-Centre Research Ethics Committee
MRI	Magnetic resonance imaging
NCI	National Cancer Institute (USA)
NCRN	National Cancer Research Network
NDMC	Non-independent Data Monitoring Committee
NHS	National Health Service
NSAID	Non-Steroidal Anti-inflammatory Drugs
ONS	Office for National Statistics
OS	Overall Survival
PI	Principal Investigator
PIS	Patient Information Sheet

po	<i>per orum</i> (orally)
PSA	Prostate Specific Antigen
QALY	Quality-adjusted Life Years
qds	<i>quater die sumendus</i> (4 times each day)
QL	Quality of Life
R&D	Research and Development
RECIST	Response Evaluation Criteria In Solid Tumours
SAE	Serious Adverse Event
sc	<i>Sub-cutaneous</i> (under skin)
SNP	Single Nucleotide Polymorphism
STAMPEDE	Systemic Therapy in Advancing and Metastatic Prostate Cancer: Evaluation of Drug Efficacy
SUSAR	Suspected Unexpected Serious Adverse Reactions
TMG	Trial Management Group
TURP	Trans-Urethral Resection of Prostate
TSC	Trial Steering Committee
ULN	Upper Limit of Normal
U+E	Urea and Electrolytes
WHO	World Health Organisation

1 SUMMARY

1.1 Lay summary

Prostate cancers depend upon the male hormone testosterone for their growth. Lowering testosterone levels (either by removing all or part of both testes, or by giving anti-hormone injections) slows the growth of prostate cancers. This type of treatment is called hormone treatment and is often used when prostate cancers have spread outside the prostate gland. Although hormone treatment is usually successful at stopping the cancer growing for a period of time, the cancer will begin to grow again in most men.

A number of newer treatments have recently become available and have shown initial promise against prostate cancer. Newer treatments are usually used in prostate cancer when hormone treatment is no longer effective and the cancer has started to grow again. The aim of this trial, which is called STAMPEDE, is to assess three of these newer treatments, given earlier in the course of the disease in combination with hormone treatment.

The three new treatments to be assessed are: -

1. Zoledronic acid: Prostate cancer cells can spread to bones and weaken them. Zoledronic acid is a drug that reduces bone destruction and hardens bones. This may make them more resistant to attack by cancer cells.

2. Docetaxel: A drug that stops cells replicating that is currently being used to treat lung, breast and ovarian cancer.

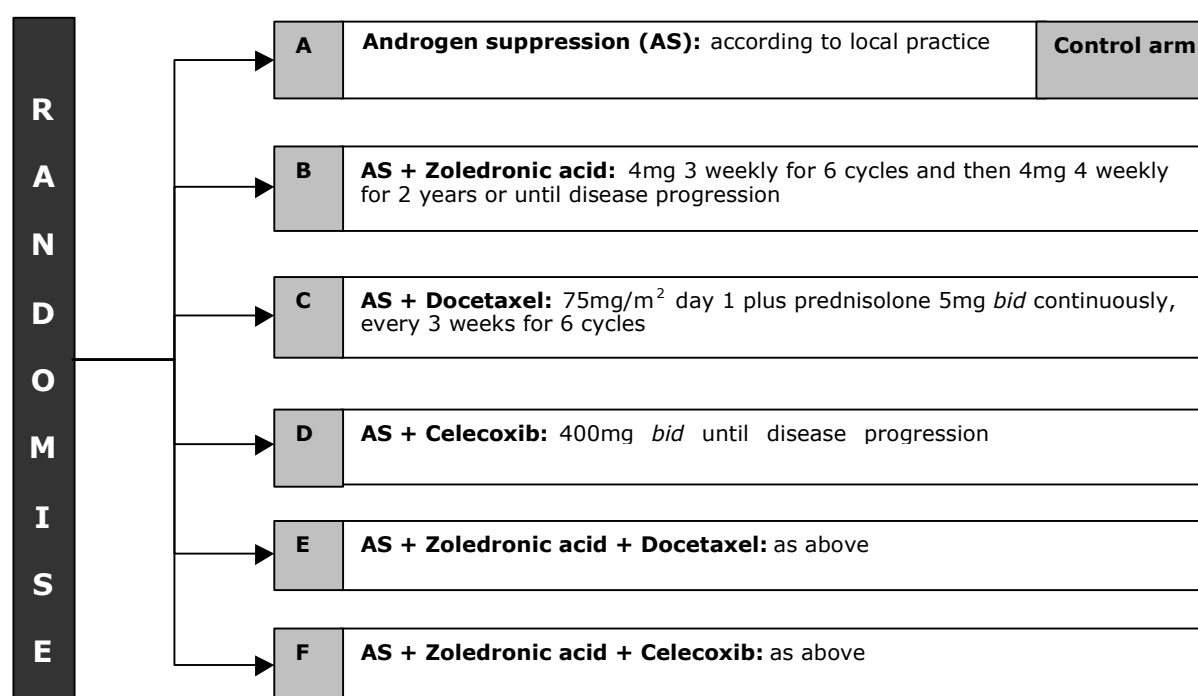
3. Celecoxib: An aspirin-like drug that is used to treat arthritis. It slows down the growth of cancer cells in the laboratory. We wish to see if it has the same effect on cancer cells in patients.

STAMPEDE will look at the effect of combining one or two of the new treatments described above with hormone treatment. A computer program will be used to allocate which treatment the patient receives, using a chance process. Two in every seven patients will receive hormone treatment only; the other five in seven patients will be split equally between the 5 newer combined treatment options. The trial will look at the effects of the combined treatments on quality of life and find out whether the new treatment combinations increase the time when the cancer is not growing and result in patients living longer. The study will also look at which treatment provides the greater value for money for the health service. Approximately 3,300 patients will participate in the trial and it will take between 6-10 years to complete.

1.2 Abstract and summary of trial design

STAMPEDE is a multi-centre, randomised controlled trial for patients with locally advanced or metastatic prostate cancer who are about to commence androgen suppression (AS) therapy. Patients can have either newly diagnosed disease, or have been previously treated with radical radiotherapy or surgery but now have a rising prostate specific antigen (PSA) (further details on eligibility see section 4). The trial will assess the effects of adding three different agents, both as single agents and in combinations, to AS therapy. The investigational agents are (i) a bisphosphonate, zoledronic acid, (ii) a cytotoxic chemotherapeutic agent, docetaxel and (iii) a cyclooxygenase (Cox-2) inhibitor, celecoxib. The trial has six arms; the control arm of the trial is AS only, achieved through the use of luteinising hormone releasing hormone (LHRH) analogues or bilateral orchidectomy according to local practice. The other trial arms are summarised in Figure 1.

Figure 1 - Arms of the STAMPEDE Trial



AS = Androgen suppression as in Arm A

The trial will be conducted in FIVE stages: a Pilot Phase, Efficacy Stages I to III and Efficacy Stage IV. The primary outcome measure of the Pilot Phase is the safety of the investigational arms and 210 patients will be recruited. Investigational arms will only proceed to recruitment in the next stage if they have been shown to be both safe and feasible, although patient data from all patients and all stages will be included in the final analyses. In Efficacy Stages I-III the primary outcome measure is failure free survival (FFS). Further patients will be recruited until 115, 225 and 355 FFS events have been observed in the control arm. Some evidence of efficacy

will be required for an research arm to proceed to further recruitment in each stage and guidelines are in place. In Efficacy Stage IV, patients will be recruited until 440 deaths have been observed in the control arm. Overall, approximately 3,300 patients will be recruited over 5-7 years although the exact number of patients and duration will depend on the observed accrual rate, observed event rate and the number of patients accruing at each stage.

Patients will be assessed 6 weekly for the first 24 weeks after randomisation and then every 12 weeks up to 2 years. Assessments will then be 3-monthly for the next 2 years and then 6-monthly, thereafter. Patients will also be asked to complete questionnaires aimed at assessing the effects of the investigational treatments on their quality of life (QL) and on their use of health care resources (HE study).

In addition, there is a molecular genetics substudy. Patients willing to participate will be asked to donate 8ml of blood which will be stored for either DNA and protein analysis in order to try to identify genes that are associated with response to therapy, side-effects or susceptibility to prostate cancer. There are separate patient information sheets for the QL and HE study, and the molecular genetics study (For further details of ancillary studies see section 16).

1.3 Trial Documentation

Figure 2 presents a summary of the timing of the required trial documentation for participating centres and their randomised patients.

Figure 2 - Summary of timing of trial documentation

Trial documentation	Timing
Centre accreditation form	Before centre participation
PIS and CF on local paper	Before centre accreditation
R&D approval	Before centre accreditation
GP letter	Before centre participation
SSA Favourable Opinion	Before centre participation – Main REC
Eligibility log	To be appended for each considered patient
Randomisation form	At randomisation
Pre-treatment form	At randomisation
Treatment forms	Every 6wks until 24wks, then with FU forms
FOLLOW-UP FORMS	EVERY 12WKS UNTIL 2YRS THEN 3 MONTHS UNTIL 5YRS, THEN 6 MONTHLY
Toxicity form	EVERY 12WKS UNTIL 2YRS THEN 3 MONTHS UNTIL 5YRS, THEN 6 MONTHLY
Quality of life forms	At 6, 12, 18, 24 and 52wks, then annually
Progression forms	<i>As needed</i>
Serious adverse event form	<i>As needed</i>
Death form	<i>As needed</i>

2 BACKGROUND

2.1 Introduction and Rationale

Prostate cancer is a major health problem world-wide and accounts for nearly one fifth of all newly diagnosed male cancers. In the UK, approximately 25,000 men are diagnosed with prostate cancer each year and in 2001 almost 10,000 men died from the disease (1).

The initial (first line) treatment for locally advanced or metastatic prostate cancer is androgen suppression (AS) achieved either surgically with bilateral orchidectomy, or medically with LHRH analogues (2). AS produces responses in up to 85% of patients (3) but it is not curative and disease recurs in virtually all patients, with a median time to progression of 18-24 months. Such disease is referred to as hormone refractory prostate cancer (HRPC).

There are several treatments, which are used 'second-line' in patients with HRPC, but no evidence as to which is associated with the best response or whether any of them might have a role in first-line treatment; these include further hormonal manipulations (4, 5), bisphosphonates (6), cytotoxic chemotherapy (3) and novel agents (7). The traditional approach to the testing and introduction of new treatments for prostate cancer is in hormone refractory disease. An alternative approach is to investigate new drugs and new approaches to treatment as first-line therapy in patients starting hormone therapy. At this point patients would be fitter and better able to tolerate treatment than when they have HRPC, and there is also the possibility of having a larger and more durable effect.

STAMPEDE (as known as MRC PR08) is an innovative, 6-arm, multi-centre, randomised controlled trial. It assesses the effects of a bisphosphonate (zoledronic acid), a cytotoxic chemotherapeutic agent (docetaxel) and a cyclooxygenase (Cox-2) inhibitor (celecoxib), as single agents or combinations, in patients commencing AS therapy for advancing or metastatic prostate cancer. The trial is divided into five stages such that, for each investigational arm, safety and activity data are generated in the first four stages; an investigational arm will only proceed to the fifth and final stage of recruitment, where it will be assessed for its effect on overall survival, if it has been shown to be safe and active. It is important to note, however, that patient data from all arms and all stages will be included in the final analyses of the primary outcome measure, even if the investigational arm did not proceed to the final stage.

2.2 Bisphosphonates

The bisphosphonates are a class of drug that act by reducing osteoclast formation, inhibiting osteoclast activity and inducing osteoclast apoptosis. They are effective at controlling hypercalcaemia and preventing skeletal complications associated with malignant disease (8).

Zoledronic acid is a new, highly potent, third generation bisphosphonate; studies comparing the efficacy of zoledronic acid to other bisphosphonates suggest that zoledronic acid has a 40-850 fold higher potency than clodronate in preclinical models of bone resorption (9). It has also been shown to be more effective than pamidronate (90mg) in controlling malignant hypercalcaemia (10). In addition, zoledronic acid has also demonstrated direct anti-cancer activity, including inhibition of proliferation of breast cancer and prostate cancer cells *in vitro* (11).

In randomised controlled trials of 1,648 patients, 4mg zoledronic acid was more effective than pamidronate in reducing the risk of skeletal complications in patients with bone metastases from breast cancer (12). Also, in metastatic prostate cancer, zoledronic acid has been shown to reduce the rate of skeletal related events compared to placebo in a trial involving 429 men (13). In April 2002, zoledronic acid received approval from the Committee for Propriety Medicinal Products for the prevention of skeletal related events (for example, fractures) in patients with any advanced malignancies involving bone.

The MRC PR05 prostate cancer trial showed that a first generation bisphosphonate (clodronate) commenced at the time of hormone therapy initiation, delayed time to progression in patients with bony metastatic disease and there was some evidence that it may also improve survival (14). There is, therefore, a good rationale for investigating a more potent bisphosphonate in patients with prostate cancer who are about to commence AS therapy.

2.3 Chemotherapy

Over recent years there has been increasing evidence of the clinical efficacy of chemotherapy in prostate cancer. One of the most active agents is docetaxel, a semi-synthetic taxane that binds microtubules and inhibits mitosis (15). In phase I and II trials, docetaxel exhibited significant activity in prostate cancer in dosing regimens in the range of 40 to 75 mg/m², administered every three weeks. Evidence of this activity includes PSA decline, objective response in bi-dimensionally measurable lesions and improvements in pain control. The safety profile was assessed as being acceptable throughout these studies with a good risk/benefit ratio in this clinical setting (16-19). In addition, the concurrent administration of prednisolone has been shown to materially decrease the toxicity associated with docetaxel (20).

Recent evidence also indicates that docetaxel compares favourably with mitozantrone which had previously been considered the most active chemotherapy agent against prostate cancer. A phase II trial reported PSA response rates of 38-46% with docetaxel with prednisolone compared to 28% for mitozantrone plus prednisolone (21).

2.4 Cyclooxygenase-2 inhibitors

Cyclooxygenase-2 (Cox-2) is an isoenzyme induced by a variety of mitogens, cytokines and growth factors that are associated with inflammation, ovulation and carcinogenesis (22-24). There is a growing body of evidence that inhibition of Cox-2 may play an important role in the prevention of cancer and the delay of progression in established cancer. A number of case-control studies have shown a reduction in risk of prostate cancer associated with the use of non-steroidal anti-inflammatory drugs (NSAID), which include inhibition of Cox-2 amongst their mode of action (25). Pathological studies show Cox-2 is upregulated in prostate carcinoma (26) and one study suggested that NSAID use may delay progression from subclinical to clinical prostate cancer (27).

Celecoxib, a Cox-2 inhibitor, is better tolerated than other NSAIDs and there is evidence that it is active as a chemoprevention agent (28). It also has important antineoplastic properties such as the ability to inhibit angiogenic factors and induce apoptosis in human cancer cells including prostate cancer (29, 30).

Given case-control data suggesting effects on prostate cancer, pathological expression of Cox-2 in prostate cancer and *in vitro* data suggesting that inhibition of Cox-2 inhibits growth and invasiveness, further investigation in prostate cancer is warranted, particularly given the very large body of safety data for this agent in the treatment of arthritis.

2.5 Treatment Combinations

Bisphosphonate and Chemotherapy

Zoledronic acid and docetaxel have different mechanisms of action. In addition to its skeletal protection activity, zoledronic acid has shown direct activity against prostate cancer cells, both *in vitro* and *in vivo* (11,31,32). There is also *in vitro* and *in vivo* evidence to suggest synergy between zoledronic acid and chemotherapy in breast cancer cells and anti-angiogenic effects in patients (33, 34).

Toxicities of the two agents are complementary and administration in combination is expected to be feasible and safe. These aspects will be evaluated in the initial Pilot Phase of the trial. Since both agents show considerable promise as single agents and there is *in vitro* evidence of synergy, we believe there is a strong rationale for evaluating these two agents in combination.

Bisphosphonate and Cyclooxygenase-2 Inhibitors

An alternative approach to combination therapy is to target the principal site of relapse and a key mode of progression and this is the rationale for combining zoledronic acid with a Cox-2

inhibitor. Bisphosphonates have already been shown to delay bone disease progression in hormone refractory disease (14). Cox-2 appears to play a crucial role in the molecular phenotype of advanced prostate cancer as outlined above, and this effect is likely to be apparent in both soft tissue and in bone. Toxicities of the two agents are likely to be complementary and there is no strong *a priori* reason to anticipate unacceptable toxicity. The Pilot Phase of the trial will evaluate tolerability and safety of the combination. Targeting both bone progression and the underlying molecular changes leading to progression can be expected to have synergistic benefits in terms of delaying development of hormone refractory disease.

3 SELECTION OF CENTRES/CENTRE ACCREDITATION

Centres who wish to participate in the trial should be registered with the Medical Research Council Clinical Trials Unit (MRC CTU). Before any patients are randomised the MRC CTU must receive a completed accreditation form (**Appendix M**) and a favourable Site Specific Assessment (SSA) for the site. Only a limited number of centres will participate in the Pilot Phase of the trial; this is to ensure that safety and feasibility data are collected expediently. The other stages of the trial will be open to any centre that wishes to participate and has fulfilled the requirements described above.

The Clinical Trials Authorisation (CTA) for this study requires that the Medicines and Healthcare Products Regulatory Agency (MHRA) be supplied with the names and addresses of all participants. Clinicians **must** therefore notify the MRC CTU of their intentions to participate **before** entering patients.

Before a patient is entered into the trial written informed consent must be obtained. Approved patient information sheets and informed consent forms are supplied in **Appendix B**. In addition, a baseline questionnaire for the QL and HE studies, must also be completed before the patient is informed which treatment has been allocated.

4 SELECTION OF PATIENTS

4.1 Patient inclusion criteria

Patients must fulfil **one** of the criteria in section 4.1.1 or **one** of the criteria in section 4.1.2. Additionally, **all** patients must fulfil the criteria in section 4.1.3.

4.1.1 High Risk Newly Diagnosed Patients

...with one of:-

- (i) Stage T3/4 N0 M0 histologically confirmed prostate adenocarcinoma with PSA $\geq$ 40ng/ml or Gleason sum score 8-10
- (ii) Stage T_{any} N+ M0 or T_{any} N_{any} M+ histologically confirmed prostate adenocarcinoma
- (iii) Multiple sclerotic bone metastases with a PSA $\geq$ 100ng/ml without histological confirmation

OR

4.1.2 Patients with histologically confirmed prostate adenocarcinoma previously treated^ψ with radical surgery or radiotherapy who are now relapsing

...with one of:-

- (i) PSA $\geq$ 4ng/ml and rising with doubling time less than 6 months
- (ii) PSA $\geq$ 20ng/ml

^ψ**Note:** Prior hormone therapy for localised disease must have been completed at least 12 months previously and have been no longer than 12 months in duration. It can have been given as adjuvant or neoadjuvant therapy.

AND

4.1.3 For all patients:-

- (i) Intention to treat with long-term androgen suppression
- (ii) Fit for all protocol treatment^φ and follow-up, WHO performance status 0-2 (see **Appendix A**)
- (iii) Have completed the appropriate investigations prior to randomisation
- (iv) Adequate haematological function: neutrophil count $\geq 1.5 \times 10^9/l$ and platelets $\geq 100 \times 10^9/l$
- (v) Adequate renal function: Serum creatinine ≤ 1.5 ULN
- (vi) Adequate liver function: ALT or AST ≤ 1.5 ULN, bilirubin $\leq$ ULN
- (vii) Normal testosterone level prior to hormone treatment
- (viii) Written informed consent
- (ix) Willing and expected to comply with follow-up schedule

^φ Note: Medical contraindications to the trial medications are given in **Appendix G**, in particular please note celecoxib is contraindicated in patients with active peptic ulceration, gastrointestinal bleeding, inflammatory bowel disease or severe congestive heart failure.

4.2 Patient exclusion criteria

- (i) Prior systemic therapy for locally advanced or metastatic prostate cancer except as listed in 4.1.2.
- (ii) Metastatic brain disease or leptomeningeal disease.
- (iii) Any other previous or current malignant disease which, in the judgement of the responsible physician, is likely to interfere with STAMPEDE treatment or assessment.
- (iv) Symptomatic peripheral neuropathy $\geq$ grade 2 (NCI CTC).
- (v) Any surgery (e.g. TURP) performed within the past 4 weeks

4.3 Screening procedures

4.3.1 Investigations Prior to Randomisation

All patients require the following examinations to have been performed within 5 weeks (35 days) prior to randomisation:-

- CT or MRI of pelvis and abdomen
- Bone Scan
- Chest X-ray and ECG

...and the following blood tests within 2 weeks (14 days) prior to randomisation:-

- PSA
- Testosterone
- Urea and Electrolytes
- Liver Function Tests
- Serum creatinine
- Serum corrected calcium
- Phosphates
- Magnesium

It is preferable that patients are not started on hormones prior to randomisation. However, if this has occurred, then AS therapy should have not have started more than 4 weeks (28 days) before randomisation and the baseline PSA measurement must be taken before the treatment was initiated and within the 4 week period prior to randomisation.

For patients who are hypercalcaemic prior to randomisation and require treatment it is recommended that they are treated with a bisphosphonate and that the treatment should be discontinued when they are stabilised.

4.3.2 Concomitant Medications

All concomitant medications should be recorded including regular consumption of NSAID and/or aspirin and use of other bisphosphonates (see **Section 4.3.1**). All concomitant medications should be continued throughout the trial unless the responsible physician decides otherwise.

4.3.3 Additional Details for Patients Participating in the Sub-studies

Baseline QL and HE forms must be completed before randomisation. Full details of these studies are given in Section 17.

An additional 8ml blood must be taken if the patient has given their consent to participate in the molecular genetics study. Full details of this study and instructions relating to the handling of the blood sample are given in Section 17 and **Appendix D**.

5 RANDOMISATION AND ENROLMENT

To enter a patient the randomisation form should be completed and the MRC CTU contacted by phone or fax: -

RANDOMISATIONS

To randomise call MRC CTU, Monday to Friday 0900-1700

Tel: 020 7670 4777

Fax: 020 7670 4818

A trial number and treatment will be allocated and given over the phone or by return fax. In addition, a letter confirming these details will be sent. The trial number will be the primary way in which the patient will be identified and should be used in all correspondence.

5.1 Co-enrolment guidelines

Ideally, patients should not be participating in any other clinical trial of prostate cancer treatment when they enter STAMPEDE and should not enter any other trials until the patient has had a failure-free survival (FFS) event reported. After this point, the patient may be entered into further, second-line treatment studies. The primary endpoint of STAMPEDE is survival. Therefore, follow-up to STAMPEDE must continue and must not be affected by co-enrolment to other studies. It is preferable that the MRC CTU should be notified in writing, with details of the trial: trial name, sponsor, randomisation arms, study endpoints and a declaration that STAMPEDE follow-up will not be impeded, before a patient is co-enrolled.

6 TREATMENT OF PATIENTS

6.1 Trial treatment

Patients will be randomised to the control arm (Arm A) or one of the five investigational arms. All patients will receive AS to achieve castration levels of testosterone. The method of AS is a local choice but must be specified for each patient prior to randomisation. The recommended methods of AS are given in section 6.1.1. **All trial treatments should commence as soon as practically possible after randomisation.** Patients having a bilateral orchidectomy should commence any additional treatment with 4 weeks of the operation unless there is a strong clinical reason not to do so.

6.1.1 Arm A (Control arm) - Androgen Suppression Only

The recommended methods of AS are bilateral orchidectomy or LHRH analogues (see below). Please note that anti-androgens alone are not permissible as hormone therapy for patients participating in STAMPEDE, but their use is recommended in the short-term to prevent tumour “flare” which may occur after commencing LHRH analogues. At the time of randomisation centres will be asked to specify the method of AS for each patient. Other methods of AS should be discussed with the Chief Investigator or the Trial Physician.

- Bilateral orchidectomy Operations should be performed by appropriately trained surgeons. A total or subcapsular orchidectomy may be performed.
- LHRH agonists LHRH analogues used according to local practice. The prophylactic use of anti-androgens to prevent tumour “flare” is recommended.

6.1.2 Arm B - Androgen Suppression + Zoledronic Acid

- Androgen suppression as described in section 6.1.1.
- Zoledronic Acid 4mg 15min IV infusion every 3 weeks, for 6 treatments followed by zoledronic acid 4mg 15min IV infusion every 4 weeks up to a maximum of 2 years from the start of the treatment or until disease (including PSA) progression (see Section 7.2). Patients should also receive an oral supplement of 500mg calcium and 400IU vitamin D daily. These doses are available as a combination tablet.

6.1.3 Arm C - Androgen Suppression + Docetaxel

- Androgen suppression as described in section 6.1.1.
- Docetaxel 75mg/m² Day 1 as 1hr IV infusion, plus prednisolone 5mg *bid* daily for 21 days. The cycle should be repeated every 3 weeks for a maximum of 6 cycles. The recommended administration schedule, anti-emetic regimen and dose modifications for docetaxel are given in **Appendix F**.

6.1.4 Arm D - Androgen Suppression + Celecoxib

- Androgen suppression as described in section 6.1.1.
- Celecoxib 400mg *bid* until disease (including PSA) progression (see Section 7.2).

6.1.5 Arm E - Androgen Suppression + Zoledronic Acid + Docetaxel

- Androgen suppression as described in section 6.1.1.
- Zoledronic Acid 4mg 15min IV infusion every 3 weeks, for 6 treatments followed by zoledronic acid 4mg 15min IV infusion every 4 weeks up to a maximum of 2 years from the start of the treatment or until disease (including PSA) progression (see Section 7.2). Patients should also receive an oral supplement of 500mg calcium and 400IU vitamin D daily. These doses are available as a combination tablet.
- Docetaxel 75mg/m² Day 1 as 1hr IV infusion, plus prednisolone 5mg *bid* daily for 21 days. The cycle should be repeated every 3 weeks for a maximum of 6 cycles. The recommended administration schedule, anti-emetic regimen and dose modifications for docetaxel are given in **Appendix F**.

6.1.6 Arm F - Androgen Suppression + Zoledronic Acid + Celecoxib

- Androgen suppression as described in section 6.1.1.
- Zoledronic Acid 4mg 15min IV infusion every 3 weeks, for 6 treatments followed by zoledronic acid 4mg 15min IV infusion every 4 weeks up to a maximum of 2 years from the start of the treatment or until disease (including PSA) progression (see Section 7.2). Patients should also receive an oral supplement of 500mg calcium and 400IU vitamin D daily. These doses are available as a combination tablet.
- Celecoxib 400mg *bid* until disease (including PSA) progression (see Section 7.2).

6.2 Administration and Dose Modifications

6.2.1 Zoledronic Acid

Zoledronic acid will be administered by IV infusion in accordance with the instructions in the summary of product characteristics at a dose of 4mg every 3 weeks for the first 6 cycles, and thereafter every 4 weeks. There may be the opportunity for the 4-weekly infusions to be administered in the patient's home (**Appendix H**).

Serum Creatinine Measurements: Serum creatinine should be measured at baseline and within 48 hours prior to every administration of zoledronic acid.

Serum electrolytes and FBC: Serum electrolytes including calcium, phosphate and magnesium should also be measured prior to each infusion. FBC should be measured at least 3 monthly.

Zoledronic acid should be discontinued if there is any evidence of hypersensitivity to the drug. Dose reductions are not anticipated. For full details administration of zoledronic acid see **Appendix F**. Contraindications, special precautions, interactions and side effects are listed in **Appendix G**.

6.2.2 Docetaxel

The use of docetaxel should be confined to units specialised in the administration of cytotoxic chemotherapy and it should only be administered under the supervision of a physician qualified in the use of anticancer chemotherapy.

Docetaxel (Taxotere) will be administered by IV infusion in accordance with the instructions in the summary of product characteristics at a dose of 75mg/m² (up to a maximum dose of 160mg) on day 1 of the study treatment period and then every 3 weeks, thereafter, for a maximum of 6 doses. Patients with a body surface area (BSA) greater than 2.13m² should be dosed as though they have a BSA of 2.13m². No ideal weight should be used for BSA calculations. Prednisolone 5mg *bid* will be given until completion of chemotherapy. Additional dexamethasone should be given pre- and post-docetaxel infusion to suppress allergic reactions. For full details of premedication schedule, recommended anti-emetic regimen and dose modifications for docetaxel see **Appendix F**.

6.2.3 Celecoxib

Celecoxib should be administered in accordance with the instructions in the summary of product characteristics at a dose of 400mg *bid* orally. Rarely this drug is poorly tolerated and in this instance should be discontinued, particular care should be taken with patients with a history of gastrointestinal disease (see **Appendix F**). Dose reductions are not anticipated.

6.3 Trial products

Details of the procedures for obtaining the drugs within the trial, dispensing and disposal of unused drug are given in **Appendix E**.

Arrangements for free or discounted drugs are given in the Finance section (section 15).

6.4 Measures of compliance/adherence

Date of treatment, dose, delays and reasons for delays or dose modifications of all study infusions (zoledronic acid and docetaxel) will be recorded. The estimated number of celecoxib tablets taken in a given time period will also be recorded as well as any dose reductions.

6.5 Treatment data collection

Data will be recorded on case report forms (CRFs); the top copy/original should be sent to the MRC CTU for data entry and a copy kept at the local centre. The data to be recorded on these can be viewed in the CRF appendix (**Appendix O**). The type of data to be recorded is detailed in the Assessments and Procedures section (Section 7).

6.6 Non-trial treatment

6.6.1 Medications permitted

Any additional treatment that the responsible physician feels is appropriate is permitted.

6.6.2 Data on concomitant medication

All concomitant medication will be recorded on the baseline form prior to randomisation and on any subsequent Serious Adverse Event forms. This should include aspirin that may be taken on a regular basis for cardiovascular disease and the use of any Non-Steroidal Anti-inflammatory Drug (NSAID).

7 ASSESSMENTS AND PROCEDURES

7.1 Flow chart/Schedule for follow-up

A detailed follow up schedule is given in **Figure 3 (see page 21)**.

7.1.1 PSA Measurements

All patients should have PSA measured pre-treatment (within two weeks of starting treatment) and at weeks 6, 12, 18 and 24 and every 12 weeks, thereafter, up to 2 years post randomisation. Following this, PSA should be measured every 3 months for the next two years and 6 monthly, thereafter. For patients who do not have a scheduled hospital visit, it would be acceptable for arrangements to be made for blood samples to be drawn either in a GP's surgery or in the patient's home.

7.1.2 Radiological Investigations

All patients should have baseline radiological examinations as detailed in section 4.3.1. In addition all patients should have scans or X-rays repeated at 24 weeks if they were abnormal at baseline, and whenever clinically appropriate.

7.2 Efficacy parameters

The primary outcome measure in Efficacy Stage I to III is failure-free survival. This will be defined as time from randomisation to the first of four possible events:

1. Biochemical failure (see **Appendix K**)
2. New Lesions (following modified RECIST criteria see **Appendix I**)
3. Increase in baseline lesions (following modified RECIST criteria see **Appendix I**)
4. Death from any cause.

Biochemical Failure: Patients who have a rising PSA and fulfil the criteria in **Appendix K**.

Clinical Failure: Patients with new lesions or lesions that have increased in size from baseline and fulfil the criteria in **Appendix I**

7.3 Follow-up

Every effort should be made to follow-up all patients who have been randomised. Patients should, if possible, remain under the care of an oncologist or urologist for the duration of the trial. If care of a patient is returned to the GP, it is the responsibility of the consultant who obtained the patient's consent to participate in the trial to ensure that the data collection forms are completed. If the patient moves from the local area, arrangements should be made for trial follow-up to be undertaken by their new local centre. Details of other participating centres can be obtained from the MRC CTU. The consent of patients should be obtained for their names to be flagged for survival information through national registries (e.g. ONS in England/Wales and GRO in Scotland). If the clinician moves, appropriate arrangements should be made to arrange for trial follow-up to continue at the centre.

7.4 Trial closure

For the purpose of complying with UK Clinical Regulations introduced on May 2004, the trial will be considered '*closed*' when the last patient has completed protocol treatment. However, further observational follow-up of all patients enrolled in the trial will continue until all randomised patients have died. This will initially be via the hospital, but in the longer term may employ national registers.

Figure 3 - Detailed schedule for completion of forms.

Timing of Assessment	Baseline		Treatment		Outcomes		Freq
	Randomis ⁿ	Pre-Treatment	Zoledronic acid	Docetaxel	Follow-up ^ψ	QL + HE	
Yr 0 Wk 0	all	all				all	
Wk 6			BEF [†]	CE [†]	all	all	6 weekly
Wk 12			BEF [†]	CE [†]	all	all	
Wk 18			BEF [†]	CE [†]	all	all	
Wk 24			BEF [†]		all	all	
Wk 36			BEF [‡]		all	all	12 weekly
Wk 48			BEF [‡]		all	all	
Wk 60			BEF [‡]		all	all	
Wk 72			BEF [‡]		all	all	
Wk 84			BEF [‡]		all	all	
Wk 96			BEF [‡]		all	all	
Yr 2 Month 24			BEF [‡]		all	all	3 monthly
Month 27					all	all	
Month 30					all	all	
Month 33					all	all	
Yr 3 Month 36					all	all	
Month 39					all	all	
Month 42					all	all	
Month 45					all	all	
Yr 4 Month 48					all	all	6 monthly
Month 54					all	all	
Yr 5 Month 60					all	all	
6 monthly...					all	all	

Key: all=arms A-F, A=AS, B=AS + Zoledronic acid, C=AS + Docetaxel, D=AS + Celecoxib, E=AS + Zoledronic acid + Docetaxel, F= Zoledronic acid + Celecoxib

^ψ Including treatment with hormone therapy and celecoxib and toxicity assessments

[†] Form records data for two cycles

[‡] Form records data for three cycles

Progression, Death and Withdrawal forms to be completed as required

Note: an individualised form with a follow-up schedule will be provided for each randomised patient. For patients who are receiving LHRH analogues it is assumed that any additional treatment will commence within two weeks of randomisation. For patients who are due to have an orchidectomy it is recognised that surgery will have to be scheduled and the scheduling of any additional treatments may be affected by post-operative recovery. All patients who had abnormal radiological investigations at baseline should have them repeated 24 weeks after randomisation.

8 WITHDRAWAL OF PATIENTS

Patients should be given every encouragement to adhere to protocol treatment and follow-up, in order to reduce biases. However, a patient has the right to withdraw consent for participation in any aspect of this trial at any time. They may refuse to take certain treatments, attend scheduled follow-up visits, or move from the area (see section 8.3). Clear distinction must be made as to whether the patient is withdrawing from trial treatments/procedures whilst allowing further follow-up, or whether the patient refuses **any** further trial treatments/procedures **and** follow-up participation. In all instances the MRC-CTU must be informed immediately.

8.1 Withdrawal from trial interventions

A patient may withdraw, or be withdrawn, from trial treatment for the following reasons:

1. Progression whilst on therapy
2. Unacceptable toxicity
3. Intercurrent illness which prevents further treatment
4. Withdrawal of consent for treatment.
5. Any alteration in the patient's condition which justifies the discontinuation of treatment in the clinician's opinion.

The reason should be recorded on the treatment and/or follow-up forms as well as the withdrawal form.

Unless a patient states otherwise, it should be assumed that consent is given to continue to record trial data.

8.2 Withdrawal from the trial completely

If a patient explicitly withdraws consent to have **any** data recorded their decision must be respected and recorded on a withdrawal form. All communication surrounding the withdrawal should be noted in the patient's records and no further STAMPEDE CRFs should be completed for that patient.

Patients can change their minds about withdrawal at any time and re-consent to participate in the trial. Follow-up data should be collected **only** from the point of when consent was re-instated.

8.3 Patient transfers

For patients moving from the area, every effort should be made for the patient to be followed-up at another participating trial centre and for this trial centre to take over responsibility for the patient. A copy of the patient STAMPEDE CRFs will need to be provided to the new site. The patient will have to sign a new consent form at the new site, and until this occurs, the patient remains the responsibility of the original centre.

9 STATISTICAL CONSIDERATIONS

9.1 Method of Randomisation

Patients will be randomised centrally using a computerised algorithm developed and maintained by the MRC CTU. Randomisation will be performed using the method of minimisation over a number of clinically important stratification factors with an additional random element. To decrease determinability, the factors are not listed here. The trial has 1 control arm and 5 research arms. As the control arm is the comparator arm for all the research arms, it is intended to recruit twice as many patients to the control arm as to each research arm. Therefore, the randomisation ratio will be 2:1:1:1:1:1.

9.2 Outcome Measures

The overall primary outcome measure for the trial is overall survival (all cause mortality). The design of the trial is such that it is important to have additional intermediate outcome measures to assess each research arm as the trial progresses. These are listed in Table 2. The reasons for different emphases in each recruitment stage is explained in section 9.3.

Table 2: Trial Outcome Measures

Trials stage	Primary outcome measures	Secondary outcome measures
Pilot phase	Safety [*]	Feasibility
Efficacy Stage I-III	Failure-free survival (FFS) [†]	Overall survival (OS) Toxicity Skeletal related events
Efficacy Stage IV	Overall survival	Quality of life Cost effectiveness Failure-free survival [†] Toxicity Skeletal related events

^{*}Based on toxicity

[†]Including biochemical failure (see **Appendix K**)

9.3 Sample Size

9.3.1 Overall sample size

The overall design for this study is a six arm, multi-centre randomised controlled trial (see Figure 4, page 26). There are five stages to the study, a Pilot Phase, Efficacy Stages I-IV. Full details of the methodology underlying the trial design are given by Royston *et al.* (35). The sample size calculations were performed using the -stage2- program (version 1.2.0, March 2002) (36) and -stagen- program (version 1.1.1, 18 May 2004), both implemented in Stata 8

(Stata Corp, TX). The trial is designed under the assumptions in Table 3, and additionally, we assume a slightly higher proportion of non-metastatic than metastatic patients such that the median FFS is two years and median OS four years.

Table 3: Hazard ratio assumptions under null and alternative hypotheses

Size of HR	Pilot	Stage I-III	Stage IV
Under null hypothesis (H_0)	n/a	HR(FFS) = 1.0	HR(OS) = 1.0
Under alternative hypothesis (H_1)	n/a	HR(FFS) = 0.75	HR(OS) = 0.75

The HR of 0.75 for any research arm relative to control would translate into an absolute improvement in FFS of 10%, from approximately 50% to 60% at two years and OS of 10%, from approximately 50% to 60% at four years. A beneficial difference of this size would be clinically worthwhile and, indeed, experience tells us it may be unrealistic to expect a larger difference. Therefore, we have adequately powered the trial to detect a HR of 0.75 for overall survival. This design gives 95% power at Efficacy Stages I-III and 90% power at Efficacy Stage IV. Further details of the sample size calculations are given in a separate Statistical Design document which is available on request.

Assuming an accrual rate of 500 patients/year then between 2800 and 3600 patients are planned to be entered into the trial over a period of 5½ and 7 years. The exact number of patients to be entered depends on the observed accrual and event rates. The primary analysis on overall survival requires 445 deaths to be observed on the control arm.

9.3.2 Pilot Phase

It is anticipated that 210 patients will be recruited to the Pilot Phase from a limited number of centres over a one year period. Approximately 60 patients will be randomised to the control arm and 30 patients to each of the five research arms each of which will be assessed for safety and feasibility. If recruitment proves infeasible or any of the research arms prove unsafe or not feasible to administer (e.g. poorly tolerated or unexpected toxicity) recruitment to these arms will be discontinued. There are already considerable safety data on the use of docetaxel and zoledronic acid in patients with malignancies including prostate cancer, and on the use of Cox-2 inhibitors (including celecoxib), although mainly from patients with musculoskeletal disorders. There are fewer data on the combination arms; but it is thought very unlikely that any of the research arms will be discontinued during the Pilot Phase. Safety data will continue to be assessed throughout the trial. When 210 patients have been on the trial for a minimum of 18 weeks, the independent Data Monitoring Committee (IDMC) will then review the data from the Pilot phase. Recruitment will continue to the trial during this period as equipoise will remain.

9.3.3 Efficacy Stages I-III

In the sample size calculations we assume that all research arms successfully pass through the Pilot Phase to Efficacy Stage I and that patients will be recruited at a rate of approximately 500 per year. This is faster than in the Pilot Phase because the trial will recruit from additional centres, both in the UK and internationally. The analysis of Efficacy Stages I, II and III are planned with 115, 225 and 355 failure-free survival events have been observed in the control arm, respectively.

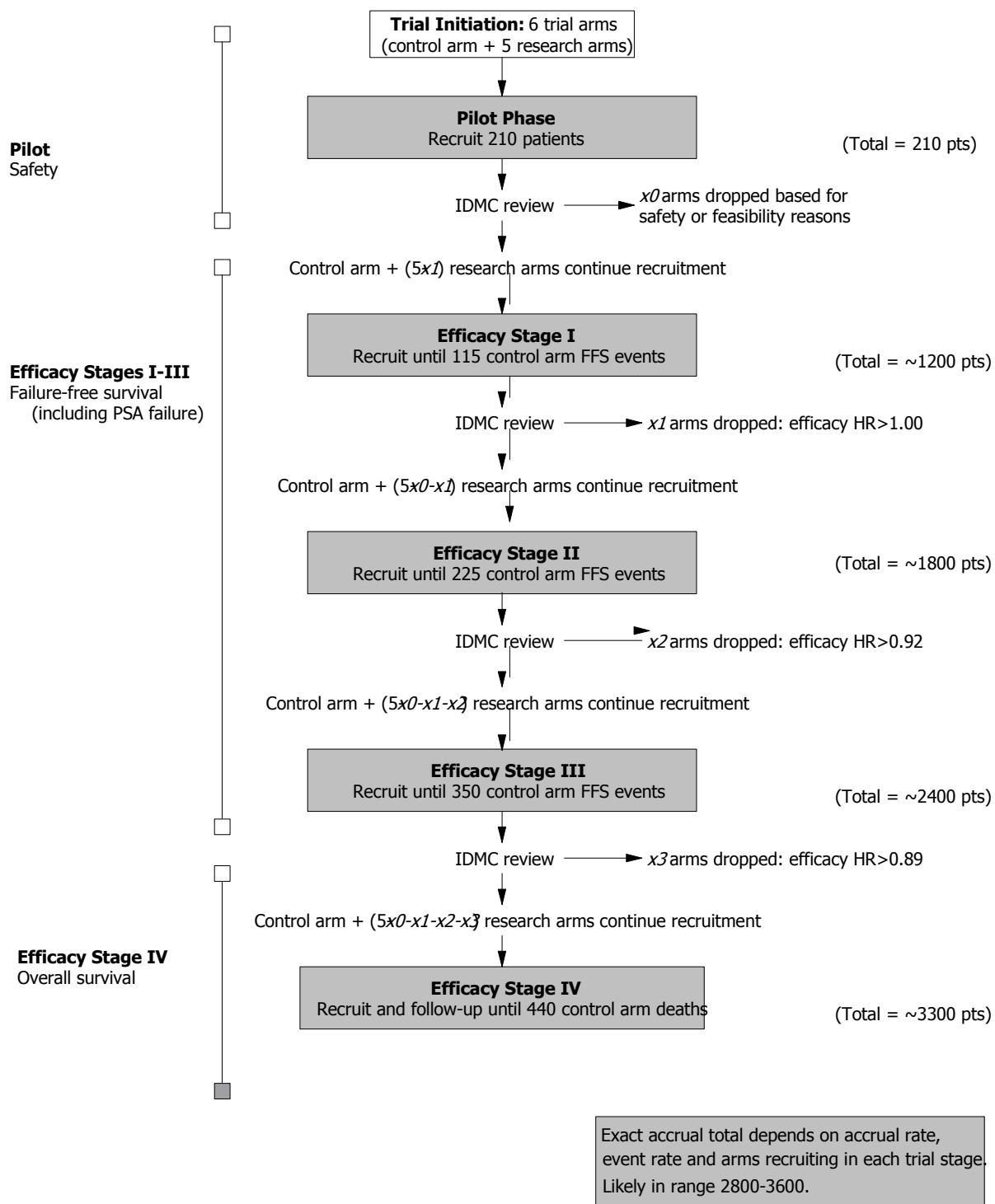
The Efficacy Stage analyses will comprise pairwise comparisons of FFS between the control arm and each of the 5 research arms ($i=1$ to 5). Let $HR_i(\text{true})$ represent the hazard ratio (HR) of the i^{th} research arm to the control arm, and $HR_i(\text{observed})$ the observed value. Discontinuation of accrual of further patients will be considered for the i^{th} research regimen at each of Efficacy Stages I-III according to the guidelines in Table 4.

Table 4: Guidelines for stopping accrual to the i^{th} research arm

Efficacy Stage	Number of control arm events	Consider discontinuation if $HR_i(\text{observed})$ is...
I	115	>1.00
II	225	>0.92
III	355	>0.89

9.3.4 Efficacy Stage IV

The analysis of Efficacy Stage IV will be performed when 445 deaths have been observed in the control arm. This would give 90% power to detect the targeted hazard ratio of 0.75 at one-sided significance level of 0.025. The actual length of this stage and the number of patients to be recruited depends on the number of arms passing through to further recruitment from Efficacy Stages I-III and the observed accrual and event rates.

Figure 4: Progress of STAMPEDE through the trial stages**Key**

IDMC = Independent Data Monitoring Committee

FFS = Failure-free survival

HR = Hazard Ratio

$$0 \leq x_0 \leq 5, 0 \leq x_1 \leq (5-x_0), 0 \leq x_2 \leq (5-x_0-x_1), 0 \leq x_3 \leq (5-x_0-x_1-x_2)$$

9.3.5 Factorial design

We note here that we have not employed a factorial design in this trial because we anticipate the possibility of synergy between hormone therapy, zoledronic acid and docetaxel and between hormone therapy, zoledronic acid and celecoxib. It would not be possible to assess any such interactions reliably in a factorial trial (see the Statistical Design document for further details).

9.4 Interim Monitoring and Analyses

Formal interim analyses of the accumulating data will be performed at regular intervals (approximately annually) for review by an Independent Data Monitoring Committee (IDMC) (see also section 16). These analyses will be performed by the trial team at the MRC CTU. The IDMC will be asked to give advice on whether the accumulating data from the trial with the guidelines for discontinuation of accrual for Efficacy Stages I-III, together with results from any other relevant trials, justifies continuing recruitment of further patients or further follow-up. A decision to discontinue recruitment, in all patients or in selected subgroups will be made only if the result is likely to convince a broad range of clinicians including those entering patients into the trial and the general clinical community. If a decision is made to continue, the IDMC will advise on the frequency of future reviews of the data on the basis of accrual and event rates. The IDMC will make recommendations to the Trial Steering Committee (TSC, see section 16) as to whether the trial should continue in its present form. While the trial is ongoing the accumulating data will remain confidential.

9.5 Outline Analysis Plan

Analyses will be performed on an intention-to-treat basis. The standard unadjusted log-rank approach will be applied to analyses of FFS and OS. The impact of potential confounders including the stratification factors used at randomisation will be considered in a Cox proportional hazard model. The χ^2 test or Mann-Whitney test will be implemented for categorical data comparisons, including toxicity, as appropriate. The primary outcome measures in Table 2 (see page 25) will be considered for all arms of the trial at each phase, but the main emphasis will be placed on the comparison of the research arms that have continued to recruit throughout the trial.

9.5.1 Pilot Phase

The Pilot Phase randomises patients between all the study arms so that the results from these patients can be included in the main trial. Feasibility is considered in terms of the acceptability of the trial randomisation and reported toxicities and adherence to trial medication. Centres participating in the Pilot Phase will be required to keep an anonymised log of all patients

assessed for trial eligibility. We will then summarise the number of patients who do not participate in the study, and also the number of eligible patients who choose to not participate in the study (reasons for non-participation will be collected where the patients are willing). On the patients who are randomised, we shall describe the incidence of expected and unexpected severe toxicities and adverse events/reactions (see Section 11) to decide whether to continue with research arms beyond the Pilot Phase. As indicated above, we do not anticipate that recruitment to the research arms will be discontinued after the Pilot Phase, as there is considerable experience with zoledronic acid and docetaxel when combined with AS, while Cox-2 inhibitors generally have a good toxicity profile. Although there are limited data on the combinations, we do not expect severe toxicity.

9.5.2 Efficacy Stages

The approach to analysis of these stages is summarised within the sample size calculations (see Section 9.3.3). Each research arm will be compared in a pairwise fashion against the control arm.

10 DATA VERIFICATION AND SITE MONITORING

10.1 Verification of data at MRC CTU

Data stored at the MRC CTU will be checked for missing or unusual values (range checks) and checked for consistency over time. If any such problems are identified, a photocopy of the problematic CRF(s) will be returned to the local site by post or fax for correction. The incorrect data should be crossed through with a single line in black biro. The correct data should be written next to the amendment in black biro and initialled and dated by the investigator, in accordance with GCP. The amended version should be returned to MRC CTU and the site's copy should also be amended. MRC CTU will send reminders for any overdue data.

10.2 Clinical site monitoring

10.2.1 Direct Access to Data

Participating investigators should agree to allow trial-related monitoring, including audits, ethics committee review and regulatory inspections by providing direct access to source data/documents as required. Patients' consent for this must be obtained.

10.2.2 Quality Assurance and Quality Control of Data

A sample of 10% of patients will be selected for source data verification throughout the trial. This may include random sampling, but may also include patients who might exhibit a reason for the data to be checked in this way.

11 ADVERSE EVENTS

11.1 Adverse events and grading of adverse events

Adverse events will be graded using the Common Toxicity Criteria (CTC) version 3.0 given in **Appendix J** and recorded on the CTC toxicity part of the treatment form (see **Appendix O**). All patients experiencing a SAE or an adverse event should be followed up as outlined in the protocol. For patients who decide to withdraw from the trial treatment after an adverse event the procedure outlined in Section 8 should be followed.

Previously reported (and, therefore, expected) reactions from zoledronic acid, docetaxel and celecoxib are listed in **Appendix G**.

11.2 Serious adverse events

A Serious Adverse Event (SAE) is defined as an event that meets at least one of these criteria: -

- (i) Fatal or life threatening (excluding haematological CTC grade 4)
- (ii) Requires or prolongs hospitalisation
- (iii) Results in a persistent or significant disability/incapacity
- (iv) Is a congenital abnormality/birth defect
- (v) Any other important medical condition not included above that may jeopardise the patient and require medical or surgical intervention to prevent one of the above. (Examples include allergic bronchospasm requiring intensive emergency treatment, seizures or blood dyscrasias which did not result in hospitalisation)

Death as a result of disease progression and other events that are primary or secondary outcome measures (see Section 9.2) are not considered to be SAEs and should be reported in the normal way, on follow-up forms (see **Appendix O**).

All SAEs, **apart from haematological events**, should be reported immediately (within 24 hours or the next working day) to the MRC CTU by fax, using the SAE form, to +44 (0)207 670 4818. All SAEs will be directed to the Chief Investigator and will be reviewed to determine whether the adverse event is expected or not. All unexpected, possibly treatment-related serious adverse events (SUSARs) will be reported to the appropriate National Regulatory

Authority (e.g. MHRA in the UK) within specified timelines and forwarded to the relevant pharmaceutical company and MREC, as agreed.

If death occurs within 4 weeks of treatment, consideration must be made, by the responsible consultant, as to whether the treatment caused or contributed to the death. Any death which is suspected of being related to trial treatment must be notified to the MRC CTU within 24 hours or the next working day, using a SAE form.

12 ETHICAL CONSIDERATIONS AND APPROVAL

12.1 Ethical considerations

This is a randomised trial therefore neither the patients nor their physicians will be able to choose the patients' treatment. Treatment will be allocated randomly using a computer-based algorithm. This is to ensure that the groups of patients receiving each of the different treatments are similar.

Five sevenths of the patients will receive one or two of the newer treatments in combination with hormone treatment; hormone treatment alone is the standard treatment for these forms of prostate cancer. These newer combined treatment options are being assessed in a detailed and systematic fashion in this trial. There is some evidence to suggest that the newer treatment options *may* have advantages over standard treatment (hormone therapy) alone with regards clinical outcome, but this is not confirmed and toxicity may be increased. This trial will follow a large group of men who have been randomly allocated to either the standard treatment (hormone therapy alone) or the newer combined treatment options in order to measure the benefits of the new treatments. The patients will also be followed-up for toxicity and safety issues, so that any benefits can be weighed against any negative aspects.

Patients participating in the trial will have some additional hospital visits and some extra blood samples taken compared to patients who are not participating in the trial. Sometimes the blood samples can be taken when the patient is attending hospital for treatment, anyway. On some of the trial arms, the patient may have to make additional visits to the hospital for the blood sample to be taken, although in some cases it may be possible for the blood sample to be taken in the GP's surgery or in the patients home. The additional visits and blood samples are to ensure that follow-up of patients is comparable in all the treatment groups.

12.2 Ethical approval

The protocol has a Favourable Opinion from a Main Research Ethics Committee (MREC), but each site must also gain a favourable Site Specific Assessment (SSA) before patients can be entered

into the trial. The patient's informed consent to participate in the trial should be obtained after a full explanation has been given of the treatment options, including the conventional and generally accepted methods of treatment. Suggested patient information sheets and patient consent forms are given in **Appendix B**.

The right of the patient to refuse to participate in the trial without giving reasons must be respected. After the patient has entered the trial, the clinician must remain free to give alternative treatment to that specified in the protocol, at any stage, if he/she feels it to be in the best interest of the patient. However, the reason for doing so should be recorded and the patient will remain within the trial for the purpose of follow-up and data analysis according to the treatment option to which he/she has been allocated. Similarly, the patient must remain free to withdraw at any time from the protocol treatment without giving reasons and without prejudicing his/her further treatment.

A statement of MRC policy on ethical considerations in clinical trials of cancer therapy, including the question of informed consent, is available from the MRC Head Office web site (<http://www.mrc.ac.uk>).

13 REGULATORY APPROVAL

This trial has been approved by the MHRA and will be conducted under a CTA in the UK.

14 INDEMNITY

The MRC and NHS are both publicly funded bodies and are not allowed to purchase advance insurance to cover indemnity because they are backed by the resources of the Treasury.

The MRC will give sympathetic consideration to claims for non-negligent harm suffered by a person as a result of trial or other work supported by MRC. This does not extend to liability for non-negligent harm arising from conventional treatment where this is one arm of a trial. The MRC acts as its own insurer and does not provide cover for non-negligent harm in advance for participants in MRC-funded studies.

Where studies are carried out in a hospital, the hospital continues to have a duty of care to a patient being treated within the hospital, whether or not the patient is participating in an MRC-supported study. MRC does not accept liability for any breach in the hospital's duty of care, or any negligence on the part of employees of hospitals. This applies whether the hospital is a NHS Trust or not.

15 FINANCE

STAMPEDE is funded by the Clinical Trials Advisory Awards Committee (CTAAC) (on behalf of Cancer Research UK, Medical Research Council, and other charities). The trial has National Cancer Research Network (NCRN) approval and, therefore, local NCRN funds may be available at each centre to support entry of patients into this trial.

Zoledronic acid is manufactured by Novartis. Novartis have agreed to provide an educational grant to support the conduct of this study. Novartis have also agreed to supply the study drug, zoledronic acid free of charge for patients participating in the study.

Docetaxel is manufactured by Aventis Pharma. They have agreed to supply the study drug, docetaxel at a discounted rate for patients that are participating in the trial and to provide an educational grant to support the conduct of the study.

Celecoxib is manufactured by Pfizer. They have agreed to supply free drug and provide funds to distribute drug to participating sites.

16 TRIAL COMMITTEES

16.1 Trial Management Group (TMG)

A Trial Management Group (TMG) has been formed comprising the Chief Investigator, other co-investigators and members of the MRC CTU. The membership of the TMG may be expanded if other groups of trialists wish to participate. The TMG will be responsible for the day-to-day running and management of the trial and will meet by teleconference at least 3 monthly and in person as needed. The TMG members are detailed in **Appendix L**.

16.2 Trial Steering Committee (TSC)

A Trial Steering Committee (TSC) will be formed to provide overall supervision for the trial and provide advice through its independent chair. The ultimate decision for the continuation of the trial lies with the TSC. The TSC will meet twice a year.

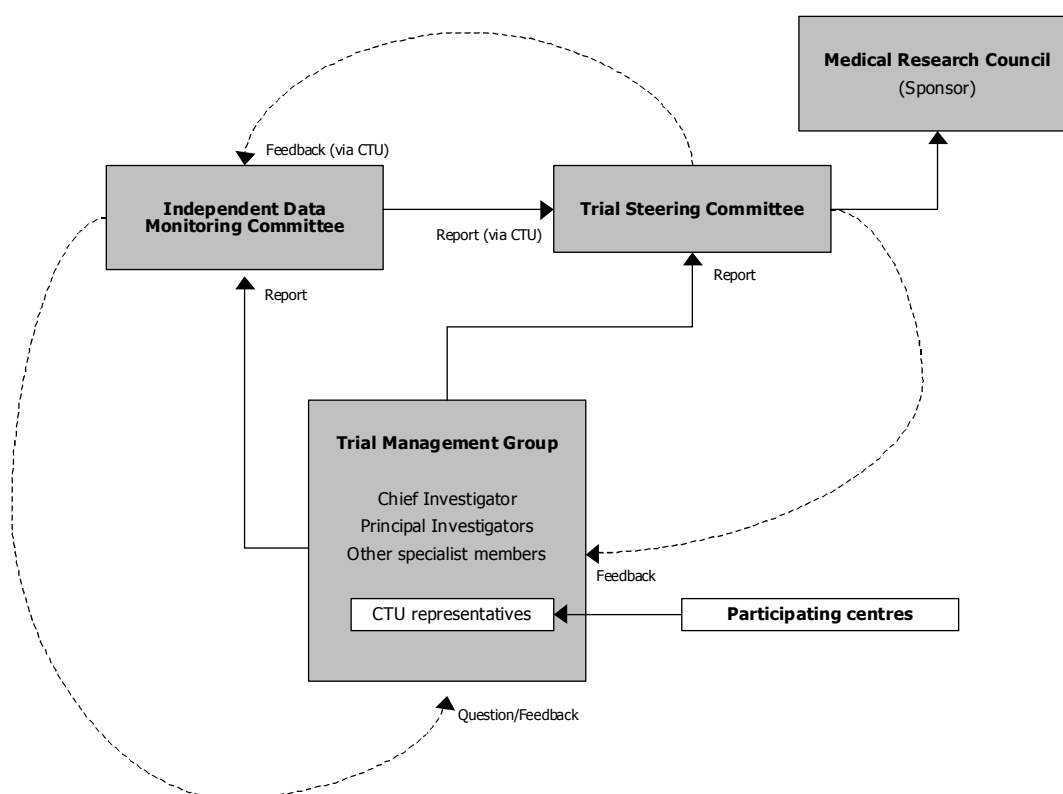
16.3 Independent Data Monitoring Committee (IDMC)

An Independent Data Monitoring Committee (IDMC) will be formed. The IDMC will be the only group who sees the confidential, accumulating data to the trial. Reports to the IDMC will be produced by the MRC CTU. The IDMC will meet within 6 months of the trial opening with the

frequency of meetings dictated by the IDMC. The IDMC will consider data in accordance with the analysis plan (see section 9.5) and will be advisory to the TSC. The IDMC can recommend premature closure or reporting of the trial, or that recruitment to any research arm be discontinued.

Further details of IDMC functioning, and the procedures for interim analysis and monitoring are provided in the IDMC charter (available on request).

Figure 5 - Diagram of relationships between trial committees



17 ANCILLARY STUDIES

17.1 Quality of Life

A quality of life (QL) study is being performed to assess the impact of each treatment arm on the quality of patient's lives and all patients should be asked if they are willing to participate. The EORTC QLQ-C30 with the prostate-specific module QLQ PR25 will be used. Key items for assessment are pain reduction for patients with metastatic disease and urinary symptoms for patients with locally advanced disease. In addition specific hypotheses will be generated for each of the research arms. The EuroQol (EQ-5D) (37) will be used in the study as a generic measure of health-related quality of life which can be linked to public preferences. These data will be used to calculate quality-adjusted life-years as part of the economic evaluation (see Section 17.2).

A patient information sheet about the QL study is given in **Appendix B** and the QL questionnaires are given in **Appendix N**. The first questionnaires should be administered within three days prior to, or on the day of randomisation, but in either case before the patient is informed which trial treatment he has been assigned. The subsequent questionnaires will be administered during the treatment and during the follow-up period. Questionnaires should be self-administered, although it is recommended that a key person (e.g. research nurse) at each centre be responsible for the data collection to optimise compliance and completeness of the data.

The QL and the HE questionnaires should be completed without conferring with friends or relatives and all questions should be answered even if the patient feels them to be irrelevant.

The responsible person should check each questionnaire for its completeness, ensuring that the correct date of completion and patient identifiers are present. The research nurse should approach patients at appropriate clinical visits to complete a questionnaire. If no clinical visit is scheduled for the patient (with a window of 4 weeks around the expected date) the nurse should organise the completion of the questionnaire, by post or by a visit to the patient at home (or in a hospice).

17.2 Health Economics

A health economics (HE) sub-study will be performed. Core resource use information will be collected, using CRFs on days in hospital (by speciality) and outpatient visits. Data being collected on concomitant medication will also be used in the economic analysis. Information on patients' use of primary care and community-based services will be collected as additional questions in the QL questionnaire. Costs will be calculated on the basis of representative UK unit

costs at the point of analysis. Health outcomes will be assessed in terms of quality-adjusted life years (QALYs). Quality adjustments will be based on patients' responses to the EQ-5D health status measure which will be administered at baseline and each point of follow-up as part of the QL questionnaire (**Appendix N**). A cost-effectiveness analysis will compare all regimens in Efficacy Stage II.

17.3 Molecular genetics

A molecular genetics sub-study is being performed. With the patient's consent, an additional 8ml blood sample will be collected and stored for either DNA and protein analysis in order to try to identify genes that control response to therapy and its side effects, and susceptibility to prostate cancer.

Blood samples should be sent directly to the central laboratory in the packs provided. Patient information sheets and consent forms **Appendix B**, while details of specimen collection, posting and contact details are given in **Appendix D**.

18 PUBLICATION

The results from different centres will be analysed together and published as soon as possible. Individual clinicians must not publish data concerning their patients that are directly relevant to questions posed by the study until the TMG has published its report. The TMG together with the STAMPEDE collaborators will form the basis of the writing committee and decide on the nature of publications. The main publication will be in the name of the STAMPEDE Trial Collaborators and the writing committee will be listed at the end of the manuscript. All publications will acknowledge the participating centres and clinicians, and these will be detailed in an appendix to the main report.

19 PROTOCOL AMENDMENTS

None - this is version 1.0 (May 2004).

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21 APPENDICES

Appendix A – Additional Definitions

A1: WHO Performance Status

Grade Performance status

- | | |
|---|--|
| 0 | Able to carry out all normal activity without restriction |
| 1 | Restricted in physically strenuous activity but ambulatory and able to carry out light work |
| 2 | Ambulatory and capable of all self-care but unable to carry out any work; up and about more than 50% of waking hours |
| 3 | Capable of only limited self-care; confined to bed or chair more than 50% of waking hours |
| 4 | Completely disabled; cannot carry on any self-care; totally confined to bed or chair |

A2: TNM Classification*

Primary Tumour

- | | |
|----|---|
| Tx | Primary tumour cannot be assessed |
| T0 | No evidence of primary tumour |
| T1 | Clinically upapparent tumour not palpable or visible by imaging <ul style="list-style-type: none">- T1a: Tumour incidental, found in 5% or less resected tissue- T1b: Tumour incidental, found in more than 5% resected tissue- T1c: Tumour identified by needle biopsy (because of high PSA) |
| T2 | Tumour confined within the prostate gland <ul style="list-style-type: none">- T2a: Tumour involves one lobe- T2b: Tumour involves both lobes |
| T3 | Tumour extends through the prostatic capsule <ul style="list-style-type: none">- T3a: Extracapsular extensions (unilateral or bilateral)- T3b: Tumour invades seminal vesicles |
| T4 | Tumour is fixed or invades adjacent structures other than seminal vesicles: bladder neck, external sphincter, rectum, levator muscles and/or pelvic wall |

Regional Lymph Nodes

- | | |
|----|---|
| Nx | Regional lymph nodes have not been assessed |
| N0 | No regional lymph node metastasis |
| N1 | Regional lymph node metastasis |

Distant metastasis

- | | |
|----|---|
| Mx | Distant metastasis has not been assessed |
| M0 | No distant metastasis |
| M1 | Distant metastasis <ul style="list-style-type: none">- M1a: Non-regional lymph nodes- M1b: Bone- M1c: Other sites |

*TNM staging of Prostate Cancer. (1997 TNM Classification of malignant tumours. Fifth edition. Wiley-Liss)

Appendix B - Patient Information Sheets & Consent Form

Guidance for Administrators

As STAMPEDE is a relatively complex trial with six different treatment arms, each of which requires a certain amount of specific patient information, the patient information sheets have been organised into two sections.

1. General Patient Information Sheet.

This is intended for use prior to randomisation. It describes the reason for the research, the design of the trial and a brief outline of the treatments that the patient may receive.

2. Treatment Arm-Specific Patient Information Sheets

These should usually be given to the patient after randomisation and before treatment starts. Some patients may wish to receive all of these before making a decision about randomisation; this is acceptable, but is not mandatory. They include more detailed information about possible unwanted effects and safety instructions. These sheets also cover important general issues such as withdrawing from the research, indemnity and confidentiality.

There is also a separate information sheet entitled **Additional Research** that covers quality of life, health economics and the molecular genetics sub-study that should be given to all patients prior to randomisation.

These forms follow:

- STAMPEDE patient information sheet – general
- STAMPEDE additional research
- STAMPEDE patient consent form
- STAMPEDE patient information sheet – treatment group A
- STAMPEDE patient information sheet – treatment group B
- STAMPEDE patient information sheet – treatment group C
- STAMPEDE patient information sheet – treatment group D
- STAMPEDE patient information sheet – treatment group E
- STAMPEDE patient information sheet – treatment group F

STAMPEDE: PATIENT INFORMATION SHEET-GENERAL

(to be printed on local hospital headed paper)

Version 1.0 (Aug 2004)

A LARGE PRINT VERSION IS AVAILABLE ON REQUEST

STAMPEDE: Systemic Therapy in Advancing or Metastatic Prostate Cancer - Evaluation of Drug Efficacy.

General Information

Your doctor has explained to you that you have prostate cancer and has invited you to participate in this clinical trial. Before you decide, it is important for you to understand why the research is being done and what it will involve. Participation is entirely voluntary. If you decide not to take part, your decision will be accepted without question, and your subsequent treatment will not be affected in any way.

Why is this research being done?

Prostate cancer often depends upon the male hormone (testosterone) to grow, and one of the main ways of treating prostate cancer is to lower the level of male hormones in the body. This is called "hormone treatment". It is usually successful at first: further growth of the cancer is prevented and it may cause the cancer to shrink. However, after a period of time the cancer can and often does begin to grow again. This trial aims to investigate whether combining newer drugs with hormone treatment can lengthen the time before the cancer starts to grow again.

There are three different, newer drugs that have shown promising activity in prostate cancer that we wish to study. These are given in Table 1 below.

Table 1: The newer drugs in this study

Zoledronic acid (Zometa®) - Prostate cancer cells can spread to bones and weaken them. Zoledronic acid reduces bone destruction and hardens them; this may make them more resistant to attack by cancer cells.

Docetaxel (Taxotere®) - This is a drug that stops cells dividing and multiplying. It is currently used to treat lung, breast and ovarian cancer. Previous studies have shown that it may also be effective in prostate cancer.

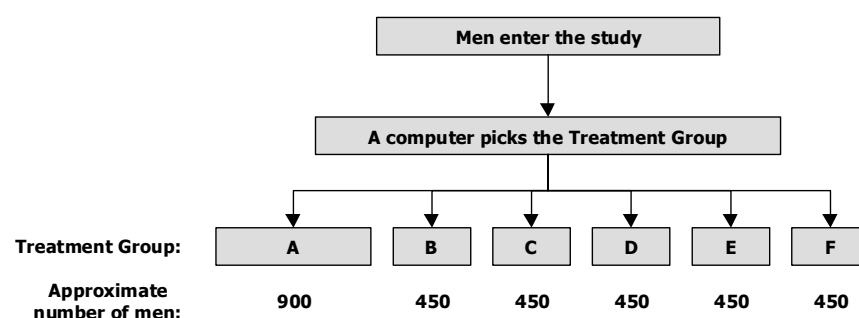
Celecoxib (Celebrex®) - This is an aspirin-like drug that is used to treat arthritis. It slows down the growth of cancer cells in laboratory tests, and is being tested for use in preventing bowel cancers in patients. We want to see if it has the same effect on prostate cancer.

This trial is called STAMPEDE and will look at the effect of combining one or two of the newer drug treatments described above with hormone treatment. In addition to studying the effect of the treatment on the growth of your cancer, we will also be asking how it has affected your quality of life, and will be aiming to find out more about how prostate cancer develops and grows. This knowledge will help us to better understand the disease and help treat patients like you in the future. Details of these additional studies are covered in the information sheet entitled "Additional Research" that you will have been given.

How is the research done?

The best way of determining whether one treatment is more effective than another is by carrying out a randomised controlled trial. This means that a computer will allocate which treatment you receive randomly and you will have a fair and equal chance of receiving any of the new treatments being studied. Neither you nor your doctor can choose the treatment and you should therefore be willing to accept whichever treatment you are allocated. We are aiming to include approximately 3300 men like you in this study (Figure 1). They will be treated at hospitals all over the UK and it is expected that it will take between 6 and 10 years to complete the study.

Figure 1: Entering the study



NB: A computer is used to allocate each of the treatments groups - A, B, C, D, E or F; these treatments are summarised in table 2. Overall twice as many men (2 out of 7) will be allocated to treatment group 'A' when compared to each of the other five treatment groups (5 out of 7: 1 per treatment group).

Which treatment might I receive if I take part?

All men taking part in the trial will receive hormone treatment which is standard treatment for your type of prostate cancer. The men who receive hormone treatment on its own, we have called Treatment Group A. The other men will receive hormone treatment and one or two of the newer drugs, and we have called these Treatment Groups B, C, D, E and F. Details of the treatments are given in Table 2 (below). Overall, two out of every seven men in the trial can expect to receive hormone treatment only. The other five out of seven men will be allocated equally between the five newer research treatments.

There are separate information sheets for each treatment group that give more details about hormone therapy and the newer drugs. If you choose to enter the trial you will be given the sheet for the treatment group you are allocated, but you are welcome to read all of these at any time.

Table 2: The Different Treatment Groups in the STAMPEDE Trial.

A. Hormone Treatment Only

If you are in Treatment Group A, you will receive hormone treatment only. This is the standard treatment for patients with your type of prostate cancer. Your study doctor will discuss the different types of hormone treatment and together you can decide on the best form of hormone treatment for you. The options within the trial are:

1. "Anti-hormone" injections: these are given once a month or once every three months.
2. An operation where all or the functioning parts of both testicles are removed. This is called a bilateral orchidectomy.

B. Hormone Treatment and Zoledronic Acid

If you are in Treatment Group B, you will receive hormone treatment as described in Treatment Group A and zoledronic acid. Zoledronic acid will be given as an injection into the vein once every three weeks, for 6 injections and then once every four weeks for up to two years.

C. Hormone Treatment and Docetaxel

If you are in Treatment Group C, you will receive hormone treatment as described in Treatment Group A and docetaxel. Docetaxel will be given as an injection into a vein once every three weeks, for a total of 6 injections. You will also be given steroid tablets to take while you are on docetaxel.

D. Hormone Treatment and Celecoxib

If you are in Treatment Group D, you will receive hormone treatment as described in Treatment Group A and celecoxib. Celecoxib will be given as tablets. You will need to take these twice a day indefinitely, but may stop them if your cancer shows signs of getting worse.

E. Hormone Treatment and Zoledronic Acid and Docetaxel

If you are in Treatment Group E, you will receive hormone treatment as described in Treatment Group A and both zoledronic acid and docetaxel. Zoledronic acid and docetaxel will be given as separate injections into a vein once every three weeks for the first 6 injections. Then only zoledronic acid will continue to be given, once every four weeks for up to two years. You will also be given steroid tablets to take while you are on docetaxel.

F. Hormone Treatment and Zoledronic Acid and Celecoxib

If you are in Treatment Group F, you will receive hormone treatment as described in Treatment Group A and both zoledronic acid and celecoxib. Zoledronic acid will be given as an injection into the vein once every three weeks for the first 6 injections and then every four weeks for up to two years. Celecoxib will be given as tablets. You will need to take these twice a day indefinitely, but may stop them if your cancer shows signs of getting worse.

What are the side-effects ?

The standard therapy, hormone treatment alone, lowers the level of the male hormone testosterone in the body. It may cause impotence, decreased sex drive (loss of libido), hot flushes, and occasionally may cause a small amount of swelling of breast tissue. Table 3 outlines the most common side-effects that might occur with the newer treatments, however most patients will not experience all of these.

Table 3: Possible side-effects from the newer treatments

Zoledronic acid - flu-like symptoms, e.g. raised temperature and body aches.

Docetaxel - hair loss, allergic reactions, fluid retention, sickness, anaemia. Also it may lower the level of other blood cells making you more susceptible to bruising, bleeding and infections.

Celecoxib - stomach discomfort, sinusitis (painful sinuses) and insomnia.

Do I have to take part?

Your participation in the STAMPEDE trial is entirely voluntary. If you decide not to take part, your decision will be accepted without question, and your subsequent treatment will not be affected in any way. The standard treatment for your type of prostate cancer is hormone treatment. All men taking part in the trial will receive hormone treatment. If you agree to take part you are free to withdraw from the trial treatment without giving your reasons. To ensure full co-operation between the hospital and the surgery we would like to inform your family doctor about the treatment and the details of the trial. If you do not want your GP to be informed about your treatment or involvement in the study please discuss this with your study doctor.

What will happen to me if I take part?

Whether you receive your treatment in hospital or at home may depend on the Treatment Group you are in. If you are to receive docetaxel, it will be given to you in a specialised ward or unit **<insert where this would be>**. If you are to receive zoledronic acid you will also need to make extra visits to your hospital for treatment though it may be possible for some of the treatments to be administered at home. Celecoxib tablets can be taken at home. If you agree to take part in the trial there will be some additional blood tests to measure the progress of your cancer (approximately 6 in the first 2 years). If you are to receive zoledronic acid and / or docetaxel (Treatment Groups B, C, E or F) there will be some more blood tests to monitor the safety of your treatment. Your study doctor will see you at least every 3 months for 3 years, then at least every 6 months as part of this trial.

What are the possible advantages of taking part in the trial?

There are a number of newer treatments that have shown early promise against prostate cancer as described above. Although the study treatments have shown promise in treating prostate cancer we don't know whether these treatments in combination with hormone treatment will be better than hormone treatment alone. Therefore your participation may not be of direct benefit to you, however it will help answer this question and help us to improve treatment for men like you in the future.

What are the possible disadvantages?

In this trial you may be asked to take additional treatment(s) as well as the hormone treatment. We do not expect there to be any disadvantage to you in terms of how your cancer responds but the additional treatments will probably have some additional side-effects (Table 3). There will also be some extra hospital visits and blood tests, as described above.

What will happen to the results of the research study?

The results will be of interest to a variety of health professionals including consultants, nurses and other healthcare professionals. Our findings will be disseminated through conference presentations and journal articles and through links already existing between the MRC and hospital sites. Your study doctor will be informed of the results and he/she can discuss the results with you and a copy of the results will be freely available to you on request. No patients will be identified in such publications.

Will my taking part be kept confidential?

If you do choose to participate we should like to tell your GP, but only with your permission. Relevant medical information about you will be sent regularly to MRC Clinical Trials Unit who are running the trial. It will be treated confidentially at all times and kept in a secure area, in a manner compliant with the Data Protection Act. Scientific and medical employees of the MRC Clinical Trials Unit and other staff involved in the trial may need to examine your medical records, but your confidentiality will be maintained. We would also like to register your name with the Office of National Statistics (ONS) if you were born or live in England or Wales, or with the General Register Office (GRO) if you were born or live in Scotland; this would allow us to check on your health status after the trial has closed, even if you are no longer seeing your hospital doctor. Again full confidentiality would be maintained.

Further Information

Whichever treatment(s) you receive, the aim is to control your cancer and help you feel well for as long as possible. If your cancer starts to grow despite the treatment or any new information becomes available or you would be better helped by an alternative treatment, your study doctor will discuss this with you. You can withdraw from the trial at anytime and it will not affect your subsequent care in any way. Similarly, your care will not be affected if you do not choose to participate.

More information about clinical trials can be found on the Cancer Research UK's patient website, Cancer Help UK (www.cancerhelp.org.uk). CancerBACUP also provide information on all aspects of cancer care including clinical trials (freephone 0808 800 1234 or www.cancerbacup.org). Consumers for Ethics in Research (CERES) publish a leaflet entitled 'Medical Research and You'. This leaflet gives more information about medical research and

looks at some questions potential recruits may want to ask. You may obtain copies from CERES, PO Box 1365, London N16 0BW.

If you have private medical insurance, you may wish to check with your company before agreeing to take part in the trial to ensure that your participation in the trial will not affect your insurance cover.

Please ask any questions you may have; take time to discuss the trial with your family, friends and family doctor. If you agree to take part in the trial, you will be asked to sign a consent form.

For more information, please contact ...

<Please insert Principal Investors Names and Address>

STAMPEDE ADDITIONAL RESEARCH

(to be printed on local hospital headed paper)

Version 1.0 (May 2004)

• QUALITY OF LIFE + HEALTH ECONOMICS STUDY - PATIENT INFORMATION SHEET**About your questionnaires**

As well as looking at the effect different treatments have on your cancer we think it is equally important to find out how you feel when you are having your treatment and after your treatment has finished. Therefore, we are asking you to complete some questionnaires that are designed to find out more about your physical and emotional feelings while you are on the trial.

These questionnaires ask you how you have been feeling during the past week and are designed to assess your day-to-day well-being, as well as any side-effects you may be experiencing. We will also ask if you have required any additional medications, or visits to your hospital or GP. Your questionnaires will be sent directly to the MRC Clinical Trials Unit where they will be treated in confidence and analysed together with questionnaires from patients in other hospitals. Your study doctor should not see the questionnaires once you have filled them in. This is to ensure that you are not influenced by what you think your study doctor may think about your responses. However, it is important that you report any concerns to your study doctor or research nurse.

We ask about a wide range of symptoms as the questionnaires are designed for use in many different areas of research, and may also be used to evaluate the impact of the newer treatments on the cost of health care in the UK.

Completing the questionnaires

If possible, you should complete the questionnaires on your own. Please make sure the correct date is written at the top of the questionnaire before you start. Try to answer all the questions but do not spend too much time thinking about each answer, as your first response is likely to be more accurate. If a question is not applicable to you, please write "not applicable" or "N/A" instead, but please do not leave any answers blank.

When you attend hospital for the first time, you will be asked to complete a questionnaire. We would like you to complete further questionnaires when you come to the hospital for an assessment at 6, 12 18 and 24 weeks after you enter the trial. Then every 3 months for the next 3 years followed by every 6 months thereafter. If you are not given a questionnaire to complete, please remind your doctor. You can, of course, decline to complete a questionnaire at any time without affecting your relationship with your doctor or your participation in the trial.

• MOLECULAR GENETICS SUB-STUDY

We would like to ask you to donate a small sample of your blood (8ml) which will be stored for future use at University College London. It is particularly useful for researchers to have access to blood samples from patients who have your type of cancer. These samples will be used to research the causes of prostate cancer and the way in which prostate cancers responds to different treatments.

The donation of a sample is entirely voluntary and you are free to withhold permission without it affecting your participation in the STAMPEDE TRIAL or your relationship with your doctor in any way.

The research may involve the analysis of components of your blood sample such as DNA or proteins. We would like to see whether certain genes are associated either with certain side effects or with beneficial results from the drugs we are using in this study. This research may help to predict which patients will benefit most from future treatments. This part of the research is unlikely to benefit you, but may help to improve treatments and be of benefit to cancer sufferers in the future.

All such work is anonymous as far as possible. Your blood sample will be coded by a number and your name will not be used. It is not intended to link the results of any genetic analysis back to you or your relatives. Neither you nor your relatives will be identified or contacted in connection with this research. Your personal information will be stored confidentially and will only be accessible to staff who have a duty of confidentiality to you as the donor of the blood sample, in accordance with the Data Protection Act.

The Medical Research Council (MRC) will specify the arrangements for access to and control of the use of the blood samples. Access to the blood samples by the commercial sector or other academic researchers is not excluded, but no one commercial company will be given exclusive rights of access. This blood sample should be considered as a gift and donors would not be entitled to share in (the unlikely event of) any financial profits from this research.

All research projects involving the stored blood samples will be approved by an independent ethics committee. If we find anything important during our research on prostate cancer, it will be published in a well-recognised scientific journal so it will be available to the whole medical and scientific community. Your personal details will not be identified in any way in any publication.

If you have any further questions please contact: Professor John Masters

3rd Floor research Laboratories
67 Riding House Street
London, W1W 7EJ
Tel: 020 7679 9543
Fax: 020 7679 9366
E-mail: j.masters@ucl.ac.uk

STAMPEDE: Patient consent form

Version 1.0 (Aug 2004) (to be printed on local hospital headed paper)

STAMPEDE: Systemic Therapy in Advancing or Metastatic Prostate cancer - Evaluation of Drug Efficacy.*Please initial the appropriate boxes:*

Agree	Disagree
-------	----------

- | | | |
|---|--------------------------|--------------------------|
| 1. I have read the patient information sheet on this clinical trial and have been given a copy to keep. I have had the opportunity to ask questions about the project and understand why the research is being done and have been told about possible side effects. | ☐ | ☐ |
| 2. I agree to participate in the Quality of Life/Health Economics study and to complete these questionnaires. <i>(This is optional)</i> | ☐ | ☐ |
| 3. I have read the patient information sheet relating to the Molecular Genetics Study and agree to donating 8ml blood which will be stored and may be used for research in the future. I understand this is a gift. <i>(This is optional)</i> | ☐ | ☐ |
| 4. I understand that sections of any of my medical notes may be looked at by responsible individuals from the Medical Research Council (MRC) Clinical Trials Unit (CTU) or from regulatory authorities where it is relevant to my taking part in research. I give permission for these individuals to have access to my records, but understand that my confidentiality will be maintained. | ☐ | ☐ |
| 5. I understand that I may not benefit directly by participating in this study but that the research may help men like me in the future. | ☐ | ☐ |
| 6. I understand that information about me will be held by the MRC CTU and that the unit is registered under the Data Protection Act to hold such information, and that my GP will be informed. | ☐ | ☐ |
| 7. I agree that my data can be stored and used for this clinical trial. | ☐ | ☐ |
| 8. I understand that my participation in all aspects of this trial is voluntary and that I am free to withdraw from the trial at any time, without giving any reason and without my medical care or legal rights being affected. | ☐ | ☐ |
| 9. I agree that I can be flagged with the Office for National Statistics for the purposes of long-term follow-up. <i>(This is optional)</i> | ☐ | ☐ |
| 10. I agree to take part in the above trial. | ☐ | ☐ |
| 11. I agree that my GP can be notified of my participation in the STAMPEDE trial. | ☐ | ☐ |

Name of patient (BLOCK CAPITALS)

Date (dd/mm/yyyy)

Signature

Name of researcher (BLOCK CAPITALS)

Date (dd/mm/yyyy)

Signature

Please sign 3 copies: 1 copy to be kept by the patient, 1 copy to be kept with hospital notes, 1 copy kept in the local investigator's file

STAMPEDE PATIENT INFORMATION SHEET – TREATMENT GROUP A

(to be printed on local hospital headed paper)

Version 1.0 (May 2004)

Information for patients receiving Hormone Treatment

You are going to receive hormone treatment alone. Details of the treatment are given below.

Hormone Treatment

Prostate cancers often depend upon the male hormone testosterone to grow. Reducing the amount of testosterone in the body usually prevents further growth of the cancer and may cause it to shrink. This is called hormone treatment and can be achieved either by the use of anti-hormone injections or an operation to remove part or all of both testicles, which produce the male hormone. Your doctor will discuss these different options with you and together you can decide which is the best form of hormone treatment for you. All forms of hormone treatment can cause the following side-effects: impotence, loss of libido (sexual drive), hot flushes, occasional swelling of breast tissue and absent-mindedness. In addition, if you receive the treatment over a long period of time you may notice an increase in weight, a reduction in your muscle tissues and your bones may be weakened.

Anti-hormone Injections: These injections (known as LHRH analogues) stop the production of the male hormone testosterone by the testicles. Depending on the type of injection, they are given once every month or once every three months into the skin of the abdomen or into the arm. Occasionally they temporarily aggravate the cancer before a benefit occurs and for this reason additional anti-hormone tablets are given for the first few weeks of the treatment. Other unwanted effects that have been reported are allergic reactions, irritation at the injection site and headaches. The injections usually have to continue indefinitely.

OR

Bilateral Subcapsular Orchiectomy: This is an operation where the functioning part of the testicles is removed. This is normally done by taking out the centre of the testicles, leaving the testicles themselves behind but reduced in size. Sometimes, instead of this operation the testicles are removed completely. Your surgeon will discuss the surgical options with you. These operations are usually straightforward but there will be some pain or discomfort in the scrotum afterwards. There may also be some swelling and bruising in the scrotum that takes a couple of weeks to subside, and as with any surgical operation an infection can occur in the wound.

Treatment Summary Table

What is my treatment?	How will I be given the treatment?	When, and for how long for?
Androgen Suppression	Orchidectomy or Hormone Treatment	According to Local Practice.

Will my doctor be paid if I participate?

There will be no payment to your doctor if you choose to participate in the trial. However, the trial is part of the National Cancer Research Network (NRCN) portfolio of trials. This means your doctor will be entitled to research support for the trial (for example, help from a research nurse to run the trial).

Will I be paid for participating?

There will be no payments to patients who agree to participate in this trial and additional travel expenses will not be reimbursed.

What if I change my mind about participating in the study?

If you change your mind about taking part in the study, you can withdraw from the trial at any time. This will not affect your relationship with the doctors and nurses, or your subsequent care, in any way. Even though you would not be taking part in the trial in this case, we would still like to monitor your progress. If you agree we would like to continue to collect some information so that the long-term effects of your treatment can be assessed.

Who is organising this trial?

The Medical Research Council (MRC) has overall responsibility for this trial. The study is funded by the Clinical Trials Advisory and Awards Committee which allocates money from MRC and Cancer Research UK. Some financial support has also been obtained from Novartis and the newer drugs have been supplied by the Pharmaceutical companies Novartis, Aventis and Pfizer free of charge or at a reduced cost. The study has received the favourable opinion of an Independent Research Ethics Committees. The trial is run from the MRC Clinical Trials Unit (CTU), London.

What if things go wrong?

The MRC is the sponsor of this trial and as such would give sympathetic consideration to claims for compensation for any non-negligent harm that you may suffer by participating in this trial. The MRC and NHS are both publicly funded bodies and are not allowed to purchase advance insurance to cover indemnity because they are backed by the resources of the Treasury. Like other publicly funded bodies, any liability arising from the MRC's activities is

underwritten by the UK Government. However, this does not extend to harm arising from receiving the “standard treatment”.

The hospital(s) you are treated in continue to have a duty of care to you, whether or not you agree to participate in this trial. Therefore, the MRC does not accept liability for negligence on the part of employees of hospital. This applies whether the hospital is an NHS Trust or not, and the MRC cannot be held liable for any breach in the hospital’s duty of care. If you wish to complain about any aspect of the way you have been approached or treated during the course of the study, the normal National Health Service complaint mechanisms will be available to you.

What if new information becomes available?

Sometimes during the course of a research project, new information becomes available about the treatment/drug that is being studied. Throughout the study, medical information from this study and any other studies will be looked at by an independent committee. If any new information becomes available that may affect your participation in this study or would affect your future care, your study doctor will tell you about it and discuss with you whether you want to continue in the study. If you decide to withdraw, your study doctor will make arrangements for your care to continue. If you decide to continue in the study you would be asked to sign an updated consent form.

Please report any unwanted effects to your cancer doctor or nurse.

If you become unwell between hospital visits, please seek advice immediately, either from your hospital team or from your GP.

Your contact numbers are:

STAMPEDE PATIENT INFORMATION SHEET – TREATMENT GROUP B

(to be printed on local hospital headed paper)

Version 1.0 (May 2004)

Information for patients receiving Hormone Treatment and Zoledronic Acid

You are going to receive hormone treatment and zoledronic acid. Details of the treatment are given below.

Hormone Treatment

Prostate cancers often depend upon the male hormone testosterone to grow. Reducing the amount of testosterone in the body usually prevents further growth of the cancer and may cause it to shrink. This is called hormone treatment and can be achieved either by the use of anti-hormone injections or an operation to remove part or all of both testicles, which produce the male hormone. Your doctor will discuss these different options with you and together you can decide which is the best form of hormone treatment for you. All forms of hormone treatment can cause the following side-effects: impotence, loss of libido (sexual drive), hot flushes, occasional swelling of breast tissue and absent-mindedness. In addition, if you receive the treatment over a long period of time you may notice an increase in weight, a reduction in your muscle tissues and your bones may be weakened.

Anti-hormone Injections: These injections (known as LHRH analogues) stop the production of the male hormone testosterone by the testicles. Depending on the type of injection, they are given once every month or once every three months into the skin of the abdomen or into the arm. Occasionally they temporarily aggravate the cancer before a benefit occurs and for this reason additional anti-hormone tablets are given for the first few weeks of the treatment. Other unwanted effects that have been reported are allergic reactions, irritation at the injection site and headaches. The injections usually have to continue indefinitely.

OR

Bilateral Subcapsular Orchiectomy: This is an operation where the functioning part of the testicles is removed. This is normally done by taking out the centre of the testicles, leaving the testicles themselves behind but reduced in size. Sometimes, instead of this operation the testicles are removed completely. Your surgeon will discuss the surgical options with you. These operations are usually straightforward but there will be some pain or discomfort in the scrotum afterwards. There may also be some swelling and bruising in the scrotum that takes a couple of weeks to subside, and as with any surgical operation an infection can occur in the wound.

Zoledronic Acid

Zoledronic acid is a drug that prevents weakening of bones. It will be given as an injection into a vein once every 3 weeks for the first 6 injections and then once every 4 weeks until 2 years. The drug will be stopped earlier if there is a reason to suggest that your cancer is not responding to the treatment. The injection will take approximately 15 minutes. Unwanted effects are unusual although it may cause flu-like symptoms such as fever, bone pain and muscle cramps. It may also lower the level of some of the salts in your blood such as calcium and phosphate; blood tests will be needed to monitor these levels.

Treatment Summary Table

What is my treatment?	How will I be given the treatment?	When, and for how long for?
Androgen Suppression	Orchidectomy or Hormone Treatment	According to Local Practice.
Zoledronic Acid	15 Minute Intravenous (IV) Infusion	Once every 3 weeks for 18 weeks then Once every 4 weeks for up to 2 years.

(A blood test will be performed before every cycle of Zoledronic Acid).

Will my doctor be paid if I participate?

There will be no payment to your doctor if you choose to participate in the trial. However, the trial is part of the National Cancer Research Network (NRCN) portfolio of trials. This means your doctor will be entitled to research support for the trial (for example, help from a research nurse to run the trial).

Will I be paid for participating?

There will be no payments to patients who agree to participate in this trial and additional travel expenses will not be reimbursed.

What if I change my mind about participating in the study?

If you change your mind about taking part in the study, you can withdraw from the trial at any time. This will not affect your relationship with the doctors and nurses, or your subsequent care, in any way. Even though you would not be taking part in the trial in this case, we would still like to monitor your progress. If you agree we would like to continue to collect some information so that the long-term effects of your treatment can be assessed.

Who is organising this trial?

The Medical Research Council (MRC) has overall responsibility for this trial. The study is funded by the Clinical Trials Advisory and Awards Committee which allocates money from MRC and Cancer Research UK. Some financial support has also been obtained from Novartis and the newer drugs have been supplied by the Pharmaceutical companies Novartis, Aventis and Pfizer free of charge or at a reduced cost. The study has received the favourable opinion of an Independent Research Ethics Committees. The trial is run from the MRC Clinical Trials Unit (CTU), London.

What if things go wrong?

The MRC is the sponsor of this trial and as such would give sympathetic consideration to claims for compensation for any non-negligent harm that you may suffer by participating in this trial. The MRC and NHS are both publicly funded bodies and are not allowed to purchase advance insurance to cover indemnity because they are backed by the resources of the Treasury. Like other publicly funded bodies, any liability arising from the MRC's activities is underwritten by the UK Government. However, this does not extend to harm arising from receiving the "standard treatment".

The hospital(s) you are treated in continue to have a duty of care to you, whether or not you agree to participate in this trial. Therefore, the MRC does not accept liability for negligence on the part of employees of hospital. This applies whether the hospital is an NHS Trust or not, and the MRC cannot be held liable for any breach in the hospital's duty of care. If you wish to complain about any aspect of the way you have been approached or treated during the course of the study, the normal National Health Service complaint mechanisms will be available to you.

What if new information becomes available?

Sometimes during the course of a research project, new information becomes available about the treatment/drug that is being studied. Throughout the study, medical information from this study and any other studies will be looked at by an independent committee. If any new information becomes available that may affect your participation in this study or would affect your future care, your study doctor will tell you about it and discuss with you whether you want to continue in the study. If you decide to withdraw, your study doctor will make arrangements for your care to continue. If you decide to continue in the study you would be asked to sign an updated consent form.

Please report any unwanted effects to your cancer doctor or nurse.

If you become unwell between hospital visits, please seek advice immediately, either from your hospital team or from your GP.

Your contact numbers are:

STAMPEDE PATIENT INFORMATION SHEET – TREATMENT GROUP C

(to be printed on local hospital headed paper)

Version 1.0 (May 2004)

Information for patients receiving Hormone Treatment and Docetaxel

You are going to receive hormone treatment and docetaxel. Details of the treatment are given below.

Hormone Treatment

Prostate cancers often depend upon the male hormone testosterone to grow. Reducing the amount of testosterone in the body usually prevents further growth of the cancer and may cause it to shrink. This is called hormone treatment and can be achieved either by the use of anti-hormone injections or an operation to remove part or all of both testicles, which produce the male hormone. Your doctor will discuss these different options with you and together you can decide which is the best form of hormone treatment for you. All forms of hormone treatment can cause the following side-effects: impotence, loss of libido (sexual drive), hot flushes, occasional swelling of breast tissue and absent-mindedness. In addition, if you receive the treatment over a long period of time you may notice an increase in weight, a reduction in your muscle tissues and your bones may be weakened.

Anti-hormone Injections: These injections (known as LHRH analogues) stop the production of the male hormone testosterone by the testicles. Depending on the type of injection, they are given once every month or once every three months into the skin of the abdomen or into the arm. Occasionally they temporarily aggravate the cancer before a benefit occurs and for this reason additional anti-hormone tablets are given for the first few weeks of the treatment. Other unwanted effects that have been reported are allergic reactions, irritation at the injection site and headaches. The injections usually have to continue indefinitely.

OR

Bilateral Subcapsular Orchiectomy: This is an operation where the functioning part of the testicles is removed. This is normally done by taking out the centre of the testicles, leaving the testicles themselves behind but reduced in size. Sometimes, instead of this operation the testicles are removed completely. Your surgeon will discuss the surgical options with you. These operations are usually straightforward but there will be some pain or discomfort in the scrotum afterwards. There may also be some swelling and bruising in the scrotum that takes a couple of weeks to subside, and as with any surgical operation an infection can occur in the wound.

Docetaxel

Docetaxel will be given as an injection into a vein once every three weeks, for a total of six injections. In addition, you will be given prednisolone (steroid tablets) to be taken twice a day while you are receiving the docetaxel. The docetaxel injection will take approximately one hour. Unwanted effects may include nausea and vomiting (although you will be given additional medication to prevent this), hair loss, fluid retention such as lower leg swelling, allergic reactions and numbness of the hands and feet. Docetaxel may also effect some cells in your blood, causing anaemia or making you more susceptible to bruising, bleeding and infections. It will be necessary to check your blood count before each injection to be sure it is safe. If you have a temperature or become unwell while you are on docetaxel you should contact your hospital cancer team immediately. The 24 hour contact details for your hospital are given below.

The prednisolone tablets may cause indigestion, weight gain, swelling of the lower legs, increased appetite, mood changes and difficulty in sleeping.

Treatment Summary Table

What is my treatment?	How will I be given the treatment?	When, and for how long for?
Androgen Suppression	Orchidectomy or Hormone Treatment	According to Local Practice.
Docetaxel	1 hour Intravenous (IV) Infusion	Once every three weeks for 18 weeks.

(Blood Tests will be taken before each session of Docetaxol treatment)

Will my doctor be paid if I participate?

There will be no payment to your doctor if you choose to participate in the trial. However, the trial is part of the National Cancer Research Network (NRCN) portfolio of trials. This means your doctor will be entitled to research support for the trial (for example, help from a research nurse to run the trial).

Will I be paid for participating?

There will be no payments to patients who agree to participate in this trial and additional travel expenses will not be reimbursed.

What if I change my mind about participating in the study?

If you change your mind about taking part in the study, you can withdraw from the trial at any time. This will not affect your relationship with the doctors and nurses, or your

subsequent care, in any way. Even though you would not be taking part in the trial in this case, we would still like to monitor your progress. If you agree we would like to continue to collect some information so that the long-term effects of your treatment can be assessed.

Who is organising this trial?

The Medical Research Council (MRC) has overall responsibility for this trial. The study is funded by the Clinical Trials Advisory and Awards Committee which allocates money from MRC and Cancer Research UK. Some financial support has also been obtained from Novartis and the newer drugs have been supplied by the Pharmaceutical companies Novartis, Aventis and Pfizer free of charge or at a reduced cost. The study has received the favourable opinion of an Independent Research Ethics Committees. The trial is run from the MRC Clinical Trials Unit (CTU), London.

What if things go wrong?

The MRC is the sponsor of this trial and as such would give sympathetic consideration to claims for compensation for any non-negligent harm that you may suffer by participating in this trial. The MRC and NHS are both publicly funded bodies and are not allowed to purchase advance insurance to cover indemnity because they are backed by the resources of the Treasury. Like other publicly funded bodies, any liability arising from the MRC's activities is underwritten by the UK Government. However, this does not extend to harm arising from receiving the "standard treatment".

The hospital(s) you are treated in continue to have a duty of care to you, whether or not you agree to participate in this trial. Therefore, the MRC does not accept liability for negligence on the part of employees of hospital. This applies whether the hospital is an NHS Trust or not, and the MRC cannot be held liable for any breach in the hospital's duty of care. If you wish to complain about any aspect of the way you have been approached or treated during the course of the study, the normal National Health Service complaint mechanisms will be available to you.

What if new information becomes available?

Sometimes during the course of a research project, new information becomes available about the treatment/drug that is being studied. Throughout the study, medical information from this study and any other studies will be looked at by an independent committee. If any new information becomes available that may affect your participation in this study or would affect your future care, your study doctor will tell you about it and discuss with you whether you want to continue in the study. If you decide to withdraw, your study doctor will make arrangements for your care to continue. If you decide to continue in the study you would be asked to sign an updated consent form.

Please report any unwanted effects to your cancer doctor or nurse.

If you become unwell between hospital visits, please seek advice immediately, either from your hospital team or from your GP.

Your contact numbers are:

STAMPEDE PATIENT INFORMATION SHEET – TREATMENT GROUP D

(to be printed on local hospital headed paper)

Version 1.0 (May 2004)

Information for patients receiving Hormone Treatment and Celecoxib

You are going to receive hormone treatment and celecoxib. Details of the treatment are given below.

Hormone Treatment

Prostate cancers often depend upon the male hormone testosterone to grow. Reducing the amount of testosterone in the body usually prevents further growth of the cancer and may cause it to shrink. This is called hormone treatment and can be achieved either by the use of anti-hormone injections or an operation to remove part or all of both testicles, which produce the male hormone. Your doctor will discuss these different options with you and together you can decide which is the best form of hormone treatment for you. All forms of hormone treatment can cause the following side-effects: impotence, loss of libido (sexual drive), hot flushes, occasional swelling of breast tissue and absent-mindedness. In addition, if you receive the treatment over a long period of time you may notice an increase in weight, a reduction in your muscle tissues and your bones may be weakened.

Anti-hormone Injections: These injections (known as LHRH analogues) stop the production of the male hormone testosterone by the testicles. Depending on the type of injection, they are given once every month or once every three months into the skin of the abdomen or into the arm. Occasionally they temporarily aggravate the cancer before a benefit occurs and for this reason additional anti-hormone tablets are given for the first few weeks of the treatment. Other unwanted effects that have been reported are allergic reactions, irritation at the injection site and headaches. The injections usually have to continue indefinitely.

OR

Bilateral Subcapsular Orchiectomy: This is an operation where the functioning part of the testicles is removed. This is normally done by taking out the centre of the testicles, leaving the testicles themselves behind but reduced in size. Sometimes, instead of this operation the testicles are removed completely. Your surgeon will discuss the surgical options with you. These operations are usually straightforward but there will be some pain or discomfort in the scrotum afterwards. There may also be some swelling and bruising in the scrotum that takes a couple of weeks to subside, and as with any surgical operation an infection can occur in the wound.

Celecoxib

The celecoxib tablets are taken twice a day and will continue indefinitely. They will only be stopped if your cancer shows signs of not responding to the treatment or if you and your doctor decide this is best. Unwanted effects may include stomach discomfort, indigestion, abdominal pain and flatulence, fluid retention, difficulty sleeping, sinusitis or a sore throat and a rash.

Treatment Summary Table

What is my treatment?	How will I be given the treatment?	When, and for how long for?
Androgen Suppression	Orchidectomy or Hormone Treatment	According to Local Practice.
Celecoxib	Tablet	Two times a day until disease progression.

Will my doctor be paid if I participate?

There will be no payment to your doctor if you choose to participate in the trial. However, the trial is part of the National Cancer Research Network (NRCN) portfolio of trials. This means your doctor will be entitled to research support for the trial (for example, help from a research nurse to run the trial).

Will I be paid for participating?

There will be no payments to patients who agree to participate in this trial and additional travel expenses will not be reimbursed.

What if I change my mind about participating in the study?

If you change your mind about taking part in the study, you can withdraw from the trial at any time. This will not affect your relationship with the doctors and nurses, or your subsequent care, in any way. Even though you would not be taking part in the trial in this case, we would still like to monitor your progress. If you agree we would like to continue to collect some information so that the long-term effects of your treatment can be assessed.

Who is organising this trial?

The Medical Research Council (MRC) has overall responsibility for this trial. The study is funded by the Clinical Trials Advisory and Awards Committee which allocates money from MRC and Cancer Research UK. Some financial support has also been obtained from Novartis and the newer drugs have been supplied by the Pharmaceutical companies Novartis, Aventis and Pfiser free of charge or at a reduced cost. The study has received the favourable opinion of an Independent Research Ethics Committees. The trial is run from the MRC Clinical Trials Unit (CTU), London.

What if things go wrong?

The MRC is the sponsor of this trial and as such would give sympathetic consideration to claims for compensation for any non-negligent harm that you may suffer by participating in this trial. The MRC and NHS are both publicly funded bodies and are not allowed to purchase advance insurance to cover indemnity because they are backed by the resources of the Treasury. Like other publicly funded bodies, any liability arising from the MRC's activities is underwritten by the UK Government. However, this does not extend to harm arising from receiving the "standard treatment".

The hospital(s) you are treated in continue to have a duty of care to you, whether or not you agree to participate in this trial. Therefore, the MRC does not accept liability for negligence on the part of employees of hospital. This applies whether the hospital is an NHS Trust or not, and the MRC cannot be held liable for any breach in the hospital's duty of care. If you wish to complain about any aspect of the way you have been approached or treated during the course of the study, the normal National Health Service complaint mechanisms will be available to you.

What if new information becomes available?

Sometimes during the course of a research project, new information becomes available about the treatment/drug that is being studied. Throughout the study, medical information from this study and any other studies will be looked at by an independent committee. If any new information becomes available that may affect your participation in this study or would affect your future care, your study doctor will tell you about it and discuss with you whether you want to continue in the study. If you decide to withdraw, your study doctor will make arrangements for your care to continue. If you decide to continue in the study you would be asked to sign an updated consent form.

Please report any unwanted effects to your cancer doctor or nurse.

If you become unwell between hospital visits, please seek advice immediately, either from your hospital team or from your GP.

Your contact numbers are:

STAMPEDE PATIENT INFORMATION SHEET – TREATMENT GROUP E

(to be printed on local hospital headed paper)

Version 1.0 (May 2004)

Information for patients receiving Hormone Treatment and Zoledronic Acid and Docetaxel

You are going to receive hormone treatment and zoledronic acid and docetaxel. Details of the treatment are given below.

Hormone Treatment

Prostate cancers often depend upon the male hormone testosterone to grow. Reducing the amount of testosterone in the body usually prevents further growth of the cancer and may cause it to shrink. This is called hormone treatment and can be achieved either by the use of anti-hormone injections or an operation to remove part or all of both testicles, which produce the male hormone. Your doctor will discuss these different options with you and together you can decide which is the best form of hormone treatment for you. All forms of hormone treatment can cause the following side-effects: impotence, loss of libido (sexual drive), hot flushes, occasional swelling of breast tissue and absent-mindedness. In addition, if you receive the treatment over a long period of time you may notice an increase in weight, a reduction in your muscle tissues and your bones may be weakened.

Anti-hormone Injections: These injections (known as LHRH analogues) stop the production of the male hormone testosterone by the testicles. Depending on the type of injection, they are given once every month or once every three months into the skin of the abdomen or into the arm. Occasionally they temporarily aggravate the cancer before a benefit occurs and for this reason additional anti-hormone tablets are given for the first few weeks of the treatment. Other unwanted effects that have been reported are allergic reactions, irritation at the injection site and headaches. The injections usually have to continue indefinitely.

OR

Bilateral Subcapsular Orchiectomy: This is an operation where the functioning part of the testicles is removed. This is normally done by taking out the centre of the testicles, leaving the testicles themselves behind but reduced in size. Sometimes, instead of this operation the testicles are removed completely. Your surgeon will discuss the surgical options with you. These operations are usually straightforward but there will be some pain or discomfort in the scrotum afterwards. There may also be some swelling and bruising in the scrotum that takes a couple of weeks to subside, and as with any surgical operation an infection can occur in the wound.

Zoledronic Acid

Zoledronic acid is a drug that prevents weakening of bones. It will be given as an injection into a vein once every 3 weeks for the first 6 injections and then once every 4 weeks until 2 years. The drug will be stopped earlier if there is a reason to suggest that your cancer is not responding to the treatment. The injection will take approximately 15 minutes. Unwanted effects are unusual although it may cause flu-like symptoms such as fever, bone pain and muscle cramps. It may also lower the level of some of the salts in your blood such as calcium and phosphate; blood tests will be needed to monitor these levels.

Docetaxel

Docetaxel will be given as an injection into a vein once every three weeks, for a total of six injections, on the same day that you receive your zoledronic acid injection. In addition, you will be given prednisolone (steroid tablets) to be taken twice a day while you are receiving the docetaxel. Docetaxel will be given as an injection into a vein once every three weeks, for a total of six injections. In addition, you will be given prednisolone (steroid tablets) to be taken twice a day while you are receiving the docetaxel. The docetaxel injection will take approximately one hour. Unwanted effects may include nausea and vomiting (although you will be given additional medication to prevent this), hair loss, fluid retention such as lower leg swelling, allergic reactions and numbness of the hands and feet. Docetaxel may also effect some cells in your blood, causing anaemia or making you more susceptible to bruising, bleeding and infections. It will be necessary to check your blood count before each injection to be sure it is safe. If you have a temperature or become unwell while you are on docetaxel you should contact your hospital cancer team immediately. The 24 hour contact details for your hospital are given below.

Treatment Summary Table

What is my treatment?	How will I be given the treatment?	When, and for how long for?
Androgen Suppression	Orchidectomy or Hormone Treatment	According to Local Practice
Zoledronic Acid	15 Minute Intravenous (IV) Infusion	Once every 3 weeks for 18 weeks then Once every 4 weeks for up to 2 years.
Docetaxel	1 hour Intravenous (IV) Infusion	Once every three weeks for 18 weeks.

(Blood Tests will be taken before each treatment)

The prednisolone tablets may cause indigestion, weight gain, swelling of the lower legs, increased appetite, mood changes and difficulty in sleeping.

To receive both the zoledronic acid and the docetaxel you will probably be in the hospital for about half a day.

Will my doctor be paid if I participate?

There will be no payment to your doctor if you choose to participate in the trial. However, the trial is part of the National Cancer Research Network (NRCN) portfolio of trials. This means your doctor will be entitled to research support for the trial (for example, help from a research nurse to run the trial).

Will I be paid for participating?

There will be no payments to patients who agree to participate in this trial and additional travel expenses will not be reimbursed.

What if I change my mind about participating in the study?

If you change your mind about taking part in the study, you can withdraw from the trial at any time. This will not affect your relationship with the doctors and nurses, or your subsequent care, in any way. Even though you would not be taking part in the trial in this case, we would still like to monitor your progress. If you agree we would like to continue to collect some information so that the long-term effects of your treatment can be assessed.

Who is organising this trial?

The Medical Research Council (MRC) has overall responsibility for this trial. The study is funded by the Clinical Trials Advisory and Awards Committee which allocates money from MRC and Cancer Research UK. Some financial support has also been obtained from Novartis and the newer drugs have been supplied by the Pharmaceutical companies Novartis, Aventis and Pfizer free of charge or at a reduced cost. The study has received the favourable opinion of an Independent Research Ethics Committees. The trial is run from the MRC Clinical Trials Unit (CTU), London.

What if things go wrong?

The MRC is the sponsor of this trial and as such would give sympathetic consideration to claims for compensation for any non-negligent harm that you may suffer by participating in this trial. The MRC and NHS are both publicly funded bodies and are not allowed to purchase advance insurance to cover indemnity because they are backed by the resources of the Treasury. Like other publicly funded bodies, any liability arising from the MRC's activities is underwritten by the UK Government. However, this does not extend to harm arising from receiving the "standard treatment".

The hospital(s) you are treated in continue to have a duty of care to you, whether or not you agree to participate in this trial. Therefore, the MRC does not accept liability for negligence on the part of employees of hospital. This applies whether the hospital is an NHS Trust or not, and the MRC cannot be held liable for any breach in the hospital's duty of care. If you wish to complain about any aspect of the way you have been approached or treated during the course of the study, the normal National Health Service complaint mechanisms will be available to you.

What if new information becomes available?

Sometimes during the course of a research project, new information becomes available about the treatment/drug that is being studied. Throughout the study, medical information from this study and any other studies will be looked at by an independent committee. If any new information becomes available that may affect your participation in this study or would affect your future care, your study doctor will tell you about it and discuss with you whether you want to continue in the study. If you decide to withdraw, your study doctor will make arrangements for your care to continue. If you decide to continue in the study you would be asked to sign an updated consent form.

Please report any unwanted effects to your cancer doctor or nurse.

If you become unwell between hospital visits, please seek advice immediately, either from your hospital team or from your GP.

Your contact numbers are:

STAMPEDE PATIENT INFORMATION SHEET – TREATMENT GROUP F

(to be printed on local hospital headed paper)

Version 1.0 (May 2004)

Information for patients receiving Hormone Treatment and zoledronic acid and celecoxib

You are going to receive hormone treatment and zoledronic acid and celecoxib. Details of the treatment are given below.

Hormone Treatment

Prostate cancers often depend upon the male hormone testosterone to grow. Reducing the amount of testosterone in the body usually prevents further growth of the cancer and may cause it to shrink. This is called hormone treatment and can be achieved either by the use of anti-hormone injections or an operation to remove part or all of both testicles, which produce the male hormone. Your doctor will discuss these different options with you and together you can decide which is the best form of hormone treatment for you. All forms of hormone treatment can cause the following side-effects: impotence, loss of libido (sexual drive), hot flushes, occasional swelling of breast tissue and absent-mindedness. In addition, if you receive the treatment over a long period of time you may notice an increase in weight, a reduction in your muscle tissues and your bones may be weakened.

Anti-hormone Injections: These injections (known as LHRH analogues) stop the production of the male hormone testosterone by the testicles. Depending on the type of injection, they are given once every month or once every three months into the skin of the abdomen or into the arm. Occasionally they temporarily aggravate the cancer before a benefit occurs and for this reason additional anti-hormone tablets are given for the first few weeks of the treatment. Other unwanted effects that have been reported are allergic reactions, irritation at the injection site and headaches. The injections usually have to continue indefinitely.

OR

Bilateral Subcapsular Orchiectomy: This is an operation where the functioning part of the testicles is removed. This is normally done by taking out the centre of the testicles, leaving the testicles themselves behind but reduced in size. Sometimes, instead of this operation the testicles are removed completely. Your surgeon will discuss the surgical options with you. These operations are usually straightforward but there will be some pain or discomfort in the scrotum afterwards. There may also be some swelling and bruising in the

scrotum that takes a couple of weeks to subside, and as with any surgical operation an infection can occur in the wound.

Zoledronic Acid

Zoledronic acid is a drug that prevents weakening of bones. It will be given as an injection into a vein once every 3 weeks for the first 6 injections and then once every 4 weeks until 2 years. The drug will be stopped earlier if there is a reason to suggest that your cancer is not responding to the treatment. The injection will take approximately 15 minutes. Unwanted effects are unusual although it may cause flu-like symptoms such as fever, bone pain and muscle cramps. It may also lower the level of some of the salts in your blood such as calcium and phosphate; blood tests will be needed to monitor these levels.

Celecoxib

The celecoxib tablets are taken twice a day and will continue indefinitely. They will only be stopped if your cancer shows signs of not responding to the treatment or if you and your doctor decide this is best. Unwanted effects may include stomach discomfort, indigestion, abdominal pain and flatulence, fluid retention, difficulty sleeping, sinusitis or a sore throat and a rash.

Treatment Summary Table

What is my treatment?	How will I be given the treatment?	When, and for how long for?
Androgen Suppression	Orchidectomy or Hormone Treatment	According to Local Practice
Zoledronic Acid	15 Minute Intravenous (IV) Infusion	Once every 3 weeks for 18 weeks then Once every 4 weeks for up to 2 years.
Celecoxib	Tablet	Two times a day until disease progression.

(Blood Tests will be taken before each session of Zoledronic Acid treatment)

Will my doctor be paid if I participate?

There will be no payment to your doctor if you choose to participate in the trial. However, the trial is part of the National Cancer Research Network (NRCN) portfolio of trials. This means your doctor will be entitled to research support for the trial (for example, help from a research nurse to run the trial).

Will I be paid for participating?

There will be no payments to patients who agree to participate in this trial and additional travel expenses will not be reimbursed.

What if I change my mind about participating in the study?

If you change your mind about taking part in the study, you can withdraw from the trial at any time. This will not affect your relationship with the doctors and nurses, or your subsequent care, in any way. Even though you would not be taking part in the trial in this case, we would still like to monitor your progress. If you agree we would like to continue to collect some information so that the long-term effects of your treatment can be assessed.

Who is organising this trial?

The Medical Research Council (MRC) has overall responsibility for this trial. The study is funded by the Clinical Trials Advisory and Awards Committee which allocates money from MRC and Cancer Research UK. Some financial support has also been obtained from Novartis and the newer drugs have been supplied by the Pharmaceutical companies Novartis, Aventis and Pfizer free of charge or at a reduced cost. The study has received the favourable opinion of an Independent Research Ethics Committees. The trial is run from the MRC Clinical Trials Unit (CTU), London.

What if things go wrong?

The MRC is the sponsor of this trial and as such would give sympathetic consideration to claims for compensation for any non-negligent harm that you may suffer by participating in this trial. The MRC and NHS are both publicly funded bodies and are not allowed to purchase advance insurance to cover indemnity because they are backed by the resources of the Treasury. Like other publicly funded bodies, any liability arising from the MRC's activities is underwritten by the UK Government. However, this does not extend to harm arising from receiving the "standard treatment".

The hospital(s) you are treated in continue to have a duty of care to you, whether or not you agree to participate in this trial. Therefore, the MRC does not accept liability for negligence on the part of employees of hospital. This applies whether the hospital is an NHS Trust or not, and the MRC cannot be held liable for any breach in the hospital's duty of care. If you wish to complain about any aspect of the way you have been approached or treated during the course of the study, the normal National Health Service complaint mechanisms will be available to you.

What if new information becomes available?

Sometimes during the course of a research project, new information becomes available about the treatment/drug that is being studied. Throughout the study, medical information from this study and any other studies will be looked at by an independent committee. If any new information becomes available that may affect your participation in this study or would affect your future care, your study doctor will tell you about it and discuss with you whether you want to continue in the study. If you decide to withdraw, your study doctor will make arrangements for your care to continue. If you decide to continue in the study you would be asked to sign an updated consent form.

Please report any unwanted effects to your cancer doctor or nurse.

If you become unwell between hospital visits, please seek advice immediately, either from your hospital team or from your GP.

Your contact numbers are:

Appendix C - GP Letter

Version 1.0 (May 2004)

Dear Dr _____

STAMPEDE: SYSTEMATIC THERAPY FOR ADVANCING OR METASTATIC PROSTATE CANCER

Your patient has agreed to participate in a clinical trial named STAMPEDE. This is an international study for patients with locally advanced or metastatic prostate cancer who are about to commence androgen suppression therapy. It is funded by the Clinical Trials Advisory and Awards Committee (CTAAC), sponsored by the Medical Research Council (MRC) and coordinated by the MRC Clinical Trials Unit. It will evaluate the effects of three different types of drugs that have shown promise against prostate cancer: a bisphosphonate (zoledronic acid), a chemotherapeutic agent (docetaxel) and a cyclooxygenase-2 inhibitor (celecoxib). Patients are randomised either to the control arm of the trial, androgen suppression only, or to one of five investigational arms that consist of androgen suppression plus one or two of the drugs described above.

Enclosed with this letter is a copy of the information that has been supplied to the patient about this trial. This includes a general information sheet that provides an overview of the trial and details of the general issues relating to the trial, and a separate sheet giving details about the specific treatment your patient will receive.

Your patient has been randomised to the marked arm:

- | | |
|--------------------------|--|
| ☐ | A: Androgen suppression |
| ☐ | B: Androgen suppression + zoledronic acid (Zometa) |
| ☐ | C: Androgen suppression + docetaxel (Taxotere) |
| ☐ | D: Androgen suppression + celecoxib (Celebrex) |
| ☐ | E: Androgen suppression + zoledronic acid (Zometa) + docetaxel (Taxotere) |
| ☐ | F: Androgen suppression + zoledronic acid (Zometa) + celecoxib (Celebrex) |

Androgen suppression will be given by ☐ Bilateral orchiectomy
☐ LHRH analogues

Patients receiving docetaxel may develop myelosuppression and therefore they are at risk of neutropenic sepsis and thrombocytopaenia. Should your patient develop a fever and/or other signs of sepsis during their treatment they have been advised to contact the hospital urgently as prompt treatment with intravenous antibiotics may be life saving.

Patient's Prostate Specific Antigen (PSA) levels will be measured regularly especially in the first 6 months of the trial, and the patient's consultant may wish to discuss your surgery's possible involvement in taking blood samples.

The preferred hospital for admissions for patients with suspected neutropenic sepsis or other emergencies is:.....

Please contact: (telephone no):.....

For non-urgent queries relating to your patients treatment please contact the consultant, as detailed on the accompanying correspondence.

Yours sincerely,

Hospital Consultant

Appendix D - Molecular Genetics Sub-study

Instructions to research nurses and/or clinicians for collecting blood samples for patients who have consented to take part in the molecular genetics study.

1. Collect 8ml of blood in 4ml EDTA tubes (lilac tops) at the time of randomisation.
2. Label tube with trial number, date and time.
3. Wrap in tissue to absorb fluid in case of leakage and seal in the plastic bag supplied in the container.
4. Place in the container (addressed blue plastic safebox) together with a completed blood sample request form.
5. Fix the lid securely as requested and send by first class mail the same day.

Professor John Masters
3rd Floor research Laboratories
67 Riding House Street
London
W1W 7EJ
Tel: 020 7679 9543
Fax: 020 7679 9366
E-mail: j.masters@ucl.ac.uk

Appendix E - Drug Supply information

The pharmacist is responsible for all aspects of managing the study drug supply. This includes ordering study drug via the MRC-CTU as needed to maintain a reasonable inventory; maintaining drug accountability records; managing the drug supply; assuring that the study drug is not accessible to unauthorized persons and that it remains stored under the correct conditions.

Study drugs can be ordered once a favourable opinion of Site Specific Assessment (SSA – formerly LREC) and Research and Development (R&D) approval have been issued for the trial to take place at that site. Copies of completed Site Accreditation Forms, SSA and R&D approval documents should be forwarded to the MRC-CTU when available. On receipt of these documents the MRC-CTU will notify the drug distributing centres of the new centre status and approve the centre for trial drug delivery. A starter pack of each drug will then be dispatched from the distributing centre to a named Pharmacist at the site.

The starter pack of drugs will comprise of –

Docetaxel: Twelve vials of 80mg

Zoledronic acid: 40 vials of 4mg

Celecoxib: 15 boxes (containing 140 capsules each) of 200mg capsules

When a new patient is randomised, the MRC-CTU will notify the relevant pharmacy of the randomisation and the treatment regimen to which the patient has been allocated and the patients treatment schedule. However, the patient will be treated with trial drugs held in stock at that site and it is the responsibility of the pharmacy to order replacement trial drugs using the Stampede trial specific order forms. Drug re-ordering will be co-ordinated via the MRC-CTU and it is recommend that sufficient drugs for the treatment of newly randomised patients are ordered in a timely manner.

Returned or out of date stock should be destroyed in accordance with local policies and procedures. All drug destruction should be documented on the 'Destruction logs' for auditing and monitoring purposes.

Appendix F - Administration of Docetaxel and Zoledronic Acid

Administration of Docetaxel

The use of docetaxel should be confined to units specialised in the administration of cytotoxic chemotherapy and it should only be administered under the supervision of a physician qualified in the use of anticancer chemotherapy.

Pre-medication Regimen

Dexamethasone 8mg *bid* for 2 days, with the first dose taken the night before the chemotherapy.

Suggested Anti-emetic Regimen

30 min prior to the administration of docetaxel:-

- Granisetron 3mg IV stat
- Dexamethasone 8mg IV stat

Followed by:-

- Granisetron 1mg orally the morning after the docetaxel
- Domperidone 20mg orally qds as required.

Dose Modifications in the Event of Toxicity

General guidance: If possible, toxicities should be managed symptomatically. If toxicity occurs, the appropriate treatment will be used to ameliorate signs and symptoms including antiemetics for nausea and vomiting, antidiarrheals for diarrhoea, and antipyretics and/or antihistamines for drug fever.

No more than 2 dose reductions will be adopted per patient. If more than 2 dose reductions are indicated the patient should be withdrawn from trial treatment.

Docetaxel dose reductions:

Doses should be adjusted according to the following:

- Standard dose: 75 mg/m²
- First level dose reduction: 60 mg/m²
- Second level dose reduction: 45 mg/m²

Doses which have been reduced for toxicity must not be re-escalated even if the toxicity has resolved.

Docetaxel dose delay:

A treatment delay >4 days should be reported in the CRF. Treatment may be delayed no more than 2 weeks to allow recovery from acute toxicity. In case of treatment delay greater than 2 weeks, patients should be withdrawn from trial treatment.

Management of Myelosuppression**(a) Neutropaenia and its complications**

Adverse event	Action to be taken
- Grade 4 neutropenia* for 7 days or more. - Grade 3-4 neutropenia with oral fever $\geq 38.5^{\circ}\text{C}$ - Infection* (i.e. documented infection with grade 3-4 neutropenia)	If the patient develops one of these adverse events, the next infusion should be given with a one-level dose reduction.

*according to NCI-CTC version 3

Neutrophil count on day of infusion	Action to be taken
$\geq 1.5 \times 10^9/\text{L}$	Treat on time
$< 1.5 \times 10^9/\text{L}$	Delay maximum 2 weeks Blood counts have to be performed until $\text{ANC} \geq 1.5 \times 10^9/\text{L}$. Then treat with a one-level dose reduction. If no recovery (ANC still $< 1.5 \times 10^9/\text{L}$) after 2 week delay: the patient will be withdrawn from trial treatment.

(b) Thrombocytopaenia

In case of grade ≥ 3 platelets (NCI-CTC), delay maximum 2 weeks until platelets recover to $\geq 100 \times 10^9/\text{L}$, then treat with a one-level dose reduction.

Allergy and hypersensitivity reactions

Hypersensitivity reactions that occur despite pre-medication are very likely to occur *within a few minutes of start of the first or of the second infusion of docetaxel*. Therefore, during the 1st and the 2nd infusions, careful evaluation of the general sense of well being and of blood pressure and heart rate monitoring will be performed *for at least the first 10 minutes*, so that immediate intervention can occur in response to symptoms of an untoward reaction.

Facilities and equipment for resuscitation must be immediately available. If a reaction occurs, the specific treatment that is medically indicated can be given (e.g. adrenaline in case of anaphylactic shock, aminophylline in case of bronchospasm, etc.). In addition, it is recommended to take the measures listed below:

Reaction	Action to be taken
Mild symptoms localized cutaneous reaction localized pruritus flushing rash	Consider decreasing the rate of infusion until recovery of symptoms, stay at bedside At subsequent cycles consider whether additional premedication is required see below.
Moderate symptoms generalized pruritus more severe flushing or rash mild dyspnoea hypotension with systolic BP $\leq$ 80mmHg	Stop study drug infusion Give IV antihistamine and IV corticosteroids [†] A clinical decision should be made as to whether it is appropriate to resume the infusion after recovery of symptoms. At subsequent cycles, antihistamines [†] and steroids [†] will be given IV, one hour before infusion, in addition to the oral dexamethasone premedication.
Severe symptoms bronchospasm generalized urticaria hypotension with systolic BP $\leq$ 80mmHg angioedema	Stop study drug infusion Give IV antihistamine and steroids [†] add adrenaline [‡] or bronchodilators and/or IV plasma expanders if indicated.

[†]Antihistamines: Chlorpheniramine IV 10-2 mg
 Promethazine IM 25-50mg, max-100mg

Corticosteroids: dexamethasone or equivalent IV [†] 5-10 mg of dexamethasone

[‡]Adrenaline: administer standard dose - 500µg sc.

If a severe hypersensitivity reaction occurs careful consideration should be given as to whether it is clinically appropriate to administer another dose and the appropriate pre-medication required (IV steroids and antihistamines, and in addition dexamethasone 20 mg orally 24, 18, 13, 7 and 1 hour before study drug infusion). If a severe reaction occurs for a second time, the patient should be withdrawn from trial treatment.

Nausea/Vomiting

A prophylactic anti-emetic treatment should be given to patients from the first cycle. If despite the appropriate medication, grade $\geq$ 3 nausea/vomiting still occur, reduce the dose of study drug by one dose level. If despite dose reduction, nausea/vomiting still occur at grade $\geq$ 3, the patient should be withdrawn from trial treatment.

Diarrhoea

No prophylactic treatment for diarrhoea is recommended from cycle one. However, following the first episode of diarrhoea, the patient should receive symptomatic treatment with loperamide: 4mg following the first episode and then 2mg following each new episode until recovery of diarrhoea (no more than 16mg daily).

If despite the use of loperamide, grade ≥ 3 diarrhoea still occurs, reduce the dose of study drug by one dose level. If despite dose reduction, diarrhoea still occurs at grade ≥ 3 , the patient should be withdrawn from trial treatment.

Stomatitis

Grade ≤ 2 : No change, study chemotherapy should be withheld until resolution to grade ≤ 1 .

Grade 3: If grade 3 stomatitis occurs, study drug should be withheld until resolution to grade ≤ 1 . Treatment may then be resumed, but the dose of study drug should be reduced by one dose level for all subsequent doses.

Grade 4: In case of grade 4 stomatitis, the patient should be withdrawn from trial treatment.

Peripheral Neuropathy

If symptoms or signs experienced by the patient, dose modification should be performed as follows:-

Grade ≤ 1 : no change

Grade 2: treat with a one-level dose reduction (no further dose reduction is planned)

Grade 3: patient should be withdrawn from trial treatment.

Skin Toxicity

Grade ≤ 2 : No change

Grade 3: delay until grade ≤ 1 , maximum two weeks then reduce dose of study drug by one dose level; if no recovery to $\leq$ grade 1 within two weeks delay, patient should be withdrawn from trial treatment.

Liver Toxicity

In case of increase of ALT and/or AST $> 1.5 \times \text{ULN}$ or bilirubin $> \text{ULN}$, delay study drug treatment for up to 2 weeks until ALT and/or AST returned to $\leq 1.5 \times \text{ULN}$ and bilirubin $\leq \text{ULN}$. Then treat at one level dose reduction.

Docetaxel-induced fluid retention

In case of fluid retention (peripheral oedema and/or effusions) during the treatment with Docetaxel (Taxotere). The patient's body weight should be recorded and followed as frequently as possible to document any weight gain, which could be related to oedema. Treatment should commence when signs and/or symptoms of fluid retention are observed, including weight gain from baseline ≥ 1 grade not otherwise explained. Based on the hypothesis of capillary damage due to Docetaxel, the following treatment is recommended in case fluid retention occurs: frusemide 20mg po once daily.

If the symptoms cannot be controlled adequately i.e. worsening of the fluid retention or spread to another area, the dose of frusemide should be increased to 40mg. The addition of metolazone po at the recommended dose together with potassium and/or magnesium supplements may be useful.

The clinical tolerance of the patient, the overall tumour response and the medical judgment of the investigator will determine if it is in the patient's best interest to continue or to discontinue the study drug. It is recommended, however, that patients with fluid retention of grade ≥ 3 severity should be withdrawn from the trial.

Docetaxel-induced hyperlacrimation

The excessive lacrimation (epiphora) seen in some patients receiving docetaxel appears to be related to cumulative dose (median $\sim 300 \text{ mg/m}^2$) and resolves rapidly after treatment discontinuation. It seems to be the result of a chemical conjunctivitis and/or chemical inflammation (with oedema) of the lacrimal duct epithelium (producing a reversible lacrimal duct stenosis). If epiphora persists patients should be referred to an Ophthalmologist.

In patients experiencing clinically significant hyperlacrimation, the following approach is recommended:

- No dose reduction planned
- Frequent instillation of artificial tears
- Prescribe a steroid ophthalmic solution (e.g. prednisolone acetate): 2 drops each eye *bid* for 3 days starting the day before docetaxel administration, in patients **without** history of herpetic eye disease.

Administration of Zoledronic Acid

Zometa® (zoledronic acid) will be provided in plastic vials containing 4 mg zoledronic acid in 5 mL concentrate solution for infusion. Each zoledronic acid plastic vial contains 4 mg zoledronic acid (anhydrous).

The zoledronic acid 4 mg/5 mL concentrate solution is not for direct infusion and has to be further diluted prior to the use. Prior to administration, the 5ml of the concentrate solution must be diluted with 100 mL calcium-free infusion solution (0.9% sodium chloride solution or 5% glucose solution). The appropriate volume of the reconstituted zoledronic acid solution is 105 mL. The necessary infusion bags/bottles containing either 100mL calcium free 0.9% sodium chloride or 5 % dextrose solution have to be used for the set up of the infusion and will be provided by the study center.

Zometa® (zoledronic acid) 4 mg/5 mL concentrate solution must not be mixed with calcium-containing solutions such as Ringer's solution.

If not used immediately after dilution with infusion media, for microbiological integrity, the final solution must be placed in a refrigerator with a temperature between 2-8 °C. The refrigerated solution should then be equilibrated to room temperature prior to administration. The total time between dilution, storage in a refrigerator and end of administration of the infusion must not exceed 24 hours. Reconstituted zoledronic acid solutions must be administered in no less than a 15-minute intravenous infusion in a line separate from all other drugs.

Patients must be evaluated prior to and following the administration of the zoledronic acid infusion to ensure that they are adequately hydrated.

Since no data are available on the compatibility of zoledronic acid with other intravenously administered substances, zoledronic acid must not be mixed with other medications or substances and should always be given through a separate infusion line.

Stability studies using diluted zoledronic acid 4 mg/5 mL concentrate solutions in glass bottles and infusion bags made from polyvinylchloride (PVC), polypropylene (PP) and polyethylene (PE) prefilled with 0.9% sodium chloride solution or 5% dextrose solution and with infusion lines made from PVC and PE showed no incompatibility.

Appendix G - Drug Safety Information for Drugs used in the Trial

ZOLEDRONIC ACID

- **Contraindications**

Zoledronic acid is contraindicated in patients with known clinically significant hypersensitivity to zoledronic acid, other bisphosphonates or any of the excipients in the formulation of Zometa.

- **Special Warnings and precautions for use**

Patients must be assessed prior to administration of zoledronic acid to ensure that they are adequately hydrated. Metabolic parameters, such as serum levels of calcium, phosphate and magnesium, should be carefully monitored after initiating zoledronic acid therapy. If hypocalcaemia, hypophosphataemia, or hypomagnesaemia occurs, short-term supplemental therapy may be necessary.

As with other bisphosphonates, zoledronic acid has been associated with reports of renal dysfunction. Factors that may increase the potential for deterioration in renal function include dehydration and pre-existing renal impairment. While the risk is reduced with a dose of zoledronic acid 4mg administered over 15 minutes, deterioration in renal function may still occur. Increases in serum creatinine also occur in some patients with chronic administration of zoledronic although less frequently.

As with other bisphosphonates renal monitoring is recommended, for instance, measurement of serum creatinine prior to each dose of zoledronic acid. In all patients the dose should be withheld if renal function has deteriorated. In clinical studies, zoledronic acid treatment was resumed only when the creatinine level returned to within 10% of the baseline value.

Overhydration should be avoided in patients at risk of cardiac failure.

- **Interaction with other medicinal products and other forms of interaction.**

In clinical studies, zoledronic acid has been administered concomitantly with commonly used anticancer agents, diuretics, antibiotics and analgesics without clinically apparent interactions occurring. Zoledronic acid shows no appreciable binding to plasma proteins and does not inhibit human P450 enzymes, but no formal clinical interaction studies have been performed. Caution is advised when bisphosphonates are administered with aminoglycosides, since both agents may have an additive effect, resulting in a lower serum calcium level for longer periods. Caution is also indicated when zoledronic acid is used with

other potentially nephrotoxic drugs. Attention should also be paid to the possibility of hypomagnesaemia developing during treatment.

- **Undesirable Effects**

Frequencies of adverse reactions for zoledronic acid 4mg are mainly based on data collection from chronic treatment. Adverse reactions to zoledronic acid are similar to those reported for other bisphosphonates and can be expected to occur in approximately one third of patients. Intravenous administration has been most commonly associated with a flu-like syndrome in about 9% of patients, including bone pain (9.1%), fever (7.2%), fatigue (4.1%) and rigors (2.9%). Arthralgia and myalgia have been reported in approximately 3%. No information is available on the reversibility of these adverse effects.

Frequently, the reduction in renal calcium excretion is accompanied by a fall in serum phosphate levels (in approximately 20% of patients), which is asymptomatic not requiring treatment. The serum calcium may fall to asymptomatic hypocalcaemic levels in approximately 3% of patients.

Gastrointestinal reactions, such as nausea (5.8%) and vomiting (2.6%) have been reported following intravenous infusion of zoledronic acid. Occasionally local reactions at the infusion site such as redness or swelling and/or pain were also observed in less than 1% of the patients.

Anorexia was reported in 1.5% of patients treated with zoledronic acid 4mg. Few cases of rash or pruritus have been observed (below 1%). As with other bisphosphonates, cases of conjunctivitis in approximately 1% have been reported. There have been some reports of impaired renal function (2.3%), although the aetiology appears to be multifactorial in many cases. Based on pooled analysis of placebo controlled studies, severe anaemia (Hb<8.0g/dl) was reported in 5.2% of patients receiving zoledronic acid 4mg versus 4.2% on placebo.

Other adverse drug reactions are listed in Table G.2.

- **Overdose**

There is no experience of acute intoxication with zoledronic acid. Patients who have received doses higher than those recommended should be carefully monitored. In the event of clinically significant hypocalcaemia, reversal may be achieved with an infusion of calcium gluconate.

DOCETAXEL

- **Contraindications**

Docetaxel is contraindicated in those patients with a known hypersensitivity to docetaxel or to any of the excipients and should not be used in patients with absolute neutrophil counts $<1.5 \times 10^9/l$ or if there is severe liver impairment.

- **Special Warnings and Precautions**

A premedication consisting of an oral corticosteroid is required (see **Appendix F**).

Neutropenia is the most frequent adverse reaction of docetaxel. Neutrophil nadirs occur at a median of seven days. Frequent monitoring of complete blood counts should be conducted on all patients receiving docetaxel.

In the case of severe neutropenia (<500 cells/mm³ for seven days or more) during a course of docetaxel therapy, a reduction in dose for subsequent courses is recommended (see **Appendix F**).

- **Hypersensitivity reactions**

Patients should be observed closely for hypersensitivity reactions especially during the first and second infusions. Hypersensitivity reactions may occur within a few minutes following the initiation of the infusion of docetaxel, thus facilities for the treatment of hypotension and bronchospasm should be available. If hypersensitivity reactions occur, minor symptoms such as flushing or localised cutaneous reactions do not require interruption of therapy. However, severe reactions, such as severe hypotension, bronchospasm or generalised rash/erythema require immediate discontinuation of docetaxel and appropriate therapy, (see **Appendix F**). Patients who have developed severe hypersensitivity reactions should not be re-challenged with docetaxel.

- **Cutaneous reactions**

Localised skin erythema of the extremities (palms of the hands and soles of the feet) with oedema followed by desquamation has been observed. Severe symptoms such as eruptions followed by desquamation which lead to interruption or discontinuation of docetaxel treatment have been reported

- **Fluid retention**

Patients with severe fluid retention such as pleural effusion, pericardial effusion and ascites should be monitored closely.

- **Nervous system**

The development of severe peripheral neurotoxicity requires a reduction of dose (see **Appendix F**)

- **Interaction with other medicinal products and other forms of interaction**

In vitro studies have shown that the metabolism of docetaxel may be modified by the concomitant administration of compounds which induce, inhibit or are metabolised by (and thus may inhibit the enzyme competitively) cytochrome P450-3A such as cyclosporine, terfenadine, ketoconazole, erythromycin and troleandomycin. As a result, caution should be exercised when treating patients with these drugs as concomitant therapy since there is a potential for a significant interaction.

Undesirable effects

- **Blood disorders**

The most commonly reported adverse reaction was neutropenia, which was reversible and not cumulative. The median time to nadir was 7 days and the median duration of severe neutropenia (<500 cells/mm³) was 7 days.

Bone marrow suppression and other hematologic adverse reactions have also been reported.

- **Hypersensitivity reactions**

Hypersensitivity reactions have generally occurred within a few minutes following the start of the infusion of docetaxel and were usually mild to moderate. The most frequently reported symptoms were flushing, rash with or without pruritus, chest tightness, back pain, dyspnoea and drug fever or chills. Severe reactions, characterised by hypotension and/or bronchospasm or generalized rash/erythema, resolved after discontinuing the infusion and appropriate therapy (see **Appendix F**).

- **Skin and subcutaneous tissue disorders**

Reversible cutaneous reactions have been observed and were generally considered as mild to moderate. Reactions were characterised by a rash including localised eruptions mainly on the feet and hands, but also on the arms, face or thorax, and frequently associated with pruritus. Eruptions generally occurred within one week after the docetaxel infusion. Less frequently, severe symptoms such as eruptions followed by desquamation which rarely lead to interruption or discontinuation of docetaxel treatment were seen. Severe nail disorders are characterised by hypo- or hyperpigmentation and sometimes pain and onycholysis. Very rare cases of bullous eruption such as erythema multiforme or Stevens-Johnson syndrome have been reported with docetaxel and other concomitant factors may have contributed to the development of these effects.

- **Fluid retention**

Events such as peripheral oedema and less frequently pleural effusion, pericardial effusion, ascites and weight gain have been reported. The peripheral oedema usually starts at the lower extremities and may become generalised with a weight gain of 3 kg or more. Fluid retention is cumulative in incidence and severity. Fluid retention has not been accompanied by acute episodes of oliguria or hypotension. Dehydration and pulmonary oedema have rarely been reported.

- **Gastrointestinal disorders**

Nausea, Stomatitis, Vomiting, Diarrhoea and Constipation may occur.

- **Nervous system disorders**

The development of severe peripheral neurotoxicity requires a reduction of dose (see **Appendix F**). Mild to moderate neuro-sensory signs are characterised by paraesthesia, dysaesthesia or pain including burning. Neuro-motor events are mainly characterised by weakness.

- **General disorders and administration site conditions**

Infusion site reactions were generally mild and consisted of hyperpigmentation, inflammation, redness or dryness of the skin, phlebitis or extravasation and swelling of the vein.

Other undesirable effects are described in table G.2.

- **Overdose**

There were a few reports of overdose. There is no known antidote for docetaxel overdose. In case of overdose, the patient should be kept in a specialised unit and vital functions closely monitored. The primary anticipated complications of overdose would consist of bone marrow suppression, peripheral neurotoxicity and mucositis. Patients should receive therapeutic G-CSF as soon as possible after discovery of overdose. Other appropriate symptomatic measures should be taken, as needed.

CELECOXIB

- **Contraindications**

Celecoxib is contraindicated if the patient has a known hypersensitivity to the active substance or to any of the excipients or sulphonamides. It is also contraindicated in patients who have experienced asthma, acute rhinitis, nasal polyps, angioneurotic oedema, urticaria or other allergic-type reactions after taking acetylsalicylic acid or NSAIDs.

It is also contraindicated if the patient has active peptic ulceration, gastrointestinal bleeding, inflammatory bowel disease, severe congestive heart failure and severe hepatic disease.

- **Special Warnings and Precautions for use**

Upper gastrointestinal perforations, ulcers or bleeds have occurred in patients, some of them fatal, treated with celecoxib. Therefore, caution should be taken in patients with a history of gastrointestinal disease, such as ulceration and inflammatory conditions and in patients at special risk.

As with other drugs known to inhibit prostaglandin synthesis fluid retention and oedema have been observed in patients taking celecoxib. Therefore, celecoxib should be used with caution in patients with history of cardiac failure, left ventricular dysfunction or hypertension, and in patients with pre-existing oedema from any other reason, since prostaglandin inhibition may result in deterioration of renal function and fluid retention. Caution is also required in patients taking diuretic treatment or otherwise at risk of hypovolaemia.

In patients on concurrent therapy with warfarin, serious bleeding events have occurred. Caution should be exercised when combining celecoxib with warfarin (See Interactions).

Celecoxib inhibits CYP2D6. Although it is not a strong inhibitor of this enzyme, a dose reduction may be necessary for individually dose-titrated drugs that are metabolised by CYP2D6 (See below).

- **Interaction with other medicinal products and other forms of interaction.**

Pharmacodynamic interactions

Anticoagulant activity should be monitored in patients taking warfarin or similar agents, particularly in the first few days after initiating or changing the dose of celecoxib. Bleeding events in association with increases in prothrombin time have been reported, predominantly in the elderly, in patients receiving celecoxib concurrently with warfarin, some of them fatal.

NSAIDs may reduce the effect of diuretics and antihypertensive drugs. As for NSAIDs, the risk of acute renal insufficiency may be increased when ACE inhibitors are combined with celecoxib.

Coadministration of NSAIDs and cyclosporin or tacrolimus have been suggested to increase the nephrotoxic effect of cyclosporin and tacrolimus. Renal function should be monitored when celecoxib and any of these drugs are combined.

Celecoxib can be used with low dose acetylsalicylic acid but is not a substitute for acetylsalicylic acid for cardiovascular prophylaxis.

Pharmacodynamic interactions

- *Effects of celecoxib on other drugs*

Celecoxib is an inhibitor of CYP2D6. During celecoxib treatment, the plasma concentrations of the CYP2D6 substrate dextromethorphan were increased by 136%. The plasma concentrations of drugs that are substrates of this enzyme may be increased when celecoxib is used concomitantly. Examples of drugs which are metabolised by CYP2D6 are antidepressants (tricyclics and SSRIs), neuroleptics, anti-arrhythmic drugs, etc. The dose of individually dose-titrated CYP2D6 substrates may need to be reduced when treatment with celecoxib is initiated or increased if treatment with celecoxib is terminated.

In vitro studies have shown some potential for celecoxib to inhibit CYP2C19 catalysed metabolism. The clinical significance of this *in vitro* finding is unknown. Examples of drugs which are metabolised by CYP2C19 are diazepam, citalopram and imipramine.

In healthy subjects, co-administration of celecoxib 200mg twice daily with 450mg twice daily of lithium resulted in a mean increase in C_{max} of 16% and in AUC of 18% of lithium. Therefore, patients on lithium treatment should be closely monitored when celecoxib is introduced or withdrawn.

- *Effects of other drugs on celecoxib*

Since celecoxib is predominantly metabolised by CYP2C9 it should be used at half the recommended dose in patients receiving fluconazole. Concomitant use of 200mg single dose of celecoxib and 200mg once daily of fluconazole, a potent CYP2C9 inhibitor, resulted in a mean increase in celecoxib C_{max} of 60% and in AUC of 130%. Concomitant use of inducers of CYP2C9 such as rifampicin, carbamazepine and barbiturates may reduce plasma concentrations of celecoxib.

Ketoconazole or antacids have not been observed to affect the pharmacokinetics of celecoxib.

• Undesirable Effects

Approximately 7,400 patients have been treated with celecoxib in controlled trials and of those approximately 2,300 have received it for 1 year or longer. The common reactions, those occurring in less than 1 in 10 patients are listed below. Other reactions are summarised in Table G.2.

Common

Body as a whole: peripheral oedema/ fluid retention.

Gastrointestinal: abdominal pain, diarrhoea, dyspepsia, flatulence.

Nervous system: dizziness.

Psychiatric: insomnia.

Respiratory: pharyngitis, rhinitis, sinusitis, upper respiratory tract infection.

Skin: rash.

• Overdose

There is no clinical experience of overdose. Single doses up to 1,200 mg and multiple doses up to 1,200 mg twice daily have been administered to healthy subjects for nine days without clinically significant adverse effects. In the event of suspected overdose, appropriate supportive medical care should be provided e.g. by eliminating the gastric contents, clinical supervision and, if necessary, the institution of symptomatic treatment. Dialysis is unlikely to be an efficient method of drug removal due to high protein binding.

Table G.2: Comparative table of symptom frequency by drug and by system

Adverse reactions listed using the following convention:

- very common (>1/10)
- common (>1/100, <1/10)
- uncommon (>1/1,000, <1/100)
- rare (>1/10,000, <1/1,000)
- very rare (<1/10,000).

System	Symptom	Docetaxel	Celecoxib	Zoledronic acid
Allergic Reactions	Anaphylactic shock	-	Very rare	-
	Angioedema	-	Very rare	Rare
	Hypersensitivity reactions	Common	-	Uncommon
	Myositis	-	Very rare	-
	Severe allergic reactions	-	Very rare	-
	Vasculitis	-	Very rare	-
Auditory	Decreased hearing	-	Very rare	-
	Tinnitus	-	Uncommon	-
Blood & Lymph	Anaemia	Very common	Uncommon	Common

System	Symptom	Docetaxel	Celecoxib	Zoledronic acid
	Febrile neutropaenia	Common	-	-
	Leukopaenia	-	Rare	Uncommon
	Neutropaenia	Very common	-	-
	Pancytopenia	-	Very rare	Rare
	Thrombocytopenia	Very common	Rare	Uncommon
Cardiac Disorders	Bradycardia	-	-	Rare
	Cardiac dysrhythmia	Common	Uncommon	-
	Congestive heart failure	-	Very rare	-
	Heart failure	-	Very rare	-
	Hypertension	-	Uncommon	Uncommon
	Hypotension	Common	-	-
	Myocardial infarction	-	Very rare	-
Gastrointestinal Disorders	Acute pancreatitis	-	Very rare	-
	Abdominal pain	-	Common	Uncommon
	Constipation	Common	Uncommon	Uncommon
	Diarrhoea	Very common	Common	Uncommon
	Dry mouth	-	-	Uncommon
	Duodenal/gastric/oesophageal ulceration	-	Rare	-
	Dyspepsia	-	Common	Uncommon
	Dysphagia	-	Rare	-
	Eructation	-	Uncommon	-
	Flatulence	-	Common	-
	Gastritis	-	Uncommon	-
	GI haemorrhage	-	Very rare	-
	Intestinal perforation	-	Rare	-
	Melaena	-	Rare	-
	Nausea	Very common	-	Common
	Oesophagitis	-	Rare	-
	Stomatitis	Very common	Uncommon	Uncommon
	Vomiting	Very common	Uncommon	Common
General Disorders	Asthenia	Very common	-	Uncommon
	Chest pain	-	-	Uncommon
	Fever	-	-	Common
	Flu-like symptoms	-	-	Common
	Injection site reactions	-	-	Uncommon
	Lethargy	Common	Uncommon	-
	Oral candidiasis	Common	-	-
	Pain	Very common	-	-
	Pain in limb	Common	Uncommon	-
	Radiation recall phenomena	Rare	-	-
	Weight increase	-	-	Uncommon
Hepatic Disorders	Abnormal hepatic function	-	Uncommon	-
	G3/4 Bilirubin increase (<2%)	Common	-	-
	Hepatitis	-	Very rare	-
	Increased SGOT	-	Uncommon	-
	Increased SGPT	-	Uncommon	-
	Jaundice	-	Very rare	-
Laboratory Abnormalities	Hyperkalaemia	-	-	Rare
	Hypernatraemia	-	-	Rare
	Hypocalcaemia	-	-	Common
	Hypokalaemia	-	-	Uncommon
	Hypomagnesaemia	-	-	Uncommon
	Hypophosphataemia	-	-	Very Common
Metabolism and Nutrition Disorders	Abnormal renal function tests (increased creatinine, BUN, hyperkalaemia)	-	Uncommon	Common
	Anorexia	Very common	-	Common
	Taste alteration	-	Rare	Uncommon
Musculoskeletal Disorders	Arthralgia	-	-	Common
	Bone pain	-	-	Common
	Generalised pain	-	-	Common
	Muscle cramps	-	-	Uncommon
	Myalgia	Common	Uncommon	Common
Nervous System Disorders	Ataxia	-	Rare	-
	Aggravated epilepsy	-	Very rare	-
	Dizziness	-	Common	Uncommon
	Headache	-	-	Common
	Hyperaesthesia	-	-	Uncommon

System	Symptom	Docetaxel	Celecoxib	Zoledronic acid
	Hypertonia	-	Uncommon	-
	Hypoaesthesia	-	-	Uncommon
	Paraesthesia	-	Uncommon	Uncommon
	Tremor	-	-	Uncommon
Ocular	Blurred vision	-	Uncommon	Uncommon
	Conjunctivitis	-	-	Common
Peripheral Oedema/Fluid Retention	Fluid retention	Very common	Common	Uncommon
Psychiatric Disorders	Anxiety	-	Uncommon	Uncommon
	Confusion	-	Very rare	Rare
	Depression	-	Uncommon	-
	Hallucinations	-	Very rare	-
	Insomnia	-	Common	Uncommon
Renal	Abnormal renal function tests (increased creatinine, BUN, hyperkalaemia)	-	Uncommon	Common
	Acute renal failure	-	Very rare	Uncommon
	Haematuria	-	-	Uncommon
	Interstitial nephritis	-	Very rare	-
	Proteinuria	-	-	Uncommon
	Renal impairment	-	-	Common
	Urinary tract infection	-	Uncommon	-
Respiratory Disorders	Bronchospasm	-	Very rare	-
	Coughing	-	Uncommon	Uncommon
	Dyspnoea	-	Uncommon	Uncommon
	Pharyngitis	-	Common	-
	Rhinitis	-	Common	-
	Sinusitis	-	Common	-
	Upper respiratory tract infection	-	Common	-
Skin/Subcutaneous Tissue Disorders	Alopecia	Very common	Rare	-
	Cutaneous reaction	Very common	-	-
	Epidermal necrolysis	-	Very rare	-
	Erythema multiforme	-	Very rare	-
	Increased sweating	-	-	Uncommon
	Nail changes	Common	-	-
	Photosensitivity	-	Rare	-
	Pruritus	-	-	Uncommon
	Rash	-	Common	Uncommon
	Stevens-Johnson syndrome	-	Very rare	-
	Urticaria	-	Uncommon	-

Appendix H - Health Care at Home Study

The possibility of Zoledronic acid infusions being administered at home if patients prefer, by trained nurses, is currently being discussed with Novartis.

Appendix I: Evaluation of Baseline Lesions and New Lesions

These guidelines are based on the RECIST criteria (38) but have been modified to include progression based on PSA measurements.

Measurable Disease

Measurable disease is defined as at least one lesion that can be accurately measured in at least one dimension (longest dimension to be recorded). Each lesion must ≥ 20 mm when measured by conventional techniques, including palpation, CT, and MRI, or ≥ 10 mm when measured by spiral CT.

Baseline documentation of “Target” and Non-Target lesions

All measurable lesions up to a maximum of 5 lesions per organ and 10 lesions in total representative of all involved organs should be identified as target lesions and will be recorded and measured at baseline. Target lesions should be selected on the basis of their size (lesions with the longest dimension) and their suitability for accurate repetitive measurements by one consistent method of assessment (either by imaging techniques or clinically). A sum of the longest dimension (LD) for all target lesions will be calculated and reported as the baseline sum LD. The baseline sum LD will be used as reference to further characterise

All other lesions (or sites of disease) should be identified as non-target lesions and should also be recorded at baseline. Measurements are not required.

All baseline evaluations of disease status should be performed as close as possible to the start of the treatment and not more than 4 weeks before the beginning of the treatment.

Definition of Progression

(for patients with measurable disease at randomisation)

Progression is defined as ANY of the following:-

- At least a 20% increase in the sum of LD target lesions taking as reference the smallest sum LD recorded since study entry
- The appearance of one or more new lesions
- Death due to disease without prior objective documentation of progression
- Global deterioration in health status attributable to the disease requiring a change in therapy without objective evidence of progression

- Unequivocal progression of non-target lesions, (other than pleural effusions without cytological proof of neoplastic origin) in the opinion of the treating physician (in this case an explanation must be provided).
- Progression will be most usually based on PSA measurements (see **Appendix K**) but tumour measurements should take precedence over PSA response. If measurable disease is shrinking during treatment, but the PSA is rising the patient should continue to receive protocol treatment.

Definition of Progression

(for patients with non-measurable disease at randomisation)

Progression (for patients with non-measurable disease at randomisation) is defined as increasing clinical or radiological evidence of disease since study entry. Progression can also be based on PSA measurements (**Appendix K**).

Appendix J - Common Toxicity Criteria

The following table shows the known side effects and how they relate to the CTCv3 Toxicity heading. The complete CTCv3 can be found at the following web site - <http://ctep.cancer.gov/reporting/ctc.html>

System	Symptom	CTC v3 Toxicity Grade Heading/Adverse Event
Allergic Reactions	Anaphylactic shock	Allergy/Immunology - Allergic reaction/hypersensitivity
	Angioedema	Dermatology - Urticaria
	Hypersensitivity reactions	Allergy/Immunology - Allergic reaction/hypersensitivity
	Myositis	Musculoskeletal - Myositis
	Severe allergic reactions	Allergy/Immunology - Allergic reaction/hypersensitivity
	Vasculitis	Allergy/Immunology - Allergic reaction/hypersensitivity
Auditory	Decreased hearing	Auditory/Ear -Hearing
	Tinnitus	Auditory/Ear - Tinnitus
Blood & Lymph	Anaemia	Blood/Bone Marrow - Haemoglobin
	Febrile neutropaenia	Infection - Febrile neutropaenia
	Leukopaenia	Blood/Bone Marrow - Leukocytes
	Bone Marrow depletion	Blood/Bone Marrow - Bone marrow cellularity
	Infection	Infection - Other (<i>Specify</i>)
	Neutropaenia	Blood/Bone Marrow - Neutrophils
	Other Haematology	Blood/Bone Marrow - Other (<i>Specify</i>)
	Pancytopenia	Blood/Bone Marrow - See Haemoglobin, Leukocytes, Platelets
	Thrombocytopenia	Blood/Bone Marrow - Platelets
Cardiac Disorders	Bradycardia	Cardiac Arrhythmia - Supraventricular and nodal arrhythmia
	Cardiac dysrhythmia/Palpitations	Cardiac Arrhythmia - Cardiac arrhythmia
	Congestive heart failure	Cardiac General - Cardiac ischemia/infarction
	Heart failure	Cardiac General - Cardiac ischemia/infarction
	Hypertension	Cardiac General - Hypertension
	Hypotension	Cardiac General - Hypotension
	Myocardial infarction	Cardiac General - Cardiac ischemia/infarction
	Pericardial effusion	Cardiac General - Pericardial effusion
	Venous thromboembolic events	Vascular - Thrombosis/thrombus/embolism
Gastrointestinal Disorders	Ascites	Gastrointestinal - Ascites
	Acute pancreatitis	Hepatobiliary/Pancreas - Pancreatitis
	Abdominal pain	Pain - Abdomen

System	Symptom	CTC v3 Toxicity Grade Heading/Adverse Event
	Colitis	Gastrointestinal - Colitis
	Constipation	Gastrointestinal - Constipation
	Dehydration	Gastrointestinal - Dehydration
	Diarrhoea	Gastrointestinal - Diarrhoea
	Dry mouth	Gastrointestinal - Dry mouth
	Ulceration (Duodenal/gastric/ Oesophageal)	Gastrointestinal - Ulcer (<i>Select site</i>)
	Dyspepsia	Gastrointestinal - Dyspepsia
	Dysphagia	Gastrointestinal - Dysphagia
	Eructation	Gastrointestinal - Dyspepsia
	Flatulence	Gastrointestinal - Flatulence
	Gastritis	Gastrointestinal - Gastritis
	GI haemorrhage	Haemorrhage/Bleeding - GI
	Intestinal perforation	Gastrointestinal - Perforation GI (<i>Select site</i>)
	Melaena	Haemorrhage/Bleeding - GI
	Nausea	Gastrointestinal - Nausea
	Neutropaenic enterocolitis	Infection - GI
	Obstruction	Gastrointestinal - Obstruction GI (<i>Select site</i>)
	Oesophagitis	Gastrointestinal - Oesophagitis
	Stomatitis	Gastrointestinal - Mucositis/Stomatitis
	Vomiting	Gastrointestinal - Vomiting
General Disorders	Asthenia	Constitutional Symptoms - Asthenia
	Back pain	Pain - Back
	Chest pain	Pain - Chest
	Chills	Constitutional Symptoms - Rigors/chills
	Fever	Constitutional Symptoms - Fever
	Flu-like symptoms (including fatigue, rigors, malaise and flushing)	Syndromes - Flu-like syndrome
	Injection/administration site reactions (including pain, irritation, swelling and induration)	Dermatology/Skin - Injection site reaction/extravasation changes
	Lethargy	Constitutional Symptoms - Fatigue
	Oral candidiasis	Infection - Oral cavity
	Pain	Pain - (<i>Select site</i>)
	Pain in limb	Pain - Extremity – limb
	Radiation recall phenomena	Dermatology/Skin - Rash (chemo/radiation)
	Weight increase (see fluid retention)	Constitutional Symptoms - Weight gain
Hepatic Disorders	Abnormal hepatic function	Hepatobiliary/Pancreas - Other (<i>Specify</i>)

System	Symptom	CTC v3 Toxicity Grade Heading/Adverse Event
	Bilirubin increase	Metabolic/Laboratory - Bilirubin
	Hepatitis	Infection - Viral hepatitis/Infection (<i>Select</i>)
	Increased SGOT	Metabolic/Laboratory - AST, SGOT (serum glutamic oxaloacetic transaminase)
	Increased SGPT	Metabolic/Laboratory - ALT, SGPT (serum glutamic pyruvic transaminase)
	Jaundice	Metabolic/Laboratory - Bilirubin
Laboratory Abnormalities	Hyperkalaemia	Metabolic/Laboratory - Hyperkalaemia
	Hypernatraemia	Metabolic/Laboratory - Hypernatraemia
	Hypocalcaemia	Metabolic/Laboratory - Hypocalcaemia
	Hypokalaemia	Metabolic/Laboratory - Hypokalaemia
	Hypomagnesaemia	Metabolic/Laboratory - Hypomagnesaemia
	Hypophosphataemia	Metabolic/Laboratory - Hypophosphataemia
Metabolism and Nutrition Disorders	Abnormal renal function tests (increased creatinine, Blood Urea Nitrogen, hyperkalaemia)	Renal/Genitourinary - Other (<i>Specify</i>) See also: Metabolic/Laboratory - Creatinine/hyperkalaemia/hyperuricaemia
	Anorexia	Gastrointestinal - Anorexia
	Taste alteration	Gastrointestinal - Taste alteration
Musculoskeletal Disorders	Arthralgia	Pain - Joint
	Bone pain	Pain - Bone
	Generalised pain	Pain - (<i>Select site</i>)
	Muscle cramps	Pain - Muscle
	Myalgia	Pain - Muscle
Nervous System Disorders	Ataxia	Neurology - Ataxia
	Aggravated epilepsy	Neurology - Seizure
	Burning pain	Neurology - Neuropathy sensory
	Convulsion	Neurology - Seizure
	Dizziness	Neurology - Dizziness
	Dysaesthesia	Neurology - Neuropathy sensory
	Headache	Pain - Headache
	Hyperaesthesia	Neurology - Neuropathy sensory
	Hypertonia	Neurology - Pyramidal tract dysfunction
	Hypoaesthesia	Neurology - Neuropathy sensory
	Loss of Consciousness (transient)	Neurology - Syncope
	Paraesthesia	Neurology - Neuropathy sensory
	Tremor	Neurology - Tremor
	Weakness	Neurology - Neuropathy motor
Ocular	Blurred vision	Ocular/Visual - Blurred vision

System	Symptom	CTC v3 Toxicity Grade Heading/Adverse Event
	Conjunctivitis	Infection - Eye
	Lacrimal duct obstruction	Ocular/Visual - Other (<i>Specify</i>)
	Visual disturbance	Ocular/Visual - Blurred vision, Diplopia, Flashing lights
Peripheral Oedema/Fluid Retention	Fluid retention	Lymphatics - Oedema, limb
Psychiatric Disorders	Anxiety	Neurology - Mood alteration
	Confusion	Neurology - Confusion
	Depression	Neurology - Mood alteration
	Hallucinations	Neurology - Psychosis
	Insomnia/sleep disturbance	Constitutional Symptoms - Insomnia
Renal	Abnormal renal function tests (increased creatinine, Blood Urea Nitrogen, hyperkalaemia)	Renal/Genitourinary - Other (<i>Specify</i>) See also: Metabolic/Laboratory - Creatinine/hyperkalaemia/hyperuricaemia
	Acute renal failure	Renal/Genitourinary - Renal failure
	Haematuria	Haemorrhage/Bleeding - GU
	Interstitial nephritis	Infection - Kidney
	Proteinuria	Metabolic/Laboratory - Proteinuria
	Renal impairment	Renal/Genitourinary - Other (<i>Specify</i>)
	Urinary tract infection	Infection - Urinary tract
Respiratory Disorders	Bronchospasm	Pulmonary/Upper Respiratory -Bronchospasm
	Chest tightness	Pain - Chest/thorax
	Coughing	Pulmonary/Upper Respiratory -Cough
	Dyspnoea	Pulmonary/Upper Respiratory - Dyspnoea
	Pharyngitis	Infection - Pharynx
	Pleural effusion	Pulmonary/Upper Respiratory - Pleural effusion
	Pulmonary oedema	Cardiac General - Left ventricular diastolic dysfunction, Left ventricular systolic dysfunction
	Rhinitis	Infection - Nose
	Sinusitis	Infection - Sinus
	Upper respiratory tract infection	Infection - Upper airway
Skin/Subcutaneous Tissue Disorders	Alopecia	Dermatology/Skin - Alopecia
	Cutaneous reaction	Dermatology/Skin - Rash
	Desquamation	Dermatology/Skin - Rash
	Epidermal necrolysis	Dermatology/Skin - Erythema multiforme etc.
	Erythema multiforme	Dermatology/Skin - Erythema multiforme etc.
	Flushing	Dermatology/Skin - Flushing
	Increased sweating	Constitutional Symptoms - Sweating

System	Symptom	CTC v3 Toxicity Grade Heading/Adverse Event
	Nail changes	Dermatology/Skin - Nail changes
	Photosensitivity	Dermatology/Skin - Photosensitivity
	Pruritus	Dermatology/Skin - Pruritis
	Rash	Dermatology/Skin - Rash
	Stevens-Johnson syndrome	Dermatology/Skin - Erythema multiforme etc.
	Urticaria	Dermatology/Skin - Urticaria

Appendix K - Definition of Biochemical Failure

Biochemical failure: The initial response to hormonal therapy for prostate cancer can be variable. A few patients exhibit only a small fall in PSA, with little evidence of clinical response. At the other extreme a rapid fall to the normal range or even undetectable levels of PSA occurs. In a group of patients the response lies between these two extremes. The rate of fall of PSA and the level of the PSA nadir are recognised to have prognostic significance. Comparison of PSA responses between the treatment groups could be used as secondary data for confirming the response rate of the trial treatments. However, in defining PSA relapse, the extent of the primary response has to be taken into account. Three groups of patients will be defined:-

- A. If the PSA nadir is more than 50% of the last pre-treatment PSA, the patient should be defined as a treatment failure (at time zero).
- B. For patients whose PSA falls by more than 50% of the last pre-treatment PSA, but remains above 4ng/ml, PSA relapse will be deemed to have occurred when PSA is confirmed as increasing by 50% above the nadir level.
- C. For patients whose PSA falls below 4ng/ml, PSA relapse will be defined by either 50% increase from their nadir or the PSA increasing above 4, whichever is the greater. For example, a nadir PSA of 3.6 would require a PSA of 5.4 to define relapse, while PSA nadir of 2.5 will be considered to have relapsed at a PSA of 4.

Note: this is a definition of PSA progression based on a previous study of strontium.

Timing of PSA tests: All patients will commence hormone therapy shortly after randomisation. The final pre-randomisation PSA must be measured within 4 weeks prior to randomisation. Once on trial, PSA tests will be performed at 6, 12, 18 and 24 weeks from randomisation then at each follow-up visit (3 monthly until 3 years, then 6 monthly).

Nadir PSA: The PSA nadir will be the lowest reported PSA level between randomisation and the 24 weeks PSA assessment (i.e. assessments at 6, 12, 18 and 24 weeks). The critical value that would constitute subsequent biochemical progression will be calculated from this nadir value. Once the 6 month PSA level has been recorded and sent promptly to the MRC CTU, a letter will be sent to the responsible clinician confirming the PSA level which would be taken as progression.

Confirming failure-free survival (including biochemical failure): In applying the definitions of Failure-free survival (including biochemical failure) above, the increase in PSA must be confirmed (i.e. at least two readings are required). The following approach will be applied:

1. Patient reaches nadir PSA and continues with 3-monthly PSA assessments. [Any PSA measurement taking place within 6 weeks of a change in hormone therapy or a manipulation (e.g. cystoscopy) that could produce a spurious increase in PSA is to be ignored].
2. At a subsequent assessment (say, assessment X), the PSA value is noted to have risen above the critical value, but patient is symptom free.
3. Patient should be recalled for a confirmatory PSA test between one week and 3 months later. If this value is at the same level or higher, the patient will be considered to have reached the biochemical failure endpoint.

Confirmatory step 3 will not be required if the patient has demonstrable local progression, new metastases or death from prostate cancer during the intervening period.

Time to failure-free survival (including biochemical failure): Patients in Group A (treatment failure) will be recorded as events at time zero. The time to FFS for patients in Groups B and C will be from randomisation to the first PSA above the critical value (i.e. assessment X).

Non-protocol treatment: No new non-protocol treatment for prostate cancer should be administered until trial progression has been confirmed.

PSA Monitoring Group: Consideration will be given to the formation of such a group, whose remit would be to monitor the accurate recording of PSA data, and independently assess failure outcomes, blinded of the treatment allocation. This would be, by definition, an independent group involving clinicians not recruiting patients to STAMPEDE.

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To be appointed

Independent Data Monitoring Committee

To be appointed

Appendix M - Participating Site Accreditation Form

The Site accreditation form will be inserted here.

Appendix N - Quality of life and health economics

1. Overview

The economic evaluation will take the form of a cost-effectiveness analysis in which the differential cost of the alternative treatments will be related to their differential benefits in terms of quality-adjusted life years (QALYs). Incremental analysis will be undertaken and cost-effectiveness acceptability curves will be used to show the probability of one option being more cost-effective than the others.

2. Estimating costs

A cost analysis will be undertaken from the perspective of the National Health Service. Resource use measurement during the trial will be divided into that relating to the hospital and NHS non-hospital. These are dealt with in turn below.

2.1 Hospital resource use

Within the trial, hospital resource use data will be collected on all patients entering the trial. Specifically, this will include in-patient nights in hospital, distinguishing intensive care from stay on a general ward. These will be collected using case record forms completed at clinical review at the follow-up points shown in Figure 1 of the main document. Some visits to, and stays in, hospital may relate to non-study hospitals. To ensure that data on this form of resource use are captured, a questionnaire will be administered to patients as part of the quality of life assessments.

These resources will be valued in monetary terms using unit costs representative of UK practice at the time of analysis. For drugs, this will be based on British National Formulary prices. For hospital procedure and hotel costs, unit costs will, if available, be based on NHS Reference Costs. Otherwise, they will be estimated from a sample of UK centres randomising patients into the trial.

2.2 NHS non-hospital resource use

Data on patients' use of community-based NHS (and complementary health) services will be collected from patients as part of the quality of life assessments. The resources will include visits to and from a GP or district nurse. Costing of community-based resources will be based on published unit costs². Other services will be costed using data available at the point of analysis.

3. Measuring effects

The clinical trial is estimating a range of clinical and health-related quality of life (HRQL) effects in trial patients. The purpose of the economic evaluation will be to set these in context of the resource costs incurred in achieving them. A cost-effectiveness analysis will

relate differential cost to an aggregated measure of effect in the form of a QALY. QALYs will be based on observed mortality and patients' responses to the EQ-5D questionnaire. The latter asks patients to categorise their health, with 3 levels of response (no problems, moderate problems, severe problems) on 5 dimensions (mobility, self care, usual activities, pain/discomfort, depression anxiety) Ref 39. Each of the 245 possible health states has been 'valued' on a zero to one 'utility' based on the preferences of 3,395 members of the UK public (ref. 40).

4. Analysis

All resource use data will be valued in monetary terms as described above such that each patient has a cost over the period of follow-up. A full stochastic analysis will be undertaken to allow for sample variation in resource use and effect data. Methods are developing quickly in this area and, by the time of the analysis, 'best practice' may have altered markedly from today. If such an analysis were to be undertaken now, the general methods would be as follows.

A QALY profile will be estimated for each patient based on their survival duration weighted by their responses to the EQ-5D HRQL questionnaire, which generates a single index value for health at each point of follow-up⁶. The profiles will assume a straight-line relationship between the index value at time t and the value at time $t+1$. The number of QALYs they experience during the period of follow-up in the trial will be the area under the QALY profile. In the primary analysis, only data collected in the trial will be used in the analysis; in other words, the estimate of QALYs for each group is likely to reflect the fact that some patients are still alive after the follow-up (i.e. the survival curve is truncated and survival analysis techniques will be used to estimate QALYs).

As a secondary analysis, extrapolation techniques will be used to estimate the final portion of the survival curve so as to provide a full estimate of differential life expectancy. A number of extrapolation techniques will be used to provide a range of estimates of differential QALYs over a lifetime time horizon.

Cost-effectiveness acceptability curves will be used to facilitate a measure of uncertainty around cost-effectiveness estimates¹. These curves show the probability of one form of management being more cost-effective than the others assuming alternative levels of the maximum amount decision-makers are willing to pay for an extra QALY.

Sensitivity analysis will be used to consider the importance of sources of uncertainty other than sample variation (e.g. unit costs, discount rates, method of extrapolation). Multiple regression techniques will be employed to provide as precise a measure of cost-

effectiveness as possible and to undertake sub-group analysis using baseline patient characteristics which will be defined in advance in the analysis plan.

Appendix O - Case report forms

The case report forms will be inserted here.

Back cover

STAMPEDE: Trial Summary

ELIGIBILITY CRITERIA

Newly diagnosed patients:

- ⇒ Histologically confirmed prostate adenocarcinoma **or**
- ⇒ multiple sclerotic bone metastases with PSA $\geq 100\text{ng/ml}$
- ⇒ stage T3/4 N0/ M0 **or** T₁₋₂ N1 M0 **or** T₁₋₂ N₁₋₂ M1

Patients previously treated with radical surgery or RT:

- ⇒ Histologically confirmed prostate adenocarcinoma
- ⇒ PSA $>4\text{ng/ml}$ **and** rising with doubling time $<6\text{months}$ **or** PSA $>20\text{ng/ml}$
- ⇒ Prior systemic therapy for **localised** disease must be completed $>12\text{m}$ previously, have been no longer than 8m duration and given as neoadjuvant or adjuvant therapy only

All Patients:

- ⇒ Intention to treat with long term androgen ablation
- ⇒ Fit for all protocol treatment and follow up, WHO performance status 0-2.
- ⇒ Written informed consent.

TRIAL ARMS

RANDOMISE

→ A	Control: Androgen suppression (AS) (according to local practice)
→ B	AS + Bisphosphonate (Zoledronic Acid): 4mg monthly for 6 months, then 4mg every 3 months until disease (PSA) progression
→ C	AS + Docetaxel: 75mg/m ² D1 as 1hr iv infusion, plus Prednisolone 5mg bd continuously, every 3 weeks for 6 cycles.
→ D	AS + COX-2 Inhibitor (Celecoxib): 400mg bd until disease (PSA) progression
→ E	AS + Zoledronic Acid + Docetaxel: as above
→ F	AS + Zoledronic Acid + Celecoxib: as above

To Randomise a patient into STAMPEDE:

- ⇒ Phone: 020 7670 4777
- ⇒ Fax: 020 7670 4814

Monday-Friday 09:00 – 17:00

TRIAL STAGES

Pilot phase:

- ⇒ Limited to a small number of centres
- ⇒ Accrual: Approx 210 patients over a one year period
- ⇒ To confirm safety and feasibility of each research arms when used in combination with AS.

Efficacy stage 1:

- ⇒ Open to all centres
- ⇒ Accrual: Approx 2700 patients (over 5.5 years)
- ⇒ Intermediate outcome measure = Failure Free Survival (FFS), including biochemical failure.

Efficacy stage 2:

- ⇒ Open to all centres
- ⇒ Accrual - to continue for approx 1 further year (to 6.5 years), bringing the total to approx 3,300 patients.
- ⇒ Patients will be randomised between control and the arm(s) that show increased FFS in Stage 1.
- ⇒ The primary endpoint would be overall survival.
- ⇒ An independent Data Monitoring Committee (DMC) will assess the trial throughout and provide recommendations on moving between trial stages

STAMPEDE

Systemic Therapy in Advancing or Metastatic Prostate Cancer: Evaluation of Drug Efficacy

**A multi-arm multi-stage randomised controlled
trial**

Version:	15.0
Date:	29 -Mar-2016
MRC CTU ID:	PR08
ISRCTN #:	ISRCTN78818544
NCT #:	NCT00268476
EUDRACT #:	2004-000193-31
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MREC #:	04/MRE07/35

Authorised by:	
Name:	PROFESSOR NICHOLAS D JAMES
Role:	CHIEF INVESTIGATOR
Signature:	

Name:	MATTHEW SYDES
Role:	TRIAL STATISTICIAN

GENERAL INFORMATION

This document was constructed using the MRC CTU Protocol Template Version 4.0. It describes the STAMPEDE trial, coordinated by the Medical Research Council (MRC) Clinical Trials Unit (CTU) at University College London (UCL), and provides information about procedures for entering patients into it. The protocol should not be used as an aide-memoire or guide for the treatment of other patients. Every care has been taken in drafting this protocol, but corrections or amendments may be necessary. These will be circulated to the registered investigators in the trial, but sites entering patients for the first time are advised to contact the Cancer And Other Non-Infectious Diseases Group, MRC CTU at UCL, London, UK, to confirm they have the most up-to-date version.

COMPLIANCE

The trial will be conducted in compliance with the approved protocol, the Declaration of Helsinki 1996, the principles of Good Clinical Practice (GCP), Commission Directive 2005/28/EC with the implementation in national legislation in the UK by Statutory Instrument 2004/1031 and subsequent amendments, the UK Data Protection Act (DPA number: Z5886415), and the National Health Service (NHS) Research Governance Framework for Health and Social Care (RGF). International sites will comply with the principles of GCP as laid down by the ICH topic E6 (Note for Guidance on GCP), Commission Directive 2005/28/EC (the European Directive 2001/20/EC [where applicable]) and applicable national regulations.

SPONSOR

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AUTHORISATIONS AND APPROVALS

The following persons are authorised to sign the final protocol and protocol amendments for the sponsor: Professor Nicholas James (Chief Investigator) and Matthew Sydes (Trial Statistician).

TRIAL REGISTRATION

This trial has been registered with the ClinicalTrials.gov Clinical Trials Register, where it is identified as NCT00268476.

RANDOMISATIONS

To randomise, call MRC CTU at UCL, Monday to Friday 0900-1700
excluding public holidays or dates when notice has been given by the Unit.
Tel: +44 (0) 20 7670 4777

SAE REPORTING

Fax to 020 7670 4818 within 24 hours of becoming aware of the event

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ABBREVIATIONS

Abbreviation	Expansion
ACE	Angiotensin-Converting Enzyme
ACTH	Adrenocorticotrophic hormone
ADT	Androgen deprivation therapy
AR	Androgen receptor
AS	Activity Stage
bid	Twice a day (bis in die)
BP	Blood pressure
BRG	Biological Research Group
BSA	Body surface area
CF	Consent Form
CI	Chief Investigator
CI	Confidence interval
Cox-2	Cyclooxygenase 2
CRF	Case Report Form
CRUK	Cancer Research UK
CRPC	Castrate Refractory Prostate Cancer
CT	Computerised tomography
CTA	Clinical Trials Authorisation
CTAAC	Clinical Trials Advisory and Awards Committee
CTC	Common Toxicity Criteria
CTU	Clinical Trials Unit
CTV	Clinical Tumour Volume
CXR	Chest X-ray
DDX	Doctors and Dentists Exemption
DHT	Dihydrotestosterone
DNA	Deoxyribonucleic Acid
DPA	Data Protection Act
ERC	Endpoint Review Committee
ES	Efficacy Stage
ICH	International Conference on Harmonization
ECG	Electro cardiogram
FBC	Full Blood Count

Abbreviation	Expansion
FFS	Failure-Free Survival
GCP	Good Clinical Practice
GP	General Practitioner
GRO	General Register Office
HbA1c	Glycated haemoglobin
Hb	Haemoglobin
HE	Health Economics
HES	Hospital Episode Statistics
Hr	Hour
HR	Hazard Ratio
HRPC	Hormone Refractory Prostate Cancer
HSCIC	Health & Social Care Information Centre
HT	Hormone Therapy
IDMC	Independent Data Monitoring Committee
IM	Intramuscular
IMRT	Intensity Modulated Radiation Therapy
ISRCTN	International Standard Randomised Controlled Trial Number
IU	International Units
IV	Intravenous
LFTs	Liver Function Tests
LHRH	Luteinising Hormone Releasing Hormone
LREC	Local Research Ethics Committee
m	Month
MHRA	Medicine and Healthcare Products Regulatory Agency
min	Minutes
MRC	Medical Research Council
MREC	Multi-Centre Research Ethics Committee
MRI	Magnetic resonance imaging
M0	Non-metastatic
M1	Metastatic
NCI	National Cancer Institute (USA)
NCRN	National Cancer Research Network
NHS	National Health Service
NSAID	Non-Steroidal Anti-inflammatory Drugs

Abbreviation	Expansion
ONS	Office for National Statistics
OS	Overall Survival
PI	Principal Investigator
PIS	Patient Information Sheet
po	per orum (orally)
PSA	Prostate Specific Antigen
pts	Patients
PTV	Planned Tumour Volume
QALY	Quality-adjusted Life Years
qds	quater die sumendus (4 times each day)
QL	Quality of Life
R&D	Research and Development
RECIST	Response Evaluation Criteria In Solid Tumours
SAE	Serious Adverse Event
SAR	Serious Adverse Reaction
sc	Sub-cutaneous (under skin)
SNP	Single Nucleotide Polymorphism
SOC	Standard-of-Care
SSA	Site Specific Assessment
STAMPEDE	Systemic Therapy in Advancing and Metastatic Prostate Cancer: Evaluation of Drug Efficacy
SUSAR	Suspected Unexpected Serious Adverse Reactions
SWOG	South West Oncology Group
TMG	Trial Management Group
TMT	Trial Management Team
TURP	Trans-Urethral Resection of Prostate
TSC	Trial Steering Committee
UCL	University College London
ULN	Upper Limit of Normal
U+E	Urea and Electrolytes
WHO	World Health Organisation

1 SUMMARY

1.1 LAY SUMMARY

Prostate cancers depend upon the male hormone testosterone for their growth. Lowering testosterone levels (either by removing all or part of both testes, or by giving anti-hormone treatment) slows the growth of prostate cancers. This type of treatment is called hormone treatment or androgen deprivation therapy (ADT) and is often used when prostate cancers have spread outside the prostate gland. Although hormone treatment is usually successful at stopping the cancer growing for a period of time, the cancer will begin to grow again in most men.

There are increasing numbers of treatments available for advanced prostate cancer. These treatments are usually used in prostate cancer when hormone treatment is no longer effective and the cancer has started to grow again. The aim of this trial, which is called STAMPEDE, is to assess some of these treatments, given earlier in the course of the disease in combination with the current standard-of-care.

The treatments that have been, or are being, assessed during the trial are:

1. Zoledronic acid (now closed to recruitment): Prostate cancer cells can spread to bones and weaken them. Zoledronic acid is a drug that reduces bone destruction and hardens bones cells. Recruitment to this treatment has been completed and the results show that the addition of zoledronic acid does not prolong survival.

2. Docetaxel (now closed to recruitment):: A drug that stops cells replicating that is currently being used to treat a range of cancers including lung, breast and ovarian cancer as well as prostate cancer. Docetaxel prolongs survival in men with relapsed metastatic prostate cancer. Recruitment to this treatment has been completed and the results show that the addition of docetaxel to hormone treatment does improve survival in men with metastatic disease and delays the time to progression for men with locally advanced and metastatic disease. Docetaxel may now be given as part of standard treatment to all men entering STAMPEDE (from Protocol version 14.0).

3. Celecoxib (now closed to recruitment):: An aspirin-like drug that is used to treat arthritis. It slows down the growth of cancer cells in the laboratory. We wished to see if it had the same effect on cancer cells in patients. Recruitment stopped early as a planned intermediate analysis failed to demonstrate sufficient effect of this drug.

4. Abiraterone (in single use included from Protocol version 8.0; in combination with enzalutamide included from Protocol 12.0): An inhibitor of steroid hormone synthesis that blocks prostate cancer cells from generating their own male hormones. This is thought to be a major way in which prostate cancer cells resume growth following anti-hormonal therapies. Abiraterone has been shown to prolonging survival in men with advanced disease when given before and after chemotherapy.

5. Prostate radiotherapy (included from Protocol version 9.0): treatment with high-energy x-rays targeted to the prostate gland. This treatment is now mandatory for patients with cancer that is confined to the prostate gland as large trials have shown it improves survival times. We are not certain whether we should give radiotherapy to the prostate if the cancer has already spread and so we are investigating this in STAMPEDE. Recruitment to this treatment group is ongoing.

6. Enzalutamide (in combination with abiraterone included from Protocol version 12.0): This is a blocker of androgen receptors. These stimulate the cancer when hormone therapies have failed. Enzalutamide may be mutually complementary to abiraterone in terms of blocking mechanisms of resistance. The agent prolongs survival when given to men following failure of docetaxel chemotherapy. Recruitment to this treatment group is now completed.

7. Metformin (included from Protocol version 15.0): This anti-diabetic medication is proposed to have both anti-cancer effects and may help prevent the adverse metabolic effects of long term ADT. STAMPEDE will investigate whether adding metformin to the current standard-of-care for non-diabetic men can improve all-cause survival.

STAMPEDE will look at the effect of combining one or two of the treatments described above with hormone treatment. A computer program will be used to allocate which treatment each participant receives, using a chance process. The trial will look at the effects of the combined treatments on quality of life and find out whether the new treatment combinations control prostate cancer growth and enable men to live longer. The study will also look at which treatment provides the greater value for money for the health service. More than 9,000 men will join the trial with answers becoming available throughout the trial.

1.2 ABSTRACT AND SUMMARY OF TRIAL DESIGN

STAMPEDE is a multi-centre, randomised controlled trial for patients with locally advanced or metastatic prostate cancer who are commencing long-term Androgen Deprivation Therapy (ADT). Participants can have either newly diagnosed disease, or have been previously treated with radical radiotherapy or surgery but now have signs of progression such as a rising prostate specific antigen (PSA) (further details on eligibility see [Section 4](#)). The trial will assess the effects of adding different agents, both as single agents and in combinations, to the standard-of-care.

When the trial opened in 2005 there were five "original comparisons" which included the following investigational agents (i) a bisphosphonate, zoledronic acid, (ii) a cytotoxic chemotherapeutic agent, docetaxel and (iii) a cyclooxygenase (Cox-2) inhibitor, celecoxib. The results of these comparisons have now been presented.

Since then, the trial has been amended to include additional research arms in order to evaluate (iv) abiraterone, a steroid synthesis inhibitor and (v) enzalutamide, an inhibitor of androgen receptor signalling. A further research arm involving prostate radiotherapy for patients with newly-diagnosed metastatic disease was added (Protocol version 9.0).

From Protocol version 15.0, (vi) metformin, an anti-diabetic medication will be evaluated (Arm K) and recruitment to the "enzalutamide and abiraterone comparison" (Arm J) has now closed. The current arms open to recruitment are therefore Arm H (Prostate RT for M1 patients) and Arm K ("metformin comparison").

The trial has multiple arms; the control arm of the trial receives standard therapy alone. When the trial started standard treatment was androgen deprivation therapy (ADT) only, achieved through the use of luteinising hormone releasing hormone (LHRH) analogues or antagonist or bilateral orchidectomy according to local practice. Since primary results from the trial "original comparisons" have emerged showing a benefit in overall survival for patients receiving docetaxel in addition to ADT, the standard treatment has changed accordingly. Standard treatment may now include

docetaxel chemotherapy for all men entering STAMPEDE. Radiotherapy is also mandated for men with node negative non-metastatic disease. The current trial design is shown in [Figure 1](#); previous trial designs can be viewed in Protocol version 13.0.

Figure 1: Arms of the STAMPEDE trial from Protocol version 15.0

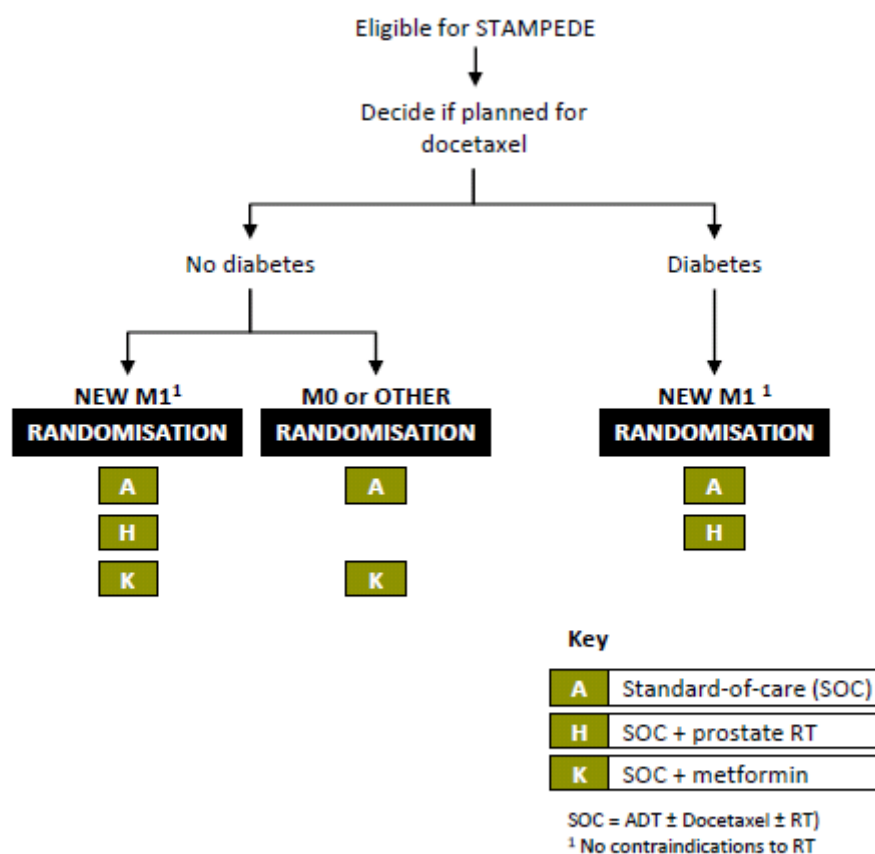
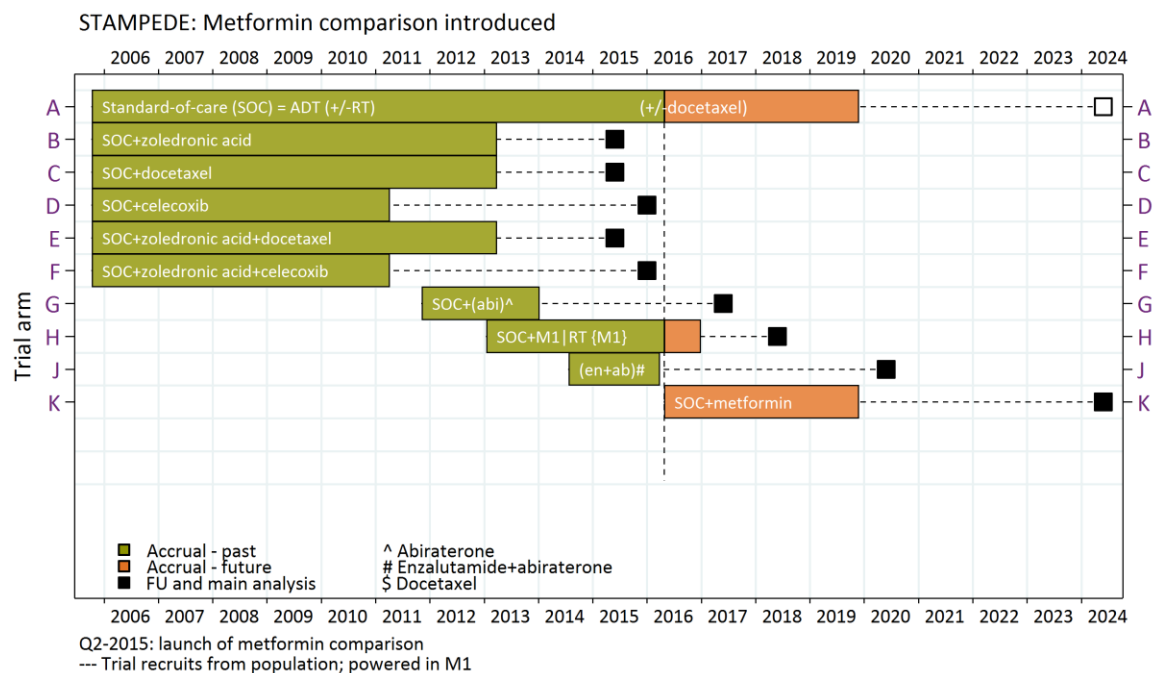


Figure 2: Arms of the STAMPEDE trial open to recruitment over time



For each comparison of research arm against control, the trial will be conducted in a number of stages: a Pilot/Safety Phase, Activity Stages and a final Efficacy Stage. The primary outcome measure of the Pilot/Safety Phase is safety, with 30-50 patients recruited to each research arm. Research arms will only continue to recruitment in the next stage if they have been shown to be both safe and feasible, although patient data from all patients and all stages will be included in the final analyses. In the Activity Stages the primary outcome measure is failure-free survival (FFS). Each Activity Stage is triggered when a pre-specified number of FFS events have been observed in the control arm of the relevant comparison (see [Section 9](#) for further detail). Recruitment to Arms D (ADT + celecoxib) and F (ADT + zoledronic acid + celecoxib) was stopped in Apr-2011 after the second planned activity analysis when the IDMC and TSC considered the lack-of-benefit guidelines.⁽¹⁾ (See to [Section 9](#) for further information regarding the guidelines for stopping accrual to research arms during the activity stages of the trial).

Some evidence of activity will be required for a research arm to continue past each stage and guidelines are in place for this assessment of activity. The Efficacy Stage will take place when a pre-specified number of deaths are observed amongst the control arm patients for that relevant comparison. This was when around 403 deaths had been reported in the control arm for the “original comparisons” (involving docetaxel and zoledronic acid) and will be when around 267 deaths are reported in the control arm for the “abiraterone comparison”, the “M1 | RT comparison” and the “enzalutamide+abiraterone comparison”. The exact number of patients randomised to, and duration of, the trial will depend on the observed accrual rate, observed event rate and the number of other research arms open to recruitment.

In Protocol version 8.0 a new Arm G (ADT + abiraterone) was added. Arm H (ADT+ prostate radiotherapy) was added in Protocol version 9.0. The trial stages remain similar to those at trial inception but will be staggered in time compared to the stages for the original Arms A-F. Protocol version 10.0 was approved following the completion of recruitment to the remaining original trial arms (B, C and E) and was a “housekeeping” change to remove references to the completed arms from the information sheets. Protocol version 11.0 was approved following the extension of the

recruitment target sample for the “abiraterone comparison” from 1,500 to around 1,800 patients. Protocol version 12.0 added a new combination therapy arm containing abiraterone with enzalutamide; for this comparison we envisage only two pre-planned interim analyses. Protocol version 13.0 was approved following the extension of the recruitment target sample for the “M1|RT comparison”, from 1,250 to around 1,800 patients, and the introduction of saliva sample collection for DNA analysis. In response to the results of the primary analysis of the “original comparisons” the standard-of-care was updated to permit docetaxel in Protocol version 14. Protocol version 15.0 includes the addition of the metformin comparison as a new research arm, open to all non-diabetic men and reflects the completion of accrual to the “enzalutamide + abiraterone comparison”.

Patients will be assessed 6 weekly for the first 24 weeks after randomisation and then every 12 weeks up to 2 years, 6-monthly until 5 years and annually, thereafter. Quality of Life (QL) and use of health care resources (Health Economics) data is being collected in all patients until first Failure Free Survival event is reached or trial treatment completed. Please see [Section 17](#) for more details.

In addition, there are translational sub-studies. Patients willing to participate will be asked at randomisation to donate a saliva sample (previously a droplet of blood), which will be stored for DNA and protein analysis to try to identify markers that are associated with response to therapy, side-effects or susceptibility to prostate cancer.

Patients will also be asked for permission to use some of their stored material (e.g. tissue samples obtained at prostate biopsy or surgery) for further studies aiming to understand the causes and nature of prostate cancer and to identify biomarkers of treatment response. For further details of ancillary studies, see [Section 17](#).

1.3 TRIAL DOCUMENTATION

Table 1 presents a summary of the required trial documentation for participating centres

Table 1: Trial documentation required for participating centres

TRIAL DOCUMENTATION	TIMING
R&D approval (including IRMER approval)	Before centre participation
Investigator Statement	Before centre participation
Signature list & delegation of responsibilities	Before centre participation
Trial personnel contact details	Before centre participation
PIS, GP & CF on local paper	Before centre participation
Signed Clinical Trial Agreement between Trust and Sponsor (or Variation if applicable)	Before centre participation
RTQA accreditation	Before centre participation
Site initiation training	Before centre participation
Pharmacy Pack acknowledgment	Before centre participation

2 BACKGROUND

2.1 INTRODUCTION AND SETTING

Prostate cancer is a major health problem world-wide and accounts for nearly one fifth of all newly diagnosed male cancers. In the UK, approximately 41,000 men are diagnosed with prostate cancer each year and in 2012 over 10,000 men died from the disease.(2)

2.1.1 LONG-TERM ANDROGEN DEPRIVATION THERAPY

The initial (first line) treatment for locally advanced or metastatic prostate cancer is androgen deprivation therapy (ADT) achieved either surgically with bilateral orchidectomy, or medically with LHRH agonists or antagonists (3). Oral anti-androgens are no longer permitted for new patients within the trial from Protocol version 8.0.

ADT produces responses in up to 95% of patients but it is not curative and disease recurs in virtually all patients treated with ADT as sole therapy, with a median time to progression of 18-24 months (3). Data from the control arm in STAMPEDE has shown that for men with newly-diagnosed metastatic disease time to progression is just 11 months. Such progressive disease is referred to as castrate resistant prostate cancer (CRPC); although this term is unpopular with patient groups due to its perceived pejorative overtones related to castration and hence terminology may yet change again in the future.

2.1.2 ROLE OF RADIOTHERAPY FOR MEN WITH M0 DISEASE

Two randomised trials, SPCG7 (4) and NCIC PR.3 / MRC PR07 (5-7) have tested the question of whether ADT alone combined with radiotherapy is the best treatment for high-risk patients with no evidence of spread outside the pelvis. Both trials demonstrated an improvement in overall and disease specific survival from the addition of radiotherapy to ADT. The size of this overall survival benefit is substantial (hazard ratio 0.68 in SPCG7 and HR 0.77 in PR07). With substantial benefit demonstrated in two mature, large, well conducted randomised trials, we now mandate that radiotherapy be standard for patients with no nodal or metastatic spread. Patients in this category will now only be allowed to enter the trial if standard radiotherapy is planned, with the exception of those for whom radiotherapy is contra-indicated. Such patients should be discussed with the Trials Unit prior to inclusion. For patients with node positive, M0 disease there are no clear data on whether radiotherapy is indicated or not. The NCIC PR.3 / MRC PR07 trial included patients with unknown nodal status who received whole pelvic radiotherapy (8). Given the large overall benefit observed in this trial, the STAMPEDE TMG recommends that pelvic nodal radiotherapy be considered for patients with node positive, non-metastatic disease at the discretion of the treating clinician (9).

2.1.3 ROLE OF DOCETAXEL FOR MEN WITH M0 OR M1 DISEASE

The primary analysis of the "original comparisons" has shown docetaxel to significantly prolong survival (HR 0.78; 95% CI 0.66-0.93). This is in support of the results of the CHARTED study which showed docetaxel improved survival in men with metastatic disease.(10) There was no evidence of heterogeneity in the treatment effect across patient groups and median survival was improved by 10 months, from 71 to 81 months. In a well powered and pre-planned sub-group analysis of men with metastatic disease at diagnosis the treatment effect was most apparent with the median survival benefit of 15 months. As a result the STAMPEDE TMG recommends that docetaxel should be strongly considered in all men with metastatic disease at presentation who are commencing ADT for the first time and are fit enough to receive chemotherapy.

Survival data for men without metastases at diagnosis is less mature but a statistically significant improvement in failure free survival is seen, therefore, docetaxel may also be considered for men with high-risk non-metastatic disease.

Therefore, docetaxel is now permitted as part of the standard-of-care for all men entering STAMPEDE at the discretion of the treating clinician and patient.

2.2 RATIONALE

There are increasing numbers of treatments which are used post relapse of first-line ADT in patients with CRPC, but there has been limited evidence as to which is associated with the best response, how they may be combined or sequenced or whether any of them might have a role as first-line treatment. An alternative approach is to investigate new drugs and new approaches to treatment, as first-line therapy in patients starting ADT. At this point, patients should be fitter and better able to tolerate treatment than when they have CRPC, and there is the possibility of having a larger and longer lasting effect. The pre-clinical and epidemiological evidence of the anti-prostate cancer effect of metformin, together with the general good tolerability of this drug underpins the rationale for evaluating metformin within the trial as a re-purposed treatment for prostate cancer. In addition metformin may counteract the metabolic effects of long-term ADT that are increasingly well recognised.

2.3 DESIGN

STAMPEDE (also known as MRC PR08) is an innovative, multi-arm multi-stage, multi-centre, randomised controlled trial. It initially assessed the effects of a bisphosphonate (zoledronic acid), a cytotoxic chemotherapeutic agent (docetaxel) and a cyclooxygenase (Cox-2) inhibitor (celecoxib), as single agents or combinations, in patients commencing long-term ADT for advancing or metastatic prostate cancer. For these questions, each comparison was divided into five stages such that, for each investigational arm, safety and activity data were generated in the first four stages; an investigational arm could only proceed to the fifth and final stage of recruitment, where it would be assessed for effect on overall survival, if shown to be sufficiently safe and active at all prior activity stages. It is important to note, however, that patient data from all arms and all stages are included in the final analyses of the primary outcome measure, even if the investigational arm did not proceed to the final stage.

A second planned interim analysis failed to demonstrate sufficient activity for celecoxib and this agent was removed from trial recruitment in April 2011; patients remaining on celecoxib treatment reverted to standard care. Protocol version 8.0 added a new drug abiraterone to the study as an additional arm (see [Section 2.7](#)). Protocol version 9.0 added a new comparison arm involving prostate radiotherapy for patients with newly-diagnosed metastatic disease (see [Section 2.8](#)). Protocol version 10.0 reflected the successful completion of recruitment to three docetaxel- and bisphosphonate-containing arms (Arms B, C and E) and removed references to these agents in the information sheets for new patients. Protocol version 11.0 extended the recruitment target for the "abiraterone comparison" (A vs G) from 1,500 to around 1,800 patients. Protocol version 12.0 added a new comparison involving the combination of abiraterone and enzalutamide. Protocol version 13.0 extended the recruitment target for the M1|RT comparison from 1,250 to around 1,800 patients. Protocol version 14.0 incorporates the permitted use of docetaxel in the standard-of-care. Protocol 15.0 sees the addition of new research (Arm K) assessing the combination of the current SOC and metformin. Protocol version 15.0 also marks the closure of Arm J (target sample size reached).

2.4 RESEARCH TREATMENT: BISPHOSPHONATES

Note: recruitment stopped to the zoledronic acid and docetaxel-containing arms in Mar-2013 as recruitment target sample was reached.

Treatment has been completed in all patients and the results reported. See Protocol version 13.0 or older for details on the rationale.

2.5 RESEARCH TREATMENT: CHEMOTHERAPY

Note: recruitment stopped to the zoledronic acid and docetaxel-containing arms in Mar-2013 as recruitment target sample was reached.

Treatment has been completed in all patients and the results reported. See Protocol version 13.0 or older for details on the rationale.

2.6 RESEARCH TREATMENT: CYCLOOXYGENASE-2 INHIBITORS

Note: recruitment completed to both celecoxib-containing arms in Apr-2011 at the end of Activity Stage II of this comparison.

Treatment has been ceased in all patients and the results to be reported in 2016. See Protocol version 13.0 or older for details on the rationale.

2.7 RESEARCH TREATMENT: STEROID SYNTHESIS INHIBITORS

Recent evidence suggests that an important mechanism for escape from tumour control by androgen ablation is the intracellular conversion of steroid precursors to androgenic steroids by prostate cancer cells. A key enzyme in this process is CYP17, which therefore represents a logical target for therapy in CRPC. (11). Abiraterone acetate (3 β -acetoxy-17-(3-pyridyl)androsta-5,16-diene, code CB7630; JNJ-212082) is rapidly converted in vivo to abiraterone (JNJ-589485; formerly code named CB7598). It is a selective, irreversible inhibitor of 17 α -hydroxylase/C17,20-lyase (cytochrome P450c17 [CYP17]), an enzyme that is critical in the production of androgens in the testes, adrenal glands and prostate tumor tissue. Inhibition of CYP17 inhibits the conversion of pregnenolone or progesterone into dehydroepiandrosterone (DHEA) or androstenedione, respectively, each of which is a precursor of testosterone. The pharmacodynamic effect is a more effective androgen depletion than can be induced by surgical castration, or medically by gonadotropin releasing (GnRH) hormone analogues used as first line hormone therapy in prostate cancer.

Approximately 2,280 prostate cancer patients participated in the two Phase 3 RCTs (COU-AA-301 and COU-AA-302), with approximately 1,335 patients receiving abiraterone acetate at 1000mg daily dose continuously, in these studies. These studies have demonstrated abiraterone to prolong survival when given post-docetaxel (HR 0.65) and pre-docetaxel (HR 0.82). As a result it is now approved use in the USA and Europe in CRPC.(12, 13)

Side-effects with abiraterone acetate are modest with the main adverse effects being elevated transaminases (usually mild), hypokalaemia and hypertension due to secondary hyperaldosteronism and fluid retention (preventable by low doses of glucocorticoids). In order to prevent secondary hyperaldosteronism, it is recommended that prednisolone (or prednisone) 10mg daily be administered in the CRPC setting. Within more recent studies in earlier stage patients, lower doses (typically 5mg of prednisone/prednisolone) are being used due to concerns about side effects of long-term exposure to glucocorticoid. Within the STAMPEDE trial, we suggest prednisolone/prednisone dose of 5mg OD which may be increased to 5mg BID at the investigator's discretion if there are any concerns about monitoring or risks for the patient with 5mg OD.

We hypothesise that the agent may be more active still when given up-front in combination with first-line ADT by preventing or delaying the development of castrate refractory disease.

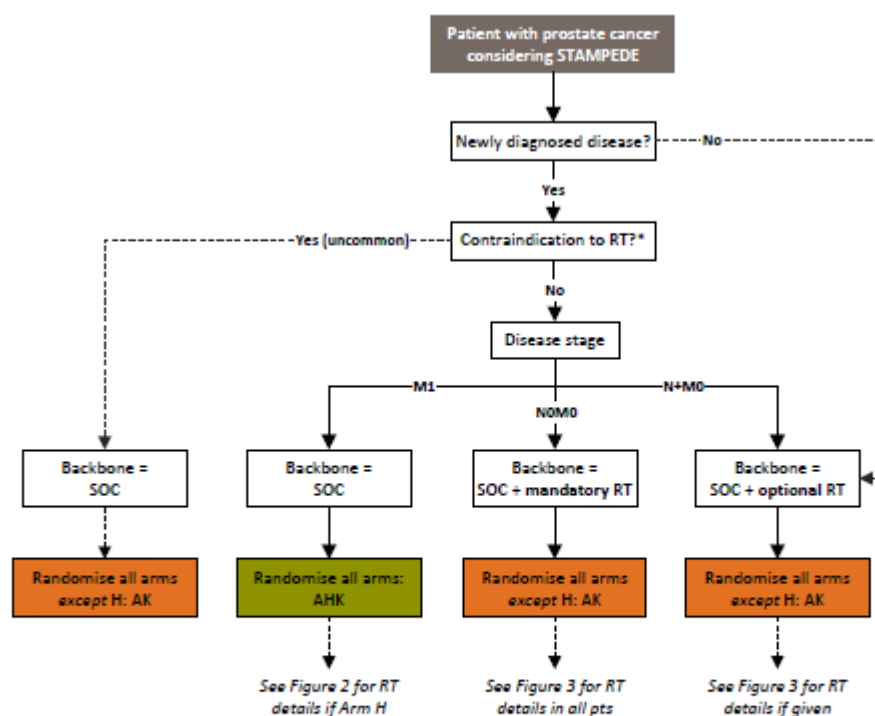
2.8 RESEARCH TREATMENT: RADIOTHERAPY TO THE PROSTATE FOR PATIENTS WITH NEWLY-DIAGNOSED METASTATIC DISEASE

Therapy directed against the primary tumour in the presence of metastatic disease has been evaluated rigorously in only one malignancy to date: renal cell carcinoma. Two cooperative groups ran randomised trials enrolling patients with previously untreated metastatic RCC whose primary tumours were amenable to surgical resection. Patients were randomised to receive the standard systemic therapy of the day, interferon-alpha, either alone or with radical nephrectomy. The combination of nephrectomy and interferon was shown to significantly improve median survival from 7 to 17 months in one trial (14) and from 8 to 11 months in the other (15). The mechanism by which nephrectomy improves survival remains obscure. In preclinical models, the primary tumour has been found to secrete molecules that prime the microenvironment in which metastases can develop. An implication of this work is that therapy directed at the primary tumour, by abrogating this endocrine signalling, could retard the formation and the growth of distant metastases.

The results of two large-scale randomised trials of prostate radiotherapy are also provocative. The Scandinavian SPCG-7 trial and the MRC PR07 trial randomised men with locally advanced prostate cancer, who were at high risk of possessing occult metastatic disease, to either ADT alone or ADT plus prostate radiotherapy (4, 8). The addition of radiotherapy dramatically improved 10-year outcomes: mortality from prostate cancer was halved. Interestingly, the benefit of radiotherapy started to emerge as early as three years from the time of randomisation. This seems improbably early if the benefit of local treatment is mediated via the prevention of subsequent disease dissemination. Rather, it is more consistent with the possibility that local treatment has a beneficial impact on the rate of progression of existing micrometastatic disease.

We hypothesise that local therapy to the primary site may retard distant disease progression and prolong survival in patients with newly-diagnosed metastatic prostate cancer.

Figure 3: Use of RT in STAMPEDE



*It is expected that only around 1% of patients will have a contraindication to RT. These cases should be discussed with the trials unit prior to randomisation (see Section 4.3).

2.9 RESEARCH TREATMENT: COMBINATIONS OF ORIGINAL RESEARCH ARMS

2.9.1 BISPHOSPHONATE AND CHEMOTHERAPY

Note: recruitment stopped to the zoledronic acid and docetaxel-containing arms in Mar-2013 at the end of Activity Stage IV. Treatment has been completed in all patients and the results reported. See Protocol version 13.0 or older for details on the rationale.

2.9.2 BISPHOSPHONATE AND CYCLOOXYGENASE-2 INHIBITORS

Note: recruitment stopped to both celecoxib-containing arms in Apr-2011 at the end of Activity Stage II.

2.10 COMBINATION OF STEROID SYNTHESIS INHIBITORS AND ANDROGEN RECEPTOR SIGNALLING INHIBITOR

The majority of patients with advanced prostate cancer who have disease progression on abiraterone or enzalutamide taken as single agents, have a rise in PSA, suggesting reactivation of androgen receptor (AR), or other steroid signalling pathways resulting in increased PSA transcription, is the pathway to the development of resistance.(16)

The primary pharmacodynamic effect of enzalutamide is inhibition of androgen binding to the AR, AR nuclear translocation in the presence of androgen and AR:chromatin association. In multiple prostate cancer cell lines that specifically model CRPC (LNCaP/AR, VCaP, W741C LNCaP), the consequences of enzalutamide treatment include inhibition of AR-induced gene transcription, reduced cell proliferation, increased cell death by apoptosis and tumor regression.

In a mouse xenograft model of CRPC using prostate cancer cells that overexpress the AR (LNCaP/AR), enzalutamide inhibits tumor growth and reduces tumor size. A major human metabolite of enzalutamide, N-desmethyl enzalutamide, demonstrates key primary pharmacodynamics of similar potency to the parent molecule, while the carboxylic acid derivative metabolite has no known pharmacodynamic effect.

The question under investigation is: can progression be delayed (and survival extended) by using a combination of abiraterone and enzalutamide?

2.10.1 SUPPLEMENTING ABIRATERONE AND PREDNISOLONE WITH ENZALUTAMIDE

Several studies have shown that the AR can become promiscuously activated by very low levels of androgens or other steroid metabolites and drugs that bind the AR (17-20). It is known that very low levels of androgens can persist in patients treated with abiraterone acetate (21). Drugs that bind the AR, may include co-administered glucocorticoids. Furthermore, AR mutations of the sort previously described in CRPC, can be activated by cortisol and other glucocorticoids at levels much lower than those reported in patients treated with abiraterone and prednisolone at a dose of 5mg bid (20, 22). Moreover, abiraterone binds the AR and, although weak antagonism of wild-type and most previously described AR mutations are observed (22), a similar mechanism to that described with classical anti-androgens, such as bicalutamide, could lead to change-of-function AR mutations associated with AR activation following abiraterone binding. Therefore, concomitant treatment with an androgen receptor signalling inhibitor could prevent “promiscuous” AR activation in patients treated with abiraterone. Enzalutamide is an androgen receptor signalling inhibitor approved for use on its own in the treatment of advanced CRPC (23), and there is evidence of activity for hormone-naïve prostate cancer (24).

2.10.2 SUPPLEMENTING ENZALUTAMIDE WITH ABIRATERONE AND PREDNISOLONE

Enzalutamide in combination with ADT is both effective and well tolerated in CRPC.(23) However, recent studies have suggested that intra-tumoral testosterone levels increase in patients treated with enzalutamide (25). The implications of this finding are that the increase in intra-tumoral testosterone could be associated with up-regulation of enzymes involved in steroid biosynthesis.(26) Although enzalutamide has a high affinity for the AR, this is several-fold lower than both the natural ligands testosterone and DHT (27), which means that enzalutamide would be out-competed at the AR ligand-binding domain if and when androgen levels rise. In vitro, a ten-fold rise in intra-cellular androgen was sufficient to prevent inhibition of AR by 30uM of enzalutamide;(22) these levels are representative of the plasma levels of enzalutamide active metabolites, which can be achieved with enzalutamide 160mg po daily (28).

A strategy for preventing the rise in intra-cellular androgens in patients treated with enzalutamide would be inhibition of CYP17A1. Abiraterone is currently the only CYP17A1 inhibitor with proven efficacy. It therefore seems logical to use the combination of enzalutamide and abiraterone to both block a rise of intra-cellular androgens and prevent promiscuous activation of the AR.

2.10.3 SUMMARY OF RATIONALE FOR THIS COMBINATION

To date, investigation has focussed on patients with CRPC but there is a strong rationale for the combination of enzalutamide and abiraterone in the hormone treatment-naïve setting in which STAMPEDE is focused.

STAMPEDE is already evaluating abiraterone plus conventional ADT but we will not assess the combination of conventional ADT plus enzalutamide; other trials by industry and other cooperative groups will address that question. The inclusion of an arm with ADT and enzalutamide in STAMPEDE was therefore considered to be a duplication of effort and was not supported by the Trial Management Group.

The combination of enzalutamide and abiraterone is a novel approach and offers considerable promise in delaying progression – it therefore represents an attractive addition to the comparisons under investigation in STAMPEDE, and one that is unlikely to be replicated in other planned trials of this size.

2.11 RATIONALE FOR THE "METFORMIN COMPARISON"

All men joining STAMPEDE are planned for long-term ADT, a treatment associated with an increased risk of insulin resistance, hyperglycaemia, dyslipidaemia and obesity. Over 50% of men receiving long-term ADT will develop Metabolic Syndrome (29) resulting in increased cardiovascular morbidity and mortality. Obesity and high bind insulin C-peptide levels, indicating insulin resistance are independent predictors of increased prostate cancer-specific mortality and the presence of metabolic syndrome and diabetes in men treated with ADT is associated with shorter survival.

Metformin, which in non-diabetic individuals has been shown to lower the incidence of diabetes, counteracts some of these side-effects of ADT, including insulin insensitivity, hyperinsulinaemia and diabetes. It also reduces the levels of cholesterol, LDLs and triglycerides by inhibiting the fatty acid synthesis via activation of Adenosine Monophosphate Activated Kinase (AMPK) and decreases the platelet aggregation factor 1, platelet aggregation, vascular adhesion molecules, CRP and leptin (30-33). Through mitigation of the cardiovascular and metabolic consequences of ADT, metformin is proposed to reduce treatment associated morbidity and improve all-cause mortality.

In addition, recent data has emerged consolidating the knowledge that cancer progression is linked integrally with metabolic modulators and that modification of this process by metformin has an important effect on cancer progression and survival. Pre-clinical data has shown that metformin is an important stimulator of AMPK which acts as the cellular “masterswitch” for energy regulation. AMPK acts to inhibit the effects of elevated insulin levels which promote metastasis, tumour growth and treatment resistance. Insulin increases mRNA and protein expression of steroidogenic enzymes leading to the up-regulation of intracellular testosterone levels, secreted androgens, thereby activating the AR (34). Metformin also influences the PI3K-AKT pathway and has an anti-proliferative effect via inhibitor of mTOR as well as targeting cancer stem cells. In vitro, metformin has been shown to inhibit androgen-induced IGF-IR up-regulation through disruption of androgen signalling (35).

Evidence in support of this includes a systematic review and meta-analysis of 13,008 men with type 2 diabetes mellitus (T2DM) and concurrent cancer which has shown improved survival in men treated with metformin compared with other anti-diabetic agents. In a systematic review of observational data from over 1 million men, there was a significant association seen between metformin and decreased risk of death from any cancer. Another systematic review found that the use of metformin in diabetic patients was associated with a significantly lower risk of cancer incidence and cancer mortality (36). In a large retrospective cohort study of 3837 diabetic men with prostate cancer, metformin was associated with a decreased risk of prostate cancer specific mortality (HR=0.76 [0.64-0.89]) and death (HR=0.76 [0.70-0.82]). In a prospective non-randomised phase II study in non-diabetic CRPC patients, 36% of patients receiving metformin were progression-free at 3 months and >50% had a prolongation of their PSA doubling time (37). In summary,

metformin is proposed to mitigate many of the adverse side effects of long-term ADT as well as having multiple potential anti-cancer effects and therefore STAMPEDE will evaluate re-purposing this treatment as a novel therapeutic approach in the management of high risk or metastatic prostate cancer.

3 SELECTION OF INSTITUTIONS AND INVESTIGATORS

Centres who wish to participate in the STAMPEDE trial should be registered with the Medical Research Council Clinical Trials Unit at University College London (MRC CTU at UCL) for this purpose. Before any patients are randomised the MRC CTU must receive a completed and signed Investigator Statement. The STAMPEDE investigator statement is signed by the Principal Investigator for that institution (download from www.stampedetrial.org). R&D approval for the site, along with a fully-signed model agreement, is also required before recruitment can begin.

In addition, and in compliance with the principles of GCP, all institutions participating in the trial will complete a delegation log and forward this to the MRC CTU. Each person working on the STAMPEDE trial must sign off a section of this log indicating their responsibilities. The MRC CTU must be notified of any changes to trial personnel and/or their responsibilities. An up-to-date copy of this log must be stored in the Investigator Site file at the institution and also at the MRC CTU.

The Clinical Trial Authorisation (CTA) for the STAMPEDE trial requires that the Medicines and Healthcare Products Regulatory Agency (MHRA) be supplied with the names and addresses of all participating investigators/institutions. Trial staff at the MRC CTU will perform this task; hence, it is vital to receive full contact details for all investigators prior to their entering patients.

Finally, before a patient is entered into the trial written informed consent must be obtained. Approved patient information sheets and informed consent forms are supplied as templates.

Only a limited number of centres participated in the initial Pilot Phase of the original trial; this was to ensure that safety and feasibility data were collected expediently. Subsequent stages of the trial are open to any centre that wishes to participate and has fulfilled the requirements described above.

3.1 RADIOTHERAPY ACCREDITATION

The introduction of the "M1 | RT comparison" in Protocol version 9.0 introduced the need for RTQA accreditation in sites giving radiotherapy. The detail of RTQA accreditation is in [Appendix K](#). However, centres that have been RTQA accredited for another multi-centre prostate radiotherapy trial in the UK (e.g. RADICALS or CHHIP) will be automatically granted STAMPEDE RTQA accreditation.

4 SELECTION OF PATIENTS

4.1 GENERAL INCLUSION CRITERIA

Participants must fulfil both of the criteria in [Section 4.1.1](#) or one criterion in [Section 4.1.2](#) or at least one criterion in [Section 4.1.3](#). Additionally, all patients must fulfil the criteria in [Section 4.1.4](#).

4.1.1 HIGH-RISK NEWLY-DIAGNOSED NON-METASTATIC NODE-NEGATIVE DISEASE

Both:

- At least two of: T category T3/4, PSA $\geq$ 40ng/ml or Gleason sum score 8-10
- Intention to treat with radical radiotherapy (unless there is a contra-indication; exemption can be sought in advance of consent, after discussion with MRC CTU)

OR

4.1.2 NEWLY-DIAGNOSED METASTATIC OR NODE-POSITIVE DISEASE

At least one of:

- Stage T_{any} N+ M0
- Stage T_{any} N_{any} M+

OR

4.1.3 PREVIOUSLY RADICALLY TREATED, NOW RELAPSING (PRIOR RADICAL SURGERY AND/OR RADIOTHERAPY)

At least one of:

- PSA $\geq$ 4ng/ml and rising with doubling time less than 6 months
- PSA $\geq$ 20ng/ml
- N+
- M+

AND

4.1.4 FOR ALL PATIENTS

- I. Histologically confirmed prostate adenocarcinoma
- II. Intention to treat with long-term androgen deprivation therapy
- III. Treating clinician and patient should have decided if docetaxel is to be part of the standard-of-care prior to randomisation
- IV. Fit for all protocol treatment¹ and follow-up, WHO performance status 0-2²
- V. Have completed the appropriate investigations prior to randomisation
- VI. Adequate haematological function: neutrophil count $>1.5 \times 10^9/l$ and platelets $>100 \times 10^9/l$
- VII. Estimated creatinine clearance $>30ml/min$
- VIII. Serum potassium $\geq 3.5mmol/L$
- IX. Written informed consent
- X. Willing and expected to comply with follow-up schedule
- XI. Using effective contraceptive method if applicable

¹ Medical contraindications to the trial medications are given in [Section 6](#)

² For WHO performance status definitions see [Appendix A](#)

4.2 GENERAL EXCLUSION CRITERIA

Patients must not fulfil any of the criteria, below.

- I. Prior systemic therapy for locally advanced or metastatic prostate cancer except as listed in [Section 4.1.3](#)
- II. Metastatic brain disease or leptomeningeal disease
- III. Abnormal liver functions consisting of any of the following:
 - Serum bilirubin $\geq 1.5 \times$ ULN (except for patients with Gilbert's disease, for whom the upper limit of serum bilirubin is $51.3 \mu\text{mol/l}$ or 3mg/dl)
 - Aspartate aminotransferase (AST) or alanine aminotransferase (ALT) $\geq 2.5 \times$ ULN
- IV. Any other previous or current malignant disease which, in the judgement of the responsible clinician, is likely to interfere with STAMPEDE treatment or assessment
- V. Symptomatic peripheral neuropathy grade ≥ 2 (NCI CTC)³
- VI. Any surgery (e.g. TURP) performed within the past 4 weeks
- VII. Patients with significant cardiovascular disease such that, in the investigator's opinion, the patient is unfit for any of the study treatments. This might include:
 - Severe/unstable angina
 - Myocardial infarction less than 6 months prior to randomisation
 - Arterial thrombotic events less than 6 months prior to randomisation
 - Clinically significant cardiac failure requiring treatment (NYHA II-IV)⁴
 - Cerebrovascular disease (e.g. stroke or transient ischaemic episode) less than 6 months prior to randomisation
 - Patients with uncontrolled hypertension defined as systolic BP greater or equal than 160 mmHg or diastolic BP greater or equal than 95 mmHg
- VIII. Prior chemotherapy for prostate cancer (excluding patients receiving docetaxel as part of the new SOC)
- IX. Prior exposure to long-term hormone therapy before randomisation (unless as described in [Section 4.5.3](#))
- X. Prior exposure to systemic treatment for prostate cancer (excluding hormone therapy) e.g. abiraterone and enzalutamide.

4.3 SELECTION CRITERIA FOR COMPARISON OF RESEARCH RT FOR METASTATIC DISEASE (M1 | RT)

All patients meeting criteria in [Section 4.1](#) and [4.2](#) are eligible for the trial, but not all can be allocated to the research (M1) radiotherapy arm. The selection criteria for the "M1 | RT comparison" are:

- Newly-diagnosed prostate cancer
- Demonstrable M1 disease
- No contraindication to radiotherapy e.g. no previous pelvic radiotherapy and no history of inflammatory bowel disease
- No previous radical prostatectomy

Any patients meeting these criteria will have a chance to be allocated to Arm H. For those rare cases where radical RT is planned for a newly-diagnosed M1 patient, the TMT and TMG will need to review and approve the inclusion of the patient for randomisation only between Arm A and K.

³ See [Appendix I](#) for common toxicity grading

⁴ NYHA classifications can be found in [Appendix A](#)

4.4 SELECTION CRITERIA FOR METFORMIN COMPARISON

All patients meeting criteria in [Section 4.1](#) and [4.2](#) are eligible for the trial, but patients with known diabetes mellitus are not eligible for randomisation to the "metformin comparison".

All non-diabetic patients require an HbA1c to be performed ideally within 8 weeks prior to randomisation to confirm their non-diabetic status.

In summary, additional inclusion criteria for the "metformin comparison" are:

- HbA1c <48 mmol/mol (equivalent to <6.5%)
- Estimated creatinine clearance ≥60mls/min
- No history of metabolic acidosis or pre-disposing conditions
- Not current or previous treatment with metformin
- No contra-indications to metformin

If following screening the patient is found to have diabetes mellitus (i.e. HbA1c is 6.5% or higher) the patient is only eligible for randomisation (between Arms A and H only) if they meet all of the selection criteria for the M1|RT comparison (see [Section 4.2](#)). M0 or previously treated patients with known diabetes mellitus, or who are found to be diabetic on screening, are currently ineligible for randomisation and will therefore be screen failures.

All patients with abnormal baseline HbA1c should be informed and investigated further according to local practice e.g. referred to their GP for further management.

If the patients is currently receiving short-term corticosteroids, ideally repeated screening should be performed once corticosteroid treatment has stopped providing this remains within 12 weeks of starting ADT (see [Section 4.5.3](#)).

4.5 SCREENING PROCEDURES

4.5.1 INVESTIGATIONS PRIOR TO RANDOMISATION

All patients should have the following examinations performed within 8 weeks to confirm eligibility prior to randomisation.

The following standard imaging is required and the latest available scans should be used:

- CT or MRI of pelvis and abdomen
- Bone Scan (or equivalent e.g. whole body MRI, choline-PET-CT, PSMA-CT-PET)
- Chest X-ray (only if chest was not included in CT or MRI which would be preferable)

Any additional imaging such as CT-PET scanning can be performed according to local practice but, for the purposes of the trial, the recorded stage should be the CT stage only; additional information on the CT-PET stage will also be collected.

The following bloods and additional measurements are required prior to randomisation:

- ECG
- Blood pressure
- PSA Test
- HbA1c
- Full blood count

- Urea and Electrolytes
- Liver function tests
- Serum creatinine
- Systolic and diastolic blood pressure

Patients who initially fail to meet the eligibility criteria can be re-screened at a later date. Of note, for patients receiving standard-of-care docetaxel at the point of screening, it is acceptable to use a full blood count measurement prior to chemotherapy to confirm eligibility.

Prior to randomisation:

- Check details of any prior treatments for prostate cancer
- Check any contraindications to radiotherapy or metformin
- Check concomitant medications

4.5.2 ADDITIONAL BASELINE INVESTIGATIONS

The following blood tests are required at baseline (within 4 weeks before or after to randomisation):

- Testosterone (pre-ADT, if available)
- Serum corrected calcium
- Phosphate
- Magnesium
- Albumin
- Fasting glucose (mandatory)
- Fasting triglycerides (mandatory)
- Lipid profile (non-fasting; total cholesterol, LDL and HDL) (mandatory)

The following additional procedures are required and mandatory at baseline

- Waist circumference measurement
- Weight and height

A trial screening log will be available to all centres; copies are not required by the STAMPEDE Trial Team unless requested.

4.5.3 ANDROGEN DEPRIVATION THERAPY PRIOR TO RANDOMISATION

If ADT has already started prior to randomisation, the first LHRHa must have been given within 12 weeks of randomisation. Additionally, if anti-androgens are being used, these must have started within 14 weeks of randomisation. A PSA level must have been taken prior to treatment starting.

Note that long-term anti-androgen monotherapy is not permitted in the trial for newly recruited patients from Protocol version 8.0; patients may change treatment to join the trial, provided that they have not had more than 12 weeks of ADT prior to randomisation. Further details on hormone therapies allowed prior to randomisation are discussed in [Appendix L](#).

Any relapsing patients treated with adjuvant or neo-adjuvant hormone therapy alongside their radical surgery or radiotherapy must have completed that period of hormone therapy at least 12 months before joining STAMPEDE and it must have been no longer than 12 months in duration.

Note that baseline testosterone measurements will not be required in patients who have already commenced hormone manipulation prior to randomisation.

4.5.4 CHOICE ABOUT STANDARD-OF-CARE DOCETAXEL

The treating clinician and patient must have decided, prior to randomisation, whether docetaxel is to be given as part of standard-of-care. Docetaxel treatment must start within 12 weeks after starting ADT, preferably within 8 weeks. Patients can have already started docetaxel treatment when randomised providing this is within 12 weeks after starting ADT.

4.5.5 STARTING TRIAL TREATMENT

For patients allocated to Arm H: For patients not receiving SOC docetaxel, RT should not be commenced until SOC docetaxel has been completed; this should start within 3-4 weeks from after the last administration of docetaxel and continued according to the predefined scheduled unless toxicity is reported. Should any docetaxel-related toxicities occur during treatment these need to have recovered to grade 1 before the allocated research treatment begins. Please discuss with the trial team if any docetaxel-related toxicities of greater severity than grade 1 persist (this excludes alopecia, nail changes and neuropathy, which do not need to have resolved prior to starting research treatment).

Investigators should discuss with the trial team any patient who may not start their allocated trial treatment or where a delay of 6 or more weeks, between last administration and starting research treatment is expected. Any delays in starting research radiotherapy should be discussed with the STAMPEDE team and recorded as appropriate in the relevant CRF.

For patients allocated to Arm K: For all patient allocated to metformin, treatment should start as soon as possible after randomisation. Metformin can be given in combination with SOC docetaxel. Investigators should aim that this is at least within 4 weeks post-randomisation and within 12 weeks of starting ADT (see [Section 6](#)).

4.5.6 CONCOMITANT MEDICATIONS

All concomitant medications should be recorded including any vitamin and mineral supplements the patient is taking, regular consumption of NSAID and/or aspirin and use of other bisphosphonates (see [Section 4.2](#)). Of particular interest are herbal preparations such as PC-SPES, Prostatol, Saw Palmetto and St John's Wort.

Caution should be exercised when starting any concomitant medications that may result in a worsening of renal function e.g. initiating anti-hypertensive therapies such as ACE inhibitors, diuretics such as frusemide, or starting a non-steroidal anti-inflammatory drug (NSAID). Please refer to [Table 15](#) for more information on drugs which may require additional monitoring of renal functions.

All concomitant medications should be continued throughout the trial unless the responsible clinician decides otherwise. If patients continue to require medication for the management of docetaxel-related toxicities, please discuss this with the trial team. Please see [Section 6](#) for more information on concomitant medications and their use with abiraterone and enzalutamide.

From Protocol version 15.0 onwards the trial will require information regarding long-term (>6 months) use of the following concomitant medications of classes of interest.

- Statins
- Metformin
- Aspirin
- Bisphosphonates or Denosumab/calcium and vitamin D

- Opiate pain killers
- ACE inhibitors or angiotension II antagonists
- Vitamin B12

This information will be collected at each follow up (see [Table 16](#)).

4.6 ADDITIONAL DETAILS FOR PATIENTS JOINING SUB-STUDIES

If the patient has given their consent to participate in the DNA analysis sub-study, a saliva sample will be collected. This replaces the droplet of blood collected in previous versions of the protocol. An additional element of DNA analysis will involve the collection of plasma samples for patients in the "enzalutamide + abiraterone comparison".

Patients have been asked to consent for the use of remaining tissue samples obtained at prostate biopsy or following surgery for use in additional research. Randomising sites will be asked to assist in the retrieval of tissue samples stored as FFPE blocks from pathology stores or referring hospitals when these are required for additional translational sub-studies. Full details of all sub-studies and instructions relating to the handling of the saliva and blood sample are given in [Section 17](#) and [Appendix D](#).

5 RANDOMISATION AND ENROLMENT

Patients will be allocated to any of the open research arms for which they are suitable. Patients with non-metastatic disease or who have had previous local therapy to the prostate or who have a contraindication to radiotherapy will not be allocated to Arm H (see [Section 4.3](#)). Patients with a known diagnosis of diabetes will not be allocated to Arm K (see [Section 4.4](#)).

To enter a patient the randomisation form should be completed carefully and the MRC CTU contacted by phone:

RANDOMISATIONS

To randomise, call MRC CTU at UCL, Monday to Friday 0900-1700
excluding public holidays or dates when notice has been given by the CTU.
Tel: +44 (0) 20 7670 4777

A trial number and treatment will be allocated and given over the phone or by return fax. In addition, a letter confirming these details will be sent. The trial number will be the primary way in which the patient will be identified and should be used in all correspondence.

5.1 CO-ENROLMENT GUIDELINES

Ideally, patients should not be participating in any other clinical trial of prostate cancer treatment when they enter STAMPEDE and should not enter any other trials until a failure-free survival (FFS) event has been experienced and reported. After this point, the patient may be entered into further, second-line treatment studies. The primary outcome measure of STAMPEDE is overall survival. Participation in post-progression studies must be reported on the Co-enrolment CRF.

Data sharing agreements with “down-stream” trials are encouraged to improve data quality in both trials and to reduce costs to both organisations.

6 TREATMENT OF PATIENTS

6.1 TRIAL TREATMENT

Patients will be randomised to the control arm (Arm A) or one of the research arms. All patients will receive androgen deprivation therapy (ADT) to achieve castration levels of testosterone. The method of ADT is a local choice but must be specified for each patient prior to randomisation. The recommended methods of ADT are given in [Section 6.2.1](#). Note that from Protocol version 8.0 onwards, bicalutamide monotherapy is no longer permitted as a trial therapy for new patients but patients may switch to a permitted therapy to join the trial. Please see [Appendix L](#) for more information on ADT timing before randomisation.

6.1.1 REQUIRED TIMELINES WHEN STARTING TRIAL TREATMENT

Allocated treatment should start promptly after randomisation. Please refer to [Section 4.5.3, 4.5.4, 4.5.5](#) for more information on starting of trial treatment.

6.2 ARM A: STANDARD-OF-CARE

The standard-of-care for this patient group is **androgen deprivation therapy** (see [Section 6.2.1](#)). For some patient groups, this should now be supplemented with standard radiotherapy (see [Section 6.2.2](#)). From Protocol version 14.0 onwards the standard-of-care includes permitted use of docetaxel for all suitable patients (see [Section 6.2.3](#)).

In summary, SOC treatments may include:

- ADT alone
- ADT + Radiotherapy
- ADT ± Radiotherapy ± Docetaxel

6.2.1 HORMONE THERAPY

The permitted methods of ADT are bilateral orchidectomy, LHRH analogues and LHRH antagonists. Patients having a bilateral orchidectomy are required to adhere to the same timelines as specified in [Section 4.4.2](#), unless there is a strong clinical reason not to do so. Other methods of ADT should be discussed with the Chief Investigator or the Trial Surgeon. The planned duration of ADT should be at least 2 years.

Bilateral orchidectomy: Operations should be performed by appropriately trained surgeons. A total or sub-capsular orchidectomy may be performed.

LHRH agonists: LHRH agonists/analogues used according to local practice. The prophylactic use of anti-androgens to prevent tumour “flare” is recommended.

LHRH antagonists: LHRH antagonists used according to local practice. The use of prophylactic use of anti-androgens to prevent tumour “flare” is not necessary.

6.2.2 STANDARD-OF-CARE (M0) RT

NOMO patients: Investigators should give standard radiotherapy (RT) to patients with node negative, non-metastatic disease (NOMO), in accordance with data from the PR07 and SPCG trials. If there is an intention to omit radiotherapy (e.g. contraindications) in patients with NOMO disease this must be

discussed with the Trials Office before consent. See [Section 6.17](#) for further details of radiotherapy administration.

N+M0 patients: the benefit of radiotherapy in this group is at present uncertain with no firm data to either support or refute its use. However, the PR07 trial included some node positive patients as cross sectional imaging was not a part of the baseline assessment in this trial, which did include whole pelvis radiotherapy (8). For patients with node positive, non-metastatic disease, radiotherapy is therefore recommended in suitable cases (9).

Investigators will be asked to state their intention with regards to planned radiotherapy in this group at randomisation. Intention to give radiotherapy (or not) for node positive patients must be stated at randomisation to ensure that there is no bias towards particular combinations of systemic therapy with radiotherapy.

Standard-of-care radiotherapy is not a core part of the trial, therefore we intend to collect minimal data about the radiotherapy administered. It is accepted that some patients will develop progressive disease before radiotherapy can be administered and if this occurs the reasons for non-delivery of treatment must be recorded on the radiotherapy detail form.

Any patient who has had a previous, definite diagnosis of inflammatory bowel disease is at increased risk of disease re-activation following radiotherapy, and the risks of this must be balanced against the potential benefits of radiotherapy on an individual basis.

6.2.3 STANDARD-OF-CARE DOCETAXEL

Investigators are strongly encouraged to consider giving docetaxel as part of the standard-of-care for patients with newly-diagnosed metastatic disease based on the survival benefit demonstrated by both STAMPEDE in the primary analysis of the "original comparisons" and CHARTED (10) (38, 39). Investigators may also consider giving docetaxel to patients with high-risk locally advanced disease given both the significant improvement in failure-free-survival and consistency of effect for prostate cancer-specific survival shown by STAMPEDE.

The treating clinician and patient must have decided if docetaxel is to be given prior to randomisation and treatment may have started when the patient is randomised. As with standard radiotherapy, minimum data collection will be required however the start and end dates of treatment are needed to ensure the appropriate timelines are met (see [Section 4.5.4](#)). A confirmation whether the treatment with docetaxel has started or not as planned will also need to be sent to the trial team.

Docetaxel is given according to local protocols as a standard non-trial treatment. The regime used previously within STAMPEDE was 75mg/m² Day 1 as 1hr IV infusion, plus prednisolone 5mg bid daily for 21 days repeated every 3 weeks for a maximum of 6 cycles.

The STAMPEDE TMG would suggest prednisolone could be omitted and data on the use of co-prescribed steroid will be collected on the relevant CRF (please see [Table 16](#) for more details)

6.3 ARM B: ADT + ZOLEDRONIC ACID

Note: recruitment stopped to the zoledronic acid and docetaxel-containing arms in Mar-2013 as recruitment target sample was reached. Please see Protocol version 13.0 for information on the administration of this trial drug. Trial treatment has now been completed for all patients.

6.4 ARM C: ADT + DOCETAXEL

Note: recruitment stopped to the zoledronic acid and docetaxel-containing arms in Mar-2013 as recruitment target sample was reached. Please see Protocol version 13.0 for information on the administration of this trial drug. Trial treatment has now been completed for all patients.

6.5 ARM D: ADT + CELECOXIB

Note: recruitment completed to both celecoxib-containing arms in Apr-2011 at the end of its Activity Stage II. Please see Protocol version 13.0 for information on the administration of this trial drug.

6.6 ARM E: ADT + DOCETAXEL + ZOLEDRONIC ACID

Note: recruitment stopped to the zoledronic acid and docetaxel-containing arms in Mar-2013 as recruitment target sample was reached. Please see Protocol version 13.0 for information on the administration of this trial drug. Trial treatment has now been completed for all patients.

6.7 ARM F: ADT + ZOLEDRONIC ACID + CELECOXIB

Note: recruitment completed to both celecoxib-containing arms in Apr-2011 at the end of its Activity Stage II. Please see Protocol version 13.0 for information on the administration of this trial drug

6.8 ARM G: SOC + ABIRATERONE

Note: recruitment to the “abiraterone comparison” completed in Jan-2014. Please note that some patients will continue treatment until all types of disease progression or up to a maximum of 2 years. All NOMO patients have now reached the maximum duration on trial treatment. Please see sections below for more information.

Standard-of-care: As described in [Section 6.2](#).

Abiraterone will be administered as a single 1000mg daily oral dose (4 tablets to be taken together once a day) together with prednisolone or prednisone 5mg daily to prevent secondary mineralocorticoid excess. Abiraterone absorption is increased by food. The tablets should be taken at least 2 hours after food, swallowed whole with some water. No food should be eaten for 1 hour afterwards.

Prednisolone (prednisone in Switzerland) should be taken as a single dose with food in the morning.

In patients with M1 disease, treatment with abiraterone will continue from randomisation until all types of progression have occurred, consistent with the COU-AA-301 trial (40) i.e., abiraterone would be given for these patients until a composite assessment based on:

- PSA progression (as defined in [Appendix J](#))
- Radiological progression (appearance of new lesions or progression of existing lesions) **and**
- Clinical progression (defined as new cancer-related symptoms)

It is accepted that these flexible criteria for stopping treatment with abiraterone are open to the investigator’s interpretation and discretion. Patients might continue treatment beyond the first failure-free survival (FFS) event; the first FFS event must be reported as per the other arms; all types of progression (PSA, radiological and clinical) need to be reported once.

In patients with NOM0 disease or N+M0 disease undergoing radical radiotherapy, treatment would continue for 2 years or all types of disease progression as defined for M1 patients, whichever is the sooner. ADT can be discontinued in this group at 2 years at the discretion of the local investigator (see [Section 6.2](#)).

For patients with N+M0 disease not planned for radical radiotherapy, treatment will continue as for patients with M1 disease until all types of disease progression.

If a patient allocated to Arm G develops only biochemical failure, the responsible clinician might switch from abiraterone + prednisolone 5mg od to abiraterone and dexamethasone 0.5mg od.

Trial treatment must stop if other systemic treatments are initiated at any time for disease progression control (including chemotherapy, radium 223 etc). Anti-androgens (i.e. bicalutamide) should not be given in combination with abiraterone due to the risk of toxicity.

See [Section 7.1.2](#) for further information on the trial definition of progression.

6.9 ARM H: SOC + PROSTATE RADIOTHERAPY IN M1 PATIENTS

Standard-of-care: androgen deprivation therapy +/- docetaxel (as described in [Sections 6.2.1](#) and [6.2.3](#)).

Radiotherapy will commence as soon as practicable. For non-docetaxel treated patients, this should ideally be within 4 weeks after randomisation. Docetaxel-treated patients should aim to start trial treatment within 3-4 weeks after the last administration of docetaxel, preferably within 6 weeks (see [Section 6.9.1](#)).

Treatment will be according to the guidelines in [Section 6.9.1](#). Two radiotherapy dose-fractionation schedules are permitted:

- 36Gy in 6 fractions of 6Gy, administered weekly over 6 consecutive weeks
- 55Gy in 20 fractions of 2.75Gy, administered daily, five days per week, over 4 consecutive weeks

Details of the recommendations for outlining, CTV and PTV are in [Section 6.9.1](#).

6.9.1 RESEARCH (M1) PROSTATE RADIOTHERAPY TREATMENT ADMINISTRATION

A treatment planning CT scan will be acquired with the patient supine, with empty rectum and comfortably full bladder.

Megavoltage equipment is required with effective photon energies $\geq 6\text{MV}$. Minimum source-to-axis distance is 100cm. Field arrangement is at the clinician's discretion: acceptable treatment techniques (field arrangement) include a 3-field (anterior, right lateral, and left lateral), 4-field (anterior, posterior, right lateral, and left lateral), or 6-field (right and left anterior oblique, right and left posterior oblique, and right and left lateral) or equivalent coplanar technique with multi-leaf collimation for all fields to adequately protect normal structures.

The Clinical Target Volume (CTV) will consist of the prostate gland alone as visualized on the treatment-planning CT scan. The base of the seminal vesicles may also be included if they are macroscopically involved. Inclusion of pelvic lymph nodes in the CTV is not permitted. The Planning Target Volume will have a 0.8 cm margin posteriorly and 1.0 cm margin in all other directions around the CTV to account for prostate gland motion and uncertainty in daily treatment setup.

Critical normal tissues should be delineated on the treatment-planning CT scan by the treating clinician:

- Rectum – inferior limit: level of ischial tuberosities; superior limit: sigmoid flexure
- Bladder – entirety

Two radiotherapy dose-fractionation schedules are permitted. In either case, radiotherapy is prescribed such that the PTV receives at least 95% of the prescribed dose:

- 36Gy in 6 fractions of 6Gy, administered weekly over 6 consecutive weeks
- 55Gy in 20 fractions of 2.75Gy, administered daily, five days per week, over 4 consecutive weeks

Dose-volume objectives for each dose-fractionation schedule are shown in [Tables 2 and 3](#) below. Values have been calculated using the formula $BED = D[1+d/(\alpha\text{-}\beta\text{ ratio})]$ assuming an alpha-beta ratio of 3 for rectum and bladder. These are provided for guidance only.

Portal imaging to verify accuracy of treatment delivery may be done according to the participating centre's local guidelines. Image-guidance technology (e.g., gold seed intraprostatic fiducial markers, cone-beam CT scanning) will be permitted according to clinician preference but is not required. Further illustration on the research radiotherapy arm schedule is shown in [Figure 4](#).

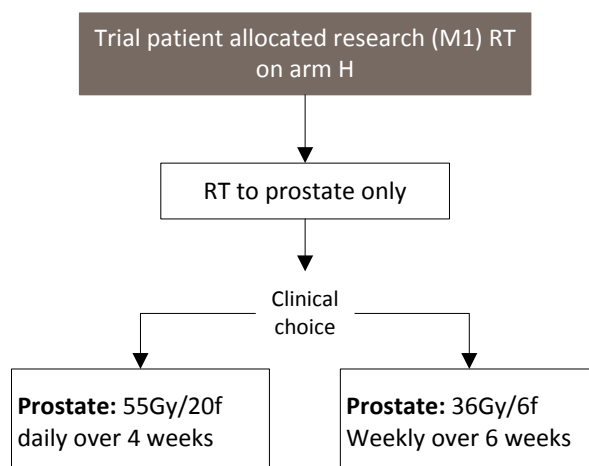
Table 2: Rectal dose volume objectives

55Gy/20F	36Gy/6F	MAX VOL (%)
52.5 Gy	33.3 Gy	50%
43.5 Gy	27.8 Gy	60%
26.1 Gy	16.7 Gy	80%

Table 3: Bladder dose-volume objectives

55Gy/20F	36Gy/6F	MAX VOL (%)
52.2	33.3	25%
43.5	27.8	50%

Figure 4: Diagram for deciding approach to research (M1) RT to the prostate



6.10 ARM J: SOC + ENZALUTAMIDE + ABIRATERONE + PREDNISOLONE

Note: recruitment to the “enzalutamide + abiraterone comparison” completed in March-2016. Please note that some patients will continue treatment until all types of disease progression or up to a maximum of 2 years. Please see sections below for more information.

Standard-of-care: androgen deprivation therapy +/- docetaxel and +/- M0 RT (as described in [Section 6.2](#)).

Abiraterone as described in [Section 6.8](#).

Prednisolone as described in [Section 6.8](#).

Enzalutamide will be administered as a 160mg oral dose (four capsules), taken together at the same time every day, with or without food.

Trial treatment must stop if other systemic treatments are initiated at any time for disease progression control (including chemotherapy, radium 223 etc). Anti-androgens (i.e. bicalutamide) should not be given in combination with enzalutamide due to the risk of toxicity.

In patients with M1 disease, treatment with both abiraterone and enzalutamide will continue until all types of progression have occurred, consistent with the approach taken for abiraterone (see [Section 6.8](#)) i.e., abiraterone and enzalutamide will be given until a composite of:

- PSA progression (as defined in [Appendix J](#))
- Radiological progression (appearance of new lesions or progression of existing lesions) **and**
- Clinical progression (defined as new cancer-related symptoms).

It is accepted that these flexible criteria for stopping treatment with abiraterone and enzalutamide are open to the investigator’s interpretation and discretion. Patients may continue treatment beyond the first failure-free survival (FFS) event; the first FFS event must be reported as per the other arms.

In patients with NOM0 disease or N+M0 disease undergoing radical radiotherapy, treatment would continue for 2 years or all types of disease progression as defined for M1 patients, whichever is the

sooner. ADT can be discontinued in this group at 2 years at the discretion of the local investigator (see [Section 6.2.1](#)).

For patients with N+M0 disease not planned for radical radiotherapy, treatment will continue as for patients with M1 disease until all types of disease progression.

If a patient has had PSA progression before commencing abiraterone and enzalutamide (i.e. on or shortly after completing docetaxel), they should still start trial treatment and should continue until radiological and/or clinical progression occurs. In the rare instances of a patient commencing abiraterone and enzalutamide having had biochemical **and** radiological progression, they should continue trial treatment whilst there is perceived evidence of benefit as judged by the local investigator. If a patient develops PSA progression only whilst on abiraterone and enzalutamide, the local investigator might consider switching from from abiraterone + prednisolone 5mg od to abiraterone and dexamethasone 0.5mg od.

See [Section 7.1.2](#) for further information on the definition of progression.

6.11 ARM K: SOC + METFORMIN

Standard-of-care: ADT ± RT (M0) ± docetaxel (as described in [Section 6.2](#))

Metformin will be given as a daily dose in addition to standard-of-care treatment. The starting daily dose is 850mg once-daily. If tolerated, this should be increased to the target dose of 850mg twice-daily after 1 month.

Metformin will be given as a daily dose in addition to standard-of-care treatment. The starting daily dose is 850mg once-daily. If tolerated, this should be increased to the target dose of 850mg twice-daily after 1 month.

In the case of **M0 patients**, if ADT is stopped after a minimum of 2 years, metformin should continue for a minimum of 3 years following randomisation and for a further 12 months after the administration of the last LHRH (which ever is longer). This is to allow for the delay in testosterone levels returning to normal following stopping ADT. In the event that ADT is re-started whilst patients remain on metformin (i.e. within 12 months of the last administration of LHRH) then metformin should continue whilst on ADT. If metformin is stopped 12 months after the last administration of LHRH it should not be re-started in the event of relapse.

For **M1 patients** metformin should continue whilst on ADT. Treatment should continue post-progression providing it is judged to be in the patients best interest. Metformin can be given together with any additional treatments started for progression excluding other IMPs i.e. investigators may choose to stop metformin treatment post progression in order to enable patients to participate in another clinical trial evaluating treatments for CRPC.

If metformin is paused for more than 3 months or >50% of doses are missed please discuss with the trial team as treatment is likely to need to be stopped.

In the event of stopping research treatment, unless a patient states otherwise, consent is assumed for continued recording trial data.

6.12 ADMINISTRATION, DOSE MODIFICATIONS AND MANAGEMENT OF TOXICITIES

6.12.1 ZOLEDRONIC ACID OR DOCETAXEL (ARMS B, C AND E)

Note: recruitment stopped to the zoledronic acid and the docetaxel-containing arms in Mar-2013 as recruitment target sample was reached. Primary survival analysis was presented at ASCO in Jun-2015. Please see Protocol version 13.0 for information on how these drugs were previously administered in the trial.

6.12.2 CELECOXIB (ARMS D AND F)

Note: recruitment completed to both celecoxib-containing arms in Apr-2011 at the end of Activity Stage II. No new patients should be receiving this agent as first-line within the trial. Primary survival analysis was presented at GU ASCO in Jan-2016. Please see Protocol version 13.0 for information on how this drug was previously administered in the trial.

6.12.3 ABIRATERONE OR ENZALUTAMIDE + ABIRATERONE (ARMS G AND J)

Abiraterone absorption is increased by food. The tablets should be taken at least 2 hours after food, swallowed whole with some water. No food should be eaten for 1 hour afterwards. Prednisolone (prednisone in Switzerland) should be taken as a single dose with food in the morning.

Enzalutamide can be taken with or without food.

6.12.3.A Abiraterone Contraindications

- Unusual or allergic reaction to past abiraterone acetate treatment
- Uncontrolled hypertension
- Uncontrolled heart failure
- Abnormal liver function or active or chronic liver disease

6.12.3.B Abiraterone Special Warnings and Precautions For Use

Timing of administration compared with meals

Administration of abiraterone acetate with food significantly increased the absorption of abiraterone acetate, it is therefore recommended that abiraterone acetate is taken on an empty stomach.

Cardiovascular history

Abiraterone acetate should be used with caution in patients with a history of cardiovascular disease. The safety of abiraterone acetate in patients with left ventricular ejection fraction <50% or New York Heart Association (NYHA) Class III or IV heart failure has not been established. Before treatment with abiraterone acetate, hypertension must be controlled and hypokalemia must be corrected.

Caution is required in treating patients whose underlying medical conditions might be compromised by increases in blood pressure, hypokalemia, or fluid retention, e.g. those with heart failure, recent myocardial infarction, or ventricular arrhythmia.

Blood pressure management

Blood pressure should be monitored **every 2 weeks until week 12 and at every follow-up visit after week 12** whilst the patients remain on treatment. For the management of abiraterone induced hypertension see [Table 5](#).

Hepatic Impairment

No dosage adjustment was necessary for patients with pre-existing mild hepatic impairment. Abiraterone acetate should not be used in patients with pre-existing moderate or severe hepatic impairment.

Hepatotoxicity

Marked increases in liver enzymes leading to drug discontinuation or dosage modification occurred in controlled clinical studies. Serum transaminase and bilirubin levels should be measured prior to starting treatment with abiraterone acetate, **every 2 weeks for the first 3 months of treatment, and monthly thereafter.**

If clinical symptoms or signs suggestive of hepatotoxicity develop, serum transaminases, in particular serum alanine aminotransferase (ALT), should be measured immediately. See [Table 4](#) for the management of abiraterone induced hepatotoxicity.

Renal Impairment

No dose adjustment are required in renal impairment; however caution is advised if patients develop severe renal impairment as there is limited clinical data in this population. Systemic exposure to abiraterone after a single oral 1000mg dose did not increase in patients with end-stage renal disease on dialysis.

6.12.3.C Abiraterone Undesirable Effects

The most common adverse drug reactions observed in the integrated safety data for those patients who received 1000mg abiraterone acetate plus prednisone or prednisolone in clinical studies (n=1,070) were fatigue, arthralgia, peripheral oedema, back pain, bone pain, nausea, constipation, hypokalemia and anaemia.

The adverse events graded as 3 or 4 and which occurred in more than 5% of patients were fatigue, peripheral oedema, anaemia and back pain see [Appendix G](#).

6.12.3.D Abiraterone Overdose

There have been no reports of overdose of abiraterone acetate during clinical studies. There is no specific antidote to abiraterone acetate. In the event of an overdose, administration of abiraterone acetate should be stopped and general supportive measures undertaken, including monitoring for cardiac arrhythmias. Liver function should also be assessed.

6.12.3.E Enzalutamide Contraindications

The full induction potential of enzalutamide may not occur until approximately 1 month after the start of treatment, when steady-state plasma concentrations of enzalutamide are reached, although some induction effects may be apparent earlier. Monitoring should continue for at least the first month of treatment and dose adjustments considered. Given the long half-life of enzalutamide (5.8 days), effects on enzymes may persist for one month or longer after stopping enzalutamide. A gradual dose reduction of the concomitant medicinal product may be necessary when stopping enzalutamide treatment.

6.12.3.F Enzalutamide Special Warnings and Precautions For Use

History of seizures

Caution should be used in administering enzalutamide to patients with a history of seizures or other predisposing factors including, but not limited to, underlying brain injury, stroke, primary brain tumors or brain metastases or alcoholism. In addition, the risk of seizure may be increased in patients receiving concomitant medications that may lower the seizure threshold. Enzalutamide should be permanently discontinued in patients who have a seizure while on treatment.

Hepatic impairment

A hepatic impairment study showed that the composite AUC of enzalutamide plus N-desmethyl enzalutamide after administration of a single dose of enzalutamide was similar in patients with baseline mild, moderate or severe hepatic impairment (Child-Pugh Class A, B or C respectively) relative to patients with normal hepatic function, and no starting dose adjustment is needed.

Renal impairment

No dose adjustment are required in renal impairment; however caution is advised if patients develop severe renal impairment as there is limited clinical data in this population.

6.12.3.G Enzalutamide Overdose

There is no antidote for enzalutamide. In the setting of an overdose, stop treatment with enzalutamide and initiate general supportive measures taking into consideration the $t^{1/2}$ of 5.8 days. Patients may be at increased risk of seizures following an overdose.

6.12.3.H Management of specific toxicities due to abiraterone and enzalutamide

The safety monitoring and toxicity management plan described below takes into account AEs based on the reported clinical safety data of abiraterone and enzalutamide given separately. There are limited reported data on the safety and toxicity of the combination of enzalutamide and abiraterone however the recommendations summarised here have been updated in light of the experience gained in STAMPEDE as recommended by the STAMPEDE TMG.

Seizures

If any patient suffers a seizure whilst on treatment, enzalutamide should be discontinued immediately. Abiraterone and prednisolone can be continued providing there are no abiraterone-specific toxicities.

Table 4: Management of Abnormal Liver Function Tests (LFTs) associated with abiraterone (given alone or in combination with enzalutamide)

TOXICITY EVENT	ACTION
Grade 1 increases in AST, ALT or bilirubin (e.g. increase in AST or ALT from ULN to 2.5X ULN; increase in total bilirubin from ULN to 1.5X ULN)	The frequency of LFT monitoring should be increased to at least weekly, if the investigator judges that the laboratory abnormalities are potentially related to study medication. No dose reduction is required. Providing LFTs are stable for 4 weeks, resume monthly checks
Grade 2 increases in AST, ALT or bilirubin (e.g. increase in AST or ALT to >2.5-5X ULN; increase in total bilirubin from >1.5-3X ULN)	Withhold abiraterone, enzalutamide and all other concomitant medications that are potentially hepatotoxic. The frequency of LFT monitoring should be increased to at least weekly until the liver function tests return to baseline value or grade 1 when all trial medication can be re-started. No dose reduction is required after one episode providing this resolved within 4 weeks but should be considered if Grade 2 derangements recurs.
Grade 3 increases in AST, ALT or bilirubin (e.g. increase in AST or ALT to >5X ULN; increase in total bilirubin to >3X ULN),	Withhold abiraterone and enzalutamide and all other concomitant medications that are potentially hepatotoxic. At least weekly monitoring is required until the LFTs return to baseline value or grade 1. Enzalutamide can be re-started with no dose reduction. See below for abiraterone re-challenge.
Grade 4 increases in AST, ALT or bilirubin (e.g. increase in AST or ALT to >20x ULN; increase in total bilirubin to >10x ULN)	Patients must discontinue abiraterone and enzalutamide immediately. At least weekly monitoring is required until the LFTs return to baseline value or grade 1 and then prednisone can be discontinued and the investigator can consider restarting enzalutamide. Abiraterone should not be re-introduced.
RE-CHALLENGE	ACTION
Recurrent G2 derangement	Reduce to 750mg once LFTs return to G1
If study treatment resumption is considered for patients who have experienced Grade 3 increases in AST,	Resume study treatment with abiraterone dose reduction to 750mg when grade 3 toxicities resolve to grade 1 or baseline.

TOXICITY EVENT	ACTION
ALT, or bilirubin	
If Grade 3 or higher increases in AST, ALT or bilirubin recur after the first dose reduction	Hold study medication and all other concomitant medications that are potentially hepatotoxic. At least weekly LFT monitoring is required, starting immediately regardless of study schedule and continued until a return to baseline values or Grade 1.
If study treatment resumption is considered for patients who have experienced Grade 3 increases in AST, ALT, or bilirubin with the first dose reduction	Resume study treatment with abiraterone dose reduction to 500mg when AST, ALT or bilirubin returns to baseline value or Grade 1.

An opinion from a hepatologist should be considered if there are any concerns or liver function derangement shows no improvement within 2 weeks of discontinuation of abiraterone.

Table 5: Management of hypertension associated with abiraterone (given alone or in combination with enzalutamide)

TOXICITY EVENT	ACTION
Grade 1-2	Management as per investigator with anti-hypertensive treatment and increase frequency of blood pressure monitoring to at least weekly. Follow local guidance for selection of anti-hypertensives but avoid thiazide diuretics to minimise risk of serum potassium derangement. Calcium channel antagonists or beta blockers are often preferred. As with other symptoms of mineralocorticoid excess, consider increasing prednisolone dose to 5mg BID.
Grade 3-4	Withhold abiraterone and enzalutamide. Adjust or add anti-hypertensive medications to mitigate the toxicity. When hypertension resolves to Grade ≤ 1 , resume both enzalutamide and abiraterone at full dose with prednisolone 5mg bid.

An opinion from a cardiologist should be considered if blood pressure control is not achieved within 4 weeks.

Table 6: Management of hypokalaemia associated with abiraterone (given alone or in combination with enzalutamide)

TOXICITY EVENT	ACTION
Grade 1 (LLN- 3.0mmol/L)	Supplement with oral potassium and monitor closely and increase prednisolone dose to 5mg BID. Exclude and manage other causes of hypokalemia.
Grade 2 (<LLN – 3.0mmol/L and symptomatic)	Pause abiraterone Supplement with oral potassium and monitor closely and increase prednisolone dose to 5mg BID. Exclude and manage other causes of hypokalemia Re-start abiraterone with close monitoring, discontinue if recurs
Grade 3	Abiraterone will be permanently discontinued and the patients will be

TOXICITY EVENT	ACTION
(<3.0–2.5mmol/L) or life-threatening Grade 4 (<2.5mM)	hospitalized for intravenous potassium replacement and cardiac monitoring. After the return of serum potassium to normal, prednisolone will be discontinued. The patient can continue on enzalutamide alone. If hypokalaemia persists consider a dose reduction of enzalutamide to 120mg once a day.

Table 7: Management of fluid retention/oedema associated with abiraterone (given alone or in combination with enzalutamide)

TOXICITY EVENT	ACTION
Grade 1-2	Increase prednisolone dose to 5mg bid.
Grade 3-4	Withhold abiraterone. Consider addition of mineralocorticoid receptor antagonist eplerenone until resolution of symptoms. Enzalutamide can be continued. When fluid retention/oedema returns to baseline or resolves to ≤Grade 1, resume abiraterone at full dose with prednisone 5mg bid, if symptoms do not resolve abiraterone should not be re-started and enzalutamide should be dose reduced to 120 mg per day.

Table 8: Management of diarrhoea (associated with abiraterone or enzalutamide)

TOXICITY EVENT	ACTION
Grade 1-2	Symptomatic management.
Grade 3-4	Withhold abiraterone. If no improvement reduce dose of enzalutamide to 120 mg per day. Once resolved to Grade 1, recommence abiraterone at 750 mg per day.

Table 9: Management of arthralgia & muscle pain

TOXICITY EVENT	ACTION
Grade 1-2	Symptomatic management
Grade 3-4	Reduce dose of enzalutamide to 120 mg per day

Table 10: Management of fatigue (associated to enzalutamide)

TOXICITY EVENT	ACTION
Grade 1-2	Consider a dose reduction to 120mg/day
Grade 3-4	Pause enzalutamide for 1 week or until the toxicity grade improves to grade 2 or lower severity. Re-started at a reduced dose (120mg/day or 80mg/day) in consultation with the study team.

6.12.3.1 Management of Specific Toxicities from Prednisolone

Prednisolone/prednisone will be started at 5mg once daily, to prevent secondary mineralocorticoid excess.

Prednisolone/prednisone dose increase of up to 5mg BID is permitted to manage mineralocorticoid-related toxicities (e.g., hypokalaemia, hypertension) which are refractory to standard management.

Patients experiencing serious symptoms of Cushing's syndrome (e.g., weight gain, muscle loss) can decrease or discontinue (temporarily or permanently) steroids at the investigator's discretion but should be closely monitored for symptoms of secondary mineralocorticoid excess. It should be noted that weight gain and muscle loss are also associated with ADT.

6.12.4 METFORMIN

The starting dose for metformin is 850mg once daily. If tolerated this should be increased to the target dose of 850mg twice daily after 1 month. Metformin should be taken around the same time each day and treatment tolerance is best if taken with or after food. For twice daily dosing, the minimum time between doses should be 8 hours, doses should not be taken closer together if forgotten or missed.

6.12.4.A Metformin Contraindications

Metformin should be permanently stopped if the eGFR falls to ≤ 60 ml/min/1.73m².

6.12.4.B Metformin Special Warnings and Precautions For Use

Renal impairment

Metformin is not nephrotoxic, but is exclusively excreted by the kidneys. Therefore treatment should only be started in participants with stable renal function (eGFR > 60 ml/min/1.73m²). Renal function should be monitored at least every 6 months in participants with stable renal function and at least two to four times a year in participants with serum creatinine levels at the upper limit of normal and > 75 years. Additional monitoring is required in any patient at risk of deteriorating renal function (see [Table 11](#)). If the eGFR is between 45-60 ml/min/1.73m² a dose reduction to 500mg BID is required in accordance with SmPC of product being used.

Lactic acidosis is a very rare (3/100,000 patient years), but serious metabolic consequence. Reported cases have occurred primarily in diabetic patients with significant renal impairment. The risk factors for lactic acidosis are: renal impairment, prolonged fasting or malnutrition, excessive alcohol intake, hepatic insufficiency or any condition associated with hypoxia e.g. cardiac or respiratory failure or circulatory shock due to any cause.

See [Table 11](#) for situations when metformin treatment should be paused due to the risk of deterioration in renal function.

Table 11: Situations when metformin treatment should be paused

	RISK FACTOR
Iodinated contrast agents	Pause metformin 24 hours prior to receiving contrast and re-start 48 hours post administration.
Anaesthesia (peridural; spinal or general)	Pause metformin 48 hours prior to procedure and re-start no earlier than 48 hours following procedure, providing oral intake re-established and renal function is stable and at baseline.
Surgery	Pause metformin 48 hours prior to procedure and re-start no earlier than 48 hours following procedure, providing oral intake re-established and renal function is stable and at baseline.
Dehydration e.g. nausea, vomiting or diarrhoea	Pause metformin and re-start only when oral intake is re-established and renal function is stable and at baseline.
Obstructive uropathy e.g. urinary retention or ureteric obstruction	Pause metformin and re-start only when renal function confirmed to be stable and at baseline.

Treatment breaks

It is anticipated that metformin treatment will be paused for approximately 72 hours around the time of contrast enhanced CT scans (see [Table 11](#)) and may need to be paused during episodes of inter-current illness. If metformin is paused for 6 days or less this information does not need to be recorded and no additional action is needed. Treatment pauses of 7 days or more should be recorded by updating the metformin treatment log.

If metformin treatment is paused for more than 2 weeks, investigators may consider re-starting at 850mg once daily for the first 4 weeks before escalating to full dose providing tolerance is acceptable. It is suggested that, providing patients have a sufficient supply of labelled IMP metformin tablets, a telephone consultation may be sufficient to assess tolerance and advice regarding dose modification in order to limit hospital visits.

If treatment is paused for more than 3 months or >50% of doses are missed for any reason the trial team should be informed as metformin may need to be discontinued.

6.12.4.C Management of Specific Toxicities from Metformin

Gastrointestinal disturbance

Gastrointestinal disturbances are very common with metformin and include nausea, vomiting, diarrhoea, abdominal pain and loss of appetite. These are most common when first starting treatment (>1/10).

Table 12: Management of metformin related gastrointestinal toxicity

TOXICITY EVENT	ACTION
Grade 1	Ensure metformin is taken with or after food. Consider switching to 750 mg sustained release preparation if available. If unavailable consider a 1 week treatment pause and re-start at 850mg once daily and attempt an escalation after 1 month. If necessary, remain at 850mg OD. If unable to tolerate 850mg OD or sustained release preparations are not available consider dose reduction to 500mg OD.
Grade 2 or higher	Reduce to 850mg OD and if symptoms improve to grade 1 or better, re-attempt dose escalation after 1 week. If symptoms recur at grade 2 or higher, pause treatment for 2 weeks. Re-start at 850mg sustained release or if not available 500mg OD. Re-attempt a dose escalation 2 months later. Continue at the maximum tolerated dose providing symptoms $\leq$ grade 1.

Other possible metformin related toxicities are taste disturbance and B12 deficiency resulting in megaloblastic anaemia and skin reactions (see [Appendix Table 7](#)). Any patient who experiences anaemia whilst taking metformin should have haematinics including vitamin B12 measured and replaced if deficient.

Lactic acidosis

This is a very rare but serious toxicity which has been observed in diabetic patients taking metformin. It is unclear whether this is due to the underlying diabetes or metformin. This is supported by a meta-analysis demonstrating similar rates of lactic acidosis in diabetics taking metformin compared with diabetics not taking metformin (41). This evidence suggests this side effect may be a complication of diabetes and may not be associated with metformin treatment. The risk of lactic acidosis should be considered in the event of non-specific symptoms such as muscle cramps, abdominal pain and/or severe weakness or lethargy. Any patient with a suspected metabolic acidosis requires immediate discontinuation of metformin and evaluation. Lactic acidosis is characterised by metabolic acidosis (decreased blood pH, high lactate above 5mmol/L and an increased anion gap and lactate/pyruvate ratio). The most effective way to remove lactate and metformin is haemodialysis.

Metformin overdose

Hypoglycaemia has not been reported with metformin doses up to 85g although lactic acidosis has rarely occurred in such circumstances. Patients should be urgently assessed in the event of an overdose and hospital admission considered. The management of metformin overdoses should be as per standard clinical care by the local team. The most effective way to remove lactate and metformin is haemodialysis.

6.13 CONCOMITANT MEDICATIONS AND DRUG INTERACTION

6.13.1 ABIRATERONE: INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

Details on drug interactions are described in [Appendix G](#). The table below provides a summary on the main interactions.

Drugs which may interact with Abiraterone

Note: Anti-androgens. Abiraterone is steroid synthesis inhibitor and should not be given together with any other anti-androgens given the risk of toxicity. Cyproterone acetate should be discontinued 10 days and finasteride stopped 48 hours before commencing abiraterone. Concomitant use of dutasteride, bicalutamide and flutamide are all contraindicated.

Table 13: Drugs which may interact with Abiraterone

DRUGS WHICH MAY INCREASE ABIRATERONE LEVELS			
Substrate	Clinical Use	Drug	Recommendation
CYP3A4 inhibitors	Macrolide antibiotics	Clarithromycin	Avoid or hold abiraterone if short term use unavoidable given increased risk of abiraterone toxicity
	Anti-fungals	Ketoconazole Itraconazole Voriconazole	Avoid or hold abiraterone if short term use unavoidable given increased risk of abiraterone toxicity
DRUGS WHICH MAY REDUCE ABIRATERONE LEVELS			
Substrate	Clinical Use	Drug	Recommendation
CYP3A4	Anti-epileptics*	Phenytoin Carbamazepine Phenobarbital Primadone	Contraindicated
	Anti-depressants	St Johns Wart	Contraindicated
	Anti-TB	Rifampicin Rifabutin	Contraindicated
	Anti-retroviral	Atazanavir Saquinavir Ritonavir Indinavir Nelfonavir	Contraindicated. Seek specialist advice and discuss with trial team
DRUGS WHICH MAY ACCUMULATE WHEN GIVEN WITH ABIRATERONE			
Substrate	Clinical Use	Drug	Recommendation
CYP2D6	Cardiac	Metoprolol Propranolol Propafenone Flecainide	Monitoring required as drug levels may increase with abiraterone use
	Anti-depressants	Desipramine Venlafaxine Citalopram	Monitoring required as drug levels may increase with abiraterone use
	Anti-psychotics	Haloperidol Risperidone	Monitoring required as drug levels may increase with abiraterone use
	Analgesia	Tramadol Codeine Oxycodone	Monitoring required as drug levels may increase with abiraterone use
	Alpha blockers	Tamsulosin	Monitoring required as drug levels may increase with abiraterone use
	Anti-diabetic	Repaglinide	Monitoring required as drug levels may increase with abiraterone use

*Please note that any history of epilepsy is an exclusion criteria.

6.13.2 ENZALUTAMIDE: INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

Details on drug interactions are described in [Appendix G](#). The table below provides a summary of the main interactions

Anti-androgens

Enzalutamide is potent androgen receptor antagonist and should not be given together with any

other anti-androgens given the risk of toxicity. Cyproterone acetate should be discontinued 10 days and finasteride stopped 48 hours before commencing enzalutamide. Concomitant use of dutasteride, bicalutamide and flutamide are all contraindicated.

Table 14: Drugs which may interact with Enzalutamide

DRUGS WHICH MAY INCREASE ENZALUTAMIDE LEVELS			
Substrate	Clinical Use	Drug	Recommendation
CYP2C8 inhibitors	Lipid-lowering	Gemfibrozil	Avoid, if no alternatives, reduce enzalutamide dose to 80mg
DRUGS WHICH MAY DECREASE ENZALUTAMIDE LEVELS			
Substrate	Clinical Use	Drug	Recommendation
CYP2C8 inducers	Anti-TB	Rifampicin Rifabutin	Avoid and switch to an alternative if possible
CYP3A4 inducers	Anti-epileptics	Phenytoin Carbamazepine Phenobarbital	Contraindicated
	Anti-depressant	St Johns Wart	Contraindicated
	Anti-retrovirals	Atazanavir Saquinavir Ritonavir Indinavir Nelfanavir	Contraindicated. Seek specialist advice and discuss with trial team
ENZALUTAMIDE MAY REDUCE DRUG LEVELS			
Substrate	Clinical Use	Drug	Recommendation
CYP2C19	Gastric protection	Omeprazole	Omeprazole AUC reduced by 70% Consider increasing dose of omeprazole for same therapeutic effect
CYP3A4	Analgesia	Fentanyl* Alfentanil* Tramadol	Monitor closely and consider alternatives
	Immunosuppressants	Sirolimus* Tacrolimus* Cyclosporine*	Monitor closely
	Anti-migraine	Ergotamine	Monitor closely
	Cardiac	Nifedipine Ivabradine	Monitor closely, consider alternatives as clinical effect may be reduced
CYP2C9	Anti-epileptics	Phenytoin*	Contraindicated
	Anti-coagulants	Warfarin*	Warfarin AUC reduced by 56% Consider switching to low molecular heparin, increase INR monitoring if this is not possible
DRUGS WHICH MAY ACCUMULATE WHEN GIVEN WITH ENZALUTAMIDE			
Substrate	Clinical Use	Drug	Recommendation
p-gp		Colchicine* Dabigatran* Digoxin*	Monitor closely

*narrow therapeutic index

6.13.3 METFORMIN: INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

Metformin does not interact with any of the other treatments for prostate cancer and can be continued during all further treatments started on progression.

Caution is needed however when initiating potential nephrotoxic drugs as metformin is renal excreted therefore may accumulate if renal function deteriorates.

As metformin is being given as an IMP in the context of a clinical trial, continued use will not be permitted if patients participate in other interventional clinical trials for prostate cancer (i.e. CRPC setting). Investigators should use their discretion and discuss discontinuing metformin with the trial team if it is felt to be in the patient's best interest.

Table 15: Drugs which require additional monitoring of renal function

Clinical use	Drug	Recommendation
Anti-hypertensives and other cardiac disease	ACE inhibitors/angiotension II receptor blockers e.g. ramipril, lisinopril, Irbesartan	Monitor renal function until confirmed to be stable and providing eGFR remains $>60\text{mls/min/m}^2$. Repeat test if necessary
	Diuretics e.g. Frusemide, budesonide	
Antibiotics	Aminoglycoside antibiotics e.g. Gentamycin or amikacin	Hold metformin during treatment and re-start providing renal function confirmed to be stable and eGFR remains $>60\text{mls/min/m}^2$
Analgesia	NSAIDS e.g. Ibuprofen, diclofenac, naproxen	Avoid if possible If no alternative increase renal monitoring to until confirmed to be stable and providing eGFR remains $>60\text{mls/min/m}^2$

6.14 TRIAL PRODUCTS

Details of the procedures for obtaining the drugs within the trial, dispensing and disposal of unused drug are given in [Appendix E](#). Arrangements for free or discounted drugs are given in the Finance section ([Section 15](#)).

6.15 MEASURES OF COMPLIANCE/ADHERENCE

Date of treatment, dose, delays and reasons for delays or dose modifications of all trial treatments will be recorded. The estimated number of abiraterone tablets and enzalutamide capsules taken in a given time period will also be recorded as well as any dose reductions.

6.16 TREATMENT DATA COLLECTION

Data will be recorded on case report forms (CRFs); the top copy/original should be sent to the MRC CTU for data entry and a copy kept at the local centre. Up-to-date versions of all CRFs can be found on the trial website (<http://www.stampedetrial.org/>) and centres will be notified of any changes throughout the course of the trial. The type of data to be recorded is detailed in the Assessments and Procedures section ([Section 7](#)).

6.17 ADMINISTRATION OF STANDARD RADIOTHERAPY TO NON-METASTATIC PATIENTS

6.17.1 TREATMENT DETAILS

Standard radiotherapy will be given to appropriate patients in each of the trial arms, following a period of neo-adjuvant ADT therapy, as is generally standard in UK practice. For patients with negative nodes on axial imaging, clinicians may choose between irradiating prostate and seminal vesicles alone or including the pelvic nodes in addition. Additional staging tests such as pelvic node sampling may be considered in making this decision. Conformal or intensity modulated radiotherapy should be used in all patients. Where patients have good clinical evidence that nodes are free of tumour or patients for whom nodal radiotherapy is contra-indicated (e.g. significant bowel disease), treatment may be given to the prostate gland and seminal vesicles only. The recommended dose is 74Gy in 37 fractions to the prostate and seminal vesicles or the equivalent using hypo-fractionated schedules. These recommendations are summarised in [Figure 5](#). Alternative dosing schedules are permitted but must be agreed with the STAMPEDE Trial Management Group.

6.17.1.A Standard-of-care RT Timing in M0 patients

If receiving docetaxel as part the standard-of-care (permitted from Protocol version 14.0), the patient must have sufficiently recovered from any docetaxel toxicity before RT can begin.

In all other patients not receiving SOC docetaxel, SOC RT may be started sooner (2-6 months post-randomisation) in line the data from the MRC PR07 trial (8).

6.17.1.B Type Of standard-of-care RT in M0 patients

Conformal or intensity modulated radiotherapy. EBRT also permitted

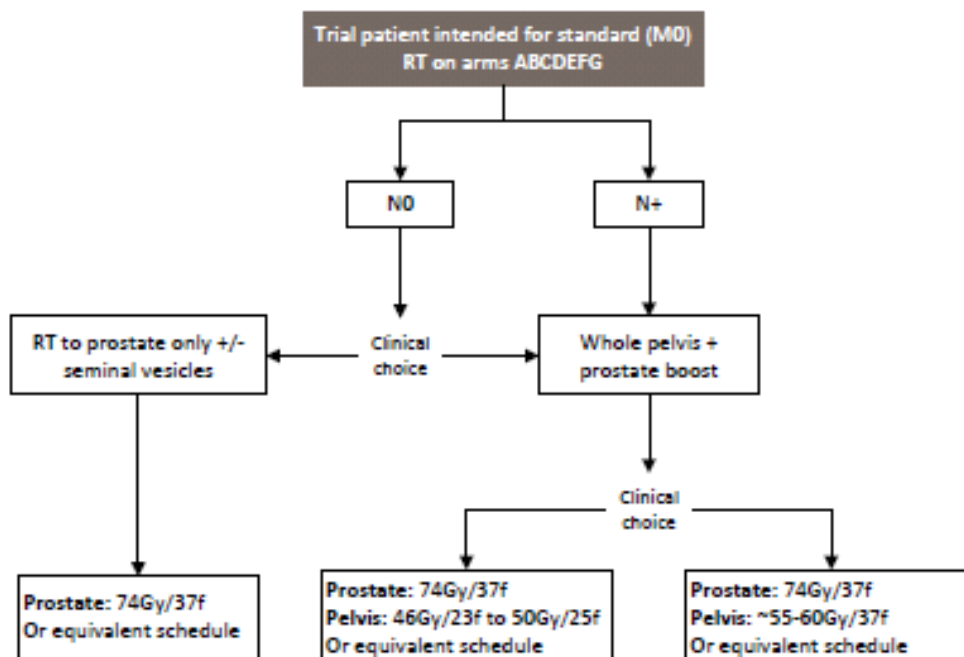
6.17.1.C Standard Clinical Target Volume in M0 patients

- **CTV1:** Prostate plus seminal vesicles
- **CTV2:** (Node positive patients) Regional lymph nodes to include internal iliac and the inferior part of the common iliac nodes as used in EORTC trial 22961 (42)
- **PTV1:** CTV1 plus 10-15 mm according to local practice
- **PTV2:** CTV2 plus 10-20mm according to local practice

6.17.1.D Standard-of-care RT Dose in M0 patients

Prostate dose of 74Gy in 2Gy fractions or equivalent, with optional dose to the pelvic nodes of 55-60Gy in 2Gy fractions or equivalent dose using IMRT is recommended. Recent evidence from the CHHIP trial has shown a dose of 60Gy in 20 fractions to be non-inferior and therefore it is now also permitted. Higher nodal doses may be considered if the department is experienced in using IMRT for nodal radiotherapy, particularly as data emerges from the PIVOTAL trial of nodal IMRT in high-risk node negative patients where a nodal dose of 60Gy in 37 fractions is being evaluated. Alternative schedules should be agreed with the STAMPEDE Trial Management Group.

Figure 5: Diagram for deciding recommended approach to standard-of-care (M0) RT in non-metastatic patients



6.18 NON-TRIAL TREATMENT

6.18.1 MEDICATIONS PERMITTED

Guidance on concomitant medications and drug interaction is detailed in [Section 6.13](#).

6.18.2 DATA ON CONCOMITANT MEDICATION

All concomitant medication will be recorded on the baseline form prior to randomisation and on any subsequent Serious Adverse Event forms. From Protocol version 15.0 onwards, details of long-term use of concomitant medications of classes of interest will be recorded on the follow up forms.

Please see [Section 6.13](#) for further details on drug interactions for abiraterone and enzalutamide.

7 ASSESSMENTS AND PROCEDURES

7.1 SCHEDULE FOR ASSESSMENTS

A detailed follow-up schedule is given in [Table 16, 17, 18 and 19](#).

7.1.1 PSA MEASUREMENTS

All patients should have PSA measured prior to starting Androgen Deprivation Therapy and at weeks 6, 12, 18 and 24 and every 12 weeks, thereafter, up to 2 years post randomisation. Following this, PSA should be measured every 6 months until 5 years and annually, thereafter. For patients who do not have a scheduled hospital visit, it would be acceptable for arrangements to be made for blood samples to be drawn at their GP surgery.

7.1.2 ASSESSMENT OF TREATMENT FAILURE (DEFINITION OF PROGRESSION)

It is not proposed to routinely assess patients for response. However, in order that objective progression can be assessed, it is necessary to have imaging taken at time of best response as judged by the treating clinician. All patients should have baseline radiological examinations as detailed in [Section 4.4.1](#). In addition, it is recommended all patients should have scans or X-rays repeated at 24 weeks (and whenever clinically appropriate) if they were abnormal at baseline, particularly if they have a low PSA value on entry in to the trial making biochemical assessment of treatment failure difficult.

The following outcomes should be reported on a Progression form & Additional Treatment CRF:

- Biochemical failure – should be reported alongside castrate levels of testosterone (see [Appendix J](#)).
- Local progression
- Lymph node progression
- Progression or development of new distant metastases, defined as lymph nodes outside the pelvis, bone or organ involvement
- Skeletal-related events (see below)

The first instance of each type of progression event should be reported. For all outcomes excluding biochemical progression, information is required on how this has been detected e.g. clinical based on symptoms or objectively based on imaging. The date of progression is defined as the earliest date of detection e.g. if clinical progression is detected and subsequently confirmed imaging please give the date of clinical progression.

Skeletal-related events (SREs) are defined as:

- Pathological Fracture
- Spinal cord compression
- Requirement for RT to bone (e.g. for pain or impending fracture)
- Requirement for surgery (e.g. for prevention or management of fracture)

SREs are a secondary outcome measure and a disease event of interest. SREs may represent disease progression but can also occur due to treatment related effects e.g. osteoporotic fracture due to treatment related bone-mineral density loss. From Protocol version 15.0 information regarding SREs will be collected at each follow up visit. All SREs should be investigated further to establish whether or not the patient has progressed and, if confirmed as progression, a Progression and Additional

Treatment CRF should be completed to record this and give details of any treatment received (e.g. palliative RT or surgery)

The summary of timing of Case Report Forms can be viewed in [Table 17](#).

7.1.2.A ADDITIONAL METABOLIC AND CARDIOVASCULAR OUTCOMES

Metformin is hypothesised to mitigate the metabolic and cardiovascular effects of long-term ADT. Additional metabolic and cardiovascular outcomes are now to be collected from Protocol version 15.0 onwards to explore this effect.

The summary of timing of Case Report Forms can be viewed in [Table 16 and 17](#).

Table 16: Collection of additional metabolic and cardiovascular outcomes

OUTCOME OF INTEREST	TIMING OF ASSESSMENT	CRF
Eligibility screening		
HbA1c	Prior to randomisation	Randomisation
Baseline		
Lipid profile (cholesterol, HDL, LDL)	Randomisation	Baseline
Fasting glucose	Randomisation	Baseline
Fasting triglyceride	Randomisation	Baseline
Weight and BMI	Randomisation	Baseline
Waist circumference	Randomisation	Baseline
Follow-up		
HbA1c	6 months 12 months 24 months	FU
Fasting glucose	6 months 12 months 24 months	FU
Fasting triglyceride	6 months 12 months 24 months	FU
Lipid profile (cholesterol, HDL, LDL)	Annual	FU
Metabolic and cardiovascular events		
New diagnosis of diabetes	As and when metabolic and cardiovascular event occurs	FU
Cardiac event: myocardial infarction or revascularization (e.g PCI or CABG)	As and when metabolic and cardiovascular event occurs	FU
Stroke or transient ischaemic event	As and when metabolic and cardiovascular event occurs	FU

7.1.3 ADDITIONAL SAFETY ASSESSMENT

Due to the risk of liver toxicity and secondary hyperaldosteronism with abiraterone, patients will require **2 weekly U+Es, LFTs** for the first **12 weeks and monthly BP checks**. It is not necessary to report these unless abnormal; in this instance, they should be reported as AEs (on the next Follow-up CRFs) and as SAEs (see [Section 11](#)) if appropriate. After 12 weeks, **monthly liver function tests** will be required on all patients receiving treatment on trial abiraterone.

Medical review and PSA measurements are repeated for all patients across all research arms (including the control arm) and follow the trial FU schedule (every 6 weeks for 6 months, every 12 weeks up to 2 years, six-monthly up to 5 years and annually thereafter). For patients who do not have a scheduled hospital visit, it would be acceptable for arrangements to be made for blood samples to be drawn either in a GP's surgery or in the patient's home.

Patients with normal and stable renal function receiving **metformin** require monitoring of **renal function (U&Es) every 6 months** whilst on treatment. More frequent monitoring is required in patients with declining renal function, or when initiating new potentially nephrotoxic medications or at times of intercurrent illness (see [Section 6.12.4](#)). Changes in renal function (eGFR, graded according to CTCAEv.4) are recorded on the follow up CRF. It is acceptable for bloods sampling to be arranged via the GP at the patient's home or local hospital.

The summary of timing of Case Report Forms can be viewed in [Table 17](#).

7.1.4 DATA COLLECTION FOR STANDARD DOCETAXEL

The decision to use docetaxel as part of the standard-of-care must be made before randomisation and should be recorded on the randomisation CRF. The date of the first cycle should be recorded at the time of randomisation; this can be a planned date when randomisation occurs prior to docetaxel commencing but must be within 12 weeks of starting ADT (see [Section 6.2.3](#)). All further details should be recorded on the standard-of-care docetaxel CRF upon completion of the final cycle.

If a patient does not receive the planned docetaxel this must also be recorded on the standard docetaxel CRF, together with the reason why.

The summary of timing of Case Report Forms can be viewed in [Table 17](#).

7.1.5. DATA COLLECTION AND NON-ADMINISTRATION OF STANDARD RADIOTHERAPY

There are CRFs to be completed for patients receiving primary radiotherapy whether this is standard-of-care radiotherapy for M0 patients (on any research arm) or research RT to the prostate for Arm H patients.

All radiotherapy and acute side effects details will be recorded on the Radiotherapy Detail and Acute Toxicity CRFs upon completion of the RT schedule; any late side effects will be recorded on the follow up form.

If RT is not given, this should be stated on the Radiotherapy Detail CRF together with the reason for non-administration of the treatment in those instances where RT was planned and not given (for example, due to early metastatic progression or patient refusal).

The summary of timing of Case Report Forms can be viewed in [Table 17](#).

7.1.6. DATA COLLECTION PALLIATIVE RADIOTHERAPY

Details of palliative radiotherapy given for progressive disease should be recorded on the progression and additional treatment CRF. This includes palliative RT for SREs e.g. bone pain and spinal cord compression, as well as salvage RT to the prostate.

7.1.7 DATA COLLECTION RESEARCH (M1) RADIOTHERAPY

All radiotherapy and acute side effects details will be recorded on the Radiotherapy Detail and Acute Toxicity CRFs upon completion of the RT schedule; any late side effects will be recorded on the follow up form.

If RT is not given as planned (for example, due to early metastatic progression or patient refusal), this should be stated on the Radiotherapy Detail CRF together with the reason for non-administration of the treatment.

The summary of timing of Case Report Forms can be viewed in [Table 17](#).

7.1.8. FOLLOW-UP SCHEDULES

An individualised form with a follow-up schedule will be provided for each randomised patient. For patients who are receiving LHRH analogues, it is assumed that any research treatment will commence within four weeks of randomisation. For patients who are due to have an orchidectomy, it is recognised that surgery will have to be scheduled and the scheduling of any additional treatments may be affected by post-operative recovery. For patients who are receiving SOC docetaxel, it is assumed that any research treatment will commence within three to four weeks from completion of the final chemotherapy cycle.

7.2 FOLLOW-UP

Every effort should be made to follow-up all patients who have been randomised. Patients should, if possible, remain under the care of an oncologist or urologist for the duration of the trial. If care of a patient is returned to the GP, it is the responsibility of the responsible clinician who obtained the patient's consent to participate in the trial to ensure that all relevant data collection forms are completed. Nurse-led follow-up is permitted and conducted in line with local practice and procedures.

If the patient moves from the local area, arrangements should be made for trial follow-up to be undertaken by their new local centre. Details of other participating centres can be obtained from the STAMPEDE Trial Team. Details on patient transfer procedures are detailed in [Section 8.2](#). If the responsible clinician moves, appropriate arrangements should be made to arrange for trial follow-up to continue at the centre.

All efforts should be made to preserve the initial patient's consent for long-term survival information to be flagged through national registries, for example the Health and Social Care Information Centre, Office of National Statistics (ONS) in England/Wales and General Register Office in Scotland, Hospital Episode Statistics (HES).

Please see [Section 8](#) for more information on early stopping of follow-up.

7.2.1 FOLLOW-UP TELEPHONE CONSULTATIONS

In certain circumstances it may be appropriate to replace hospital visits with telephone consultations providing that it is still possible to collect all the necessary follow-up information. Situations where this may be considered include at the point where patients would normally be discharged from oncology or urology services. In these instances, it is acceptable to alternate appointments with telephone consultations providing the required blood results are available to the research team. All

necessary information required to complete the follow up CRF is still required. All details on the telephone consultation must be recorded in the patients' notes as per in person assessments

Other circumstances where it may be appropriate to use telephone consultations is when assessing metformin tolerance and advising regarding dose modification. This is relevant when commencing treatment for the first time or after a treatment break of 2 weeks or more when a dose escalation is advised.

7.3 TRIAL CLOSURE

For the purpose of complying with UK Clinical Regulations introduced on May 2004, the trial will be considered 'closed' when the follow-up point for the primary analysis of the final comparison has been reached. However, further observational follow-up of all patients enrolled in the trial will continue until all randomised patients have died. This will initially be via the hospital, but in the longer term may employ national registers.

Table 17: Summary of timing of case report forms

CASE REPORT FORMS	TIMING OF ASSESSMENT AND CRF
Baseline	
Randomisation	At randomisation
Baseline	At randomisation
Cardiovascular Assessment	At randomisation
Bone Density Risk Factor	At randomisation
Pathology	At randomisation. When pathology sample has been taken and sent to Sponsor's designated laboratory.
Treatment	
Standard-of-care 1 st line docetaxel	Complete for all patients 20 weeks after randomisation
SOC Hormone Therapy	Form to be sent with corresponding follow-up form if there is a change in hormone therapy to report
Abiraterone and Enzalutamide Treatment	To be completed when treatment first started and updated when reported dose changes, treatment pauses and re-starting.
Metformin Treatment	To be completed when treatment first started and updated when reported dose changes, treatment pauses and re-starting.
RT detail	• When standard-of-care radiotherapy is completed or if planned RT is no longer to be given • Arm H when research RT completed • Arm A (M1) at 3 months Once primary RT is completed for Arm H patients or those receiving RT as standard-of-care For patients who did not receive primary RT (since Protocol version 9.0 regardless of being planned) this should be completed 10 months after randomisation or 3 months for newly-diagnosed M1 patients to confirm RT was not given
RT Acute Toxicity	For all patients who receive primary RT.
Assessments	
Follow-Up	Every 6 weeks for 6 months, then every 12 weeks until 2 years, then every 6 months until 5 years and annually thereafter. (See Table 7 for more information.)
End of Treatment	When each treatment is completed (either at end of scheduled treatment or at early cessation of treatment).
Progression & Additional Treatment	At the first occurrence of each progression event, including skeletal-related events
Additional Treatment Update	Whenever a patient who has previously progressed received additional treatment but has not experienced a new type of progression
Serious Adverse Event	Following any Serious Adverse Event
Death	At Death
Administration	
Patient Transfer	When a patient is transferred to a different hospital for the administration of trial treatment and follow up
Co-enrolment	When a patient is co-enrolled in any other clinical trial. Please see Section 5.1 for more information

Table 18: Schedule for completion of treatment and outcome forms by arm.

TIMING FROM RANDOMISATION			TREATMENT LOG (IF REQUIRED)	RT	OUTCOME FORMS	
YEARS	MONTHS	WEEKS			FOLLOW-UP ^ψ	QL + HE [¥]
6-Weekly						
-	-	6	G, J, K	-	All arms	All arms
-	-	12	G, J, K	-	All arms	All arms
-	-	18	G, J, K	-	All arms	All arms
-	6	24	G, J, K	Arm H	All arms	All arms
12-Weekly						
-	9	36	G, J, K	-	All arms	All arms
1	12	48	G, J, K	All arms	All arms	All arms
-	15	60	G, J, K	-	All arms	All arms
-	18	72	G, J, K	-	All arms	All arms
-	21	84	G, J, K	-	All arms	All arms
-	-	96	G, J, K	-	All arms	All arms
6-Monthly						
2	24	104	G, J, K	-	All arms	All arms
	30	130	G, J, K	-	All arms	All arms
3	36	156	G, J, K	-	All arms	All arms
	42	182	G, J, K	-	All arms	All arms
4	48	208	G, J, K	-	All arms	All arms
	54	234	G, J, K	-	All arms	All arms
5	60	260	G, J, K	-	All arms	All arms
Annual						
6	72	-	G, J, K	-	All arms	All arms
7	84	-	G, J, K	-	All arms	All arms
Etc	-	-	G, J, K	-	All arms	All arms

Key:

A = SOC
B = SOC + zoledronic acid
C = SOC + docetaxel
D = SOC + celecoxib
E = SOC + zoledronic acid + docetaxel
F = SOC + zoledronic acid + celecoxib
G = SOC + abiraterone
H = SOC + M1 research RT to the prostate
J = SOC + enzalutamide + abiraterone
K = SOC + metformin

Notes:

ψ See Table 16 for information required at follow-up
† Form records data for two cycles
‡ Form records data for three cycles
¥ 1st 700 patients and those recruited from Protocol version 8.0 onwards only

8 STOPPING OF TREATMENT OR FOLLOW-UP

Patients should be given every encouragement to adhere to their allocated protocol treatment and follow-up schedule, in order to reduce bias. However, a patient has the right to withdraw consent for participation in any aspect of this trial at any time.

8.1 STOPPING RESEARCH INTERVENTIONS

A patient may stop **any trial treatment** for the following reasons:

- Unacceptable toxicity
- Intercurrent illness which prevents further treatment
- Withdrawal of consent for treatment
- Any alteration in the patient's condition which justifies the discontinuation of treatment in the clinician's opinion

For patients randomised to **Arm G or J**, **trial treatment** should also be discontinued for the following reasons:

- Disease progression whilst on therapy (please refer to [Section 6.8](#))
- Intention to commence a new anti-cancer treatment due to evidence of relapse

As detailed in [Section 6.8](#), the disease event for stopping treatment may be after the first reportable failure-free survival event. The reason should be recorded on the relevant treatment and the End of Treatment form.

For patients randomised to **Arm K**, treatment duration is detailed in [Section 6.11](#).

Reasons for early stopping of metformin can be:

- Decline in renal function (eGFR <60ml/min/1.73m²)
- Decline in performance status (WHO PS >2)
- Unacceptable toxicity
- Patient refusal
- Intercurrent illness preventing continued metformin treatment
- Investigator decision e.g. administration of IMP within a CTIMP in CRPC setting

If metformin is paused for more than 3 months or >50% of doses are missed please discuss with the trial team as treatment is likely to need to be stopped.

In the event of stopping research treatment, unless a patient states otherwise, consent is assumed for continued recording trial data.

8.2 PATIENT TRANSFERS

For patients moving from the area, every effort should be made for the patient to be followed-up at another participating trial centre and for this trial centre to take over responsibility for the patient. To document the transfer process the main contact person at both the current and receiving hospitals should complete and sign the Patient Transfer Confirmation form.

A fully completed form must be returned to the CTU prior to the patient transfer and any outstanding data queries for the patient should be completed prior to transfer.

On receipt of the completed transfer form a member of the STAMPEDE team will confirm the database has been updated and request confirmation of the name of the patient's new clinician. Photocopies of the following documents may then be sent to the new hospital to complete the transfer and copies must be also retained at the original site for monitoring purposes:

- Consent form
- Completed CRFs
- Any documentation relating to the patient's participation in STAMPEDE (patient names must be removed from any documentation).

8.3 EARLY CESSATION OF TRIAL PARTICIPATION

If a patient explicitly withdraws consent to have any further trial data recorded their decision must be respected and the CTU must be informed in writing. All communication surrounding the early cessation of trial participation should be noted in the patient's records. Please note data prior to this decision will still be required.

In the majority of cases, patients give permission for their data and information on their health to continue to be collected via clinical notes and national registries. Any information on the follow-up status, however minimal, would be helpful. Investigators are encouraged to facilitate ongoing collection of follow up data for example, through considering telephone consultations (see [Section 7.2.1](#)).

Early stopping of follow-up should not be undertaken lightly and the site must consider the implications for the trial and the patient in reaching such a decision. Without long-term data, the efficacy of trial treatments would be less reliable and could lead to inconclusive results. The early stopping of trial treatment should not lead to the early cessation of trial participation and in such cases follow-up assessments should be continued as per trial protocol.

Patients can change their minds about withdrawal at any time and re-consent to participate in the trial. Follow-up data should be collected only from the point of when consent was re-instated.

9 STATISTICAL CONSIDERATIONS

9.1 METHOD OF RANDOMISATION

Patients will be randomised centrally using a computerised algorithm developed and maintained by the CTU. Randomisation will be performed using the method of minimisation over a number of clinically important stratification factors with an additional random element. To decrease determinability, the factors are not listed here but can be found in the Statistical Design Document.

Protocol version 15.0 introduces a new allocation, Arm K: SOC + metformin, and marks the closure of Arm J: SOC + enzalutamide + abiraterone with accrual completed on 31-Mar-2016. Allocation to Arm K is only available to non-diabetic patients with no contraindication to metformin.

Therefore there are three possible groups at present:

- Newly-diagnosed, non-diabetic patients with M1 disease and no contraindication to RT or metformin will be randomised A:H:K
- Newly-diagnosed patients with M1 disease and no contraindication to RT but with diabetes or with a contra-indication to metformin will be randomised A:H
- Non-diabetic patients with no contra-indication to metformin and with either newly-diagnosed M1 disease but RT contraindicated or without newly-diagnosed M1 disease will be randomised A:K.

Note that the following groups of patients are currently not eligible for randomisation:

- Patients with either diabetes or with a contra-indication to metformin and without newly-diagnosed M1 disease
- Patients with diabetes and with newly-diagnosed M1 disease but who have a contraindication to RT

See [Appendix Q Table 10](#) for the allocation weighting of each arm by previous protocol version; this also shows allocation weighting for previously closed research arms.

9.2 OUTCOME MEASURES

The overall, definitive primary outcome measure for the trial for each comparison is overall survival (all-cause mortality).

The design of the trial is such that it is important to have additional intermediate outcome measures to assess activity in each research arm as the trial progresses. For comparisons involving research arms B to J the intermediate primary outcome measure is failure-free survival; this and other outcome measures are listed in [Table 20](#). For the “metformin comparison” the intermediate primary outcome measure is survival; see [Table 23](#) for full details of all outcome measures for that comparison.

The reasons for different emphases in each recruitment stage are explained in [Section 9.3](#).

Table 19: Trial Outcome Measures by Comparison Stage (Arms B-J)

COMPARISON STAGE	PRIMARY OUTCOME MEASURE	SECONDARY OUTCOME MEASURES
Pilot phase	Safety*	Feasibility
Activity Stages (AS)	Failure-free survival (FFS) [†]	Overall survival (OS) Toxicity Skeletal-related events
Efficacy Stage (ES)	Overall survival	Quality of life Cost effectiveness Failure-free survival [†] Toxicity Skeletal-related events

*Based on toxicity

[†]Including biochemical failure (see Appendix J)

9.3 SAMPLE SIZE: PRINCIPLES

The design is a multi-arm multi-stage, multi-centre, platform, randomised controlled trial. There are a number of stages for each research arm: a Pilot/Feasibility/Safety Phase, Activity Stages and a final Efficacy Stage. Full details of the methodology underlying the trial design are given by Royston et al. (43, 44) The original sample size calculations were performed using the stage2 (version 1.2.0, March 2002) and stagen (version 1.1.1, 18 May 2004) programs, both implemented in Stata (Stata Corp, TX) and updated using the later nstage program (version 1.0.3, 13-jun-2007; version 2.1.0, 28-jun-2009; version 3.0.1, 29-Sep-2014). (45)

We have adequately powered each comparison to detect an appropriate improvement in overall survival, with high power at each of the planned interim Activity Stages and at the final Efficacy Stage. For example, in a cohort with 2 years median FFS and 4 years median survival a target HR of 0.75 for research arm relative to control would translate into an absolute improvement in FFS of 10%, from approximately 50% to 60% at two years and OS of 10%, from approximately 50% to 60% at four years. As each comparison is powered to detect a difference in relative improvement, the analyses will be performed when the pre-planned number of events has been reported in the control arm, rather than after a certain number of patients have been recruited or a certain amount of time elapsed. Further details of the sample size calculations and varying assumptions for each comparison are summarised in the relevant [Sections 9.5-8](#) and detailed in a separate Statistical Design Document which are available on request.

As with all trials, changes in both the standard-of-care and second line therapies over time are possible which improve outcomes and thus will affect the observed control arm event rates and associated reporting timelines. In particular, from Protocol version 8.0, standard-of-care RT was mandated for all patients with N0 M0 disease and no RT contraindication (this is likely to improve outcomes for this subgroup) and docetaxel from Protocol version 14.0. Further agents are starting to be licensed for patients with castrate-refractory disease which may also improve survival rates. Improved FFS rates would delay the intermediate analyses, for comparisons where FFS is the intermediate primary outcome measure; whilst improved survival rates would delay the definitive analyses. For each comparison event rates are estimated based on data which are publicly available

at the time of design. The Statistical Design Document includes models where median survival is varied around such estimated rates.

9.4 SAMPLE SIZE ISSUES AND TRIAL STAGES: RESEARCH ARMS B-F

For details on those five research arms opened at trial inception in 2005 please see Protocol version 13.0

9.5 SAMPLE SIZE ISSUES AND TRIAL STAGES: ADDITIONAL RESEARCH ARM G

This is the “abiraterone comparison” and includes patients allocated to research Arm G and patients contemporaneously allocated to the control Arm A.

9.5.1 PILOT PHASE: ADDITIONAL RESEARCH ARM G

A similar approach was followed for the additional research Arm G as for the original research arms. The IDMC reviewed safety data, in the context of data from the control arm, when the first 30 patients allocated to Arm G had been on trial for at least 18 weeks.

Furthermore, an additional review of safety was performed when 30 patients with newly-diagnosed non-metastatic disease, allocated to Arm G, had been on trial for at least 18 weeks. Both of these milestones were successfully completed.

9.5.2 ACTIVITY STAGES I-III: ADDITIONAL RESEARCH ARM G

The same principles were applied to this new comparison as to the original research comparisons. The notable difference was in the accrual rate to this comparison which was anticipated to be higher. There were two reasons for this. First, STAMPEDE initially started recruitment slowly in only a limited number of pilot sites. As more sites have been activated, including internationally, accrual has increased. At the time of adding Arm G (protocol version 8.0), monthly accrual to the trial was averaging around 60 patients/month (over 700 patients/year). Second, there was an equal allocation ratio for the abiraterone arm compared to the control arm. It was this different allocation ratio which meant that the number of control arm events required to trigger the intermediate analyses was lower for the assessment of abiraterone than the assessment of the original research arms. This is shown in the table below:

Table 20: Guidelines for stopping accrual to additional research Arms G and H

ACTIVITY STAGE	SIG LEVEL	POWER	TARGETED HR	NUMBER OF CONTROL ARM EVENTS	CONSIDER DISCONTINUATION IF HR(OBSERVED) IS...
I	0.50	95%	0.75	~75	>1.00
II	0.25	95%	0.75	~142	>0.92
III	0.10	95%	0.75	~221	>0.89

9.5.3 EFFICACY STAGE IV: ADDITIONAL RESEARCH ARM G

The analysis of Efficacy Stage IV for this comparison will be performed when around 267 deaths have been observed in the control arm. This will give 90% power to detect the targeted hazard ratio of 0.75 at a one-sided significance level of 0.025.

9.5.4 SAMPLE SIZE FOR ADDITIONAL RESEARCH ARM G

Up to around 1,800 patients were targeted join the abiraterone comparison, with half allocated to the research arm G; the observed allocation was 1,917. Consideration was given to ceasing further randomisations to Arm G if it was not showing sufficient evidence of activity at the interim analyses, just as was done for research Arms B to F.

The original plan intended for accrual to be halted either when 1,500 patients had been recruited or after 3 years, whichever was the sooner, providing the accrual rate remained above 50 patients/months.

The total number of patients joining this comparison depended not just on the same issues as the "original comparisons" (notably, observed accrual and event rates), but also the length of time that the original research arms co-recruited alongside this additional research arm; it was originally assumed that this would be for approximately 1 year, but it was closer to 1.5 years. The sample size calculations and projected durations are fairly robust to changes in the length of co-recruitment with the original research arms and future co-recruitment with any further research arms which the Trial Management Group may introduce. Many scenarios are detailed in the Statistical Design Document.

In Sep-2013, the target sample size for the "abiraterone comparison" was increased from around 1,500 patients to around 1,800 patients, with note that the efficacy analysis remains unchanged and is still to be triggered by around 267 control arm deaths. This increase in sample size was primarily because of an increase in the proportion of non-metastatic patients joining the comparison; this related to the activation of Arm H which only recruits patients with newly-diagnosed metastatic disease and thereby reduces the numbers of metastatic patients randomised to the "abiraterone comparison". Non-metastatic patients have a lower event rate than the metastatic patients and maintaining the same overall sample size would lead to a delay in time to the primary analysis. The increase in sample size was achievable because recruitment rates to the trial had been substantially higher than the anticipated 50 patients/month for the 6 months preceding the increase.

9.6 SAMPLE SIZE ISSUES AND TRIAL STAGES: ADDITIONAL RESEARCH ARM H

This is the "M1|RT comparison" and includes patients allocated to research Arm H and newly-diagnosed M1 patients with no contraindication to RT allocated to the control Arm A whilst Arm H is open to recruitment. Suitability for allocation to the comparison is assessed before randomisation to ensure comparability with contemporaneous control arm patients.

9.6.1 PILOT PHASE: ADDITIONAL RESEARCH ARM H

The IDMC reviewed safety data, in the context of data from the control arm, when the first 30 patients allocated to Arm H had been on trial for around six months.

9.6.2 ACTIVITY STAGES I-III: ADDITIONAL RESEARCH ARM H

The same principles were applied to this new comparison as to previous comparisons and an equal allocation ratio of control arm patients to patients allocated to Arm H was employed; as for Arm G. The number of control arm events required to trigger the intermediate analyses were the same as for the "abiraterone comparison" (see [Table 21](#)).

9.6.3 EFFICACY STAGE IV: ADDITIONAL RESEARCH ARM H

The analysis of Efficacy Stage IV for this comparison will be performed when around 267 deaths have been observed in the relevant control arm patients. This will give 90% power to detect the targeted hazard ratio of 0.75 at one-sided significance level of 0.025.

9.6.4 SAMPLE SIZE FOR ADDITIONAL RESEARCH ARM H

Consideration was given to ceasing further randomisations to Arm H if it was not showing sufficient evidence of activity on the intermediate primary outcome measure (FFS), just as for the other research arms. This research comparison is relevant to around 60% of patients joining STAMPEDE. At the point of the scientific approval, accrual was averaging around 80 patients per month to the trial. If accrual to the trial was slower at 70 patients per month, then accrual to this comparison could be between 18 and 42 patients per month, depending on which other trial arms are open to recruitment at the time.

We are targeting a 25% relative improvement in overall survival following local radiotherapy to the prostate in this patient group. This is the same size of effect targeted with the other research arms in STAMPEDE. This relative improvement can be further justified in the light of MRC PR07 which demonstrated an improvement of this magnitude for adding radiotherapy to ADT in locally advanced disease, with a hazard ratio for overall survival of 0.77 (95% CI 0.61 to 0.98). In that trial, fewer than half of the deaths were from prostate cancer, whereas in newly-diagnosed metastatic patients nearly all men will die of their disease. Therefore, it is relevant to note the relative benefit of radiotherapy in PR07 in terms of prostate cancer-specific survival, where the hazard ratio was 0.54 (95% CI 0.27 to 0.78). Long-term survival-based data, with a median follow-up of ~10 years, were presented orally at the American Society of Clinical Oncology 2012 which confirmed these findings.⁽⁷⁾

We anticipated that around 1250 patients were required over 4 years to observe 267 control arm deaths after 5.25 years. This assumed that (i) recruitment was constantly 70 pts/m to the trial overall; (ii) the original research arms stopped accrual within 6 months after activation of the RT arm; (iii) the abiraterone arm stops accrual around 24 months after activation of the RT arm; and (iv) a further new research arm with an equal allocation ratio was introduced 18 months after activation of the RT arm. In protocol version 13.0, we reflected on these four points: (i) recruitment to the trial has been faster; (ii) the original research arms completed accrual 2 months after activation of the RT arm; (iii) the abiraterone arm stopped accrual 12 months after activation of the RT arm; and (iv) Arm J was activated 18 months after activation of the RT arm, Arm H.

Of patients joining STAMPEDE during this time, 60% have been eligible for the “M1|RT comparison”. Prior to randomisation, a RT schedule must be nominated: Weekly or Daily (see [Section 6.9](#)). We have observed that around half of patients in the comparison are nominated for RT with the Daily schedule and half for the Weekly schedule, primarily chosen by trial site with patient groups nominated for each schedule observed to be comparable at baseline. There will likely be interest to know the effect of each RT schedule when the main results are reported. This will be explored by “within schedule” comparisons of patients randomised to research vs control (arms H vs A) within each nominated RT schedule.

Therefore, in protocol version 13.0, the target sample size was increased from 1,250 patients up to around 1,800 patients, resulting in an approximate increase in the split by planned RT schedule from 625 to 900 in each “within schedule” analysis. A FFS analysis “within schedule” will be carried out at the time of the “main analysis”; predicted to have ~300 control arm FFS events by schedule (FFS “within schedule” analysis parameters: target HR=0.75, power 90%, 1-sided $\alpha=0.015$). For either of the RT schedules showing evidence of an effect on FFS, a comparative “within schedule” analysis will be carried out on survival when ~199 control arm deaths are observed in that schedule comparison. This is a closed test with OS only formally compared within schedule if there is an advantage in FFS for that RT schedule at the main analysis. Thus, extending recruitment enables a secondary analysis

of the impact of RT on survival by planned “RT schedule” to happen within around 18 months from the first main analysis.

All sample scenarios are documented in the Trial Master File.

All patients joining the trial will be starting long-term ADT for the first time. The focus of this comparison will be on the newly-diagnosed, metastatic patients (with no contraindications to RT), which is the largest subgroup of patients in the trial and the group of patients at highest risk of death from prostate cancer. Patients with non-metastatic disease will be excluded from this particular comparison as there are already randomised data demonstrating the survival benefit from radiotherapy in patients with locally advanced disease. Radiotherapy is now mandatory in node negative patients; it is also recommended in the node-positive, non-metastatic (N+ M0) group. Relapsing patients are also excluded from this comparison.

For the control arm of the whole trial, we constructed sample size scenarios median failure-free survival being 18, 24 or 30 months and constructed sample size scenarios around each of these options; the event rate would depend on the patient mix. We now know that around 60% of patients have M1 disease at trial entry and we have reported that FFS at 24 months is 51% across the whole of the control arm.⁽¹⁾

For the updated sample size calculation for this comparison, we based our estimates on the subgroup of patients with newly-diagnosed M1 disease in the control arm. Therefore, we estimate median FFS to be 1 year and estimate that median overall survival will be around 3.5 years.

9.7 SAMPLE SIZE ISSUES AND TRIAL STAGES: ADDITIONAL RESEARCH ARM J

This is the “enzalutamide + abiraterone comparison” and included patients allocated to research Arm J and patients contemporaneously allocated to the control Arm A.

9.7.1 PILOT PHASE: ADDITIONAL RESEARCH ARM J

The IDMC first reviewed safety data for this combination when the first 50 patients allocated to Arm J had been on trial around 6 weeks (i.e. to the first follow-up visit).

Furthermore, an additional review of safety was performed when these 50 Arm J patients had been on trial for around 6 months. Safety is routinely reviewed at regular intervals and additional safety reviews will be performed if the IDMC raises any concerns.

Direct comparison will be available with contemporaneously randomised patients on Arm A (hormones alone). Contextual data will be provided from Arm G (hormones plus abiraterone). Indicative safety data may also be available on the combination from other studies in CRPC.

9.7.2 ACTIVITY STAGES I-II: ADDITIONAL RESEARCH ARM J

The principles of intermediate analyses will be applied to this new comparison as to previous comparisons, but some of the details will be different, and an equal allocation ratio of control arm patients to patients allocated to Arm J was employed; as for Arms G and H. Owing to the expected accrual rate to the trial (>100 pts/m) and the expected slower event rate, only two activity stages are planned before accrual is completed. These are set out in [Table 22](#).

Table 21: Guidelines for stopping accrual to the additional research Arm J

ACTIVITY STAGE	SIG LEVEL	POWER	TARGETED HR	NUMBER OF CONTROL ARM EVENTS	CONSIDER DISCONTINUATION IF HRJ(OBSERVED) IS...
I	0.40	95%	0.70	~66	>0.957
II	0.12	95%	0.70	~139	>0.869

9.7.3 EFFICACY STAGE III: ADDITIONAL RESEARCH ARM J

The analysis of the final Efficacy Stage for this comparison will be performed when around 267 deaths have been observed in the control arm. This would give 90% power to detect the targeted hazard ratio of 0.75 at a one-sided significance level of 0.025.

9.7.4 SAMPLE SIZE FOR ADDITIONAL RESEARCH ARM J

Consideration was given to ceasing further randomisations to Arm J if it was not showing sufficient evidence of activity on the intermediate primary outcome measure (FFS), just as for the other research arms.

The patient mix for this comparison is likely to represent a more favourable prognosis on average than in the original research trial's other arms, due to concurrent recruitment of M1 but not M0 patients, to Arm H.

We anticipate that around 1,800 patients are required within 3.5 years to observe ~267 control arm deaths within 6 years. This time will be dependent on the observed overall survival. The default scenario assumes that (i) recruitment is constantly 70pts/m to the trial overall, (ii) the M1|RT arm H accrues throughout and (iii) a further new research arm with an equal allocation ratio is introduced 18 months after activation of Arm J. The stopping date for Arm G is no longer an assumption.

Variations on these factors are documented in a Statistical Design Document. If accrual rates to the trial are at 150pts/m (as observed during Summer 2013), accrual of around 1,800 patients to the comparison could be achieved within 2 years. These sample scenarios will also be documented in the Trial Master File.

Updating the standard-of-care to include docetaxel has minimal impact on the projected time to maturity of the "enzalutamide + abiraterone comparison".

9.7.5 FURTHER SAMPLE SIZE ISSUES FOR ADDITIONAL RESEARCH ARM J

Careful consideration will be given to the implications of any emerging data from the "abiraterone comparison". This would have minimal to no effect on the "enzalutamide + abiraterone comparison" because the recruitment target will be reached before any data are available from the "abiraterone comparison".

Indirect comparisons to understand the contribution from each agent may be possible if this research arm is demonstrably superior to the standard-of-care. These plans will be developed and documented elsewhere, but a higher number of patients will help with the power to the indirect comparison.

9.8 SAMPLE SIZE ISSUES AND TRIAL STAGES: ADDITIONAL RESEARCH ARM K

This is the “metformin comparison” and includes patients allocated to research Arm K and the equivalent non-diabetic patients with no contraindication to metformin contemporaneously allocated to the control Arm A whilst Arm K is open to recruitment. Suitability for allocation to the comparison is assessed before randomisation to ensure comparability with contemporaneous control arm patients

9.8.1 IMPLEMENTATION: ADDITIONAL RESEARCH ARM K

The implementation of the MAMS principles are different in this comparison for the following reasons:

- Although all non-diabetic patients will be eligible for allocation to the “metformin comparison”, the timing of the analyses will be driven only by the M1 patients.
- Failure-free survival will not be used as the intermediate primary outcome measure; overall survival will be used as both the intermediate and definitive primary outcome measure. This is because we are not convinced that any comment on metformin’s usefulness should be determined from an ability to act on a PSA-driven outcome measure. Furthermore, treatment with metformin is intended to continue throughout long-term hormone therapy which may include going well beyond an FFS event, particularly in M1 patients.
- The target HR is 0.80 for overall survival (a 20% relative improvement). This is a smaller relative improvement in survival than targeted for previous comparisons because of metformin’s known low toxicity profile, the low cost of the drug and the potential positive effects on metabolic parameters and morbidity; a smaller impact may still have clinical benefit.

9.8.2 OUTCOME MEASURES: ADDITIONAL RESEARCH ARM K

Table 22 lists the outcome measures for this comparison and can be compared with the outcome measures for the other comparisons in **Table 19**.

Table 22: Trial outcome measures by stage for the “metformin comparison”

COMPARISON STAGE	PRIMARY OUTCOME MEASURES	SECONDARY OUTCOME MEASURES
Pilot phase	Safety*	Feasibility Metabolic effects including: :: Changes in BMI :: Changes in haemoglobin A1c (HbA1c) :: Changes in waist circumference :: New diagnosis of diabetes mellitus :: Cardiovascular event: major adverse cardiac events‡
Activity Stage (AS) I	Overall survival	Metabolic effects Toxicity Symptomatic skeletal events Failure-free survival† (FFS)
Efficacy Stage (ES) II	Overall survival	Metabolic effects Toxicity Symptomatic skeletal events (SSE) Failure-free survival† (FFS) Quality of life Cost effectiveness Correlative outcomes [¶]

*Based on toxicity

‡MACE; nonfatal MI, nonfatal stroke, and death from cardiovascular causes

†Including biochemical failure (see Appendix J)

¶Plasma lipid and fasting triglyceride levels, fasting plasma glucose

Sarcopenia and/or radiological progression free survival (rPFS)

Plasma insulin

AMP Kinase

Note: All arms are unblinded so primary outcome measures for this comparison are objectively measured with caution to be taken around interpretation of more subjective secondary outcome measures such as symptomatic skeletal events

9.8.3 PILOT PHASE: ADDITIONAL RESEARCH ARM K

The IDMC will review safety data for this comparisons when the first 50 patients allocated to Arm K have been on trial around 12 months. Furthermore, analyses will be conducted on metabolic parameters (see [Table 23](#)). If there is harm observed in metabolic effects, or any serious concerns regarding the toxicity profile, recruitment would be stopped; there are no formal criteria to guide this.

Safety is routinely reviewed at regular intervals and additional safety reviews will be performed if the IDMC raises any concerns.

9.8.4 ACTIVITY STAGE I: ADDITIONAL RESEARCH ARM K

The principles of intermediate analyses will be applied to this new comparison as to previous comparisons, but some of the details will be different, and an equal allocation ratio of control arm patients to patients allocated to Arm K is employed; as for Arms G, H and J. Owing to the expected accrual rate to the trial overall (>100 pts/m) and the interim primary outcome being overall survival, only one intermediate activity stage is planned before accrual is completed; this is set out in the [Table 24](#).

Although analyses are triggered by events in M1 patients, they will include all patients in the “metformin comparison”; this will have high power. A separate subgroup analysis in M1 patients (conventionally-powered) and M0 patients (limited power) will then look at consistency of effect; there will be few deaths in M0 patients at this time. The IDMC recommendation will be based on the totality of the available data, including safety, metabolic and compliance data.

Table 23: Guidelines for stopping accrual to the additional research Arm K

ACTIVITY STAGE	SIG LEVEL	POWER	TARGETED HR	NUMBER OF CONTROL ARM EVENTS	CONSIDER DISCONTINUATION IF HR_K (OBSERVED) IS...
I	0.40	90%	0.80	~104 M1 deaths	>0.965

9.8.5 EFFICACY STAGE II: ADDITIONAL RESEARCH ARM K

The analysis of the final Efficacy Stage for this comparison will be performed when around 374 deaths have been observed for M1 patients randomised contemporaneously to the control arm. This would give 85% power to detect the targeted hazard ratio of 0.80 at a one-sided significance level of 0.025.

As with the intermediate activity, this analysis will include all patients in the comparison, with a separate subgroup analysis in M1 and M0 patients looking at consistency of effect. At this time point we predict <60 control arm M0 deaths will be observed. Further subgroup analyses, defined by the stratification factors, are planned to check for consistency of effect at intermediate and final analyses.

9.8.6 SAMPLE SIZE FOR ADDITIONAL RESEARCH ARM K

Consideration will be given to ceasing further randomisations to Arm K if it is not showing sufficient evidence of improvement on overall survival at the intermediate analysis.

We anticipate that around 1,800 patients, including around 1,100 M1 patients, are required over 3 years to observe ~374 control arm M1 deaths over around 8 years. This number and time will be dependent on the observed overall survival. The default scenario assumes (i) recruitment is constantly 100pts/m to the trial overall, (ii) co-recruitment throughout of the equivalent of one other research arm, and (iii) the majority of metastatic patients will also have docetaxel but non-metastatic patients will not.

Variations on these factors are documented in a Statistical Design Document. If accrual rates to the trial are at 150pts/m (as observed during summer 2013), accrual of around 1,800 patients to the comparison could be achieved within 2 years. These sample scenarios will also be documented in the Trial Master File.

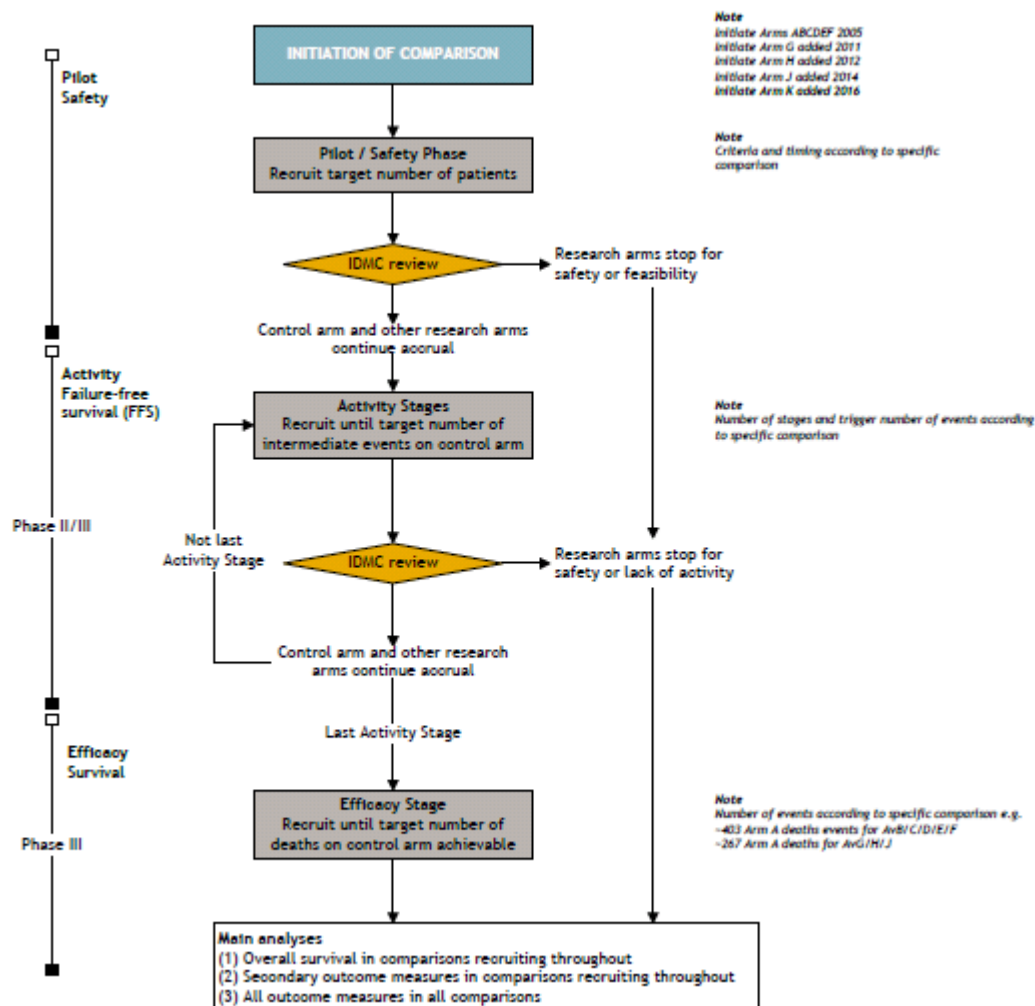
Updating the standard-of-care to permit first-line use of docetaxel was assumed within the sample size scenarios and is reflected in the projected time to maturity of the “metformin comparison”.

9.8.7 FURTHER SAMPLE SIZE ISSUES FOR ADDITIONAL RESEARCH ARM K

Careful consideration will be given to the emerging data from the “abiraterone comparison” when this reports in 2017.

Analyses for the “metformin comparison” will be timed from randomisation. The point of randomisation compared to the start of hormone therapy may differ, depending on the planned use of docetaxel. This practical information will be reviewed by the TMG and IDMC.

- **Schema of progress of STAMPEDE through the trial**



9.9 FURTHER NOTES ON TRIAL DESIGN

9.9.1 OVERALL SAMPLE SIZE

Given the adaptive nature of the study, there is no formal overall sample size target, but the numbers of patients required for each comparison are detailed in [Sections 9.5-8](#). To date, more than 8,000 patients have been recruited overall.

9.9.2 FACTORIAL DESIGN

We note here that we have not employed a factorial design in the original design of this trial because we anticipate the possibility of synergy between SOC, zoledronic acid and docetaxel and between SOC, zoledronic acid and celecoxib.

It would not be possible to assess any such interactions reliably in a factorial trial (see the Statistical Design Document for further details).

9.10 INTERIM MONITORING AND ANALYSES

The accumulating data will be reviewed at regular intervals (approximately annually) by an Independent Data Monitoring Committee (IDMC), including pre-specified formal intermediate analyses of activity data (see also [Section 16](#)). These analyses will be performed by the trial team at the CTU. Only patients randomised contemporaneously, and eligible for that comparison, will be included in the comparison of each research arm against control e.g. patients allocated to the control arm prior to Protocol version 12.0 will not contribute to the "enzalutamide + abiraterone comparison" (Arm A vs Arm J).

The IDMC will be asked to give advice on whether the accumulating data from the trial justifies continuing recruitment of further patients or further follow-up; guidelines for discontinuation of accrual for the relevant Activity Stages, together with results from any other relevant trials will aid them in this. A decision to discontinue recruitment, either in all patients or in selected subgroups, will be made only if the result is likely to convince a broad range of clinicians including those entering patients into the trial and the general clinical community. The intermediate stopping guidelines apply to the intermediate primary outcome measure. To stop accrual early for benefit in any comparison would require convincing data in terms of the definitive primary outcome measure, overall survival. For example, this could be $p < 0.001$ as proposed by Haybittle-Peto.(46, 47) The use of such a guideline for stopping for benefit has a minimal impact on the operating characteristics.

If a decision is made to continue without change, the IDMC will advise on the frequency of future reviews of the data on the basis of accrual and event rates. The IDMC will make recommendations to the Trial Steering Committee (TSC, see [Section 16](#)) as to whether the trial should continue in its present form. While the trial is ongoing the accumulating data will generally remain confidential, unless the TSC and IDMC agree that the data should be made public.

9.11 OUTLINE ANALYSIS PLAN

Analyses will be performed on an intention-to-treat basis. The standard unadjusted log-rank approach will be applied to analyses of FFS and OS. The impact of potential confounders including the stratification factors used at randomisation will be considered in a Cox proportional hazard model. Flexible parametric models will be used to calculate the absolute differences between the arms to show treatment differences over time and to estimate restricted mean "survival" times (RMST). The estimated difference in restricted means survival time (RMST) will be used preferentially to compare treatment arms if the proportional hazards assumptions required for

hazard ratios cannot be supported. The χ^2 test or Mann-Whitney test will be implemented for categorical data comparisons, including toxicity, as appropriate. Where relevant the primary outcome measures (see [Section 9.2](#)) will be considered for all arms of the trial at each phase, but the main emphasis will be placed on the comparison of the research arms that have continued to recruit throughout the trial.

9.11.1 PILOT / SAFETY PHASES

The Pilot Phase randomised patients between all the trial arms so that the results from these patients can be included in the main trial. Feasibility is considered in terms of acceptability of the trial randomisation and reported toxicities and adherence to trial medication. Centres participating in the Pilot Phase for the original research arms were required to keep an anonymised log of all patients assessed for trial eligibility (see Protocol version 2.0) so that the number of patients who did not participate in the study and the number of eligible patients who chose to not participate in the study could be summarised (reasons for non-participation were collected where the patients was willing). The anonymised logs are no longer needed for new research arms (since Protocol version 8.0).

For the patients who are randomised, we shall describe the incidence of expected and unexpected severe toxicities and adverse events/reactions (see [Section 11](#)) to decide whether to continue with research arms beyond the Pilot Phase.

9.11.2 ACTIVITY AND EFFICACY STAGES

The approach to analysis of these stages is summarised within the sample size calculations (see earlier subsections of [Section 9.4.3](#)). Each research arm will be compared in a pairwise fashion against the contemporaneously recruited control arm.

Full details are available in the Statistical Analysis Plan. See [Figure 6](#) for an overview of the schema of progress.

10 MONITORING AND QUALITY ASSURANCE

10.1 MONITORING AT MRC CTU

Data provided to the MRC CTU will be checked for missing or unusual values (range checks) and consistency over time. If missing or questionable data are identified, staff at the MRC CTU will request that the data be clarified. The exact procedures for data clarification and the amendment of CRFs will be described in the trial Data Management Plan and instructions will be sent to all STAMPEDE institutions as soon as they have been approved to participate in the trial. The MRC CTU will also send reminders for any overdue data.

Anonymised copies of the initial patient's consent form and any subsequent re-consent should be sent to the STAMPEDE team at the MRC CTU.

10.2 DIRECT ACCESS TO DATA

Collaborating institutions should be aware that direct access to patient data by MRC CTU staff may be required for trial-related monitoring or audit. Patient consent for this will be obtained as part of the general trial consent process.

10.3 VISITS TO INVESTIGATOR SITES

A selection of institutions will be visited at least once during the course of the STAMPEDE trial. The MRC CTU will give the responsible investigator adequate notice of the monitoring visit to allow adequate time, space and staff for these visits. The standard operating procedures (SOP) for monitoring are available from the MRC CTU.

After the monitoring visit the monitor will complete a site visit report. This report may be circulated to the TMT for comment. Once the TMT have reviewed the report and agreed on any recommendations the monitor will finalise the report and send a copy to the Principal Investigator (PI) at the site. A copy will be kept in the MRC CTU STAMPEDE Trial Master File.

10.4 CONFIDENTIALITY

All information collected during the course of the research will be kept strictly confidential. In addition, all procedures for handling, processing, storage and destruction of data are compliant with the Data Protection Act 1998. No individual patients will be identified when results from the trial are published.

Patients will be asked for permission for information about their health status to be obtained from the Office of National Statistics (ONS) or via the HSCIC or similar by the Medical Research Council, if necessary. In addition, patients will be asked for permission to inform their GP of their involvement in the STAMPEDE trial.

11 SAFETY REPORTING

ICH GCP requires that both investigators and sponsors follow specific procedures when reporting adverse events/reactions in clinical trials. These procedures are described in this section of the protocol. Further information on the expected toxicities for the trial interventions (docetaxel, zoledronic acid, abiraterone, radiotherapy, enzalutamide and metformin) can be found in [Appendix G](#).

11.1 DEFINITIONS

The safety reporting definitions from ICH GCP apply in this trial protocol. These definitions are given in [Table 26](#).

Table 24: Event Terms and Definitions

TERM	DEFINITION
Adverse Event (AE)	Any untoward medical occurrence in a patient or clinical trial patient to whom a medicinal product has been administered including occurrences which are not necessarily caused by or related to that product.
Adverse Reaction (AR)	Any untoward and unintended response to an investigational medicinal product related to any dose administered.
Unexpected Adverse Reaction (UAR)	An adverse reaction, the nature or severity of which is not consistent with the information about the medicinal product in question set out in the summary of product characteristics (or Investigator brochure) for that product.
Serious Adverse Event (SAE) or Serious Adverse Reaction (SAR) or Suspected Unexpected Serious Adverse Reaction (SUSAR)	Respectively any adverse event, adverse reaction or unexpected adverse reaction that: • results in death • is life-threatening* • requires hospitalisation or prolongation of existing hospitalisation** • results in persistent or significant disability or incapacity • consists of a congenital anomaly or birth defect • Other important medical condition***

Clarifications and Exceptions

*The term 'life-threatening' in the definition of 'serious' refers to an event in which the patient was at risk of death at the time of the event; it does not refer to an event which hypothetically might have caused death if it were more severe.

**Hospitalisation is defined as an inpatient admission, regardless of length of stay, even if the hospitalisation is a precautionary measure for continued observation. Hospitalisations for a pre-existing condition (including elective procedures that have not worsened) do not constitute an SAE.

***Medical judgement should be exercised in deciding whether an AE/AR is serious in other situations. Important AE/ARs that are not immediately life-threatening or do not result in death or hospitalisation but may jeopardise the patient or may require intervention to prevent one of the other outcomes listed in the definition above, should also be considered serious.

Pregnancy occurring in a STAMPEDE patient's partner during the patient's participation in the trial, must be reported to the MRC CTU within the same timelines as an SAE and classified as an 'other important medical condition' on the SAE form. The outcome of a pregnancy should be followed up carefully and any abnormal outcome to the mother or child should be reported.

Patients who develop any new primary carcinomas should have the event reported on a SAE CRF as "other important medical condition".

11.1.1 TRIAL-SPECIFIC EXEMPTIONS

Disease progression or death as a result of disease progression are not considered to be SAEs and should be reported on the STAMPEDE Progression Form or Death Form only.

The following situations that fulfil the definition of an SAE are excluded from expedited notification on an SAE form and should be reported only on the STAMPEDE follow-up form:

- Elective hospitalisation and surgery for treatment of locally advanced or metastatic prostate cancer or its complications
- Elective hospitalisation to simplify treatment or procedures
- Elective hospitalisation for pre-existing conditions that have not been exacerbated by trial treatment

Furthermore, given the solid accumulated evidence on its safety profile (38, 48), hospitalisation or prolongation of hospitalisation for the following SOC docetaxel-related events is exempted from reporting:

- Febrile neutropenia
- Thrombocytopenia

However, it is expected that investigators continue to report the abovementioned expected reactions via the MHRA Yellow Card Scheme (<https://yellowcard.mhra.gov.uk/>).

The exemption does **not** apply for events resulting in death; these should be reported as per procedures detailed in **Section 11.2.1.D**.

11.2 INSTITUTION/INVESTIGATOR RESPONSIBILITIES

All non-serious AEs/ARs, whether expected or not, should be recorded in the toxicity (symptoms) section of the Follow-up CRF and sent to the MRC CTU within one month of the form being due. SAEs/SARs should be notified to the MRC CTU as described below.

The severity (i.e. intensity) of all AEs/ARs (serious and non-serious) in this trial should be should be graded using Common Terminology Criteria for Adverse Events (CTCAE) v3.0 (ctep.cancer.gov/reporting/index.html). Any questions concerning this process should be directed to the MRC CTU in the first instance.

11.2.1 INVESTIGATOR ASSESSMENT

11.2.1.A Seriousness

When an AE/AR occurs the investigator responsible for the care of the patient must first assess whether the event is serious using the definitions given in **Table 25**. If the event is serious and not exempt from expedited reporting, then an SAE form must be completed and the MRC CTU notified.

11.2.1.B Causality

The Investigator must assess the causality of all serious events/reactions in relation to the trial therapy using the definitions in [Table 25](#). There are 5 categories: unrelated, unlikely, possible, probable and definitely related. If the causality assessment is unrelated or unlikely to be related the event is classified as a SAE. If the causality is assessed as either possible, probable or definitely related then the event is classified as a SAR.

Table 25: Assigning type of SAE through causality

RELATIONSHIP	DESCRIPTION	EVENT TYPE
Unrelated	There is no evidence of any causal relationship	Unrelated SAE
Unlikely	There is little evidence to suggest there is a causal relationship (e.g. the event did not occur within a reasonable time after administration of the trial medication). There is another reasonable explanation for the event (e.g. the patient's clinical condition, other concomitant treatment).	Unrelated SAE
Possible	There is some evidence to suggest a causal relationship (e.g. because the event occurs within a reasonable time after administration of the trial medication). However, the influence of other factors may have contributed to the event (e.g. the patient's clinical condition, other concomitant treatments).	SAR
Probable	There is evidence to suggest a causal relationship and the influence of other factors is unlikely.	SAR
Definitely	There is clear evidence to suggest a causal relationship and other possible contributing factors can be ruled out.	SAR

11.2.1.C Expectedness

If the event is a SAR the Investigator must assess the expectedness of the event. Please see [Appendix Table 7](#) for a list of expected toxicities associated with the drugs being used in this trial. If a SAR is assessed as being unexpected it becomes a SUSAR.

11.2.1.D Notification

Investigators must notify the MRC CTU of all SAEs occurring from the time of randomisation until 30 days after the last protocol treatment administration for any research arm in the trial, including standard-of-care treatments, HT and docetaxel. Similarly, SAEs occurring in patients randomised to Arm A must be reported until 30 days after last injection or progression (whichever is sooner). SARs and SUSARs must be notified to the MRC CTU indefinitely for all arms (i.e. no matter when they occur after randomisation). See [Table 26](#) for a summary of the SAE reporting timelines by treatment

SAE REPORTING

Fax to 020 7670 4818 within 24 hours of becoming aware of the event

Table 26:SAE reporting timelines by treatment

TREATMENT	SAE	SAR	SUSAR
ADT	Up to 30 days after last injection or progression (whichever is sooner)	Indefinitely	Indefinitely
Docetaxel (Research)	Up to 30 days after last treatment	Indefinitely	Indefinitely
Docetaxel (SOC)	Up to 30 days after last treatment	Indefinitely	Indefinitely
Zoledronic Acid	Up to 30 days after last treatment	Indefinitely	Indefinitely
Celecoxib	Up to 30 days after last treatment	Indefinitely	Indefinitely
Research RT (Arm H)	Up to 30 days after last treatment	Indefinitely	Indefinitely
Abiraterone	Up to 30 days after last treatment	Indefinitely	Indefinitely
Enzalutamide	Up to 30 days after last treatment	Indefinitely	Indefinitely
Metformin	Up to 30 days after last treatment	Indefinitely	Indefinitely

11.2.2 NOTIFICATION PROCEDURE

The SAE form must be completed by the Investigator (consultant named on the signature list and delegation of responsibilities log who is responsible for the patient's care), with due care being paid to the grading, causality and expectedness of the event as outlined above. In the absence of the responsible investigator the form should be completed and signed by a member of the site trial team. The responsible investigator should subsequently check the SAE form, make changes as appropriate, sign and then re-fax to the MRC CTU as soon as possible. The initial report shall be followed by detailed, written reports as appropriate.

Send the SAE form by fax to the MRC CTU. Fax Number: + 44 (0) 20 7670 4818. The STAMPEDE trial team will confirm receipt of the SAE report to the main point of contact via email. Contact the STAMPEDE trial team If receipt is not received within 24 hours.

Follow-up: Patients must be followed-up until clinical recovery is complete and laboratory results have returned to normal or baseline, or until the event has stabilised. Follow-up should continue after completion of protocol treatment if necessary. Follow-up information can be updated on the original SAE form by ticking the box marked 'follow-up' and faxing to the MRC CTU as information becomes available. Extra, annotated information and/or copies of test results may be provided separately. The patient must be identified by trial number, date of birth and initials only. The patient's name should not be used on any correspondence.

11.3 MRC CTU RESPONSIBILITIES

Medically qualified staff at the MRC CTU and/or the Chief Investigator (or a medically qualified delegate) will review all SAE reports received. The causality assessment given by the local Investigator at the hospital cannot be overruled and in the case of disagreement, both opinions will be provided in any subsequent reports.

The MRC CTU is undertaking the duties of trial sponsor and is responsible for the reporting of SUSARs and other SARs to the regulatory authorities (MHRA and competent authorities of other European member states and any other countries in which the trial is taking place) and the research ethics committees as appropriate.

The MRC CTU will also keep all investigators informed of any safety issues that arise during the course of the trial.

12 ETHICAL CONSIDERATIONS AND APPROVAL

12.1 ETHICAL CONSIDERATIONS

This is a randomised trial therefore neither the patients nor their physicians will be able to choose the patients' treatment. Treatment will be allocated randomly using a computer-based algorithm. This is to ensure that the groups of patients receiving each of the different treatments are as similar as possible.

All patients will receive standard treatment which will include ADT and may include radiotherapy and/or docetaxel. Patients may be randomised to one (or two; dependent on allocated arm) of the newer treatments in combination with standard-of-care treatment. The trial has employed an unequal allocation ratio for some comparisons to maximise efficiency; this was explained in detail in the patient information sheet.

The newer combined treatment options are being assessed in a detailed and systematic fashion in this trial. There is some evidence to suggest that the newer treatment options may have advantages over standard treatment alone with regards to clinical outcome, but this is not confirmed and toxicity may be increased. This trial will follow a large group of men who have been randomly allocated to either the standard treatment (androgen deprivation therapy alone) or the newer combined treatment options in order to measure the benefits of the new treatments. The patients will also be followed-up for toxicity and safety issues, so that any benefits can be weighed against any negative aspects.

Patients participating in the trial will have some additional hospital visits and some extra blood samples taken compared to patients who are not participating in the trial, with the amount varying according to the allocated treatment. Sometimes the blood samples can be taken when the patient is attending hospital for treatment, anyway. On some of the trial arms, the patient may have to make additional visits to the hospital for the blood sample to be taken, although in some cases it may be possible for the blood sample to be taken in the GP's surgery. The additional visits and blood samples are to ensure that follow-up of patients is comparable in all the treatment groups. The blood samples will also be used for genetic and serum marker studies, where this information will be considered with clinical data. Blood samples will be link-anonymised. There will be no feedback to individual patients.

If new information emerges during the course of the trial which may affect the treatment or follow-up of patients who have joined the trial, information will be provided through by the trial team to all Principal Investigators. PIs therefore have the duty to inform the patients in their care of any new information emerging using any appropriate channel (e.g. letter, communication at follow up clinic, etc).

The introduction of the "metformin comparison" means that all patients will be screened for diabetes prior to trial entry. This is to enable the effect of metformin to be studied in non-diabetic patients. All patients in whom screening bloods are abnormal will be referred for confirmatory tests and further management according to local guidelines e.g. via their GP. Screening is expected to lead to a small proportion of potential trial participants receiving a new diagnosis of diabetes but will ensure appropriate management of both conditions.

12.2 ETHICAL APPROVAL

The protocol has a Favourable Opinion from an appropriate Research Ethics Committee, according to national guidelines. Additionally, each site must also obtain management permission for research (R&D approval) from the relevant host organisations before patients can be entered into the trial. The patient's informed consent to participate in the trial should be obtained after a full explanation has been given of the treatment options, including the conventional and generally accepted methods of treatment. Patient information sheets and patient consent forms are available on the STAMPEDE website (www.stampededtrial.org).

The right of the patient to refuse to participate in the trial without giving reasons must be respected. After the patient has entered the trial, the clinician must remain free to give alternative treatment to that specified in the protocol, at any stage, if he feels it to be in the best interest of the patient. However, the reason for doing so should be recorded and the patient will remain within the trial for the purpose of follow-up and data analysis according to the treatment option to which he has been allocated. Similarly, the patient must remain free to withdraw at any time from the protocol treatment without giving reasons and without prejudicing his further treatment.

A statement of MRC policy on ethical considerations in clinical trials of cancer therapy, including the question of informed consent, is available from the MRC Head Office web site (<http://www.mrc.ac.uk>).

13 REGULATORY APPROVAL

This trial has been approved in the UK by the MHRA and will be conducted under a CTA (Ref: 00316/0026/001-0001) in the UK.

The trial has been approved in Switzerland by Swissmedic (Ref: 2009 DR 3235).

14 INDEMNITY

University College London holds insurance against claims from participants for injury caused by their participation in this clinical trial. Participants may be able to claim compensation if they can prove that UCL has been negligent. However, as this clinical trial is being carried out in a hospital, the hospital continues to have a duty of care to the participant of the clinical trial. University College London does not accept liability for any breach in the hospital's duty of care, or any negligence on the part of hospital employees. This applies whether the hospital is an NHS Trust or otherwise.

Participants may also be able to claim compensation for injury caused by participation in this clinical trial without the need to prove negligence on the part of University College London or another party. Participants who sustain injury and wish to make a claim for compensation should do so in writing in the first instance to the Chief Investigator, who will pass the claim to the managing organisation's Insurers, via the managing organisation's office.

Hospitals selected to participate in this clinical trial must provide clinical negligence insurance cover for harm caused by their employees and a copy of the relevant insurance policy or summary can be provided on request.

15 FINANCE

STAMPEDE is funded by the Clinical Trials Advisory Awards Committee (CTAAC) on behalf of Cancer Research UK; it is also funded by the MRC through the MRC Clinical Trials Unit. The trial has National Institute for Health Research Clinical Research Network (NIHR CRN) approval and, therefore, local NCRN funds may be available at each centre to support entry of patients into this trial.

Funding arrangements for research arms now closed to recruitment can be found in [Protocol version 13.0](#)

ADT will be administered as per routine clinical care using local NHS supplies.

M1|RT will be administered as per trial protocol using NHS RT equipment following successful RTQA by trial team.

Abiraterone is manufactured by Janssen Pharma PV (pharmaceutical companies of Johnson & Johnson). They have agreed to provide free drug and funds to distribute drug to participating sites and to help support the conduct and management of the trial.

Enzalutamide is manufactured by Astellas Pharma. They have agreed to provide free drug and funds to distribute drug to participating sites and to help support the conduct and management of the trial.

Metformin will be administered using local NHS supplies

16 TRIAL COMMITTEES

16.1 TRIAL MANAGEMENT GROUP (TMG)

A Trial Management Group (TMG) has been formed comprising the Chief Investigator, other co-investigators and members of the MRC CTU. The membership of the TMG may be expanded if other groups of trialists wish to participate. It will also be amended during the trial if other circumstances require e.g. retirement.

The TMG will be responsible for the day-to-day running and management of the trial. They will meet by teleconference at least 3-monthly and in person as needed. The TMG members are detailed in [Appendix M](#).

Further details of TMG functioning are provided in the TMG charter (available on request).

16.2 TRIAL STEERING COMMITTEE (TSC)

A Trial Steering Committee (TSC) has been formed to provide overall supervision for the trial and provide advice through its independent chair. The ultimate decision for the continuation of the trial lies with the TSC. The TSC will meet regularly.

Further details of TSC functioning are provided in the TSC charter (available on request).

16.3 INDEPENDENT DATA MONITORING COMMITTEE (IDMC)

An Independent Data Monitoring Committee (IDMC) has been formed. The IDMC will be the only group who sees the confidential, accumulating data to the trial. Reports to the IDMC will be produced by the MRC CTU. The IDMC will meet within 6 months of the trial opening with the frequency of meetings dictated by the IDMC. The IDMC will consider data in accordance with the analysis plan (see [Section 9](#)) and will be advisory to the TSC. The IDMC can recommend premature closure or reporting of the trial, or that recruitment to any research arm is discontinued.

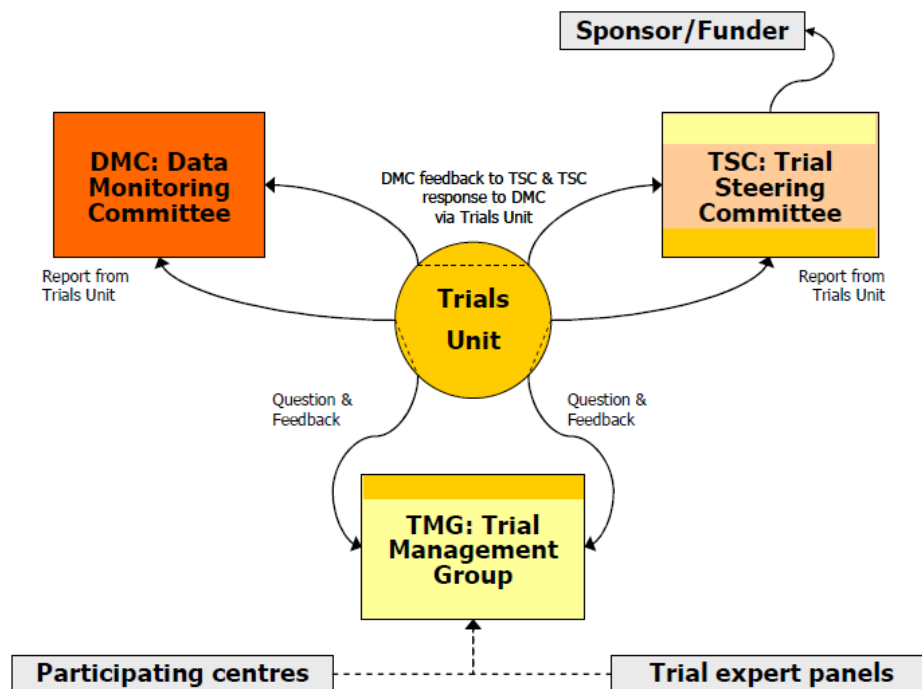
From Protocol version 8.0 onwards, any recommendation from the IDMC to stop recruitment to one or more trial arms will be acted upon immediately, pending ratification from the TSC. As this period between meetings should be very short, sites would not be notified until after the TSC have made a decision. IDMC recommendations based on emerging safety issues will be discussed with sites promptly.

Further details of IDMC functioning and the procedures for interim analysis and monitoring are provided in the IDMC charter (available on request).

16.4 TRIAL EXPERT PANELS

The trial has two established translational expert groups chaired by TMG members the Biological Research Group (BRG) and the Metabolic Translational Group (MTG). Both groups input and provide expert oversight of relevant translational aspects of the trial and associated sub-studies.

Figure 6: Diagram of relationships between trial committees



17 ANCILLARY STUDIES

17.1 QUALITY OF LIFE

A quality of life (QL) study is being performed to assess the impact of each treatment arm on the quality of patient's lives. Initial participation in this study was limited to the first 700 patients recruited (this was reached in Sep-2008) patients. The QL study re-opened from the implementation of version 8.0 of the protocol. The EORTC QLQ-C30 with the prostate-specific module QLQ PR25 will be used. Key items for assessment are pain reduction for patients with metastatic disease and urinary symptoms for patients with locally advanced disease. In addition specific hypotheses will be generated for each of the research arms. The EuroQol (EQ-5D) (49) will be used in the study as a generic measure of health-related quality of life which can be linked to public preferences. These data will be used to calculate quality-adjusted life-years as part of the economic evaluation (see [Section 17.2](#)). Patients recruited into the QL study, should continue on the study throughout the trial. Questionnaires should be self-administered, although it is recommended that a key person (e.g. research nurse) at each centre be responsible for the data collection to optimise compliance and completeness of the data.

The QL and the HE questionnaires should be completed without conferring with friends or relatives and all questions should be answered even if the patient feels them to be irrelevant.

The responsible person should check each questionnaire for its completeness, ensuring that the correct date of completion and patient identifiers are present. The research nurse should approach patients at appropriate clinical visits to complete a questionnaire. If no clinical visit is scheduled for the patient (with a window of 4 weeks around the expected date) the nurse should organise the completion of the questionnaire, by post or by a visit to the patient at home (or in a hospice).

17.2 HEALTH ECONOMICS

A health economics (HE) sub-study will be performed. Core resource use information will be collected, using CRFs on days in hospital (by speciality) and outpatient visits. Data collected on concomitant medication will also be used in the economic analysis. Information on patients' use of primary care and community-based services will be collected as additional questions in the QL questionnaire. Costs will be calculated on the basis of representative UK unit costs at the point of analysis. Health outcomes will be assessed in terms of quality-adjusted life years (QALYs). Quality adjustments will be based on patients' responses to the EQ-5D health status measure which will be administered at baseline and each point of follow-up as part of the QL questionnaire. A cost-effectiveness analysis will compare all regimens that continue to recruit into their final Efficacy Stage IV.

17.3 TRANSLATIONAL SUB-STUDIES

17.3.1 DNA ANALYSIS

Blood samples from as many patients as possible have been collected for future translational research. With patient consent, an additional droplet of blood sample has been collected using FTA Elute cards and stored for DNA and protein analysis in order to try to identify molecular features of clinical significance.

FTA Elute cards supplies have not been available since Dec-2013 and the STAMPEDE TMG has pursued an alternative method for genomic DNA collection using the Oragene® DNA kits for saliva sampling.

Oragene kits are widely used for collection of DNA from patients participating in clinical trials and they have been demonstrated to be a suitable alternative to DNA collection from whole blood providing a non-invasive, painless method of high quality sample collection.

A subset of patients may be asked if they would like to donate a blood sample for additional genetic research analysis.

Details of specimen collection, posting and contact details are given in [Appendix D](#).

17.3.2 ANALYSIS OF TUMOUR TISSUE

All patients joining the trial have been asked to provide their consent for the use of remaining biological tissue obtained through prostate core biopsy or following prostate surgery (e.g. TURP). These samples are stored as FFPE samples at the site at which the procedure was performed or in storage repository. As the clinical outcome data is known analysis of archival tumour tissue is planned in order to address several translational research aims including the development of predictive biomarkers for the treatment evaluated within the trial. The trial's translational committee, the biological research group will act as a review and access committee for such trial sub-studies and where these are proposed by external collaborators this will require additional ethical approval. Randomising sites will be asked to facilitate and assist in sample retrieval and transfer to trial designated laboratories in support of this. Providing sufficient material remains following analysis, samples will be available for return to sites on request.

18 PUBLICATIONS

The results from different centres will be analysed together and published as soon as possible. Individual clinicians must not publish data concerning their patients that are directly relevant to questions posed by the study until the TMG has published its report. The TMG together with the STAMPEDE collaborators will form the basis of the writing committee and decide on the nature of publications. Any release, of efficacy or safety data, presentation or publication will be agreed with the TSC according to the terms of their charter.

All publications will acknowledge the participating centres and clinicians, and these will be detailed in an appendix to the main report. Papers will have named authors determined by the TMG according to the following principles:

- To be as inclusive as possible where this is practicable
- To ensure that there is justification for anyone to be named as an author
- Reasons for nomination for authorship may include: trial design; grant holding; day-to-day trial oversight (TMG membership); analysis; discussion and interpretation of data; representation for key groups; active participation at large recruiting sites. It should be accepted that the people qualifying for authorship will vary over time. In addition, key positions will vary depending on the nature of the publication: clinical lead for clinical papers, statistician lead for methodology papers, translational papers may be led by authors not on the main TMG if appropriate (e.g., the bone sub-study). In the event of any dispute related to authorship or data release, the TSC will be responsible for making the executive decision.

In the manuscript, a full list of sites and the number of patients recruited will be provided. In the presentations, this list of sites will also be shown. The term “the STAMPEDE investigators” will clearly be stated and relevant names included in the presentation credits.

A detailed Publication Plan is documented separately.

19 PROTOCOL AMENDMENTS

19.1 PROTOCOL

19.1.1 AMENDMENTS MADE TO SECTIONS IN PROTOCOL VERSION 1.0 (MAY 2004)

Administrative changes such as typos, word change etc.

Name additions/changes to:

TMG members

TSC members

IDMC members

'General Information' Section – additional information re. Abridged version of protocol

Section 1.2 – Figure 1, Celecoxib duration amended

Section 1.3 – Figure 2, addition of cardiovascular assessment form, name and timings amended

Section 2.3 – Docetaxel information updated

Section 2.4 – Additional text re dose and duration justification for Celecoxib use.

Section 3 – Title change and content updated

Section 4.2 – New exclusion criteria added

Section 4.3.1 – New investigations added and additional text re testosterone measurements and additional text re. prior celecoxib treatment

Section 6.1.4 – Celecoxib duration amended

Section 6.1.5 – Additional text re. Co-administration of docetaxel and bisphosphonates

Section 6.1.6 – Celecoxib duration amended

Section 6.2.2 – additional docetaxel information

Section 6.2.3 – addition of CV event history

Section 11 – Safety reporting updated

Section 12.1 – Additional text re. the collection of blood for genetic and serum marker studies

Section 15 – Additional information re. Central Subvention for docetaxel arms

19.1.2 AMENDMENTS MADE TO SECTIONS IN PROTOCOL VERSION 1.1 (MAY 2005)

Section 6.2 Administration and Dose Modifications, subsection 6.2.1 Zoledronic Acid

19.1.3 AMENDMENTS MADE TO SECTIONS IN PROTOCOL VERSION 2.0 (JUN 2005)

General Information section – SAE reporting fax number and timeframe added.

Section 1.2 – Addition of anti-androgen use for M0 patients as a method of HT

Section 1.2 – Increase in amount of blood needed & addition tissue sample request.

Section 1.3 Trial Documentation updated to include new table detailing trial documentation ahead of accreditation, the inclusion of the radiotherapy forms and correct case report form timings

Section 2.1 – Addition of anti-androgen use for M0 patients as a method of HT

Section 4.1.3 – Inclusion criteria Vii "Normal testosterone prior to hormone treatment" removed.

Section 4.1.3 - ¶note has been omitted and moved to section 4.2 (see number 8)

Section 4.2 – Exclusion criteria added to exclude patients with active peptic ulceration, gastrointestinal bleeding and inflammatory bowel disease.

Section 4.2 – Exclusion Criteria added to exclude patients with planned major dental work

Section 4.3.1 - All blood test timelines changed from 14 days to 28 days.

Section 4.3.1 – Hormone Therapy pre-randomisation deadline extended from 4 weeks to 12 weeks.

Section 4.3.1 – Additional information regarding the use of NSAIDs and cox-2-inhibitors before coming on to the STAMPEDE study and once commenced on study treatment

Section 4.3.2 – Updated to ask for all vitamins and minerals the patient is taking to be recorded.

Section 4.3.3 – Updated to include the extra blood required and the request for consent of patients' tissue samples.

Section 6.1.1 – Addition of anti-androgen use for M0 patients as a method of HT

Section 6.1.6 – Addition of the calcium & vitamin name "calcichew".

Section 6.6.2 – asking also to collect vitamins and minerals under concomitant medication.

Section 6.6.3 – New section to inform investigators that patient's, who they wish to give radiotherapy to, are also eligible for STAMPEDE

Section 6.6.4 – New section to detail what data is being collected on the radiotherapy given to patients.

Section 7.1; figure 4 – Addition of radiotherapy form and in note, addition of AA alone

Section 7.1.2 – omission of repeated scans and x-rays at 24 weeks, also omitted in note under figure 4.

Chapter 11 – Safety reporting section updated

Section 17.3 – Increase in amount of blood needed & additional tissue sample request.

19.1.4 AMENDMENTS MADE TO SECTION IN PROTOCOL VERSION 3.0 (JUL 2006)

Front Cover - NCRN logo added for accuracy

Front Cover - Clarification that protocol developed with NCRI rather than on behalf of

Front Cover - Clarification that it is a 6 arm trial

General Information section - MRC CTU staff section updated

Section 1.2 – Statistics section updated.

Section 1.2 - Additional research paragraph updated to reflect additional studies and for clarification of terms

Section 1.2 - Blood collection volume changed to reflect new technique used

Section 1.3 (figure 3) - Table showing case report form schedule updated to reflect clarification of follow-up schedule and addition of new CRF (End of Treatment)

Section 2.2 - AS changed to HT (clarification of terms)

Section 2.3 - Updated in information in regard to use of docetaxel added to reflect up to date practice

Section 2.5 - Sub-headings numbered for consistency

Section 3.0 - Information in regard to the Pilot Phase now written in past tense as Pilot Phase has now been completed

Section 4.1.1 - Inclusion criteria extended so that patients who fulfil 2 out of the three of the first inclusion criteria can be eligible.

Section 4.3.1 - Change in time scales by which baseline investigations need to be completed.

Section 4.3.1 - Clarification that chest X-ray is only required if chest is not included in the CT

Section 4.3.1 - Removal of 12 week timeline for baseline PSA test to be performed. (Stipulation that it must be performed before start of HT)

Section 4.3.2 – Information added in regard to time allowed from randomisation to start of treatment

Section 4.3.3 - Additional research paragraph updated to reflect additional studies and for clarification of terms

Section 4.3.3 - Blood collection volume changed to reflect new technique used

Sections 6.1.2-6.1.6 - Androgen Suppression replaced with hormone therapy for consistency of terms

Section 6.2.2 - '(Taxotere)' Removed for consistency

Section 6.2.2 _ information added in regard to the need to closely monitor liver function prior to docetaxel administration

Section 7.1 - Page number reference updated

Section 7.1.1 - PSA measurement timings updated to accurately reflect follow-up schedule

Section 7.3 (Table 4) - Table and key updated to accurately reflect follow-up schedule and to include information about new CRFs and removal of withdrawal CRF

Section 8 - Rewording for clarification of definition of trial withdrawal

Section 8.1 - Instruction that withdrawal from trial treatment should be recorded on End of Treatment Form rather than withdrawal form

Section 8.1 - Information updated to emphasise that trial treatment must be discontinued following a progression

Section 8.2- Information added in regard to patient transfers

Section 8.3 - Instruction that withdrawal from trial completely must be notified in writing to the MRC CTU rather than included on withdrawal form

Section 9 and Summary – Target event numbers updated to reflect the slightly revised numbers obtained by using –nstage- which is the new, recommended program for MAMS trials

Sections 11.1 and 11.2 - Form numbers removed to allow for future changes in numbering

Section 11.2 – Reference to toxicity grading website added

Section 11.2.1 - Reference to table in appendix G added

Section 12.2 - 'Suggested' removed from 'Suggested patient information sheets'

Section 13 - CTA reference added

Section 17.3 - Information added to reflect new blood collection method for DNA analysis and in regard to additional translational studies for which funding has recently been approved

19.1.5 AMENDMENTS MADE TO PROTOCOL VERSION 4.0 (DEC 2007)

General Information Section - Randomisation and SAE reporting details sections clarified

Section 1.2 and throughout protocol - Efficacy Stages 1-111 renamed to Activity Stages 1-111 for accuracy and clarity

Section 1.2 - Follow schedule corrected

Section 4.1.2 - Inclusion criteria widened to include high risk relapsing patients, that would not have met the previous PSA based criteria

Section 4.1.3 - Note added to reference location of WHO performance status definitions

Section 4.2 - Notes added to reference locations of toxicity gradings and NYHA classifications

Section 4.3.1 - Timings of baseline scan information changed to accurately reflect most common current practice

Section 6.1.1 - Information about use of LHRH antagonists to ensure that the protocol accurately reflects current and future practice

Section 6.1.1 - Information about suggested duration of hormone therapy added to ensure that the protocol accurately reflects current practice

Section 6.2.2 - Additional information added about the timing of liver function tests prior to docetaxel administration added for clarity

Section 6.6.4 - Information on radiotherapy data collection added

Section 7.1.1 - Erroneous information about the timing of PSA measurements removed

Figure 3 - Moved to new section in protocol for clarity and extended to include current information on data collection

Figure 3b - Added to describe how extent of data collection during follow-up should change, post treatment and post progression

Figure 4 - Notes added to explain the changes in data collected at follow-up and to information that the quality of life study will be applicable to the first 700 patients randomised only

Figure 4 - Note added to include palliative radiotherapy CRF

Section 11.3 - SAE reporting information updated

Section 19 - Protocol amendments list updated

19.1.6 AMENDMENTS MADE TO PROTOCOL VERSION 5.0 (AUG 2008)

1. General Information Section – Randomisation phone line number updated – non UK extension added
2. Section 3 – Information about QL study removed to reflect closure of QL study after first 700 patients
3. Section 4.2 – Exclusion criteria clarified to explain that only patients with severe poor cardiovascular history should be excluded
4. Section 4.3.1 – Information on co-administration of NSAIDS with celecoxib changed based on clinical advice.
5. Section 5 - Randomisation phone line number updated – non UK extension added
6. Section 6.2.1. – Information added to clarify that patients who develop an osteonecrosis of the jaw should stop zoledronic acid treatment
7. Section 6.2.3 – 'severe' text added to accurately reflect which patients should be excluded based on their cardiovascular history
8. Section 7.1.2 – Definition of disease progression extended for clarity

9. Figure 3 – Updated to include reference to newly created skeletal related event form
10. Figure 4 – Previous error in table amended to show that the 4th Zoledronic Acid form that is submitted contains information about 3 cycles rather than 2 as previously indicated
11. Table 4 – ‘Other important medical condition’ added to definition of serious in the SAE section, to accurately reflect SAE form and current practice
12. Section 11.1 – Information added on reporting or pregnancies
13. Section 17 - Information about QL study removed to reflect closure of QL study after first 700 patients

19.1.7 AMENDMENTS MADE TO PROTOCOL VERSION 6.0 (JUL 2009)

1. General Information Section – Trial Pharmacist removed and changes of:

Co-Investigator

Patient Representatives

Trial Manager

Data Manager

General Information Section - Coordinating Centre – address change

General Information Section – change of Sponsor address

Section 1.1 – ratio of patients randomised to the investigational arms updated

Section 1.2 – figure 1b added to clarify trial design from Apr-2011 onwards

Section 1.2 – paragraph added to explain trial changes after the second activity analysis

Section 1.2 – wording added to clarify that QL data only collected for first 700 patients randomised

Section 1.3 – SSA Favourable Opinion removed from list of trial documentation required ahead of site accreditation

Section 2.1 – Amount of men diagnosed with prostate cancer annually updated

Section 2.4 –note added to explain completion of recruitment to celecoxib- containing arms

Section 2.5.2 - note added to explain completion of recruitment to celecoxib- containing arms

Section 3 – SSA Favourable Opinion removed

Section 4.2 – Exclusion criterion xiii greyed out

Section 4.3.1 – paragraph removed regarding potential randomisation to celecoxib-containing arms

Section 5 – Randomisation instructions expanded to exclude public holidays or dates when notice has been given by the CTU

Section 6.1.4 – formatting changed to grey font to reflect recruitment completion for arm D

Section 6.1.6 - formatting changed to grey font to reflect recruitment completion for arm F

Section 6.2.3 – recruitment note added

Section 6.6.3 – radiotherapy statement changed to reflect data from recent trials

Section 7.1.2 – removal of reference to SRE- specific CRF

Section 7.3 – Figure 3 - Addition of Bone Density Risk Factor Form and BMD sub-study assessment forms to summary of timing table

Section 7.3 – Figure 4 – Weeks added to timings of assessments post 2 years

Section 7.3- Figure 4 – note added to explain recruitment completion for arms D and F

Section 12.1 – Wording changed to reflect change to randomisation allocation ratio

Section 12.1 – Addition of statement regarding new information emerging during the trial

Section 12.2 – Reference to SSA removed

Section 16.3 – Statement added regarding actioning IDMC recommendation ahead of TSC ratification

19.1.8 AMENDMENTS MADE TO PROTOCOL VERSION 7.0 (JUL 2011)

1. General Information Section- SAE reporting fax number corrected

2. Section 11- SAE reporting fax number corrected

19.1.9 AMENDMENTS MADE TO PROTOCOL VERSION 7.1 (JUL 2011)

Throughout protocol – numbering has been updated in some sections new accommodate new information that has been added.

General Information Section – contact details updated

General Information Section – Funding information updated to include involvement from additional company

General Information Section – Wording on compliance and regulations updated to reflect current MRC CTU standard wording

General Information Section – Abbreviations list updated

Section 1.1 – The number of investigational agents being studied updated from three to four

Section 1.1 – Information regarding celecoxib updated to reflect that recruitment to these arms was discontinued in Apr-2011

Section 1.1 – Information about new IMP, Abiraterone inserted

Section 1.1 – Sample size and trial duration information updated to reflect changes brought about by additional trial arm

Section 1.2 – Summary information updated to reflect the discontinuation of recruitment to celecoxib arms and the addition of abiraterone

Figures 1a, b and c - Updated to reflect the discontinuation of recruitment to celecoxib arms and the addition of abiraterone

Section 1.2 – Information on trial stages updated to reflect changes brought about by additional trial arm

Section 1.2 – Information updated regarding the re-opening of the quality of life sub-study from implementation of protocol version 8.0

Section 2.1 – Wording related to hormone therapy updated for clarity

Section 2.1 - Updated to reflect the discontinuation of recruitment to celecoxib arms and the addition of abiraterone

Section 2.2 – Updated references added

Section 2.3 – Updated references added

Section 2.5 – Section added to give background information on new IMP, abiraterone

Section 2.6.1 – Updated references added

Section 2.7 – Section added to give information regarding radiotherapy which is to be given as part of standard care following recently published trial data.

Section 3 – Wording updated regarding selection of investigators to reflect current MRC CTU practice

Section 4.1 – Inclusion criteria updated with new criterion regarding radiotherapy use

Section 4.1 - Inclusion criteria updated with new criterion regarding contraceptive use

Section 4.1 – Wording of inclusion and exclusion criteria updated for clarity

Section 4.1 – Exclusion criteria updated with new criterion regarding acceptable liver function for trial entry

Section 4.1 – Exclusion criteria updated with specifics related to blood pressure levels

Section 4.1 - Exclusion criteria updated with new criterion regarding concomitant medications

Section 4.1 - Exclusion criteria updated with new criterion regarding prior treatment with abiraterone

Section 4.1 - Exclusion criteria updated with new criterion regarding prior treatment with chemotherapy

Section 4.1 - Exclusion criteria updated with new criterion regarding prior treatment with zoledronic acid

Section 4.3 – Wording updated to reflect that patients who initially fail screening can be re-screened at a later date

Section 4.3.2 – Wording updated regarding prior anti-androgen and LHRH use updated for clarity

Section 5.1 – Co-enrolment guidelines information updated to describe newly created co-enrolment CRF

Section 6.1 – Trial treatment information updated to reflect the fact that anti-androgens alone will be no longer permitted as hormone therapy

Section 6.1.1 – Updated to describe patients for whom radiotherapy should be given as standard practice

Section 6.1.1 a and b - Sections added to give information regarding radiotherapy treatment

Section 6.1.1-6.1.6 – References to further sections updated

Section 6.1.7 – Section added to describe abiraterone treatment

Section 6.2.4 - Section added to describe abiraterone treatment

Section 6.6 - Section added to give information regarding radiotherapy treatment

Section 7.1.1 – Reference to blood being taken at patient’s home removed as this does not occur in practice

Section 7.1.2 – Wording updated regarding the reporting of biochemical failures for clarity

Section 7.1.2 – Wording updated regarding skeletal-related events for clarity

Section 7.1.3 – Section added to describe additional assessments required related to abiraterone treatment

Section 7.1.4 – Section added to provide information on when treatment should commence

Figure 4 – Updated for clarity regarding return of BMD sub-study forms, the addition the co-enrolment CRF and the description of the re-opening of the QoL Sub-study.

Figure 5 – Updated with reference to abiraterone and co-enrolment form

Section 7.3 - Wording on trial closure updated to reflect current MRC CTU standard wording

Section 8.1 – Additional criteria for definition of progression added for clarity

Section 8.1 – Definition of progression for abiraterone patients added.

Section 9 – Statistical information updated to describe the addition of the new trial arm

Section 11 – Safety reporting wording updated for clarity

Section 11 – SAE reporting fax number updated

Section 12 – Ethical information updated to describe the unequal randomisation allocation ratio

Section 12 – Ethical information updated to describe that the visit schedule will vary according to trial arm

Section 12.2 – Wording updated to reflect international participation in the trial

Section 13 – Wording updated to reflect international participation in the trial

Section 14 – Wording updated to reflect international participation in the trial

Section 15 - Updated to reflect the discontinuation of recruitment to celecoxib arms and the addition of abiraterone

Section 16 – Reference to trial committee charters added for information

Section 17.1 – Information added to reflect re-opening of quality of life sub-study

Section 17.2 – Timing of health economics analysis updated to previous error

Section 18 – Information on publication policy expanded for clarity

Section 19 – Information regarding amendments to protocol appendices moved to the separate appendices document

Section 20 – References extensively updated

19.1.10 AMENDMENTS MADE TO PROTOCOL VERSION 8.0 (SEP 2011)

Throughout protocol – numbering, sections headings, tables, figures and bibliographical references have been updated in some sections to accommodate new information that has been added

Throughout protocol – Androgen Deprivation Therapy has replaced Hormone Therapy as deemed more representative of the type of hormone therapy used in the study

General Information Section – New staff members of the MRC CTU and Co-Investigators added and contact details updated

General Information Section – Abbreviations list updated

Section 1.1 – Information regarding the new research radiotherapy treatment inserted

Section 1.1 – Information regarding docetaxel updated

Section 1.2 – Wording updated to reflect the addition of the new research comparison arm

Section 1.3 – Additional criteria for the re-accreditation of participating centres (for protocol version 9.0 only)

Section 2.1.1 – Wording updated to clarify the use of anti-androgen in trial patients

Section 2.1.2 – Information added to describe the rationale for the RT comparison arm

Section 2.8 – Information added to describe research RT treatment to prostate for patients with newly diagnosed metastatic disease

Section 3.1 – Information added to describe RT Quality Assurance procedures and centre accreditation

Section 4.1.1 to 4.1.3 – Wording updated to clarify inclusion criteria for all patients groups (newly diagnosed non-metastatic, metastatic and relapsing patients)

Section 4.2 – Clarification added on cardiovascular exclusion criteria

Section 4.2 – New exclusion criterion added concerning patients with prior exposure to hormone therapy

Section 4.2 – New exclusion criterion added to reflect the addition of the new RT comparison arm

Section 4.4.1 – Clarification added regarding pre-randomisation checks

Section 4.4.2 – Clarification added regarding permissible hormone therapy duration prior to randomisation

Section 4.4.5 – Information added regarding starting research radiotherapy treatment

Section 4.4.6 – Information updated on concomitant medications

Section 5 – Clarification regarding randomisation allocation added to reflect the addition of the new RT research arm

Section 6.1.8 – Information added to describe the administration of research radiotherapy

Section 6.2.1 – Clarification added regarding the measurement of serum creatinine levels prior to the administration of zoledronic acid

Section 6.2.3 – Clarification regarding the completion of recruitment to the celecoxib containing arms

Section 6.25 – Information added regarding the administration of research radiotherapy treatment

Section 6.6 – Clarification incorporated to describe the administration of standard-of-care radiotherapy

Section 7.1.4 – Information added regarding data collection and non-administration of standard radiotherapy

Section 7.2 – Section updated to include new treatment specific CRFs and timing of CRFs

Section 8.1 – Clarification added for the criteria to stop treatment for patients randomised to arm G

Section 8.2 – Section expanded to include additional details on study patient transfer to different centres

Section 8.3 – Additional sentence inserted to reinforce the importance of compliance with follow up assessments

Section 9.1 – Additional paragraph inserted to clarify the method of randomisation and allocation distribution in the light of the introduction of the new RT arm

Section 9.4 – Wording updated to clarify the assessment of safety data

Section 9.5.4 – Wording updated concerning the end of randomisations to arm G

Section 9.6 to 9.6.4 – Section added describing sample size issues and trial stages for arm H

Section 9.8 – Clarification on intermediate stopping guidelines

Section 9.9 – Clarification on the outline analysis plan

Section 11 – Information on safety reporting updated to reflect the addition of the research RT comparison arm

Section 11 – Clarification added regarding arm A safety reporting timelines

Section 12.1 – Clarification added regarding the Principal Investigator's responsibilities

Section 14 – Indemnity section updated to reflect current MRC policy

Section 16 – Clarification regarding TMG membership

Section 17.3 – Section on Bone Mineral Density sub-study removed

Section 19 – Information regarding amendments to protocol appendices moved to the separate appendices document

Section 20 – References updated

19.1.11 AMENDMENTS MADE TO PROTOCOL VERSION 9.0 (OCT-2012)

Throughout protocol – numbering, sections headings, tables, figures and bibliographical references have been updated in some sections to accommodate the completion of recruitment to original research arms B, C and E.

Throughout protocol – Tenses have been changed to reflect activities that were in the future and which have now been passed.

Section 1 – Figure added and clarifications added to each figure

Section 2 – Previous reference 8 removed

Section 4 – Clarification of acceptable alternatives to bone scans

Section 6.2.5 – Correction of an error defining the PTV: the wording has been reordered

Table 4 – Dose-volume objectives corrected: order swapped

Table 5- Correction CRFs names

Section 17.3.2 – Clarification that DNA may be extracted

19.1.12 AMENDMENTS MADE TO PROTOCOL VERSION 10.0 (APR-2013)

Throughout protocol – numbering, sections headings, tables, figures and bibliographical references have been updated in some sections

Throughout protocol – typos have been corrected

Section 4 – Clarification of exclusion criteria V (now V and VI)

Section 6 – Timing of orchidectomy prior to randomisation extended to 12 weeks

Section 6 – Clarification of hypokalaemia, blood pressure and fluid retention management

Section 9 – Statistical considerations amended in light of the recruitment extension for the abiraterone comparison

Section 14 - Section updated to reflect the changes in the structure of the MRC CTU (now MRC CTU at UCL) and indemnity arrangements

19.1.13 AMENDMENTS MADE TO PROTOCOL VERSION 11.0 (SEP-2013)

Throughout protocol – numbering, sections headings, tables, figures and bibliographical references have been updated in some sections

Throughout protocol – typos have been corrected

Co-investigators list updated to reflect the addition of the “enzalutamide + abiraterone comparison” lead

Section 1.2 – Enzalutamide added as trial treatment

Section 1.2 – Protocol version 12.0 added to the list of amendments

Section 2.10 – Rationale for the combination of enzalutamide and abiraterone

Section 4.2 – Eligibility criteria amended to reflect the addition of enzalutamide + abiraterone arm

Section 4.4.2 – Wording clarified

Section 6.8 – Clarification regarding end of trial treatment after starting trial therapy

Section 6.10 – Section added to describe enzalutamide and abiraterone treatment for the new research arm (Arm J)

Section 6.11.4.A – Section added to describe the management of toxicities from trial abiraterone

Section 6.11.4.B - Section added to describe the management of toxicities from trial enzalutamide

Section 9.1.4 – Section added to describe the statistical considerations concerning the introduction of Arm J

Section 9.3 – Principles and assumption for the introduction of Arm J added

Section 9.7 and sub-sections – Sample size issues and trial stages for Arm J

Section 9.9 – Details on interim monitoring and analyses for Arm J added

Section 11.2.1.D – Wording clarified regarding safety reporting requirements for control arm

Section 12.1 – Wording clarified

Section 15 – Details on funding for the “enzalutamide + abiraterone comparison” added

Section 19 - Amendments made to protocol updated

Reference list updated

19.1.14 AMENDMENTS MADE TO PROTOCOL VERSION 12.0 (JAN-2014)

Throughout protocol – typos have been corrected

Section 4.4.2. Wording clarified

Section 4.3. Wording clarified for eligibility to M1|RT comparison

Section 6.10. Addition of use of dexamethasone post-biochemical progression for Arm J patients

Section 6.11.4.A. Correction of CTCAE version

Section 6.11.4.C. Clarification on enzalutamide dose modification to be in line with current SmPC

Section 9.6. Sample size increase for M1|RT comparison

Section 11. Correction of safety reporting timelines for Arm A patients

Section 17. Addition of saliva samples collection for DNA analysis

Table 4, 5 and 6. Clarification on Case Report Forms and Follow-up schedule

19.1.15 AMENDMENTS MADE TO PROTOCOL VERSION 13.0 (FEB-2014)

Throughout protocol – typos have been corrected

Throughout protocol – clarification on the new definition of standard-of-care

Table of contents updated to reflect any changes to the protocol

Section 1.1. Wording added throughout section to include reference to survival results from “original comparisons”

Section 2.1.1. Section improved to include reference to survival results from “original research comparisons”

Section 2.1.2. Section improved to include reference to survival results from “original research comparisons”.

Section 2.1.3. Additional section added to describe the role of docetaxel for men with M0 or M1 disease

Section 2.9. Clarification on treatment completion and primary results for “original research comparisons”

Section 4.2. Clarification of Exclusion criteria XIII and XVI

Section 4.4.2. Clarification on HT prior to randomisation

Section 4.4.3. New section to clarify standard-of-care docetaxel treatment prior to randomisation

Section 4.4.7. Clarification on concomitant medication and contra-indicated concomitant medications

Section 4.5. Clarification provided on tissue block collection

Section 6. Inclusion of docetaxel into the standard-of-care

Section 6.2.3 New section to describe standard-of-care docetaxel administration

Section 6.11. Improvement throughout sections and sub-sections for abiraterone and enzalutamide-related toxicity management

Section 6.12. Section improved throughout to incorporate clearer details on concomitant medications and drug-to-drug interactions

Section 7.1.4. New section to describe data collection for standard-of-care docetaxel

Section 9.7.4. Clarification provided about implications for “enzalutamide+ abiraterone comparison” following change of standard-of-care treatment

Section 11.2.1.D Clarification on SAE notification timelines to reflect change in standard-of-care treatments (addition of docetaxel)

Figure 1. Figure updated to reflect change in standard-of-care

Figure 2. Figure updated to reflect trial history and recruitment over time

Figure 3. Figure updated to reflect changes in standard-of-care and recruiting arms

Table 1. Table updated to remove repetition

Table 13. Table updated to include new CRF to report standard-of-care docetaxel treatment

Table 15. Table updated to include only active trial treatments

19.1.16 AMENDMENTS MADE TO PROTOCOL VERSION 14.0 (OCT-2015)

Throughout protocol – typos have been corrected

Throughout protocol – clarification on the new definition of standard-of-care

Table of contents updated to reflect any changes to the protocol

Section 1. Wording added throughout section to include reference “metformin comparison”

Section 2. Section updated to include reference “metformin comparison”

Section 4.2. Exclusion criteria review to reflect Arm J closure and instruction of “metformin comparison”

Section 4.3. Clarification of comparison specific eligibility (M1|RT and metformin)

Section 4.5.7. Clarification on concomitant medication and contra-indicated concomitant medications

Section 6. Treatment sections improved throughout

Section 6.11. Section updated to include details on metformin treatment

Section 6.12. Amendment throughout sections and sub-sections for metformin treatment

Section 6.13. Amendment throughout sections and sub-sections for metformin treatment

Section 6.13. Improvement throughout sections and sub-sections for abiraterone and enzalutamide treatment

Section 7.0. Amendment throughout sections and sub-sections to include assessment and procedures specific to “metformin comparison”

Section 9.0. Section updated and streamlined to capture statistical considerations on each comparison

Section 9.0. Details on “metformin comparison” added

Section 11. Safety processes updated and clarified

Section 16.0 Membership to oversight groups updated

Section 11.2.1.D Clarification on SAE notification timelines to reflect change in standard-of-care treatments (addition of docetaxel)

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MRC

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STAMPEDE

APPENDICES

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APPENDIX A: ADDITIONAL DEFINITIONS

Table 1: WHO Performance Status

GRADE	PERFORMANCE STATUS
0	Able to carry out all normal activity without restriction
1	Restricted in physically strenuous activity but ambulatory and able to carry out light work
2	Ambulatory and capable of all self-care but unable to carry out any work; up and about more than 50% of waking hours
3	Capable of only limited self-care; confined to bed or chair more than 50% of waking hours
4	Completely disabled; cannot carry on any self-care; totally confined to bed or chair

Table 2: TNM Classification: Primary Tumour

CATEGORY	DESCRIPTION
Tx	Primary tumour cannot be assessed
T0	No evidence of primary tumour
T1	Clinically unapparent tumour not palpable or visible by imaging - T1a: Tumour incidental, found in 5% or less resected tissue - T1b: Tumour incidental, found in more than 5% resected tissue - T1c: Tumour identified by needle biopsy (because of high PSA)
T2	Tumour confined within the prostate gland - T2a: Tumour involves one lobe - T2b: Tumour involves both lobes
T3	Tumour extends through the prostatic capsule - T3a: Extracapsular extensions (unilateral or bilateral) - T3b: Tumour invades seminal vesicles
T4	Tumour is fixed or invades adjacent structures other than seminal vesicles: bladder neck, external sphincter, rectum, levator muscles and/or pelvic wall

Table 3: TNM Classification: Regional Lymph Nodes

CATEGORY	DESCRIPTION
Nx	Regional lymph nodes have not been assessed
N0	No regional lymph node metastasis
N1	Regional lymph node metastasis

Table 4: TNM Classification: Distant Metastasis

STAGE	DESCRIPTION
Mx	Distant metastasis has not been assessed
M0	No distant metastasis
M1	Distant metastasis • M1a: Non-regional lymph nodes • M1b: Bone • M1c: Other sites

Table 5: NYHA classification of heart failure

CLASS	DESCRIPTION
Class I: asymptomatic	No limitation in physical activity despite presence of heart disease. This can be suspected only if there is a history of heart disease which is confirmed by investigations - for example, echocardiography
Class II: mild	Slight limitation in physical activity. More strenuous activity causes shortness of breath - for example, walking on steep inclines and several flights of steps. Patients in this group can continue to have an almost normal lifestyle and employment
Class III: moderate	More marked limitation of activity which interferes with work. Walking on the flat produces symptoms
Class IV: severe	Unable to carry out any physical activity without symptoms. Patients are breathless at rest and mostly housebound

Reference

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APPENDIX B: PATIENT INFORMATION SHEETS AND CONSENT FORM

Patient information sheets and consent forms are provided to participating centres as separate documents. Current copies can be downloaded from the trial website www.stampedetrial.org

APPENDIX C: GP LETTER

The GP letter is provided to participating centres as separate documents. Current copies can be downloaded from the trial website www.stampede-trial.org

APPENDIX D: DNA ANALYSIS SUB-STUDIES

Saliva Collections

From protocol version 13.0, a new method of DNA sample collection has been introduced to the STAMPEDE trial. This change has been motivated by the positive results of a pilot study in collaboration with the Institute of Cancer Research demonstrating that DNA from saliva is equivalent to DNA from blood.

Saliva samples will be collected using the Oragene® DNA saliva kit technology manufactured by DNA Genotek and are provided by the Sponsor.

Saliva sampling has been chosen as a preferred method of sample collection by the STAMPEDE TMG as it is a non-invasive and painless method which provides of high quality, high quantity DNA collection suited for use in the clinical trial setting.

The Oragene® DNA saliva kit does not require refrigeration or freezing and can be easily transported in the post within leak proof envelopes.

The kits are easy to use and do not require medically trained staff for oversight of collection. In addition a subset of patients will be asked to donate a blood sample for research (please see below for further details of the plasma collection study).

Instructions to research nurses and/or clinicians for collecting blood and saliva samples for patients who have consented to take part in the DNA analysis sub-studies are detailed in a separate manual available on the STAMPEDE trial website (www.stampedetrial.org).

Saliva samples should be sent to the address below upon collection following the instructions details in the trial manual:

Professor Ros Eeles
STAMPEDE SALIVA
Institute of Cancer Research
15 Cotswald Road
Surrey
SM2 5NG

Plasma Collections: Sequential blood sampling for circulating cell-free DNA analysis

From Protocol version 13.0, novel pilot biomarker study has been introduced to the STAMPEDE trial that will collect serial blood sampling from STAMPEDE patients participating in Arm J ("abiraterone +enzalutamide" comparison) and its control patients.

Genetic material arising from cancer circulates in the blood of prostate cancer patients. By collecting DNA from serial blood samples from STAMPEDE patients, we aim to identify and assess genetic changes associated with treatment failure.

The aims of the proposed sample collections are to:

- Identify biomarkers that can select patients most likely to benefit from particular prostate cancer therapies.

- Establish biomarkers that can identify participants most likely to experience serious toxicity on particular treatments.
- Investigate mechanisms underlying treatment failure/cancer relapse.

With patient consent, a blood sample will be collected from M1 patients at 48 weeks after randomisation, 72 weeks and 84 weeks. For all patients, a blood sample will be collected at progression, at discontinuation of abiraterone and/or enzalutamide and immediately prior to starting second line treatment. See table 6 and the translational manual for further details.

Table 6: Plasma sampling schedule

	Arm A and J patients	
	M0	M1
Pre-progression		
48 weeks (+/- 12 weeks)		X
72 weeks (+/- 12 weeks)		X
84 weeks (+/- 12 weeks)		X
At progression¹		
Biochemical progression	X	X
Radiological progression	X	X
Clinical progression	X	X
End of treatment		
End of first line treatment in absence of progression (M0 only)	X	X
Immediately prior to starting second line treatment ²	X	X

¹ If only one type of progression occurs before end of treatment, only collect one progression sample.

² If second line treatment is started at the same visit as progression is confirmed, only collect a progression sample and no further samples required.

For all participating sites blood tubes and packaging will be provided and no onsite processing of the blood will be required: invert tube on blood collection and post to lab within 5 days (store at room temperature). Prepaid postage will be provided. Full training will be given to all participating sites. Sites interested in participating in this study are encouraged to contact the MRC CTU for further information.

Further details and instructions are available in a separate manual available on the STAMPEDE trial website (www.stampedetrial.org).

Plasma samples should be sent to the address below upon collection following the instructions details in the trial manual:

STAMPEDE BLOOD

Institute of Cancer Research
15 Cotswald Road
Surrey
SM2 5NG

APPENDIX E: DRUG SUPPLY INFORMATION

The pharmacist is responsible for many aspects of managing the study drug supply. This includes, where indicated in the pharmacy information sheet, ordering study directly from the drug distribution centres as needed to maintain a reasonable inventory; maintaining drug accountability records; managing the drug supply; assuring that the study drug is not accessible to unauthorised persons and that it remains stored under the correct conditions.

Study drugs can be ordered once Research and Development (R&D) approval has been issued for the trial to take place at that site. Copies of completed Site Accreditation Forms and R&D approval documents should be forwarded to the MRC CTU when available. On receipt of these documents the MRC CTU will notify the drug distributing centres of the new centre status and approve the centre for trial drug delivery.

Specific information regarding the ordering process for each drug can be found in the pharmacy information sheet and on the trial website (<http://www.stampedetrial.org/>).

When a new patient is randomised, the MRC CTU will notify the relevant pharmacy of the randomisation and the treatment regimen to which the patient has been allocated along with the patient's treatment schedule.

Patient need to finish each bottle of Arm G and J abiraterone and Arm J enzalutamide before starting a new monthly supply.

Returned or out of date stock should be destroyed in accordance with local policies and procedures. All drug destruction should be documented on the 'Destruction logs' for auditing and monitoring purposes.

Details on pharmacy procedures for the purpose of the trial are detailed in the Pharmacy Instructions, available on the STAMPEDE website (<http://www.stampedetrial.org/>).

APPENDIX F: ADMINISTRATION OF DOCETAXEL AND ZOLEDRONIC ACID

F.1 ADMINISTRATION OF DOCETAXEL

Note: recruitment completed to both docetaxel in Mar-2013. Patients randomised to docetaxel containing arms have now completed treatment. Docetaxel is permitted as part of the standard of care and should be given according to local protocols.

Please refer to Appendices version 11.0 for details on the administration of docetaxel as trial treatment.

F.2 ADMINISTRATION OF ZOLEDRONIC ACID

Note: recruitment completed to zoledronic acid in Mar-2013. Patients randomised to zoledronic containing arms have now completed treatment.

Please refer to appendices version 11.0 for details on the administration of zoledronic acid as trial treatment.

F.3 CO-ADMINISTRATION OF DOCETAXEL AND ZOLEDRONIC ACID

Note: recruitment completed to the combination of zoledronic acid and docetaxel in Mar-2013. Patients randomised to zoledronic containing arms have now completed treatment.

Please refer to appendices version 11.0 for details on the co-administration of docetaxel and zoledronic acid as trial treatments.

APPENDIX G: DRUG SAFETY INFORMATION FOR ONGOING RESEARCH TREATMENTS

ABIRATERONE: INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

Drug Interactions

In vitro, abiraterone was shown to inhibit the hepatic drug-metabolizing enzymes CYP1A2, CYP2D6 and CYP2C8. In a clinical study to determine the effects of abiraterone acetate (plus prednisone) on a single dose of the CYP1A2 substrate theophylline, no increase in systemic exposure of theophylline was observed.

In the same study to determine the effects of abiraterone acetate (plus prednisone) on a single dose of the CYP2D6 substrate dextromethorphan, the systemic exposure (AUC) of dextromethorphan was increased by approximately 200%. The AUC₂₄ for dextromethorphan, the active metabolite of dextromethorphan, increased approximately 33%.

Caution is advised when abiraterone acetate is administered with drugs activated by or metabolized by CYP2D6, particularly with drugs that have a narrow therapeutic index. Dose reduction of narrow therapeutic index drugs metabolized by CYP2D6 should be considered.

In a CYP2C8 drug-drug interaction trial in healthy subjects, the AUC of pioglitazone was increased by 46% and the AUCs for M-III and M-IV, the active metabolites of pioglitazone, each decreased by 10%, when pioglitazone was given together with a single dose of 1,000 mg abiraterone acetate. Patients should be monitored closely for signs of toxicity related to a CYP2C8 substrate with a narrow therapeutic index if used concomitantly with abiraterone acetate.

In vitro, abiraterone and its 2 major metabolites (abiraterone sulfate and N-oxide abiraterone sulfate), inhibited OATP-mediated substrate uptake in hepatocytes. There are no clinical data available to confirm transporter-based interaction.

In a clinical PK interaction study of healthy subjects pretreated with a strong CYP3A4 inducer (rifampicin, 600 mg daily for 6 days) followed by a single dose of abiraterone acetate 1,000 mg, the mean plasma AUC_∞ of abiraterone was decreased by 55%. Strong inducers of CYP3A4 (eg, phenytoin, carbamazepine, rifampicin, rifabutin, rifapentine, phenobarbital) during treatment with abiraterone acetate are to be avoided.

In a separate clinical PK interaction study of healthy subjects, coadministration of ketoconazole, a strong inhibitor of CYP3A4, had no clinically meaningful effect on the PK of abiraterone.

Abiraterone acetate may cause hypertension, hypokalemia, and fluid retention as a consequence of increased mineralocorticoid levels resulting from CYP17 inhibition. Co-administration of a corticosteroid suppresses ACTH drive, resulting in a reduction in the incidence and severity of these adverse reactions.

ENZALUTAMIDE: INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

Drugs that Inhibit or Induce CYP2C8

Co-administration of a strong CYP2C8 inhibitor (gemfibrozil) increased the composite AUC of enzalutamide plus N-desmethyl enzalutamide in healthy volunteers by 2.2-fold. Co-administration of enzalutamide with strong CYP2C8 inhibitors should be avoided if possible. If co-administration of enzalutamide with a strong CYP2C8 inhibitor cannot be avoided, reduce the dose of enzalutamide to 80 mg once daily. If co-administration of the strong inhibitor is discontinued, then the enzalutamide dose should be returned to the dose used prior to initiation of the strong CYP2C8 inhibitor. The

effects of CYP2C8 inducers on the pharmacokinetics of enzalutamide have not been evaluated in vivo. Co-administration of enzalutamide with strong or moderate CYP2C8 inducers (e.g., rifampin) may alter the plasma exposure of enzalutamide and should be avoided if possible. Selection of a concomitant medication with no or minimal CYP2C8 induction potential is recommended.

Drugs that Inhibit or Induce CYP3A4

Co-administration of a strong CYP3A4 inhibitor (itraconazole) increased the composite AUC of enzalutamide plus N-desmethyl enzalutamide by 1.3-fold in healthy volunteers. The effects of CYP3A4 inducers on the pharmacokinetics of enzalutamide have not been evaluated in vivo.

Coadministration of enzalutamide with strong CYP3A4 inducers (e.g., carbamazepine, phenobarbital, phenytoin, rifabutin, rifampin, rifapentine) may decrease the plasma exposure of enzalutamide and should be avoided if possible. Selection of a concomitant medication with no or minimal CYP3A4 induction potential is recommended. Moderate CYP3A4 inducers (e.g., bosentan, efavirenz, etravirine, modafinil, nafcillin) and St. John's Wort may also reduce the plasma exposure of enzalutamide and should be avoided if possible.

Effect of Enzalutamide on Drug Metabolizing Enzymes

Enzalutamide is a strong CYP3A4 inducer and a moderate CYP2C9 and CYP2C19 inducer in humans.

At steady state, enzalutamide reduced the plasma exposure to midazolam (CYP3A4 substrate), warfarin (CYP2C9 substrate) and omeprazole (CYP2C19 substrate) by 86%, 56% and 70%, respectively. Concomitant use of enzalutamide with narrow therapeutic index drugs that are metabolized by CYP3A4 (e.g., alfentanil, cyclosporine, dihydroergotamine, ergotamine, fentanyl, pimozide, quinidine, sirolimus and tacrolimus), CYP2C9 (e.g., phenytoin, warfarin) and CYP2C19 (e.g., S-mephenytoin) should be avoided, as enzalutamide may decrease their exposure. If coadministration with warfarin cannot be avoided, conduct additional INR monitoring.

Transporters

In vitro data show that enzalutamide and N-desmethyl enzalutamide are potential inhibitors, but not substrates, of the efflux transporter P-gp; and it cannot be excluded that enzalutamide may be an inhibitor of BCRP, MRP2 and OAT3 at clinically relevant systemic concentrations or in the gastrointestinal wall during absorption. Medicinal products with a narrow therapeutic range that are substrates for P-gp (e.g., colchicine, dabigatran etexilate, digoxin), BCRP, MRP2 or OAT3 (e.g., methotrexate) should be used with caution when administered concomitantly with enzalutamide and may require dose adjustment to maintain optimal plasma concentrations. Enzalutamide and its major metabolites are unlikely to inhibit the in vivo transport of OATP1B1, OATP1B3, OCT1, OCT2 or OAT1 substrates at clinically relevant concentrations.

Table 7: Comparative table of Undesirable effects of Docetaxel, Zoledronic Acid, Abiraterone and Enzalutamide

Adverse reactions listed using the following convention:

- Very common (>1/10)
- Common (>1/100, <1/10)
- Uncommon (>1/1,000, <1/100)
- Rare (>1/10,000, <1/1,000)
- Very rare (<1/10,000)

	SYMPTOM	DOCETAXEL	ZOLEDRONIC ACID	ABIRATERONE ACETATE	ENZALUTAMIDE	METFORMIN
Allergic Reactions	Anaphylactic shock or severe allergic reaction	---	---	---	---	---
	Angioedema	---	Rare	---	---	---
	Hypersensitivity reactions	Common	Uncommon	---	---	---
Blood & Lymph	Anaemia	V. Common	Common	V. Common	Common	---
	Febrile neutropenia	Common	---	---	Uncommon	---
	Leukopenia	---	Uncommon	---	---	---
	Neutropenia	V. Common	---	---	Rare	---
	Pancytopenia	---	Rare	---	Uncommon	---
	Thrombocytopenia	V. Common	Uncommon	---	Common	---
Cardiac Disorders	Bradycardia	---	Rare	Uncommon	---	---
	Angina pectoris	--	--	Common	--	---
	Atrial fibrillation	Common	--	Common	--	---
	Tachycardia	--	--	Common	--	---
	Cardiac dysrhythmia	Common	---	Common	---	---
	Heart failure	---	Uncommon	Common	Uncommon	---
	Hypertension	---	Uncommon	V. Common	Common	---
	Syncope	Common	---	Common	Uncommon	---
	Stroke	---	---	Uncommon	Uncommon	---
	Peripheral Oedema/ Fluid Retention	V.common	Uncommon	V.common	V.common	---
	Myocardial infarction	---	---	Common	Uncommon	---
GI Disorders	Abdominal pain	---	Uncommon	Common	Common	V. common
	Constipation	Common	Uncommon	V. Common	V. Common	---
	Diarrhoea	V. Common	Uncommon	V common	V. Common	V. common
	Dry mouth	---	Uncommon	---	---	---
	Duodenal/gastric/ oesophageal ulceration	---	---	Rare	---	---
	Dyspepsia	---	Uncommon	Common	---	---
	Dysphagia	---	---	Uncommon	---	---

	SYMPTOM	DOCETAXEL	ZOLEDRONIC ACID	ABIRATERONE ACETATE	ENZALUTAMIDE	METFORMIN
	Nausea	V. Common	Common	V. Common	V. Common	V. common
	Stomatitis	V. Common	Uncommon	---	---	---
	Ascites	--	--	Rare	--	---
	Vomiting	V. Common	Common	V. Common	V. Common	V. common
	Taste alteration	-v.common	Uncommon	---	Common	Common
General Disorders	Asthenia	V. Common	Uncommon	V. Common	V. Common	---
	Non-cardiac chest pain	---	Uncommon	--	---	---
	Fever	---	Common	Common	Common	---
	Flu-like symptoms	---	Common	---	---	---
	Injection site reactions	---	Uncommon	---	---	---
	Lethargy/Fatigue	Common	---	V. Common	V. Common	---
	Oral candidiasis	Common	---	---	---	---
	Radiation recall phenomena	Rare	---	---	---	---
	Weight increase	---	Uncommon	---	---	---
	Weight decrease	---	---	Common	V. Common	---
Hepatic/ Pancreatic disorders	Hyperbilirubinaemia	Common	---	Uncommon	---	---
	Hepatitis	---	---	Rare	---	V.rare
	Jaundice	---	---	Rare	---	---
	Increased AST	---	---	Common	---	V.rare
	Increased ALT	---	---	Common	Uncommon	V.rare
	Aspartate aminotransferase increase	---	---	Common	Uncommon	V.rare
	Hypertriglyceridemia	---	---	Common	---	---
Laboratory Abnormalities	Hyperkalaemia	---	Rare	Common	---	---
	Hypernatraemia	---	Rare	---	---	---
	Hypocalcaemia	---	Common	Rare	---	---
	Hypokalaemia	---	Uncommon	V. Common	---	---
	Hypomagnesaemia	---	Uncommon	---	---	---
	Hypophosphataemia	---	V. Common	Uncommon	---	---
Musculoskeletal Disorders	Arthralgia	common	Common	V. Common	V. Common	---
	Bone pain	---	Common	V. Common	V. Common	---
	Generalised pain	---	Common	V. Common	---	---
	Muscle cramps/spasms	---	Uncommon	V.common	---	---
	Myalgia	Common	Common	Common	Common	---
	Fracture	---	---	Common	Common	---

	SYMPTOM	DOCETAXEL	ZOLEDRONIC ACID	ABIRATERONE ACETATE	ENZALUTAMIDE	METFORMIN
	Myopathy	---	---	Uncommon	---	---
	Joint swelling	--	--	Common	--	---
	Rhabdomyolysis	---	---	Uncommon	---	---
	Pain in limb	Common	---	V. Common	---	---
Nervous System disorders	Spinal cord compression	---	---	Common	Common	---
	Dizziness	---	Uncommon	Common	Common	---
	Headache	---	Common	V.common	V. Common	---
	Hyperaesthesia	---	Uncommon	---	---	---
	Hypoaesthesia	---	Uncommon	---	---	---
	Paraesthesia	---	Uncommon	---	Common	---
	Tremor	---	Uncommon	---	---	---
	Tetany	---	V. Rare	---	---	---
	Seizure	---	V. Rare	---	Uncommon	---
	Peripheral sensory neuropathy	v.common	---	---	---	---
	Encephalopathy	---	---	---	V. Rare	---
Ocular	Blurred vision/Tearing	Common	Uncommon	---	Rare	---
	Conjunctivitis	---	Common	---	---	---
Psychiatric Disorders	Anxiety	---	Uncommon	---	Common	---
	Confusion	---	Rare	Uncommon	---	---
	Depression	---	---	Rare	---	---
	Hallucinations	---	---	Rare	---	---
	Insomnia	---	Uncommon	v.common	Common	---
Renal/ Genitourinary Disorders	Acute renal failure	---	Uncommon	Uncommon	Rare	---
	Haematuria	---	Uncommon	Common	Common	---
	Proteinuria	---	Uncommon	---	---	---
	Renal impairment	---	Common	Uncommon	---	---
	Urinary tract infection	---	---	V. Common	Common	---
Respiratory	Cough	---	Uncommon	V. Common	Common	---
	Dyspnoea	---	Uncommon	V.common	Common	---
	Nasopharyngitis	---	---	Common	---	---
	Allergic alveolitis	--	--	V.rare	--	---
	Respiratory tract infection	---	---	Common	Common	---
Skin/ Subcutaneous	Alopecia	V. Common	---	---	---	---
	Skin reactions	V. Common	---	---	---	V.rare

	SYMPTOM	DOCETAXEL	ZOLEDRONIC ACID	ABIRATERONE ACETATE	ENZALUTAMIDE	METFORMIN
Tissue Disorders	Erythema multiforme	---	---	---	---	---
	Nail changes	V.common	---	---	---	---
	Pruritus	---	Uncommon	---	---	V.rare
	Rash	Common	Uncommon	Rare	---	V.rare
Endocrine	Increased sweating/flushing	---	Uncommon	V. Common	V. Common	---
	Gynaecomastia	---	---	---	Common	---
	Erectile dysfunction	---	---	---	Uncommon	---
	Adrenal insufficiency	---	---	Uncommon	---	---
Metabolic and Nutritional	B12 deficiency	--	--	--	--	V.rare
	Anorexia	V. Common	Common	Common	V. Common	V. common
	Lactic acidosis	--	--	--	--	V.rare

G6. RADIOTHERAPY

G.6.1 CONTRAINDICATIONS

- Previous pelvic radiotherapy
- Inflammatory bowel disease
- Inherited radiosensitivity syndrome

G.6.2 SPECIAL WARNINGS AND PRECAUTIONS FOR USE

Patients with irreducible inguinal hernia, symptomatic haemorrhoids or diverticulitis should have their bowel pathology treated appropriately before prostate radiotherapy.

Patients with severe obstructive urinary symptoms, who are at risk of acute urinary retention, should be considered for TURP prior to prostate radiotherapy.

Patients with extensive local disease (e.g. with rectal infiltration) may not be amenable to radiotherapy at full dose. If it is not possible to meet dose-volume constraints, then radiotherapy dose reduction should be considered. For example, patients receiving weekly treatment with fractions of 6Gy may need to receive fewer than six fractions.

Table 8: Associated undesirable effects for RT

SYSTEM	SYMPTOM
Gastrointestinal Disorders	Increase in stool frequency Loose stool Proctitis Proctalgia Rectal bleeding Rectal urgency
Musculoskeletal Disorders	Bone fractures
Renal	Dysuria Hesitancy Increase in micturition frequency Urethral obstruction or stricture Urinary urgency

APPENDIX H: EVALUATION OF BASELINE AND NEW LESIONS

These guidelines are based on the RECIST criteria (1) but have been modified to include progression based on PSA measurements.

H.1 MEASURABLE DISEASE

Measurable disease is defined as at least one lesion that can be accurately measured in at least one dimension (longest dimension to be recorded). Each lesion must >20mm when measured by conventional techniques, including palpation, CT, and MRI, or >10mm when measured by spiral CT.

H.2 BASELINE DOCUMENTATION OF “TARGET” AND NON-TARGET LESIONS

All measurable lesions up to a maximum of 5 lesions per organ and 10 lesions in total representative of all involved organs should be identified as target lesions and will be recorded and measured at baseline. Target lesions should be selected on the basis of their size (lesions with the longest dimension) and their suitability for accurate repetitive measurements by one consistent method of assessment (either by imaging techniques or clinically). A sum of the longest dimension (LD) for all target lesions will be calculated and reported as the baseline sum LD. The baseline sum LD will be used as reference to further characterise

All other lesions (or sites of disease) should be identified as non-target lesions and should also be recorded at baseline. Measurements are not required.

All baseline evaluations of disease status should be performed as close as possible to the start of the treatment and not more than 4 weeks before the beginning of the treatment.

H.3 DEFINITION OF PROGRESSION (FOR PATIENTS WITH MEASURABLE DISEASE AT RANDOMISATION)

Progression is defined as **any** of the following:

- At least a 20% increase in the sum of LD target lesions taking as reference the smallest sum LD recorded since study entry
- The appearance of one or more new lesions
- Death due to disease without prior objective documentation of progression
- Global deterioration in health status attributable to the disease requiring a change in therapy without objective evidence of progression
- Unequivocal progression of non-target lesions, (other than pleural effusions without cytological proof of neoplastic origin) in the opinion of the treating physician (in this case an explanation must be provided).
- Progression will be most usually based on PSA measurements (see [Appendix J](#)) but tumour measurements should take precedence over PSA response. If measurable disease is shrinking during treatment, but the PSA is rising the patient should continue to receive protocol treatment.

H.4 DEFINITION OF PROGRESSION (FOR PATIENTS WITHOUT MEASURABLE DISEASE AT RANDOMISATION)

Progression (for patients with non-measurable disease at randomisation) is defined as increasing clinical or radiological evidence of disease since study entry. Progression can also be based on PSA measurements ([Appendix J](#)).

Reference

(1) P Therasse, SG Arbuck, EA Eisenhauer et al. New Guidelines to Evaluate the Response to Treatment in Solid Tumors. J Natl Cancer Inst (2000) 92 (3): 205-216.

APPENDIX I : COMMON TOXICITY CRITERIA

The known side effects for the drugs being used in STAMPEDE are classified using the CTCv4.0 Toxicity heading. The CTC v4.0 toxicity grade heading/adverse event column states first the broad classification of AEs based on anatomy and/or pathophysiology (category) and then the adverse event term the symptom falls within.

The complete CTCv4 .0 can be found at the following web site:

http://evs.nci.nih.gov/ftp1/CTCAE/CTCAE_4.03_2010-06-14_QuickReference_5x7.pdf

APPENDIX J: DEFINITION OF BIOCHEMICAL FAILURE

J.1 BIOCHEMICAL FAILURE

The initial response to hormonal therapy for prostate cancer can be variable. A few patients exhibit only a small fall in PSA, with little evidence of clinical response. At the other extreme a rapid fall to the normal range or even undetectable levels of PSA occurs. In a group of patients the response lies between these two extremes. The rate of fall of PSA and the level of the PSA nadir are recognised to have prognostic significance. Comparison of PSA responses between the treatment groups could be used as secondary data for confirming the response rate of the trial treatments. However, in defining PSA relapse, the extent of the primary response has to be taken into account. Three groups of patients will be defined:-

- A. If the PSA nadir is more than 50% of the last pre-treatment PSA and the PSA nadir is more than 4ng/ml, the patient should be defined as a treatment failure (at time zero).
- B. For patients whose PSA nadir is less than or equal to 50% of the last pre-treatment PSA, but remains above 4ng/ml, PSA relapse will be deemed to have occurred when PSA is confirmed as increasing by 50% above the nadir level.
- C. For patients whose PSA falls below or to 4ng/ml, PSA relapse will be defined by either 50% increase from their nadir or the PSA increasing above 4, whichever is the greater. For example, a nadir PSA of 3.6 would require a PSA of 5.4 to define relapse, while PSA nadir of 2.5 will be considered to have relapsed at a PSA of 4.

J.2 TIMING OF PSA TESTS

All patients will commence hormone therapy shortly after randomisation. The pre-randomisation PSA must be measured within 4 weeks prior to randomisation. Once on trial, PSA tests will be performed at 6, 12, 18 and 24 weeks from randomisation then at each follow-up visit (3 monthly until 2 years, then 6 monthly).

J.3 NADIR PSA

The PSA nadir will be the lowest reported PSA level between randomisation and the 24 weeks PSA assessment (i.e. assessments at 6, 12, 18 and 24 weeks). The critical value that would constitute subsequent biochemical progression will be calculated from this nadir value. Once the 6 month PSA level has been recorded and sent promptly to the MRC CTU, a letter will be sent to the responsible clinician confirming the PSA level which would be taken as progression.

J.4 CONFIRMING FAILURE-FREE SURVIVAL (INCLUDING BIOCHEMICAL FAILURE)

In applying the definitions of Failure-free survival (including biochemical failure) above, the increase in PSA must be confirmed (i.e. at least two readings are required). The following approach will be applied:

- Patient reaches nadir PSA and continues with 3-monthly PSA assessments. [Any PSA measurement taking place within 6 weeks of a change in hormone therapy or a

manipulation (e.g. cystoscopy) that could produce a spurious increase in PSA is to be ignored].

- At a subsequent assessment (say, assessment X), the PSA value is noted to have risen above the critical value, but patient is symptom free.
- Patient should be recalled for a confirmatory PSA test between one week and 3 months later. If this value is at the same level or higher, the patient will be considered to have reached the biochemical failure endpoint.
- If a patient has received intermittent hormone therapy then a reported biochemical progression must be accompanied by a testosterone measurement confirming a castrate level of testosterone at the time of the progression.

These last two confirmatory steps will not be required if the patient has demonstrable local progression, new metastases or death from prostate cancer during the intervening period.

J.5 TIME TO FAILURE-FREE SURVIVAL (INCLUDING BIOCHEMICAL FAILURE)

Patients defined as being in Group A above (treatment failure) will be recorded as events at time zero. The time to FFS for patients defined as being in Groups B and C will then be from randomisation to the first PSA above the critical value (i.e. assessment X).

J.6 NON-PROTOCOL TREATMENT

No new non-protocol treatment for prostate cancer should be administered until trial progression has been confirmed.

J.7 PSA MONITORING GROUP

Consideration will be given to the formation of such a group, whose remit would be to monitor the accurate recording of PSA data, and independently assess failure outcomes, blinded of the treatment allocation. This would be, by definition, an independent group involving clinicians not recruiting patients to STAMPEDE.

Please contact the STAMPEDE team for advice on CRF completion to report progression.

APPENDIX K: RADIOTHERAPY QUALITY ASSURANCE

The STAMPEDE Radiotherapy Quality Assurance (RTQA) Group includes clinical oncologists and medical physics experts who provide guidance on the development and implementation of the protocol as well as the RTQA accreditation process.

RTQA is required for all STAMPEDE centres administering radiotherapy for the purpose of the trial. However, centres that have received RTQA for other multi-centre large randomised trials such as RADICALS (MRC PR10) and CHHIP won't require any further accreditation. The MRC CTU will be responsible for identifying previously accredited centres before the activation of Arm H.

Those centres that have not been involved in RADICALS or CHHIP will be required to complete the following documentation:

1. Accreditation Form
2. RT Credentialing Form:
 - Centres should identify two anonymised examples of patients as if to be treated in STAMPEDE.
 - Forms will need to be sent to the STAMPEDE trial office where the RTQA group will review and accredit the centre as appropriate.
3. Three copies (on CD is acceptable) of the CT scan of the PTV and in particular:
 - The PTV will have a 0.8 cm margin posteriorly and 1.0 cm margin in all other directions around the CTV to account for prostate gland motion and uncertainty in daily treatment setup
 - Critical normal tissues should be delineated on the treatment-planning CT scan by the treating clinician:
 - Rectum – inferior limit: level of ischial tuberosities; superior limit: sigmoid flexure
 - Bladder – entirety

Plans should be sent to:

STAMPEDE trial
MRC Clinical Trial Unit at UCL
Aviation House
125 Kingsway
London
WC2B 6NH

APPENDIX L: HORMONE THERAPY PRIOR TO RANDOMISATION

As detailed in [Section 4.4.2](#) of the trial protocol, it is preferable that patients are not started on hormones prior to randomisation. However, if hormone therapy has already started, the table below illustrates within what time window prior hormone therapy is allowed for the purpose of the trial. Please note that Maxium Androgen Blockade is not deemed a suitable primary hormone therapy for patients randomised to Arm J and these patients should switch to LHRH therapy after randomisation.

Table 9: Permissible start times for trial hormone therapy

WEEK	ORCHIDECTOMY (No AA)	LHRH ANALOGUES (WITH SHORT AA)	MAB: LHRH ANALOGUES (WITH LONG AA)	AA ONLY	LHRH ANTAGONISTS	NOTES
-17	No	No	No	No	No	No HT this far back
-16	No	No	No	No	No	No HT this far back
-15	No	No	No	No	No	No HT this far back
-14	No	AA only	AA only	No	No	Only AA before LHRH allowed
-13	No	AA only	AA only	No	No	Only AA before LHRH allowed
-12	Yes	Yes	Yes	Yes (change pending)	Yes	Any HT can start now
-11	Yes	Yes	Yes	Yes (change pending)	Yes	Any HT can start now
-10	Yes	Yes	Yes	Yes (change pending)	Yes	Any HT can start now
-9	Yes	Yes	Yes	Yes (change pending)	Yes	Any HT can start now
-8	Yes	Yes	Yes	Yes (change pending)	Yes	Any HT can start now
-7	Yes	Yes	Yes	Yes (change pending)	Yes	Any HT can start now
-6	Yes	Yes	Yes	Yes (change pending)	Yes	Any HT can start now
-5	Yes	Yes	Yes	Yes (change pending)	Yes	Any HT can start now
-4	Yes	Yes	Yes	Yes (change pending)	Yes	Any HT can start now
-3	Yes	Yes	Yes	Yes (change pending)	Yes	Any HT can start now
-2	Yes	Yes	Yes	Yes (change pending)	Yes	Any HT can start now
-1	Yes	Yes	Yes	Yes (change pending)	Yes	Any HT can start now
Rand	Yes	Yes	Yes	Swap to LHRH	Yes	Any ADT can start now (except AA monoth'py)
1	Yes	Yes	Yes	Swap to LHRH	Yes	Any ADT can start now (except AA monoth'py)
2	Yes	Yes	Yes	Swap to LHRH	Yes	Any ADT can start now (except AA monoth'py)
3	Yes	Yes	Yes	Swap to LHRH	Yes	Any ADT can start now (except AA monoth'py)
4	Yes	Yes	Yes	Swap to LHRH	Yes	Any ADT can start now (except AA monoth'py)
5	Yes	Yes	Yes	Swap to LHRH	Yes	Any ADT can start now (except AA monoth'py)

APPENDIX M: TRIAL CONTACTS

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Molecular Genetics Advisor

Health Economics Advisor

Director

Trials Unit Lead, Trial Statistician

Clinical Project Manager

Trial Manager

Trial Manager

Trial Manager

Trial Manager

Trial Statistician

Data Manager

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Representatives from the drug companies involved
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APPENDIX N: PARTICIPATING SITE ACCREDITATION FORM

Current copies of the institution statement, signature list, delegation of responsibilities and personnel list can be downloaded from the trial [website http://www.stampededtrial.org/](http://www.stampededtrial.org/).

APPENDIX O: QUALITY OF LIFE AND HEALTH ECONOMICS

O.1. OVERVIEW

The economic evaluation will take the form of a cost-effectiveness analysis in which the differential cost of the alternative treatments will be related to their differential benefits in terms of quality-adjusted life years (QALYs). Incremental analysis will be undertaken and cost-effectiveness acceptability curves will be used to show the probability of one option being more cost-effective than the others.

O.2. ESTIMATING COSTS

A cost analysis will be undertaken from the perspective of the National Health Service. Resource use measurement during the trial will be divided into that relating to the hospital and NHS non-hospital. These are dealt with in turn below.

O.2.1 HOSPITAL RESOURCE USE

Within the trial, hospital resource use data will be collected on all patients entering the trial. Specifically, this will include in-patient nights in hospital, distinguishing intensive care from stay on a general ward. These will be collected using case record forms completed by clinical review at the follow-up points shown in the main document. Some visits to, and stays in, hospital may relate to non-study hospitals. To ensure that data on this resource use are captured, a questionnaire will be administered to patients as part of the quality of life assessments.

These resources will be valued in monetary terms using unit costs representative of UK practice at the time of analysis. For drugs, this will be based on British National Formulary prices. For hospital procedure and hotel costs, unit costs will, if available, be based on NHS Reference Costs. Otherwise, they will be estimated from a sample of UK centres randomising patients into the trial.

O.2.2 NHS NON-HOSPITAL RESOURCE USE

Data on patients' use of community-based NHS (and complementary health) services will be collected from patients as part of the quality of life assessments. The resources will include visits to and from a GP or district nurse. Costing of community-based resources will be based on published unit costs. Other services will be costed using data available at the point of analysis.

O.3. MEASURING EFFECTS

This clinical trial is estimating a range of clinical and health-related quality of life effects within trial patients. The purpose of an economic evaluation will be to set these in context of the resource costs incurred in achieving them. A cost-effectiveness analysis will relate differential cost to an aggregated measure of effect in the form of a QALY. QALYs will be based on observed mortality and patients' responses to the EQ-5D questionnaire. The latter asks patients to categorise their health, with 3 levels of response (no problems, moderate problems, severe problems) on 5 dimensions (mobility, self care, usual activities, pain/discomfort, depression anxiety). Each of the 245 possible health states has been 'valued' on a zero to one 'utility' based on the preferences of 3,395 members of the UK public.

O.4. ANALYSIS

All resource use data will be valued in monetary terms, as described above, such that each patient has an associated cost over their period of follow-up. A full stochastic analysis will be undertaken to allow for sample variation in resource use and effect data. Methods are developing quickly in this area and, by the time of the analysis, 'best practice' may have altered markedly from today. If such an analysis were to be undertaken now, the general methods would be as follows.

A QALY profile will be estimated for each patient based on their survival duration weighted by their responses to the EQ-5D Health Related QL questionnaire, which generates a single index value for health at each point of follow-up⁶. The profiles will assume a straight-line relationship between the index value at time t and the value at time $t+1$. The number of QALYs they experience during the period of follow-up in the trial will be the area under the QALY profile.

In the primary analysis, only data collected in the trial will be used in the analysis; in other words, the estimate of QALYs for each group is likely to reflect the fact that some patients are still alive after the follow-up (i.e. the survival curve is truncated and survival analysis techniques will be used to estimate QALYs).

As a secondary analysis, extrapolation techniques will be used to estimate the final portion of the survival curve so as to provide a full estimate of differential life expectancy. A number of extrapolation techniques will be used to provide a range of estimates of differential QALYs over a lifetime time horizon.

Cost-effectiveness acceptability curves will be used to facilitate a measure of uncertainty around cost-effectiveness estimates. These curves show the probability of one form of management being more cost-effective than the others assuming alternative levels of the maximum amount decision-makers are willing to pay for an extra QALY.

Sensitivity analysis will be used to consider the importance of sources of uncertainty other than sample variation (e.g. unit costs, discount rates, method of extrapolation). Multiple regression techniques will be employed to provide as precise a measure of cost-effectiveness as possible and to undertake sub-group analysis using baseline patient characteristics, which will be defined in advance in the analysis plan.

Reference

Dolan, P, Gudex, C, kind, P, and et al. A Social Tariff for EuroQol: Results from a UK General Population Survey. Centre for Health Economics Discussion Paper 138. Centre for Health Economics, University of York: CHE. 1995

APPENDIX P: BONE MINERAL DENSITY SUB- STUDY

Note: Recruitment to the Bone Mineral Density sub-study was stopped in May-2012. For further details on the sub-study please refer to protocol version 8.0

APPENDIX Q: HISTORY ALLOCATION AT RANDOMISATION BY PROTOCOL VERSION

Table 10: History of allocation at randomisation by protocol version

		NEWLY-DIAGNOSED M1 PATIENTS														OTHER PATIENTS															
PROTOCOL	PROTOCOL	NON-DIABETIC										DIABETIC				NON-DIABETIC										DIABETIC					
VERSION	CHANGE	A	B	C	D	E	F	G	H	J	K	A	B	C	D	E	F	G	H	J	K	A	B	C	D	E	F	G	H	J	K
V1	Not used Activated	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-	-
V2		2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	-	-	-	-	
V3		2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	-	-	-	-	
V4		2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	-	-	-	-	
V5		2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	-	-	-	-	
V6		2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	-	-	-	-	
V7	DF stopped	2	1	1	~	1	~	-	-	-	-	2	1	1	~	1	~	-	-	-	-	2	1	1	~	1	~	-	-	-	-
V8	G added	2	1	1	~	1	~	2	-	-	-	2	1	1	~	1	~	2	-	-	-	2	1	1	~	1	~	2	-	-	-
V9 ¹	H added	2	1	1	~	1	~	2	2	-	-	2	1	1	~	1	~	2	2	-	-	2	1	1	~	1	~	2	x	-	-
V10	BCE stopped	2	~	~	~	~	~	2	2	-	-	2	~	~	~	~	~	2	2	-	-	2	~	~	~	~	~	2	x	-	-
V11	G stopped	2	~	~	~	~	~	~	2	-	-	2	~	~	~	~	~	~	2	-	-	x	~	~	~	~	~	~	x	-	-
V12	J added	2	~	~	~	~	~	~	2	2	-	2	~	~	~	~	~	~	2	2	-	2	~	~	~	~	~	~	x	2	-
V13		2	~	~	~	~	~	~	2	2	-	2	~	~	~	~	~	~	2	2	-	2	~	~	~	~	~	~	x	2	-
V14	SOC change	2	~	~	~	~	~	~	2	2	-	2	~	~	~	~	~	~	2	2	-	2	~	~	~	~	~	~	x	2	-
V15 ²	K added	2	~	~	~	~	~	~	2	~	2	2	~	~	~	~	~	~	2	~	x	2	~	~	~	~	~	~	x	~	x

- 1 or 2 Allocation weight while open to recruitment
- Before allocation started (future accrual)
~ After allocation ended (past accrual)
x Not applicable as an allocation

Historically in the trial:

Approximately 60% of men were newly-diagnosed and metastatic (to v14)

Approximately 10% have diabetes (to v14)

¹ Introduction of “M1|RT comparison” meant allocation had to be split: question not relevant to men without newly-diagnosed metastases or with a contraindication to RT

² Introduction of “metformin comparison” mean allocation to had to be split: question not suitable for patients with diabetes

APPENDIX R: AMENDMENTS MADE TO APPENDICES

R.1 AMENDMENTS MADE TO APPENDICES VERSION 1.0 MAY 2004

Appendix A – Addition of NYHA classifications

Appendix B – General PIS - Docetaxel information updated, additional information explaining cardiovascular risk related to celecoxib and celecoxib duration amended

Appendix B – PIS C – Docetaxel information updated

Appendix B – PIS D – Additional information explaining cardiovascular risk related to celecoxib and celecoxib duration amended

Appendix B – PIS E – Docetaxel information updated

Appendix B – PIS F – Additional information explaining cardiovascular risk related to celecoxib and celecoxib duration amended

Appendix C – GP letter – product name changed from Celebrex to Onsenal

Appendix E – Drug supply/ordering procedures amended

Appendix F – ‘Administration of Zoledronic acid’ – updated dose modification section

Appendix F – Additional section added ‘Co-administration of docetaxel + zoledronic acid’

Appendix G – Celecoxib – updated from revised celecoxib SPC

Appendix G – Table G.2 - Comparative table of undesirable effects of docetaxel, Zoledronic Acid and Celecoxib – updated information from revised celecoxib SPC

Appendix J – Common Toxicity Criteria – additional clarification and change in table structure

Appendix M – Accreditation documents – addition of new documents

Appendix P – Assessing and notifying CTU of adverse events flow diagram added

R.2 AMENDMENTS MADE TO APPENDICES VERSION 1.1 MAY 2005

Appendix F – ‘Administration of Zoledronic acid’ – updated section on recommend dose reduction in patients with mild to moderate kidney dysfunction (defined as a creatinine clearance of 30-60ml/min) who are receiving zoledronic acid.

Appendix G – Drug Safety Information for Drugs used in the Trial; section on zoledronic acid. Additional text informing clinicians that rare cases of Osteocronosis (primarily of the jaws) have been reported in patients treated with bisphosphonates.

R.3 AMENDMENTS MADE TO APPENDICES VERSION 2.0, 08 JUNE 2005

Appendix B – Patient information sheets & consent form – Increase in amount of blood needed & additional tissue sample request.

Appendix B – General PIS – Addition of the use of bicalutamide alone as a Hormone treatment.

Appendix B – General PIS – paragraph detailing to patients that they may receive radiotherapy along side their allocated treatment in STAMPEDE.

Appendix B – General PIS – Side effects of osteoporosis added to “What are the side effects”

Appendix B – General PIS – Will my taking part be kept confidential – now asking for NHS number and mention of NHS Strategic Tracing Service.

Appendix B – STAMPEDE Additional Research – Increase in amount of blood needed & addition tissue sample request.

Appendix B – Consent form – additional consent asked for urine and tumour samples, name to be stored on computer database and an extra copy of consent form to be taken and stored at the MRC CTU.

PIS A, B, C, D, E & F – addition of information about side effects of osteoporosis

PIS A, B, C, D, E & F – Addition of bicalutamide alone as an option for hormone therapy for M0 patients

PIS B, E & F – New side effects added, kidney failure and Osteoncrosis of the jaw

Appendix E– drug ordering - number of bottles of celecoxib sent to pharmacies changed from 15 to 24

Addition of Appendix P – Detailing radiotherapy guidelines for investigators.

R.4 AMENDMENTS MADE TO APPENDICES VERSION 3.0 JULY 2006

Appendix B - Patient Information Sheets (PIS) & Consent Forms - all translational additional now research referred to as 'translational sub-studies'

Appendix B – Information added that there will be different additional research PIS and consent forms for centres taking part in the bone mineral density study (BMD) compared with those that are not

Appendix B - All PIS - 'family doctor' replaced by GP to ensure consistency of terms

Appendix B - General PIS section 12 – Information about collection of patient names re-worded for clarity

Appendix B Additional Research PIS – Type A (for centres NOT participating in the BMD study) and Type B (for centres that are participating in the BMD study) created

Appendix B -Additional Research PIS section 2 - Information on when to complete questionnaires updated to accurately reflect follow up schedule

Appendix B - Additional Research PIS - Information added to reflect new blood collection method for DNA analysis and in regard to additional translational studies for which funding has recently been approved

Appendix B - Consent Form - Type A (for centres NOT participating in the BMD study) and Type B (for centres that are participating in the BMD study) created

Appendix B Consent form - Information changed in section L in regard to volume of blood sample required to reflect new collection practice

Appendix B -Consent Form – Information about allowing consent form to be sent to MRC CTU and GP, moved to the optional section

Appendix B -All Arm Specific PIS – Wording changed in HT section to more clearly explain how different treatments may effect bone in different ways.

Appendix C -GP letter - 'study' replaced by 'trial' and 'androgen suppression' replaced by 'hormone therapy' to ensure consistency of terms

- Appendix C – GP letter – Specification on number of calcichew tablets removed to allow for variations in local practice
- Appendix D - DNA analysis sub-study sample collection information changed to reflect new technique (FTA elute cards) being used for blood collection
- Appendix E – Specificity on the contents of the drug starter pack removed to allow for changes in the future
- Appendix F - Clarification that all docetaxel treatment delays should be reported on the CRF.
- Appendix F – Information added in regard to the need to closely monitor liver function prior to administration of docetaxel
- Appendix L - MRC CTU staff updated and IDMC members updated
- Appendix M - Investigator Statement updated to reflect current version
- Appendix P - 'AS' replaced with 'HT' to ensure consistency of terms
- Appendix P - Description of hormone therapy clarified
- Appendix P – Recommended dose of radiotherapy amended to better reflect current practice
- Appendix Q – New appendix added to give information in regard to BMD study.

R.5 AMENDMENTS MADE TO APPENDICES VERSION 4.0 DEC 2007

- Appendices split for main protocol for ease of use
- Appendix B - General PIS - Information about osteonecrosis of the jaw risk added to zoledronic acid section
- Appendix B - All arm specific PIS - Information about LHRH antagonists added to the hormone therapy section
- Appendix K - Instructions in regard to confirming biochemical failure amended
- Appendix L - Trial Steering Committee contact details updated

R.6 AMENDMENTS MADE TO APPENDICES VERSION 5.0 AUG 2008

- Appendix B – Guidance for administrators – QL Information removed
- Appendix B – General PIS – QL information removed
- Appendix B – Additional Research PIS (type A) – QL Information removed
- Appendix B – Additional Research PIS (type B) – QL Information removed
- Appendix B – Consent form (type A) – QL Information removed
- Appendix B – Consent form (type B) – QL Information removed
- Appendix F - Information added to clarify that patients who develop an osteonecrosis of the jaw should stop zoledronic acid treatment
- Appendix L – TMG Contact details updated

R.7 AMENDMENTS MADE TO APPENDICES VERSION 6.0 JUL 2009

Appendix B – patient information sheets for arms D and F greyed out

Appendix B – General PIS updated to v7.0

Appendix C - GP letter updated to v7.0

Appendix E – reference to SSA favourable assessment removed

Appendix L – trial contacts updated

Appendix P – radiotherapy guidelines updated

R.8 AMENDMENTS MADE TO APPENDICES VERSION 7.0 JUN 2011

Appendix B – General Patient Information Sheet – Updated to reflect addition of new trial arm

Appendix B – General Patient Information Sheet – Updated with text to better describe randomisation as per current MRC CTU standard practice

Appendix B – Arm specific information patient information sheet for arm G added

Appendix B – Wording in all patient information sheets regarding hormone therapy updated for clarity and to reflect that anti-androgen monotherapy is now longer permitted

Appendix B - All patient information sheets updated with information regarding who reviews the trial, as per current MRC CTU standard practice

Appendix B – Wording in all patient information sheets updated for clarity as per current MRC CTU standard practice

Appendix C – GP Letter - Updated to reflect addition of new trial arm

Appendix E – Drug Supply Information - Updated to reflect addition of new trial arm

Appendix G – Drug Safety Information – Updated to include information on abiraterone

Appendix F – Zoledronic Acid dose reduction information clarified

Appendix H – Previous appendix H removed to reflect that the treatment at home with zoledronic acid is not an option in the trial

Appendix L – Reference to CTCAE toxicity grading website updated

Appendix M – Contact details updated

Appendix O – Previous appendix O containing a diagram relating to flow of safety data has been removed as this has been deemed not useful for centres. Subsequent appendices renamed accordingly.

Appendix P - – Previous appendix P regarding radiotherapy removed and added to the main protocol

Appendix O – Added to summarise changes to appendices in each version

R.9 AMENDMENTS MADE TO APPENDICES VERSION 8.0 SEP 2011

Throughout appendices – numbering and sections headers have been updated in some sections to accommodate new information that has been added

Appendix A – Wording updated

Appendix B – Patient Information Sheets removed and submitted as separate documents

Appendix C – GP letter removed and submitted as separate document

Appendix F.4 – Information added on administration of RT

Appendix G.3 – Details n Celecoxib removed

Appendix G.4 Information on abiraterone updated to reflect changes in the Investigator Brochure

Appendix G.3 – Information on undesirable effects for abiraterone updated to reflect changes in the Investigator Brochure

Appendix G.5 – Section on RT contraindications, special warnings and precautions added

Appendix K – Section on RT Quality Assurance added

Appendix L – Information on hormone therapy prior to randomisation added

Appendix M – Trial Management Group contact details updated

Appendix N – Institution statement, signature and delegation of responsibilities log and site personnel list removed

Appendix P – Section on Bone Mineral Density sub-study removed

Appendix Q – Amendments made appendices updated

R.10 AMENDMENTS MADE TO APPENDICES VERSION 9.0 OCT 2011

Appendix J – Typo corrected in section J2

Appendix M - Trial Management Group contact details updated

R.11 AMENDMENTS MADE TO APPENDICES VERSION 10.0 APR 2013

Appendix G.4 – Section updated to reflect the new abiraterone IB

Appendix G.5 –Section added to include the combination of enzalutamide and abiraterone

Appendix M - Trial Management Group, Trial Steering Committee, Independent Data Monitoring Committee and trial team contact details updated

Appendix Q - Amendments made appendices updated

R.12 AMENDMENTS MADE TO APPENDICES VERSION 11.0 JAN 2014

Typos corrected throughout the Appendices

Appendix D. Addition of saliva sample collection

Appendix G.4.3. Update of abiraterone interactions with other medicinal products

Appendix L. Clarification on use of MAB for Arm J patients

R.13 AMENDMENTS MADE TO APPENDICES VERSION 12.0 FEB 2015

Typos corrected throughout the Appendices

Appendix F. Details on primary analysis for zoledronic acid and docetaxel containing arms included

Appendix G. Details on trial treatments removed and incorporated into main Protocol

Appendix J. Wording improved to clarify definition of biochemical progression

Appendix L. Clarification on use of MAB for Arm J patients

Appendix M. Details on TMG, TSC and IDMC members updated

R.14 AMENDMENTS MADE TO APPENDICES VERSION 13.0 OCT 2015

Typos corrected throughout the Appendices

Appendix G. Table of undesirable effects updated to reflect inclusion of metformin

Appendix I. Table remove and reference to CTCAE updated

Appendix M. Details on TMG, TSC and IDMC members updated

Appendix Q. Table included to describe history of allocation at randomisation



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STAMPEDE

**Systemic Therapy in Advancing or Metastatic Prostate
Cancer: Evaluation of Drug Efficacy
A multi-arm multi-stage randomised controlled trial**

MRC PR08

ISRCTN number: ISRCTN78818544
EUDRACT number: 2004-000193-31
CTA number: 00316/0026/001-0001
NCT number: NCT00268476

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Replaces Version date: 26-Mar-2015 V1.0

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This template and all preceding versions will be stored in the Statistical Analysis Master File for this trial held in S:\Stampede_Stats\SAP

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1 ABBREVIATIONS

Abi	Abiraterone
ADT	Androgen-deprivation therapy
AMP	Adenosine monophosphate
AS	Activity Stage
BMD	Bone mineral density
CHF	Congestive heart failure
Cel	Celecoxib
CI	Chief Investigator
CRF	Case Report Form
CTU	Clinical Trials Unit
CV	Cerebrovascular
DMP	Data Management Plan
Doc	Docetaxel
Enza	Enzalutamide
ES	Efficacy Stage
FFS	Failure-free survival
FPM	Flexible parametric models
HE	Health Economics
HT	Hormone therapy
IDMC	Independent Data Monitoring Committee
ITT	Intention-to-treat
KM	Kaplan-Meier
LHRH	Luteinising hormone-releasing hormone
LOB	Lack-of-benefit
M0	Non-metastatic
M1	Metastatic
MAB	Maximum Androgen Blockade
MACE	Major adverse cardiac event
MAMS	Multi-arm multi-stage
MCAR	Missing completely at random
MI	Myocardial infarction
MRC	Medical Research Council
N+	Node positive
N0	Node negative
NSAID	Non-steroidal anti-inflammatory drug
ONS	Office of National Statistics
OS	Overall survival
PH	Proportional hazards
PSA	Prostate specific antigen
QL	Quality of Life
RMST	Restricted mean survival time
rPFS	Radiological progression-free survival
RT	Radiotherapy
RTOG	Radiation Therapy Oncology Group
SAE	Serious adverse event
SAP	Statistical Analysis Plan
SOC	Standard-of-care
SOP	Standard Operating Procedures
TMG	Trial Management Group
TSC	Trial Steering Committee
WHO PS	WHO Performance Status
ZA	Zoledronic acid

2 BACKGROUND AND DESIGN

This section gives a brief summary of the trial, including the trial aims/objectives; treatment/randomisation arms; main outcomes; and the patient eligibility criteria.

Full details of the background to the trial and its design are presented within the Protocol (current Version 15.0; Sep-2016).

2.1 Trial Summary

STAMPEDE is a multi-centre, platform, randomised controlled trial for patients with locally advanced or metastatic (M1) prostate cancer who are starting long-term androgen-deprivation therapy (ADT) for the first time. Patients can have either newly-diagnosed disease or have been previously treated with radical radiotherapy or surgery but now have a rising prostate specific antigen (PSA). Further details on eligibility can be found in [Section 2.2](#).

The trial uses multi-arm, multi-stage (MAMS) methods to simultaneously assess a number of different research treatments. The trial aims to assess the effects of adding one or two approaches to the standard-of-care (SOC).

The investigational agents to date are:

- (i) a bisphosphonate, zoledronic acid
- (ii) a cytotoxic chemotherapeutic agent, docetaxel
- (iii) a cyclooxygenase (Cox-2) inhibitor, celecoxib
- (iv) a CYP-17 inhibitor, abiraterone
- (v) radiotherapy to the prostate amongst newly-diagnosed M1 patients only
- (vi) an androgen receptor signalling inhibitor, enzalutamide
- (vii) metformin

Patients on the control arm receive the standard-of-care; the research arms have this standard supplemented with other potential treatments.

The standard-of-care is based on ADT, achieved through the use of luteinising hormone-releasing hormone (LHRH) analogues or antagonists, Maximum Androgen Blockade (MAB: long-term anti-androgens in combination with LHRH agonist) or bilateral orchidectomy according to local practice (bicalutamide for non-metastatic (M0) patients was allowed in some versions of the Protocol). This is also the backbone of therapy for the research arms.

Standard-of-care radiotherapy (RT) is encouraged for all M0 patients and was mandated for all N0M0 patients from Protocol v8.0 (Nov-2011) onwards; whilst the M1|RT research Arm H was recruiting, radiotherapy was

not permitted as SOC for newly-diagnosed M1 patients and should only be received through allocation to Arm H for patients corresponding to this sub-set.

Standard-of-care docetaxel was permitted from Protocol version 14.0 (Dec-2015) onwards, following the results of STAMPEDE's original comparisons¹, along with other external trial results combined within a meta-analysis².

Planned use of docetaxel, as reported at randomisation, is included as a stratification factor from this time.

The trial is summarised in **Figures 1-9**; with the trial initially starting as Arms A to F. Additional research arms have been included in the trial over time, at times as defined below:

- 15-Nov-2011: abiraterone arm (Arm G)
- 22-Jan-2013: metastatic radiotherapy (M1|RT) arm (Arm H)
- 29-Jul-2014: combination of enzalutamide plus prednisolone with abiraterone (Arm J)
- 05-Sep-2016: metformin arm (Arm K)

As research arms have passed through interim analyses and/or reached their target sample size they have closed to recruitment. Timelines for these are defined as follows:

- 06-Apr-2011: The celecoxib-containing arms (Arms D & F) closed to recruitment due to lack of sufficient activity following the Activity Stage 2 analysis.
- 31-Mar-2013: The remaining original arms (Arms B, C & E) closed to recruitment having reached an acceptable sample size (based on time to analysis projections).
- 17-Jan-2014: The abiraterone arm (Arm G) closed to recruitment having reached its revised target sample size (1800 pts).
- 31-Mar-2016: The enzalutamide plus abiraterone arm (Arm J) closed to recruitment having reached its target sample size.
- 02-Sep-2016: The M1|RT arm (Arm H) closed having reached its revised target sample size.

At the time a new research arm is activated, the strata totals in the randomisation system are reset; the team checks for imbalances before this occurs. Consideration is given to applying a small weight to the reset total to help correct this only if there is a major imbalance, defined as a difference of 12 patients or more in any given strata (NB to date this has not been required).

¹ [http://www.thelancet.com/pdfs/journals/lancet/PIIS0140-6736\(15\)01037-5.pdf](http://www.thelancet.com/pdfs/journals/lancet/PIIS0140-6736(15)01037-5.pdf)

² [http://www.thelancet.com/pdfs/journals/lanonc/PIIS1470-2045\(15\)00489-1.pdf](http://www.thelancet.com/pdfs/journals/lanonc/PIIS1470-2045(15)00489-1.pdf)

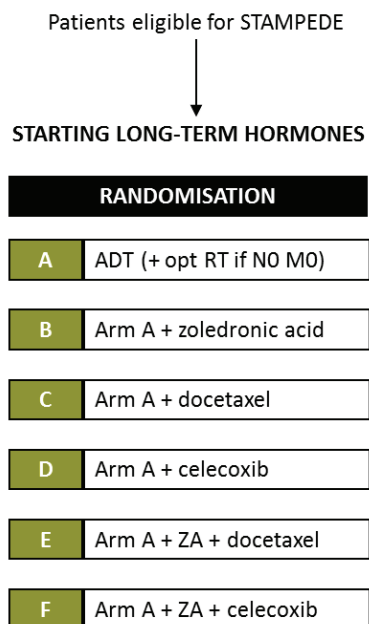
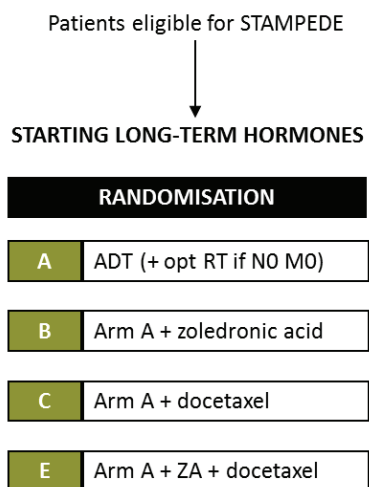
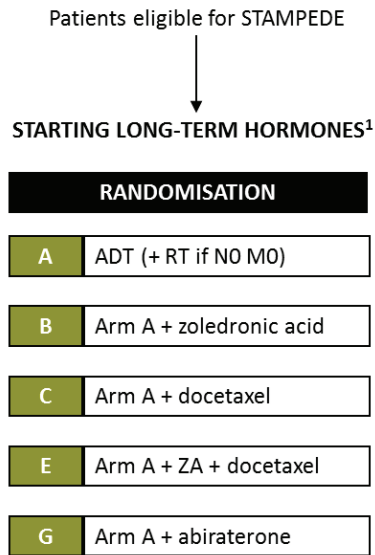
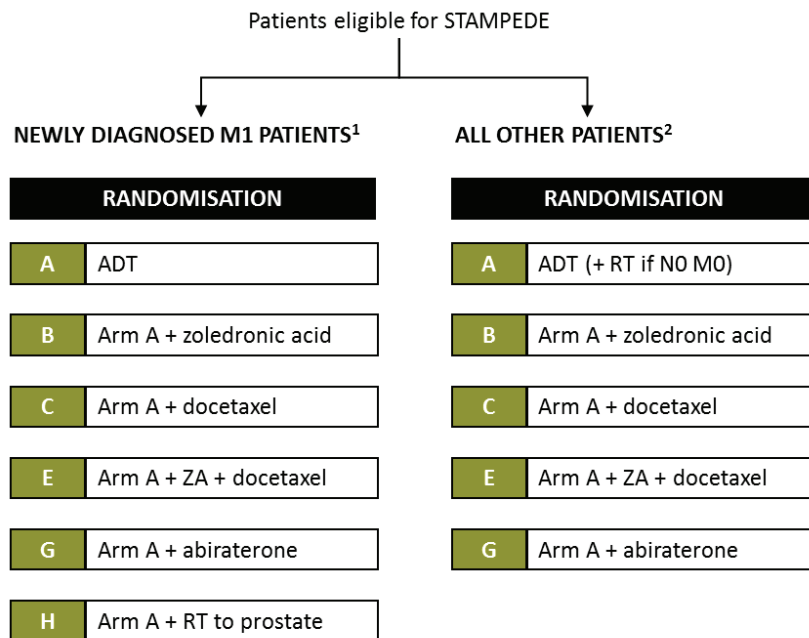
Figure 1: Recruiting arms from trial initiation to April 2011**Figure 2: Recruiting arms from Apr-2011 to Nov-2011 (Protocol version 7.0)**

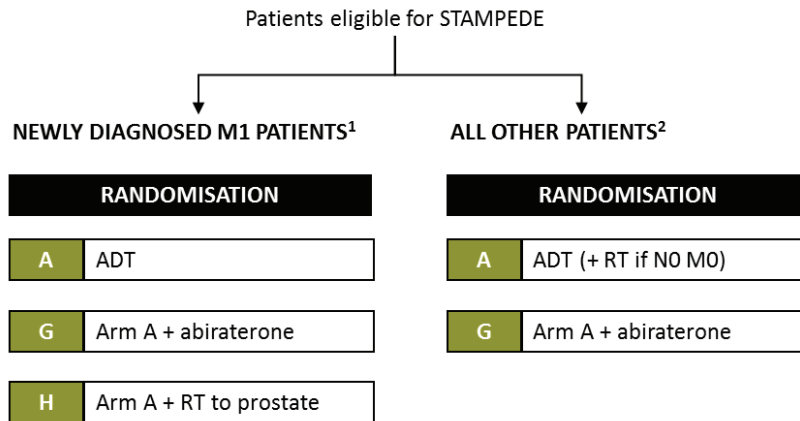
Figure 3: Recruiting arms from Nov-2011 to Jan-2013 (Protocol version 8.0)

Note: SOC includes mandated RT for N0M0 patients from Nov-2011

Figure 4: Recruiting arms from Jan-2013 to Mar-2013 (Protocol version 9.0)

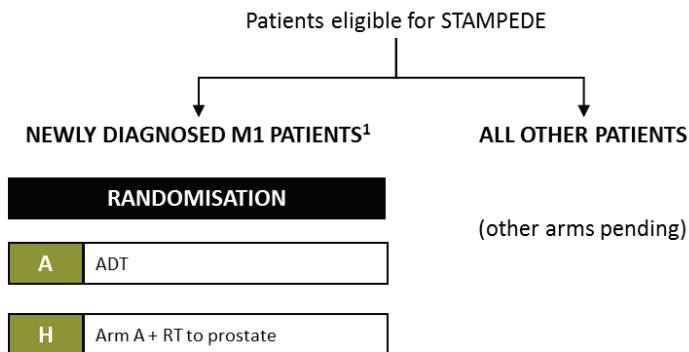
¹ except pts with a contra-indication to RT

² all suitable pts with newly diagnosed locally advanced disease should also have RT¹

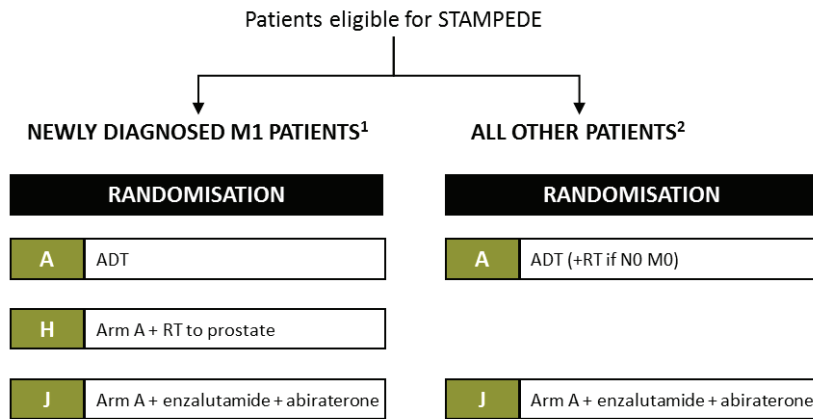
Figure 5: Recruiting arms from Apr-2013 to Jan-2014 (Protocol version 10.0)

¹ except pts with a contra-indication to RT

² all suitable pts with newly diagnosed locally advanced disease should also have RT¹

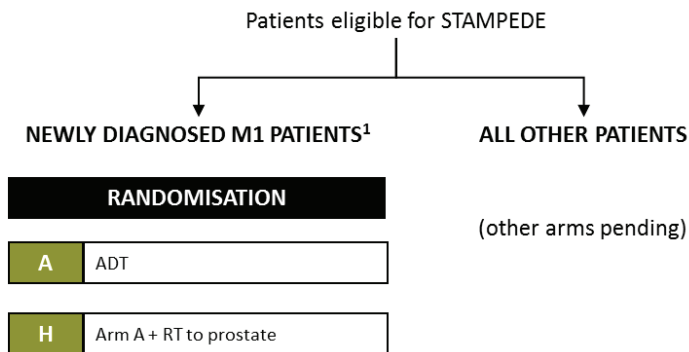
Figure 6: Recruiting arms from Jan-2014 to Jul-2014 (Protocol version 11.0)

¹ except pts with a contra-indication to RT

Figure 7: Recruiting arms from Jul-2014 to Mar-2016 (Protocol version 12.0)

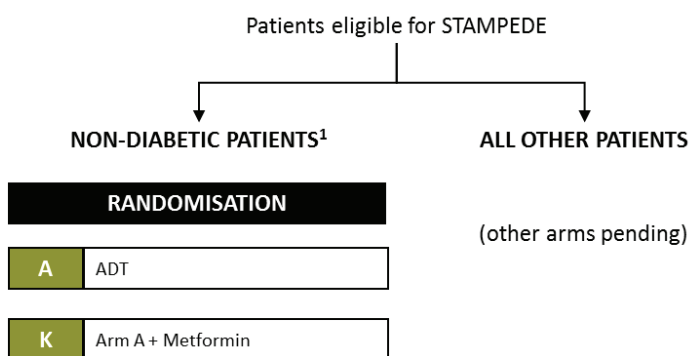
¹ except pts with a contra-indication to RT

² all suitable pts with newly diagnosed locally advanced disease should also have RT¹

Figure 8: Recruiting arms from Apr-2016 to Sep-2016 (Protocol version 14.0)

¹ except pts with a contra-indication to RT

Note: SOC includes docetaxel, where planned, from Dec-2015

Figure 9: Recruiting arms from Sep-2016 (Protocol version 15.0)

¹ except pts with a contra-indication to metformin

Note: SOC includes RT for N0M0 patients from Nov-2011, and docetaxel, where planned, from Dec-2015

2.2 Eligibility criteria

The eligibility criteria describe a broad population of patients unified by the need to start long-term ADT for the first time.

In the broadest terms, this includes patients who have had previous local therapy and now have high-risk relapse, along with patients with newly-diagnosed disease from the following groups:

- High-risk locally advanced disease (at least two of: Stage T3/4 N0 M0 histologically confirmed prostate adenocarcinoma, PSA $\geq$ 40ng/ml or Gleason sum score 8-10)
- Nodal involvement
- Metastatic disease

All patients need to give written consent, be sufficiently fit for any protocol treatment and follow-up and also meet other detailed eligibility criteria.

The details of the eligibility criteria have changed a little over time; this is shown by protocol version in a specific "[eligibility by version](#)"³ document which will continue to be updated throughout the trial.

³ S:\Stampede\10 Audit and QC documents\10.2 Monitoring_Visits_Reports\Monitoring_tools

2.3 Method of randomisation

Patients are randomised centrally using a computerised algorithm developed and maintained by the CTU. Randomisation is performed using the method of minimisation over a number of clinically important stratification factors with an additional random element.

These factors will be:

- | | |
|--|--|
| • randomising centre | each centre |
| • metastases | M0 vs M1 |
| • nodal involvement | N0 vs NX vs N+ |
| • age at randomisation | up to 69yrs vs 70yrs and over |
| • WHO performance status | PS=0 vs PS=1-2 |
| • method of ADT ⁴ | Orchidectomy vs LHRH agonist vs LHRH antagonist vs Maximum Androgen Blockade (MAB) |
| • regular aspirin or NSAID use at baseline | yes vs no |
| • radiotherapy planned ⁵ | yes vs no |
| • docetaxel planned ⁶ | yes vs no |

When implementing the additional random element of the randomisation, an 80% probability of allocation will be split between the (one or more) arms with the lowest strata totals (i.e. 80% probability of being allocated to one of the minimising arms); and the remaining 20% probability of allocation would be split amongst the remaining (one or more) arms. This method should provide simplicity of reporting and implementation.

2.4 Allocation Ratio

As of Sep-2016 (Protocol version 15.0), the trial had one control arm and nine research arms.

When the trial was initiated in Oct-2005 as the control arm was the comparator arm for five contemporaneous research arms, it was intended to recruit twice as many patients to the control arm as to each research arm. Therefore, the randomisation ratio for the five original research comparisons was 2:1:1:1:1.

For research arms G, H, J, K, added in over time, the allocation ratio has been 1:1 for control:research.

⁴ Method of ADT options have changed over time, from LHRH vs orchidectomy, to then include bicalutamide, then specify LHRH agonist or antagonist and more recently exclude bicalutamide but include MAB; see [Stratification Factors OverTime.docx](#)

⁵ "Radiotherapy planned" was added as a stratification factor at the start of recruitment to Efficacy Stage I (Mar-2008)

⁶ Docetaxel planned was added as a stratification factor from 17-Dec-2015 following publication of the "original comparisons" results indicating docetaxel improved overall survival

3 OUTCOME MEASURES

The trial seeks to improve overall survival and therefore overall survival (all-cause mortality) forms the definitive primary outcome measure for all comparisons within the trial.

The multi-arm multi-stage approach assumes that a failure-free survival event is on the causal pathway to death from PCa and that any benefit in overall survival would be preceded by a benefit (at least as large) in failure-free survival. Therefore, where it is considered to be a suitable outcome, failure-free survival is selected as the intermediate primary outcome measure.

For all outcome measures the comparison is between patients on the relevant research arm and those contemporaneously randomised control arm patients eligible for the research arm of interest.

The fully specified outcome measures for the different comparisons within STAMPEDE are listed in **Tables 1 & 2**.

3.1 Arms B-J

For the original research comparisons, along with the first three added arms (Arms G-J), the pre-specified outcome measures are detailed in **Table 1**. The reasons for different emphases in each recruitment stage are explained above.

Table 1: Trial outcome measures by Comparison Stage (Arms B-J)

Comparison Stage	Primary outcome measure	Secondary outcome measures ^T
Pilot phase(s)	Safety [*]	Feasibility
Activity Stages	Failure-free survival (FFS) [†]	Overall survival (OS) ^x Biochemical Failure Progression-free survival Disease-specific survival Non-PCa death Lymph node progression Distant metastases Toxicity Symptomatic skeletal events Therapy for progression
Efficacy Stage	Overall survival	Failure-free survival [†] Other 2 ⁰ OMs as for Activity Stages plus: Quality of life Cost effectiveness

^T Presented at activity stages where data is mature

^{*} Based on toxicity

[†] Including biochemical failure (see Protocol Section 7.1.2)

^x Cause of death with a view to cause-specific survival from PCa (with death from other causes as a competing risk) is considered under this.

Note: All arms are un-blinded so primary outcome measures for all comparisons are objectively measured, with caution to be taken around interpretation of more subjective secondary outcome measures such as symptomatic skeletal events.

3.2 Arm K (“Metformin comparison”)

For the addition of research arm K the implementation is slightly different to previous comparisons. Failure-free survival will not be used as the intermediate primary outcome measure; overall survival will be used as both the intermediate and definitive primary outcome measure. This is because we are not convinced that any comment on metformin’s usefulness should be determined from an ability to act on a PSA-driven outcome measure. Furthermore, treatment with metformin is intended to continue throughout long-term hormone therapy which may include going well beyond an FFS event, particularly in M1 patients. **Table 2** shows the outcome measures for the “metformin comparison”.

Table 2: Trial outcome measures by Comparison Stage (Arm K)

Comparison stage	Primary outcome measure	Secondary outcome measures ^T
Pilot phase	Safety [*]	Feasibility Metabolic effects including: :: Changes in BMI :: Changes in haemoglobin A1c (HbA1c) :: Changes in waist circumference :: New diagnosis of diabetes mellitus :: Cardiovascular event: major adverse cardiac events [‡]
Activity Stage	Overall survival	Metabolic effects Toxicity Symptomatic skeletal events Failure-free survival [†] (FFS) Biochemical Failure Progression-free survival Disease-specific survival Non-PCa death Lymph node progression Distant metastases Therapy for progression
Efficacy Stage	Overall survival	2 ⁰ OMs as for Activity Stages plus: Quality of life Cost effectiveness Correlative outcomes ^A

^T Presented at activity stages where data is mature

^{*} Based on toxicity

[‡] MACE; nonfatal MI, nonfatal stroke, and death from cardiovascular causes

[†] Including biochemical failure (see Protocol Section 7.1.2)

^A Plasma lipid and fasting triglyceride levels, fasting plasma glucose

Sarcopenia and/or radiological progression free survival (rPFS)

Plasma insulin

AMP Kinase

4 DATA COLLECTION AND PROCESSING

4.1 Case Report Forms and variables

Full details of data collection and timings are described in the [current trial protocol](#)⁷.

A copy of the Case Report Forms (CRFs) and Quality of Life (QL) questionnaires are presented in the Statistical Master File and the Trial Master File. Details of the variables, and any corresponding validations, are presented in the metadata which forms part of the Trial Master File.

4.2 Data completion schedule for Baseline, treatment & follow-up

The following table gives detail around the expected timing of those scheduled CRFs. Please note that detail has been removed for treatment forms relating specifically those to research arms which have reported their formal final efficacy analysis; please see the [SAP version 1.0](#) for detail of these timings.

Table 3: Timings for completion of scheduled CRFs

Timing of Assessment		Baseline		Treatment		Outcomes		Freq
		Rand ⁿ	Pre-Trt	RT	SOC Doc	Follow-up	QL HE	
Yr 0	Wk 0	all	all				all	
	Wk 6			M1: A, H		all	all	6 weekly
	Wk 12					all	all	
	Wk 18					all	all	
	Wk 24				all	all	all	
	Wk 36			M0: A, B, C, E, G, J, K		all	all	12 weekly
	Wk 48					all	all	
	Wk 60					all	all	
	Wk 72					all	all	
	Wk 84					all	all	
	Wk 96					all	all	
Yr 2	Month 24					all	all	6 monthly
	Month 30					all	all	
Yr 3	Month 36					all	all	
	Month 42					all	all	
Yr 4	Month 48					all	all	
Yr 5	Month 60					all	all	
Yr 6+						all	all	12 monthly

Key:

all = Arms A-K

A = SOC alone

G = SOC + abiraterone

H = SOC + M1 research RT to the prostate

J = SOC + enzalutamide + abiraterone

K = SOC + metformin

Text with a ~~strikethrough~~ indicates form no longer relevant

⁷ S:\Stampede\01 Study Protocol_Key Documents\1.1 Signed Protocol

Note:

- Pre-Trt assessment includes the following CRFs: baseline, cardiovascular, bone density risk factor questionnaire (for patients randomised from Protocol v6.0; Jul-2009).
- QL & HE data were collected for the first 700 patients randomised and then for all patients randomised from Protocol v8.0 onwards (Nov-2011) who opt-in.
- QL & HE data collected until disease progression.
- RT forms were not required for M0 patients on Arms D & F as they closed to recruitment prior to RT being mandated within NOMO standard-of-care.
- SOC Doc forms are required for all patients randomised from 17-Dec-2015, when the change in SOC to permit upfront docetaxel use was implemented, and for any patients randomised before this date and receiving upfront docetaxel.

4.3 Data completion schedule for other 'Non-Scheduled' CRFs

The following CRFs should be completed as required:

- Hormone therapy log (All Arms)
- Abiraterone & enzalutamide treatment log (Arms G & J)
- Metformin treatment log (Arm K)
- Toxicity form (All Arms)
- Progression log (All Arms)
- Additional treatment log (All Arms)
- Saliva pathology form (All Arms)
- Blood form (All Arms)
- SAE form (All Arms)
- End of research treatment form (Research Arms only)
- Co-enrolment form (All Arms)
- Death form (All Arms)

For all CRFs, detail of associated timings and requirements are given within their guidance.

4.4 CRFs no longer required

There are several CRFs which, over the time of the trial, have been either replaced by an updated CRF, the data collected has been moved to another CRF, or the CRF no longer required.

Such CRFs include:

- Late radiotherapy toxicity CRF
 - RTOG late side-effects were moved to be collected on the follow-up CRF from Jan-2013
- Withdrawal CRF
 - Replaced by the end of trial participation CRF; subsequently discontinued from Sep-2016
- Hormone therapy and abiraterone & enzalutamide treatment CRFs
 - Replaced by logs from Protocol version 15.0 (Sep-2016)
- Additional treatment update CRF
 - Replaced by a log, linked to the progression CRF from Protocol version 15.0 (Sep-2016)
- Palliative radiotherapy

- No longer of interest as radiotherapy for progression should be reported on the additional treatment log (discontinued from Sep-2016)
- Bone mineral density (BMD) sub-study
 - Sub-study discontinued from Protocol version 8.0 (Sep-2011).

4.5 Management of datasets

The datasets will be managed in accordance with CTU Standard Operating Procedures (SOPs) and in accordance with the [Data Management Plan](#)⁸.

4.5.1 Data Cleaning

In preparation for an IDMC, interim or final efficacy analysis the database will be frozen twice using the following procedure:

- The Statistician will perform an initial data freeze and use the data to identify any data queries or discrepancies within the database. All data queries and discrepancies will be recorded in Excel spreadsheets which will be sent to the relevant Trial Manager and/or Data manager.
- The Trial/Data Manager will contact centres and resolve as many of the data queries and discrepancies as possible. All received answers to the queries will be entered into MACRO.
- The Statistician will perform a second data freeze (final data freeze) during which they will extract the data to be used in the IDMC, interim or primary analysis report.
- Additionally, for the final efficacy analysis the process will be formally documented with the exact details agreed prior to the analysis.

4.5.2 Extracting Data

A link to details of timelines for data downloads and checks prior to analysis can be found in Section 5 of the Statistical Master File (SMF)

In summary, at the time of each data freeze:

- The Trial Manager(s), Data Managers and Statisticians will agree a date to extract data from MACRO.
- Overnight at the end of this agreed date a copy of the data will be set to be created and saved by a senior member of Data Services.
- On the following day, the statistician exports the required copied data using the MACRO Query Module to a secure area of the CTU network where access is restricted to the Statistician(s) working on the trial. Note the MACRO dataset is not truly frozen; the extracted dataset is only a snapshot of the data at that time. It is the responsibility of the Statistician to accurately record the date of freezing and ensure all data are retrieved.

⁸ S:\Stampede\08 Data Management\8.6 Data Management Plan

- As all the required data are taken from a copy of the databases, new data can continue to be entered onto the 'live' dataset.
- If any outstanding data queries are resolved whilst the analysis files are being prepared (ie when only a practice dataset has been copied), the changes need only be made to the 'live' dataset before the final copy is made by the Statistician.
- If any outstanding data queries are resolved during the analysis that relate to data in the final copied dataset (eg problems that are found during analysis or amended CRFs that are returned to CTU), and it is desired to include these changes in the report or analysis being run, corrections should be made in MACRO as usual but the data should not be exported again. Rather, the corrections should be made in the Statistician's analysis files and appropriately highlighted and annotated (notably any confirmed changes should only be made within cr_fieldupdates.do). For the primary analysis the Senior Statistician will decide on a case-by-case basis whether a correction warrants exporting the data again.
- In circumstances where the majority of queries require CRFs (for example unreported progressions) to be sent in from sites, the data will be re-extracted at a later date, as decided by the Statistician(s) and Data Manager(s).

Datasets for all analyses will be constructed and analyses performed using Stata programs (*.do).

4.6 Data verification

Data verification, consistency and range checks will have been performed at the data entry stage by the CTU, as well as checks for missing data, as documented in the trial working practices (copies of these checks can be found in the Trial Master File). Additional range, consistency and missing data checks will be performed automatically within the MACRO databases, as programmed in the database. Copied of the data chase and query report forms are stored electronically with completed and returned forms stored in patient files. Some of the checks may be duplicated and others will be performed when the datasets for analysis are constructed by the statistician. All outcome variables (primary and secondary) including treatment details will be examined for unusual, outlying, unlabelled or inconsistent values by the Statistician.

In general, given the thorough nature of our follow-up procedure we expect the issue of missing data to be relatively minimal. We anticipate high compliance with initial data collection as this is close to the time of patient registration. If any data is missing imputation will not be done, but this may be reconsidered if recommended.

Data from sites on the definitive primary outcome, overall survival, and potential other outcome measures may be supplemented by linked data from the Office of National Statistics (ONS) or a similar system. This would be used to ascertain the final survival status of those not reported as having died on the most recent follow-up, and additionally to help in the ascertainment of causes of death.

Any problems with trial data will be queried with the Trial Managers, Data Managers, or Statistician(s), as appropriate. Where possible, data queries will be resolved, although it is accepted that due to administrative reasons and data availability, a small level of problems will continue to exist; this will be minimised.

Prior to formal final primary analyses an additional set of checks will be completed. These include a Trial Manager checking MACRO data against CRFs for a percentage of primary outcome data (including date of death). These checks are detailed in Section 8.10 of the Trial Management File, kept by the STAMPEDE Trial Managers.

4.7 Data coding

The coding of the raw variables can be found in the [Macro Metadata document](#)⁹. The latest version has the largest version number "X" at the end of the document name.

Coding of variables for analysis can be found in the [Analysis File Specification documentation](#).

4.7.1 Free-text variables

Review and (re-)coding of free-text variables is carried out by a clinical member of the TMG.

⁹ S:\Stampede\08 Data Management\8.4 Metadata and Data Handling\Metadata\STAMPEDE Metadata vX

Such free-text variables to be reviewed and (re-)coded include:

- Those items reported as "other, specify", including reason for treatment action; toxicity; additional treatment; and place of death.
- Concomitant medications (collected as free-text at baseline; up to Protocol v15.0, Sep-2016)

5 DEFINITION OF TERMS

A number of terms will require definition and clarification. The table, below, aims to do this and will be updated as required.

Table 4: Definitions of certain terms (ordered by primary outcome, secondary outcome & other terms)

Term	Definition
Overall survival (OS)	Time from randomisation until death from any cause. For surviving patients, censor date is used (see below).
Failure-free Survival (FFS)	Time from randomisation until first of the following events: - Biochemical failure - Local progression - Skeletal Related Event (where confirmed disease progression) - Lymph node progression - Distant metastases - Death from prostate cancer For patients who have not had an event, censor date is used (see below). If a suspicious event is reported for any of the: local progression, lymph node progression, distant metastases progression this will be counted as a FFS event.
Progression-free Survival (PFS)	Time from randomisation until first of: - Local progression - Skeletal Related Event (where confirmed disease progression) - Lymph node progression - Distant metastases - Death from prostate cancer For patients who have not had an event, censor date is used (see below). If a suspicious event is reported for any of the: local progression, lymph node progression, distant metastases progression this will be counted as a FFS event.
Skeletal Related Event	- Bone pain requiring radiotherapy and/or surgery - Pathological fracture - Metastatic spinal cord compression
Disease-specific survival	Time from randomisation until death from prostate cancer (see below). For patients who have not had an event, censor date is used (see below).
Death from prostate cancer	All deaths are reviewed following the Death Review Process¹⁰ in due course. Deaths will be classified as the following categories: - Definitely prostate cancer - Probably prostate cancer - Possibly prostate cancer - Probably not prostate cancer - Definitely not prostate cancer Further information will be sought where there is insufficient information for the reviewer(s) to make a judgement. Where the review is not completed for any reason, the local investigator's opinion will be taken. For the main analyses, death from prostate cancer is taken as categories 1 and 2. In sensitivity analyses, these are to be supplemented by category 3.

¹⁰ S:\Stampede\08 Data Management\8.8 Death Review

Term	Definition
Censor date	Date taken from the latest of: • Date of randomisation (Form 1) • BMD assessment date (scan, blood sample, urine sample) • Date of treatment cycle (as taken from the bisphosphonate, docetaxel; Forms 4, 5, 6) • Date bloods taken (as taken from the bisphosphonate Forms 4, 5) • Date of last SOC docetaxel cycle (Form 21) • Date of any treatment action (Forms 7, 7B, 7C, 7D) • Dates reported on the Follow-up CRF (including date of PSA tests, date of any surgical interventions, date of any SRE, date of any metabolic or cardiovascular event; Forms 7, 7A) • Date of any reported progression event (Form 8) • Date additional treatment started or stopped (Forms 8, 8A) • Date of first/last RT fraction (Form 9A) • Date of late RT toxicity assessment (Form 10) • Date HT/research treatment ended (Form 11) • SAE date (onset, resolved, recent HT or trial treatment administration, start/end date of other treatment, test date) (Form 14) • Date of palliative RT fraction (Form 19) • Date blood or saliva sample obtained as reported on the pathology form (Form 18) • Date of Coenrolment to another trial (Form 15) • Date trial participation ended (Form 20) • Date last known alive (Form 7 from Version 13.0) • Date of death (Censoring date only for outcomes other than overall survival and disease-specific survival; Form 12) Notes: • Dates from the QoL forms are no longer used as a censor date as these are completed by the patient and cannot be queried for errors. • Dates of form completion are no longer used as the CRF may have been completed retrospectively. • Any date pre-randomisation is ignored within the calculation. • Unusual dates which have not yet been resolved or dates after the date of the corresponding data freeze will be ignored for the purposes of calculating this censor date.
Stratification factors ¹¹	• Randomising centre • Metastases at randomisation (Not "M1 RT comparison") • Nodal involvement • Age at randomisation • WHO performance status • Method of hormones • Regular aspirin or NSAID • Radiotherapy planned (from Mar-2008; Not "M1 RT comparison") • Docetaxel planned (from 17-Dec-2015)
Frozen Dataset	A copy of the data used for an analysis and to which no <i>new</i> CRFs are added. Data queries on the frozen copy should still be resolved, wherever possible, up to the point where the analysis is actually performed. Any changes will be hard-coded into the analysis files and highlighted and annotated as well as updating the live database (for future analyses).

¹¹ Details of the stratification factors over time: "S:\Stampede_Stats\SAP\Stratification_Factors_OverTime.doc"

6 SAMPLE SIZE CALCULATIONS

The first five research arms (B-F) have completed recruitment and reported primary overall survival results. Details of the sample size calculations for these arms can be found in the [Statistical Design Document](#).

This section gives details of the sample size calculations for the "abiraterone comparison", "M1|RT comparison", "enzalutamide + abiraterone comparison" and the "metformin comparison". The Statistical Design Document should be referred to for further detail on the design of these additional comparisons.

All do-files, STATA logs, and related presentations can be found within the relevant subfolder in [S:\Stampede Stats\STAMPEDE sample size](#).

6.1 Sample Size: "abiraterone comparison" (AvsG)

This comparison includes patients allocated to research Arm G and all patients contemporaneously allocated to the control Arm A.

6.1.1 Overall sample size

The sample size calculations were performed using the latest update of -nstage- available at the time (version 1.1.0, 28-Aug-2008), with an allocation ratio of 1:1.

This comparison was added in Nov-2011 and was designed under the same assumptions as the original research comparisons.

Up to around 1,800 patients were targeted to join the "abiraterone comparison", with half allocated to the research arm G; the observed allocation was 1,917. Consideration was given to ceasing further randomisations to Arm G if it was not showing sufficient evidence of activity at the interim analyses, just as was done for research Arms B to F.

The original plan intended for accrual to be halted either when 1,500 patients had been recruited or after 3 years, whichever was the sooner, providing the accrual rate remained above 50 patients/months.

The total number of patients joining this comparison depended not just on the same issues as the "original comparisons" (notably, observed accrual and event rates), but also the length of time that the original research arms co-recruited alongside this additional research arm; it was originally assumed that this would be for approximately 1 year, but it was closer to 1.5 years. The sample size calculations and projected durations are fairly robust to changes in the length of co-recruitment with the original research arms and future co-recruitment with any further research arms which the TMG may introduce. Many scenarios are detailed within [S:\Stampede Stats\STAMPEDE sample size](#).

In Sep-2013, the target sample size for the "abiraterone comparison" was increased from around 1,500 patients to around 1,800 patients, with note that the efficacy analysis remains unchanged and is still to be triggered by around 267 control arm deaths. This increase in sample size was primarily because of an increase

in the proportion of non-metastatic patients joining the comparison; this related to the activation of Arm H which only recruits patients with newly-diagnosed metastatic disease and thereby reduces the numbers of metastatic patients randomised to the "abiraterone comparison". Non-metastatic patients have a lower event rate than the metastatic patients and maintaining the same overall sample size would lead to a delay in time to the primary analysis. The increase in sample size was achievable because recruitment rates to the trial had been substantially higher than the anticipated 50 patients/month for the 6 months preceding the increase. The primary analysis on overall survival requires ~267 deaths to be observed on the control arm; this remains the same even with the increased sample size.

6.1.2 Reporting

The first survival results for this comparison are projected to be observed around Feb-2017.

6.2 Sample Size: "M1 | RT comparison" (AvsH)

This comparison includes patients allocated to research Arm H and newly-diagnosed M1 patients with no contraindication to RT allocated to the control Arm A whilst Arm H is open to recruitment.

Suitability for allocation to the comparison is assessed before randomisation to ensure comparability with contemporaneous control arm patients.

6.2.1 Overall sample size

The sample size calculations were performed using the latest update of *nstage* available at the time (version 1.1.0, 28-Aug-2008), with an allocation ratio of 1:1.

This comparison was added in Jan-2013 and was designed under the same assumptions as the original research comparisons.

Consideration was given to ceasing further randomisations to Arm H if it was not showing sufficient evidence of activity on the intermediate primary outcome measure (FFS), just as for the other research arms. This research comparison is relevant to around 60% of patients joining STAMPEDE. At the point of the scientific approval, accrual was averaging around 80 patients per month to the trial. If accrual to the trial was slower at 70 patients per month, then accrual to this comparison could be between 18 and 42 patients per month, depending on which other trial arms are open to recruitment at the time.

We are targeting a 25% relative improvement in overall survival following local radiotherapy to the prostate in this patient group. This is the same size of effect targeted with the other research arms in STAMPEDE. This relative improvement can be further justified in the light of MRC PR07 which demonstrated an improvement of this magnitude for adding radiotherapy to ADT in locally advanced disease, with a hazard ratio for overall survival of 0.77 (95% CI 0.61 to 0.98). In that trial, fewer than half of the deaths were from prostate cancer, whereas in newly-diagnosed metastatic patients nearly all men will die of their disease. Therefore, it is relevant to note the relative benefit of radiotherapy in PR07 in terms of prostate cancer-specific survival, where the

hazard ratio was 0.54 (95% CI 0.27 to 0.78). Long-term survival-based data, with a median follow-up of ~10 years, were presented orally at the American Society of Clinical Oncology 2012 which confirmed these findings.

We anticipated that around 1250 patients were required to be recruited over 4 years to observe 267 control arm deaths after 5.25 years. This assumed that (i) recruitment was constantly 70 pts/m to the trial overall; (ii) the original research arms stopped accrual within 6 months after activation of the RT arm; (iii) the abiraterone arm stopped accrual around 24 months after activation of the RT arm; and (iv) a further new research arm with an equal allocation ratio was introduced 18 months after activation of the RT arm. In Protocol version 13.0, we reflected on these four points: (i) recruitment to the trial has been faster; (ii) the original research arms completed accrual 2 months after activation of the RT arm; (iii) the abiraterone arm stopped accrual 12 months after activation of the RT arm; and (iv) Arm J was activated 18 months after activation of the RT arm, Arm H.

Of patients joining STAMPEDE during this time, 60% have been eligible for the "M1|RT comparison". Prior to randomisation, a RT schedule must be nominated: Weekly or Daily. We have observed that around half of patients in the comparison are nominated for RT with the Daily schedule and half for the Weekly schedule, primarily chosen by trial site with patient groups nominated for each schedule observed to be comparable at baseline. There will likely be interest to know the effect of each RT schedule when the main results are reported. This will be explored by "within schedule" comparisons of patients randomised to research vs control (arms H vs A) within each nominated RT schedule.

Therefore, in Protocol version 13.0, the target sample size was increased from 1,250 patients up to around 1,800 patients, resulting in an approximate increase from 625 to 900 patients in each "within schedule" analysis. A FFS analysis "within schedule" will be carried out at the time of the "main analysis"; predicted to have ~300 control arm FFS events by schedule (FFS "within schedule" analysis parameters: target HR=0.75, power 90%, 1-sided $\alpha=0.015$). If one or both of the RT schedules show evidence of an effect on FFS, a comparative "within schedule" analysis will be carried out on overall survival when ~199 control arm deaths are observed in that schedule comparison. This is a closed test with OS only formally compared within schedule if there is an advantage in FFS for that RT schedule at the main analysis. Thus, extending recruitment enables a secondary analysis of the impact of RT on survival by planned RT schedule to happen within around 18 months from the first main analysis.

All sample scenarios are documented in the Trial Master File.

All patients joining the trial will be starting long-term ADT for the first time. The focus of this comparison will be on the newly-diagnosed, metastatic patients (with no contraindications to RT), which is the largest subgroup of patients in the trial and the group of patients at highest risk of death from prostate cancer. Patients with non-metastatic disease will be excluded from this particular comparison as there are already randomised data demonstrating the survival benefit from radiotherapy in patients with locally advanced disease. Radiotherapy is now mandatory in node negative patients; it is also recommended in the node-positive, non-metastatic (N+ M0) group. Relapsing patients are also excluded from this comparison.

For the control arm of the whole trial, we constructed sample size scenarios for median failure-free survival being 18, 24 or 30 months; the event rate would depend on the patient mix. We now know that around 60% of patients have M1 disease at trial entry and we have reported that FFS at 24 months is 51% across the whole of the control arm.

For the updated sample size calculation for this comparison, we based our estimates on the subgroup of patients with newly-diagnosed M1 disease in the control arm. Therefore, we estimate median FFS in the Arm H control group to be 1 year and estimate that median overall survival will be around 3.5 years.

6.2.2 Reporting

The first survival results for this comparison are projected to be observed in 2018.

6.3 Sample Size: “enzalutamide + abiraterone comparison” (AvsJ)

This comparison includes patients allocated to research Arm J and all patients contemporaneously allocated to the control Arm A.

6.3.1 Overall sample size

The sample size calculations were performed using the latest update of –nstage- available at the time, with an allocation ratio of 1:1.

This comparison was added in Jul-2014 and was designed under similar assumptions as the original research comparisons; however a smaller interim HR of 0.70 was targeted with only 2 pre-planned activity analyses.

Consideration was given to ceasing further randomisations to Arm J if it was not showing sufficient evidence of activity on the intermediate primary outcome measure (FFS), just as for the other research arms.

The patient mix for this comparison is likely to represent a more favourable prognosis on average than in the original research trial’s other arms, due to concurrent recruitment of M1 but not M0 patients, to Arm H.

We anticipate that around 1,800 patients are required to be recruited within 3.5 years to observe ~267 control arm deaths within 6 years. This time will be dependent on the observed overall survival. The default scenario assumes that (i) recruitment is constantly 70pts/m to the trial overall, (ii) the M1|RT Arm H accrues throughout and (iii) a further new research arm with an equal allocation ratio is introduced 18 months after activation of Arm J. The stopping date for Arm G is no longer an assumption.

Variations on these factors are documented in [..\STAMPEDE sample size\adding arms 2013 Enzalutamide](#). If accrual rates to the trial are at 150pts/m (as observed during Summer 2013), it was anticipated that accrual of around 1,800 patients to the comparison could be achieved within 2 years. These sample scenarios are also documented in the Trial Master File.

Updating the standard-of-care to include docetaxel has minimal impact on the projected time to maturity of the “enzalutamide + abiraterone comparison”.

Careful consideration will be given to the implications of any emerging data from the "abiraterone comparison". This would have minimal to no effect on the "enzalutamide + abiraterone comparison" because the recruitment target for the latter will be reached before any data are available from the "abiraterone comparison".

Indirect comparisons to understand the contribution from each agent may be possible if this research arm is demonstrably superior to the standard-of-care. These plans will be developed and documented elsewhere, but a higher number of patients will help with the power of the indirect comparison.

6.3.2 Reporting

The first survival results for this comparison are projected to be observed around Nov-2018.

6.4 Sample size: "Metformin Comparison" (AvsK)

This comparison includes patients allocated to research Arm K and the equivalent non-diabetic patients with no contraindication to metformin contemporaneously allocated to the control Arm A whilst Arm K is open to recruitment. Suitability for allocation to the comparison is assessed before randomisation to ensure comparability with contemporaneous control arm patients.

Implementation of the MAMS principles in this comparison is different to previous comparisons for the following reasons:

- Although all non-diabetic patients will be eligible for allocation to the "metformin comparison", the timing of the analyses will be driven only by the M1 patients.
- Failure-free survival will not be used as the intermediate primary outcome measure; overall survival will be used as both the intermediate and definitive primary outcome measure. This is because we are not convinced that any comment on metformin's usefulness should be determined from an ability to act on a PSA-driven outcome measure. Furthermore, treatment with metformin is intended to continue throughout long-term hormone therapy which may include going well beyond an FFS event, particularly in M1 patients, and is allowed to continue alongside subsequent therapies.
- The target HR is 0.80 for overall survival (a 20% relative improvement). This is a smaller relative improvement in survival than targeted for earlier comparisons: a smaller impact may still have clinical benefit. Because of metformin's known low toxicity profile, the low cost of the drug and the potential positive effects on metabolic parameters and morbidity.

6.4.1 Overall sample size

The sample size calculations were performed using the latest update of -nstage- available at the time, with an allocation ratio of 1:1.

This comparison was added in Sep-2016 and was designed under different assumptions to those previously added comparisons.

Consideration will be given to ceasing further randomisations to the comparison if it is not showing sufficient evidence of improvement on overall survival at the intermediate activity analysis.

We anticipate that around 1,800 patients, including around 1,100 M1 patients, are required to be recruited over 3 years to observe ~374 control arm M1 deaths over around 8 years. This number and time will be dependent on the observed overall survival. The default scenario assumes:

- (i) median OS in M1 patients is 54 months; assuming the majority of metastatic patients will also have docetaxel but non-metastatic patients will not
- (ii) allocation ratio is 1:1
- (iii) interim one-sided alpha of 0.40 for one look
- (iv) recruitment is constantly 100pts/m to the trial overall
- (v) co-recruitment throughout of the equivalent of one other research arm

Variations on these factors are documented in the corresponding sample size folder¹². The six month co-recruitment with Arm J, which was originally anticipated, is no longer an assumption. If accrual rates to the trial are at 150pts/m (as observed during summer 2013), accrual of around 1,800 patients to the comparison could be achieved within 2 years. These sample scenarios will also be documented in the Trial Master File.

Updating the standard-of-care to permit first-line use of docetaxel was assumed within the sample size scenarios and is reflected in the projected time to maturity of the "metformin comparison".

Careful consideration will be given to the emerging data from the "abiraterone comparison" when this reports in 2017.

Analyses for the "metformin comparison" will be timed from randomisation. The point of randomisation compared to the start of hormone therapy may differ, depending on the planned use of docetaxel. This practical information will be reviewed by the TMG and IDMC.

6.5 Further details in sample size calculations and design

For further details relating to the sample size calculations and design for each research comparison see the Statistical Design Document.

¹² S:\Stamperde_Stats\STAMPEDE_sample_size\adding_arms_2014_met_rad\Metformin

7 ANALYSIS PRINCIPLES

7.1 Outline Analysis Plan

The full details of the analyses are presented in the following sections.

All efficacy analyses will be performed on an intention-to-treat basis. The standard unadjusted log-rank approach will be applied to analyses of FFS and OS. The impact of potential confounders including the stratification factors used at randomisation will be considered in a Cox proportional hazard model. The χ^2 test or Mann-Whitney test will be implemented for categorical data comparisons, as appropriate, including the main known toxicities for the research treatment.

7.2 Comparisons

A research comparison is defined by those patients allocated to the research arm, along with the corresponding contemporaneously randomised, eligible control arm patients. See [Table 5](#) for the definition of each research comparison within STAMPEDE to date.

Table 5: STAMPEDE Research Comparisons

Comparison name	Arms	Eligible patients	Accrual		Time Period(s)
			Start Date	End Date	
"Zoledronic Acid Comparison"	A, B	All patients	05-Oct-2005	31-Mar-2013	1-4
"Docetaxel Comparison"	A, C	All patients	05-Oct-2005	31-Mar-2013	1-4
"Celecoxib comparison"	A, D	All patients	05-Oct-2005	06-Apr-2011	1
"Zoledronic Acid + Docetaxel Comparison"	A, E	All patients	05-Oct-2005	31-Mar-2013	1-4
"Zoledronic Acid + Celecoxib Comparison"	A, F	All patients	05-Oct-2005	06-Apr-2011	1
"Abiraterone Comparison"	A, G	All patients	15-Nov-2011	17-Jan-2014*	3-5
"M1 RT Comparison"	A, H	Newly-diagnosed M1 patients No contraindication to RT	22-Jan-2013	02-Sep-2016	4-9
"Enzalutamide + Abiraterone Comparison"	A, J	All patients	29-Jul-2014	31-Mar-2016	7-9
"Metformin Comparison"	A, K	Non-diabetic patients No contraindication to metformin	05-Sep-2016	TBD	10-TBD

***Note:** One patient was manually randomised to Arm G after the cut-off of 17-Jan-2014

For the analysis of a research comparison, the relevant research arm will be compared against the corresponding control arm and the relative effect of treatment estimated.

In a closed test approach, research arms will be formally compared against one another only if they show an advantage over the control arm; each comparison of such effective research arms will include only those patients randomised contemporaneously who were eligible for both research arms (i.e. could have been allocated to either research arm).

Each efficacy analysis will be performed on an intention-to-treat (ITT) basis. All statistical tests will be implemented using a 2-sided p-value of 0.05, unless otherwise specified (e.g. by the trial design), but

compared to an equivalent 1-sided p-value of 0.025. There will be no formal adjustment of p-values because of the interim analyses performed, as this has been pre-considered in the design, however note will be made of the familywise error rate (FWER) associated with a comparison.

7.3 Time Periods

The main analysis for all comparisons using Cox PH models will be stratified by each time the randomisation, or other fundamental aspect which may affect the patient population being randomised, within STAMPEDE was changed. As of Sep-2016, there are ten time-defined strata (see figure below).

Figure 10: Arms active over time (time strata)

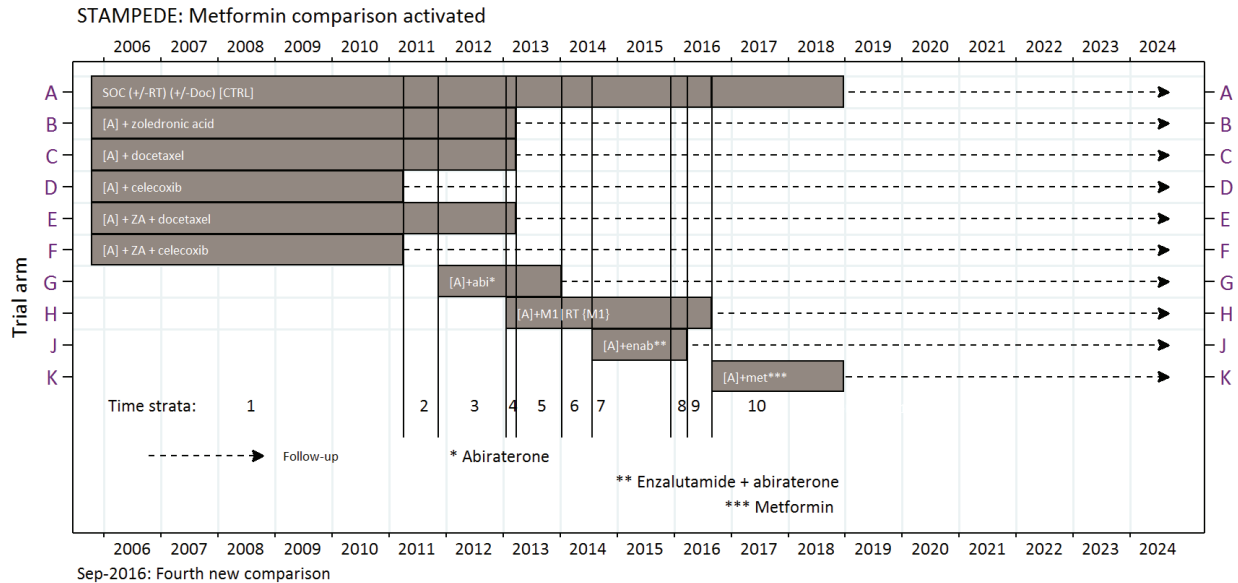


Table 6: Time periods within STAMPEDE

Time Period	Definition	Accrual		Co-recruiting research arms
		Start date	End date	
1	From the start of the trial up to the stopping of the celecoxib-containing research Arms D & F	05-Oct-2005	06-Apr-2011	B C D E F
2	Post-closure of Arms D & F up to the opening of the abiraterone research Arm G	06-Apr-2011	14-Nov-2011	B C E
3	Post-opening of Arm G up to the opening of the M1 radiotherapy research Arm H	15-Nov-2011	21-Jan-2013	B C E G
4	Post-opening of Arm H up to the closure of the remaining original research Arms B, C & E	22-Jan-2013	31-Mar-2013	B C E G H
5	Post-closure of Arms B, C & E up to the closure of abiraterone research Arm G	01-Apr-2013	17-Jan-2014*	G H
6	Post-closure of Arm G up to the opening of the enzalutamide+abiraterone research Arm J	18-Jan-2014	28-Jul-2014	H
7	Post-opening of Arm J up to the update in SOC to permit planned use of docetaxel as first line treatment	29-Jul-2014	16-Dec-2015	H J
8	Post-update of SOC up to the closure of enzalutamide+abiraterone research Arm J	17-Dec-2015	31-Mar-2016	H J
9	Post closure of Arm J up to the close of M1 RT research Arm H and opening of metformin research Arm K	01-Apr-2016	02-Sep-2016	H
10	Post-opening of Arm K onwards	05-Sep-2016	TBD	K

***Note:** One patient was manually randomised to Arm G after the cut-off of 17-Jan-2014

7.4 Period of Data Collection

For each formal data analysis, the period of data collection will be defined by the comparison of interest at that time. The period start date will be the date the comparison opened to recruitment, whilst the cut-off date will be the date of the most recent data freeze.

This detail will be clearly presented within any associated IDMC or final analysis report.

7.5 Pilot and Feasibility Phases

Each research comparison will go through an initial pilot safety phase; a report will be prepared for the IDMC to review both early safety data and feasibility for each comparison. Feasibility of a comparison is considered in terms of the acceptability of the trial randomisation and reported toxicities and adherence to trial medication. Patients randomised during this pilot phase will contribute to all future analyses of the research comparison.

If, following such a report, recruitment was determined to be unfeasible or the research arm proved unsafe or not feasible to administer, recruitment to that research arm would be discontinued.

The timing of, and trigger for, this reporting has evolved with the trial and is such that:

- Original arms - data was reviewed when 210 patients (60 patients on the control arm and 30 patients on each of the research arms) had been on trial for a minimum of 18 weeks.
- Arm G - data was reviewed when 30 patients had been allocated to Arm G and on trial for at least 18 weeks; a second safety analysis was also performed when 30 newly-diagnosed non-metastatic patients had been allocated to Arm G and on trial for at least 18 weeks.
- Arm H - data was reviewed when 30 patients had been allocated to Arm H and on trial for around six months.
- Arm J - data was reviewed when 50 patients had been allocated to Arm J and on trial for around 6 weeks (i.e. to their first follow-up visit); a second safety analysis took place when 50 patients allocated to Arm J had been on trial for 6 months. A third safety analysis was proposed for when 50 Arm J patients had completed planned RT; this was incorporated within the second activity stage analysis (Mar-2016) where 62 Arm J patients had completed planned RT (70 pts had reported RT in total for Arm J).
 - Direct comparison will be available with contemporaneously randomised patients on Arm A (hormones alone). Contextual data will be provided from Arm G (hormones plus abiraterone).
 - Indicative safety data may also be available on the combination from other studies in castrate-resistant prostate cancer.
- Arm K – data will be reviewed when 50 patients have been allocated to Arm K and on trial for around 12 months. At this time as well as reviewing safety and early feasibility data, the IDMC will consider metabolic parameters. If there is harm observed in metabolic effects, or any serious concerns regarding the toxicity profile, recruitment would be stopped; there are no formal criteria to guide this.

For such safety reports information on accrual to the trial as whole and data return rates for the research comparison, is to be presented along with baseline characteristics, treatment details, information on toxicity and events. This data is to be presented for the research comparison under review, split by arm.

Safety data are further assessed throughout the trial, with additional analyses carried out as per the request of the IDMC.

7.6 Interim Monitoring and Analyses

For each research comparison, a series of formal intermediate analyses (Activity Stages; AS) are planned where activity “hurdles” are determined on an intermediate primary outcome measure (see [Section 3](#) for detail); each research arm should need to pass this in order to continue recruitment in the next part of the trial. These are comparison-specific and will initially be timed according to the observed number of interim events required for a formal interim analysis; beyond these activity stages data will be reviewed for each comparison at regular intervals (approximately annually). These analyses will be performed by the trial team at the MRC CTU at UCL.

The un-blinded data will be reviewed by an Independent Data Monitoring Committee (IDMC) who will advise whether or not the comparison, and indeed therefore the trial, should continue on in the same way or be amended. The IDMC is advisory to a Trial Steering Committee (TSC, see Protocol Section 16) which includes some independent members and which is the executive body for the trial. Their roles are set out further in the IDMC and TSC Charters, respectively. While a comparison is ongoing the associated accumulating data will remain confidential, unless formally released by the IDMC.

For formal interim analyses, the IDMC will be guided by the intermediate hurdles but will make recommendations based on the totality of data available to them. The IDMC will be asked to give advice on whether the accumulating data from the trial justifies continuing recruitment of further patients or further follow-up; the guidelines for discontinuation of accrual for interim Activity Stages, together with results from any other trials relevant to the comparison will be used by the IDMC in this decision process. Where the sample size for a comparison is pre-specified, and as such recruitment may have completed prior to a formal intermediate analysis, any recommendations from the IDMC will consider the continuation of treatment only, as opposed to the continuation of recruitment, and make recommendations to the TSC appropriately. Where a decision is made to continue, the IDMC will advise on the frequency of future reviews of the data on the basis of accrual (if the comparison is still recruiting) and event rates.

Each intermediate analysis will be triggered by a pre-planned (approximate) number of intermediate primary outcome events being reported for patients in the control arm of each comparison; these are set out in [Tables 7 - 10](#).

Table 7: Guidelines for stopping accrual and trigger for analysis in research Arms B-F

Stage	AR	Target HR	Alpha	Power	HR cutpoint	Control arm events
AS1	2:1	0.75	0.50	95%	>1.00	~113 FFS
AS2	2:1	0.75	0.25	95%	>0.92	~216 FFS
AS3	2:1	0.75	0.10	95%	>0.89	~334 FFS
ES4	2:1	0.75	0.025	90%	---	~403 deaths

Table 8: Guidelines for stopping accrual and trigger for analysis in research Arms G&H

Stage	AR	Target HR	One-sided Alpha	Power	HR cutpoint	<u>Control arm events</u>
AS1	1:1	0.75	0.500	95%	>1.00	~75 FFS
AS2	1:1	0.75	0.250	95%	>0.92	~142 FFS
AS3	1:1	0.75	0.100	95%	>0.89	~221 FFS
ES4	1:1	0.75	0.025	90%	---	~267 deaths

Table 9: Guidelines for stopping accrual and trigger for analysis in research Arm J

Stage	AR	Target HR	Alpha	Power	HR cutpoint	<u>Control arm events</u>
AS1	1:1	0.70	0.400	95%	>0.96	~66 FFS
AS2	1:1	0.70	0.120	95%	>0.87	~139 FFS
ES3	1:1	0.75	0.025	90%	---	~269 deaths

Table 10: Guidelines for stopping accrual and trigger for analysis in research Arm K

Stage	AR	Target HR	Alpha	Power	HR cutpoint	<u>Control arm events</u>
AS1	1:1	0.80	0.400	90%	>0.96	~104 M1 deaths
ES2	1:1	0.80	0.025	85%	---	~374 M1 deaths

7.6.1 Stopping Early

The IDMC may recommend stopping recruitment early to any of the research arms, for toxicity or lack-of-sufficient-activity ("lack-of-benefit"). There is no formal stopping rule for efficacy. Any decision to discontinue recruitment/treatment, either in all patients or in selected subgroups, will be made only if the result is likely to convince a broad range of clinicians, including those entering patients into the trial, and the general clinical community. The intermediate stopping guidelines apply only to the intermediate primary outcome measure and determine only stopping early for lack-of-sufficient-activity. To stop accrual early for benefit in any comparison would require convincing data in terms of the definitive primary outcome measure, overall survival; for example, this could be $p < 0.001$ as proposed by Haybittle-Peto. The use of such a guideline for stopping for benefit has minimal impact on the operating characteristics.

7.7 Main Analysis

For each research comparison that continues (recruitment) into the final Efficacy Stage (ES), the final analysis will be triggered by a pre-specified number of definitive primary outcome events being reported for the relevant patients on the control arm of that comparison; indeed having 96% of such events would be considered sufficient to trigger this final analysis.

7.8 Factorial Thoughts

7.8.1 Factorial Analysis (Pre-specified)

Analysis of the original arms B, C and E, and also the celecoxib-containing arms, as a factorial design will be reported as a secondary analysis using a model containing both main effects. The analysis took the form of one Cox PH model including both the main treatment effects. Simulations may be carried out to assess the power to detect the presence of a treatment interaction in a further *exploratory* model with both main effects and an interaction term. The prior discussions around this type of analysis can be found in the document

["Factorial Analysis Notes 2014 05 06 v1.pdf"¹³](#).

Survival data for those other research arms which were initiated at the same time will be reported at this point.

7.9 Statistical Methods

Full details of all planned analyses are reported in [Section 8](#).

7.9.1 Time-to-event analyses

Time-to-event data will be presented using Kaplan-Meier plots. In all time-to-event analyses, patients that have not experienced the event in question (e.g. progression, death) censor date will be used, as defined in [Section 5](#). For KM plots all patients randomised to the comparison being analysed will be included. Those patients, who have no reported event and contribute no information, such that they are censored at the date of randomisation, will be censored with a time of 0.1 days.

The HR will be taken from a Cox proportional hazards model, adjusted for those stratification factors used at randomisation (see [Section 5](#)) except for centre and method of hormones, and stratified by the relevant time period/strata for when the comparison was recruiting (see [Section 7.3](#)).

Differences in time-to-event outcomes will be assessed using the log-rank χ^2 test and expressed using a hazard ratio (HR) with both a 1-sided 90% confidence interval and also a 2-sided 95% confidence interval; where the design alpha differs from 0.025 the upper limit of corresponding relevant 1-sided confidence interval will be presented (for example, for an alpha of 0.25 this would be the upper limit of a 1-sided 75% confidence interval). The number of events observed and the log-rank expected number of events will be presented.

¹³ S:\Stampede_Stats\Original Comparisons\Factorial Analysis\Factorial_Analysis_Notes_2014_05_09_v1.pdf

Alternative estimation of the treatment effect will use exploratory methods e.g. log-rank hazard ratio. Flexible parametric models, modelling the difference between treatment groups over time, will be fitted to the time-to-event data with and without including time-dependent treatment effects.

7.9.2 Proportional and Non-Proportional hazards

The assumption of proportional-hazards (PH) will be tested on the basis of Schoenfeld residuals after fitting the cox model. Application in Stata will be using the command `-estat phtest-`. Appropriate methods will be applied to the data in the case of any violation, and presented alongside the primary analysis to aid interpretation; see details below.

If there is no evidence of non-PH in the treatment effect, the HR for the adjusted Cox model takes primacy.

If there is evidence of non-PH in the treatment effect, the HRs are difficult to interpret and the restricted mean survival time¹⁴ (RMST) constructed from a flexible parametric model with time-varying treatment effects will take primacy. This analysis will use a flexible parametric model, adjusted for the stratification factors used at randomisation (except for centre and method of hormones) and relevant time strata, to determine the time-dependent treatment effect and then predict values needed for subsequent RMST analysis. The application of this methodology will be particularly useful in the case where there is adequate evidence that the proportional hazard assumptions are violated at the 5% level (i.e. $P < 0.05$). Application in Stata will be using the command `-strmst-`.

RMST will be calculated and reported for the main treatment effects regardless of whether the PH assumption is violated, as this is recognised as a useful measure. The time within which RMST will be computed will be determined by the timing of events in the control arm (using the command `-maturity_rmst-` in Stata). For this purpose, t^* (the time up to when we want to compute the outcome measure) will be defined as the maximum available follow-up time if clinically meaningful, where there is most power for the analysis, or a suitable salient time point otherwise, for each research comparison.

For formal intermediate analyses, the level of confidence around the HR will be noted; the same level confidence interval as for that stage will be constructed around the difference in RMST with the appropriate confidence interval level determined according to the alpha calculated for that intermediate analysis. If the lower bound of this confidence interval includes zero (no difference) the research arm will be considered insufficiently active.

7.9.3 Competing Risks

Analysis of those outcomes where there are considered to be competing risks will be done in addition to an adjusted Cox model with a competing risk defined as any event the patient would likely have experienced had they not experienced the outcome of interest first.

To date those outcomes with competing risks are:

¹⁴ Restricted mean time-to-event time

- Disease-specific survival: competing risk is death from other causes
- Non-PCa death: competing risk is death from PCa
- Lymph node progression: competing risk is distant metastatic progression or all cause mortality
- Distant metastatic progression: competing risk is all cause mortality

7.10 Patients with no data

All estimates of treatment effect will not include those individuals with no reported outcome event and contributing no censoring information such that they are censored at their date of randomisation. This is because Stata can struggle to fit models around such data.

7.11 Maturity

Median follow-up time will be calculated using a 'reversed' Kaplan-Meier approach, taking censor date (if alive) to be an event and death as the time of censorship. The median time will be detailed for any sub-groups for the life tables. In patients who were last known to be alive, the median time to last follow-up will be calculated using standard summary statistics. Date last seen (if alive) is as defined within the censor date in [Section 5](#).

7.12 Deaths and Cause of Death

A review of cause of death (with reviewers blinded to treatment allocation) will be conducted as a quality control process in order to ensure that the causes of death attributed by Investigators on site are as accurate as possible (see [Section 5](#)).

7.12.1 Data linkage for death

Data from ONS or similar sources will be used, when and/or where available, to confirm deaths, detect unreported deaths and determine cause of death (if unknown and/or unreported). Where such external data sources are used for analyses of survival or disease-specific survival, patients who are not confirmed by this data source as having died will be censored close to the time of receipt of this external data; the exact time will be determined in discussion with the provider of such data and will be reported explicitly. For example, for two previous CTU-led trials in prostate cancer, survival data were available from ONS and, in discussion with the ONS Chief Statistician, it was reasonable to assume that patients were still alive 8 weeks before data transfer if they were not reported as having died. This approach accepted that deaths referred to a coroner for review would have been inappropriately classified as censored rather than dead (with the time of censoring possibly later than the actual time of death).

A sensitivity analysis for linked deaths will be performed excluding those events and censoring at last contact as being event-free.

7.13 Populations

We define two populations for analysis; the intention-to-treat population and the safety population.

1. **Intention-to-treat (ITT) population**

- Comprised of all randomised patients, whether or not they actually received the allocated trial treatment.
- In ITT analyses by treatment arm, patients will be included in the treatment arm to which they were randomised.

2. **Safety population**

- Comprised of patients who have been administered at least one dose of their allocated trial treatment.
- Patients will be classed as having started trial treatment if they report a treatment start date, report an SAE which is treatment related, or report death as being within 4 weeks of trial treatment (& related to trial treatment)
- In safety population analyses, patients will be included in the treatment arm corresponding to the treatment they actually started; for example an Arm B patient not starting zoledronic acid will be included with Arm A patients, whilst an Arm E patient not starting docetaxel but starting zoledronic acid will be included with Arm B patients.
- Two sensitivity safety analyses will be conducted around this classification, wherein patients with unclear treatment starting status will be included firstly in allocated treatment arm (making the assumption they start trial treatment) and secondly in the control arm (making the assumption they don't start trial treatment).

The ITT population will be included in all analyses unless specified. The safety population will be included in analyses of adverse events and other safety data (safety analyses).

For visual illustration a CONSORT flow diagram will clearly identify any patients found to be ineligible post-randomisation or stopping trial follow-up early; these patients will be included in relevant analyses where possible. For reference, a template flow diagram can be found in [S:\Stampede Stats\SAP](#).

7.14 Safety Analyses

Safety analyses will be performed and presented on the safety population; for headline numbers reporting toxicity, these will be repeated on the ITT population as defined in [Section 7.13](#).

All safety analyses will focus on adverse events experienced during treatment and up to 30 days after the end of treatment (see Protocol section 11.2.1.D).

Safety will be evaluated by tabulation of adverse events at or up to pre-defined follow-up time points. Adverse events will be classified using the NCI Common Toxicity Criteria for Adverse Events (CTCAE v3.0; updated to CTCAE v4.0 for all assessments dated from Protocol v15.0 (05-Sep-2016) onwards) and summarised for each treatment arm. Reported grading is "0 = toxicity not experienced" up to "5 = fatal".

Adverse events (AEs) may be detected through several sources:

1. Follow-up CRF – routinely reported symptoms and "toxicities" (severity not seriousness)
 - AEs reported here up to Sep-2016.
2. Toxicity CRF – routinely reported symptoms and "toxicities" (severity not seriousness)
 - AEs reported here from Sep-2016 onwards.
 - Linked to routine follow-up visits, treatment actions and permanent stopping of treatment.
3. SAE CRF – spontaneously reported serious adverse events (severity and seriousness)

Not all serious events are severe and vice versa. "Seriousness" is a term specific to the reporting of events to regulatory bodies. We have prioritised the consideration of "severity" for balancing against activity data. SAE forms and follow-up forms both request the severity of events. Therefore, these sources can be merged to form one dataset for reporting "severe" data. The focus will be on toxicities with grade 3, 4 or 5 (fatal), however all toxicity grades will be reported for completeness.

For toxicity data reported **at set time points** from randomisation this will be reported only for patients who have not progressed by the set time point; this is because for patients on either the control arm or research arms where trial treatment is to be stopped at progression toxicity data is only expected up to the time of first reported progression. For patients on research arms where trial treatment can continue beyond first progression (namely current Arms G, J and K), an additional report of toxicity data at specific time points from randomisation will be presented only for research arm patients still on treatment at that time. Reporting windows will be defined around any such set time points which will be as close to the time of interest whilst accepting that clinical practice means that most patients will not be reviewed on a specific day.

For toxicity data reported **up to pre-specified time points** from randomisation this will include all patients with follow-up/toxicity/SAE data available within that time frame.

All patients receive ADT as standard-of-care and so interest will be in the 'additional toxicity reported' for patients on research arm relative to control arm, compared informally. Interest will also be in any proportion of known treatment toxicity viewed to be above that which is expected in this population.

7.15 Missing Data

Missing data will be assumed to be missing completely at random (MCAR) in intention-to-treat analyses; exploratory analyses may be undertaken to further understand any missing data.

7.16 Pre-specified Subgroup Effects

STAMPEDE recruits patients at high risk of dying from prostate cancer and who are starting long-term hormone therapy for the first time (hormone-naïve disease). This is a broad spectrum of patients including both men with metastases at randomisation and men without metastases, and men with de novo disease or disease relapsing after previous local therapy.

7.16.1 Stratification Factors

The stratification factors considered at the point of randomisation will form subgroups in which to assess treatment effect, with an interaction p-value of less than 0.1 used to suggest evidence of a difference across the relevant subgroups; as with all subgroups it is accepted that there is limited power both for the interaction and within group analysis. In addition the time periods the relevant comparison was recruiting across (see **Table 5**) and also whether the patient entered the trial with recurrent disease will be considered as subgroups in which to assess treatment effect, along with pre-hormone therapy PSA as a continuous variable.

7.16.2 Metastatic status

For all research comparisons where both metastatic and non-metastatic patients are eligible metastatic status at randomisation will be a pre-specified subgroup. In such instances the non-metastatic patients are a smaller subset of patients within STAMPEDE and their event rate is lower, so the TMG accepts that there will be limited power to make reliable comment on those patients. Tests for heterogeneity by metastatic status will be presented, but an effect in the subgroups will be considered as a pre-planned analysis.

In the future a further subgroup analysis will be conducted within those patients with metastatic disease at randomisation, by low volume vs high volume disease; for the “abiraterone comparison” this is planned for Q3-2017 to Q1-2018. Classification into low volume or high volume disease will be based on data from a planned sub-study of the bone scans of patients with metastatic disease at presentation; data to be collected retrospectively on patients on the “docetaxel comparison”, the “zoledronic acid + docetaxel comparison” and the “abiraterone comparison”.

Further comparison-specific subgroups are detailed in the following sections.

7.16.3 The “Original Comparisons”

Since the trial commenced, survival data have been reported from two previous CTU-coordinated trials looking at oral sodium clodronate, a first generation bisphosphonate. These trials showed a survival advantage in metastatic patients (PR05) but not in non-metastatic patients (PR04), albeit in a lower risk group than the non-metastatic patients in STAMPEDE; zoledronic acid is a much more potent bisphosphonate than clodronate.

Other data have emerged on zoledronic acid and docetaxel which also suggest that gradation in treatment effect might be expected according to disease burden. Therefore, in addition to looking in the overall population, using the approaches described in [Section 7.9](#), we looked specifically at the treatment effect in metastatic patients in the docetaxel and zoledronic acid comparisons, in light of the data from the CHAARTED¹⁵, CALGB 90202¹⁶, GETUG-12¹⁷ and GETUG-15¹⁸ trials. At the time of the original comparison primary analyses this will have good power (likely to be at least 80% for the comparison in metastatic patients alone) although reduced power compared to the overall comparison.

7.16.4 The “Abiraterone Comparison”

There are no additional pre-specified comparison-specific subgroup analyses other than those detailed in [Section 7.16.1](#) & [7.16.2](#)

7.16.5 The “M1|RT Comparison”

For the “M1|RT comparison”, RT schedule was nominated for all patients at randomisation with roughly half of patients nominating each of the two options. At the time of the primary analysis a FFS analysis “within schedule” will be carried out; predicted to have ~300 control arm FFS events by schedule (FFS “within schedule” analysis parameters: target HR=0.75, power 90%, 1-sided $\alpha=0.015$). If one or both of the RT schedules show evidence of an effect on FFS, a comparative “within schedule” analysis will be carried out on overall survival when ~199 control arm deaths are observed in that schedule comparison. This is a closed test with OS only formally compared within schedule if there is an advantage in FFS for that RT schedule at the main analysis.

7.17 IDMC analysis requests

This sub-section describes any analyses of data requested by the IDMC in order to provide further detail. To date these are as described below and comparison-specific.

7.17.1 The “abiraterone comparison”

For the “abiraterone comparison” the requests are:

- Consider whether SARs and SUSARs may be linked to men staying on abiraterone treatment for longer.
- Report the numbers of patients who develop metastases without PSA failure.

¹⁵ Sweeney CJ, Chen Y-H, Carducci M, et al. Chemohormonal Therapy in Metastatic Hormone-Sensitive Prostate Cancer. *The New England journal of medicine*. 2015;373(8):737-746. doi:10.1056/NEJMoa1503747.

¹⁶ Smith MR, Halabi S, Ryan CJ, et al. Randomized Controlled Trial of Early Zoledronic Acid in Men With Castration-Sensitive Prostate Cancer and Bone Metastases: Results of CALGB 90202 (Alliance). *Journal of Clinical Oncology*. 2014;32(11):1143-1150. doi:10.1200/JCO.2013.51.6500.

¹⁷ Fizazi K, Faivre L, Lesaunier F. Androgen deprivation therapy plus docetaxel and estramustine versus androgen deprivation therapy alone for high-risk localised prostate cancer (GETUG 12): a phase 3 randomised controlled trial. *Lancet Oncol*. 2015;16:787-794

¹⁸ Gravis G, Fizazi K, Joly F. Androgen-deprivation therapy alone or with docetaxel in non-castrate metastatic prostate cancer (GETUG-AFU 15): a randomised, open-label, phase 3 trial. *Lancet Oncol*. 2013;14:149-158.

- Consider the timing of progression compared to stopping abiraterone amongst those patients who stop for reasons other than progression.

8 ANALYSIS DETAILS

The results of the analyses will be reported following the principle of the ICH E3 guidelines on the Structure and Content of Clinical Study Reports¹⁹.

For the final efficacy analyses, the flow of patients through the trial during the time the relevant comparison was recruited, will be presented in a CONSORT diagram.

8.1 Pre-planned data checks

- ☐ Current ADT on randomisation form *vs* baseline form (patients randomised before 05-Sep-2016)
- ☐ Broad disease category *vs* [pre-HT PSA value, metastatic status, nodal status, gleason score, previous treatment reported]
- ☐ Disease progression on FU form *vs* progression form
- ☐ Death on progression form *vs* death form
- ☐ Death as SAE form *vs* SAE form with death reported
- ☐ Death reported as within 4 weeks of trial drug *vs* calculation of death within 4 weeks of trial drug
- ☐ Observed PSA at failure *vs* expected minimum PSA value at failure

8.2 Planned analyses

Within each research comparison all analyses and tabulations will be presented across all relevant trial arms and sometimes by specific subgroups.

This is a living document and changes may be required throughout the trial, including changes that could be driven by changes in practice at MRC CTU at UCL. Changes will be reflected by changes in date and version of this document.

8.2.1 Recruitment

- ☐ Graph of observed recruitment to the relevant comparison (include line of expected recruitment if sample size pre-specified)
- ☐ Table of recruitment by year & centre
- ☐ Graph of monthly summary of observed patients and active centres:
 - Interim activity analyses & annual reviews - for the past year of recruitment
 - Final efficacy analyses – for the months the comparison was open for recruitment
- ☐ Table of recruitment across relevant time strata, by treatment arm

8.2.2 Baseline Characteristics

The following baseline characteristics will be presented, broken down by treatment arm unless otherwise stated, either as n (%) or median (IQR; min-max) as applicable:

¹⁹ http://www.ich.org/fileadmin/Public_Web_Site/ICH_Products/Guidelines/Efficacy/E3/E3_Guideline.pdf

- All stratification factors (see ""S:\Stampede_Stats\SAP\Stratification_Factors_OverTime.doc" & **Section 5**)
- Randomisation CRF data:
 - Age at randomisation (years)
 - PSA at randomisation (ng/ml; defined as PSA pre-HT)
 - Ln (PSA) at randomisation (defined as Ln (PSA) pre-HT)
 - Time from diagnosis to randomisation (days)
 - Pain from prostate cancer at randomisation: Absent; Present
 - Broad disease category: N0M0 new; N+M0 new; M1 new; Local treatment now relapsing
 - T-stage at randomisation
 - N-stage at randomisation
 - Any metastases at randomisation
 - Bone metastases at randomisation
 - Liver metastases at randomisation
 - Lung metastases at randomisation
 - Nodal metastases at randomisation
 - Other metastases at randomisation
 - Use of aspirin
 - Use of NSAIDs
 - Use of short-term bisphosphonates (subgroup: broad disease category)
 - Planned type of HT (randomisation CRF; subgroup: broad disease category)
 - Planned use of long-term anti-androgens
 - Planned use of radiotherapy (subgroup: nodal & metastatic status)
 - Cumulative randomisations (subgroup: nodal, metastatic & planned RT status)
 - Participation in QL study
 - Time from randomisation to starting current HT (times negative if pt starts HT pre-randomisation)
 - Previous HT type: None; LHRH (agonist or antagonist if known); AAs alone; MAB
 - Previous local therapy type (if known): None; radical prostatectomy; radical radiotherapy; radical prostatectomy with post-operative radical radiotherapy; other
 - Duration of previous HT (days)
 - Months between end of previous HT and randomisation (subgroup: broad disease category)
- Baseline CRF data:
 - Gleason sum score at presentation
 - T-stage at presentation
 - N-stage at presentation
 - M-stage at presentation
 - PSA at first presentation (ng/ml)
 - Ln(PSA) at first presentation
 - Concomitant medications (to be clinically recoded)
- Cardiovascular assessment data
 - Smoking status (subgroup: broad disease category)

- Diabetes and type (subgroup: broad disease category)
- History of MI, CV disease, CHF, angina or hypertension (subgroup: broad disease category)

8.2.3 Standard-of-care Treatment

For all standard-of-care treatments the following data will be presented, broken down by treatment arm:

- Hormone therapy details (from the FU CRF, HT CRF or the HT Log)
 - Numbers reporting treatment action of changing or stopping (N, %)
 - Time to treatment action
 - Reason for treatment action
- Docetaxel treatment details (from the SOC Docetaxel Treatment CRF; planned details from Randomisation CRF)
 - Reported vs planned docetaxel
 - Time from randomisation to first cycle of docetaxel (days)
 - Time from starting ADT to first cycle of docetaxel (days)
 - Number of cycles administered
 - Reason for less than 6 cycles
 - Daily steroid formulation
 - Daily steroid dose
- Radiotherapy treatment details (from the RT detail CRF; planned details from the Randomisation CRF)
 - Reported vs planned use of RT
 - Timing from randomisation to first fraction of RT in:
 - All patients
 - Patients planned for RT
 - NOM0 patients
 - N+M0 patients

8.2.4 Trial Treatment

Please see [SAP v1.0](#) for previous analyses planned for docetaxel, zoledronic acid and celecoxib research treatment data.

For **abiraterone, enzalutamide and metformin** research treatment data the following analyses are planned:

- Number of patients starting allocated trial treatment (N, %)
- Details of patients confirmed as not starting allocated trial treatment (as reported on the End of Research Treatment CRF)
 - Include note of any patients found to be ineligible post-randomisation with action taken if treatment was initially started.
- Time from randomisation to start of treatment (in all pts; censor those who don't report starting)
- Starting dose
- Number of treatments administered (where multiple allocated; i.e. Arm J enzalutamide + abiraterone)

- Treatment compliance/tolerance
 - Numbers reporting treatment action of changing or pausing or restarting (N, %)
 - Time to treatment action (only in pts who report starting)
 - Reason for treatment action (including associated toxicity details if relevant)
 - Length of each treatment break
 - Proportion of missed days
- Frequency and reason for dose modification and delays, treatment termination
 - Include detail of any protocol deviations in relation to overdoses.
- Permanent cessation of trial treatment (as reported on the End of Research Treatment CRF)
 - Numbers (N, %)
 - Time from randomisation to stopping (Overall & subgroup: metastatic status)
 - Censor at randomisation if explicitly reported as not starting
 - Censor at last contact if reported as starting but no report of permanent stopping
 - Time from starting treatment to stopping in patients who report starting (Overall & subgroup: metastatic status)
 - Censor at last contact if reported as starting but no report of permanent stopping
 - Reason (including associated toxicity details if relevant)
- Additional treatments given (as reported on the Additional Treatment CRF)
 - Include detail of any patients receiving non-protocol disease-directed interventions prior to study outcomes

For **Arm H radiotherapy** treatment data the following analyses are planned:

- Number of patients starting RT
- Details of patients confirmed as not receiving RT
- Time from randomisation to first fraction of RT (in all pts; censor those who don't report starting)
- Planned vs reported RT schedule (planned details on the Randomisation CRF)
- Details of patients with different RT dose reported to that planned
- Graph of planned vs reported RT dose
- Details of patients with RT reported as being delayed
- RT treatment details; time to stopping (days from randomisation) and reason for stopping
- Graph of time from randomisation to last RT fraction
 - Censor at randomisation if explicitly reported not having RT as allocated
 - Censor at last contact if RT not yet reported
- Graph of duration of RT (first to last RT fraction; in pts who started)
- Details of patients who report RT to the pelvic nodes

8.2.5 Trial Events

All trial outcomes will be analysed on an ITT basis, as per randomised allocation regardless of treatment received.

8.2.5.1 Primary Outcome Measure

For the primary outcome, overall survival (see [Section 5](#) for definition) the following data will be presented for the research comparison, overall and split by metastatic status (high/low volume disease), unless specified:

- ☐ Incidence of death, by treatment arm
- ☐ Estimate of survival by year from randomisation, by treatment arm (%, 95% CI)
- ☐ Cause of death, by treatment arm (death form and death review process)
- ☐ Reported cause of death *vs* review cause of death
- ☐ Place of death, by treatment arm
- ☐ Death within 4 weeks of trial drug, by treatment arm
- ☐ Death related to trial treatment, by treatment arm
- ☐ Time from randomisation to death from any cause, by treatment arm
 - ☐ KM survival plot and median from FPM
 - ☐ Censor individuals at last contact if not died
- ☐ Test for proportional hazards assumption
- ☐ Fitting of flexible parametric model
- ☐ Estimation of differences between fitted FPM curves
- ☐ Comparisons of research *vs* control from Cox model
 - ☐ Adjusted for all stratification factors except centre and method of hormones
- ☐ Calculation of restricted mean survival time
 - ☐ Suitable t-star to be calculated at time of analysis
- ☐ Explicit statement of whether adjusted Cox model or RMST takes primacy
- ☐ Test for heterogeneity of treatment effect across stratification factors, Gleason score and new/previously treated
- ☐ The following sensitivity analyses will be undertaken for comparisons of research *vs* control:
 - ☐ Log-rank HR unstratified/stratified..., 95% CI
 - ☐ Unadjusted Cox model
 - ☐ Flexible-parametric model
 - ☐ Adjusted Cox with time-varying WHO performance score

8.2.5.2 Secondary Outcome Measures

For all secondary outcomes (see [Section 3](#)) the following analyses will be performed for the research comparison, overall and split by metastatic status (high/low volume disease), unless specified:

- ☐ Incidence of the outcome, by treatment arm
- ☐ Estimate of (*freedom from*) outcome by year from randomisation, by treatment arm (%, 95% CI)
- ☐ First reported progression event, by treatment arm [FFS only]
- ☐ Time from randomisation to outcome, by treatment arm

- KM survival plot and median from FPM
- Censor individuals at last contact if outcome not reported
- Test for proportional hazards assumption (see **Section 7.9.2**)
- Fitting of flexible parametric model
- Estimation of differences between fitted FPM curves [FFS only]
- Comparisons of research vs control from Cox model adjusted for stratification factors (excluding centre)
- Calculation of restricted mean survival time; using the t-star from OS analysis [FFS only]
- Note of whether adjusted Cox model or RMST takes primacy [FFS only]
- Sensitivity analyses [FFS only] =
 - Comparisons of research vs control: log-rank HR, 95% CI; unadjusted Cox model; flexible-parametric model; adjusted Cox with time-varying WHO performance score
- Presentation of competing risks analysis for those outcomes specified in **Section 7.9.3**
- For distant metastases present overall and by high/low volume disease (when available):
 - Site of first distant metastases, by treatment arm
 - Incidence of each of bone, liver, lung, distant node mets, by treatment arm
 - Freedom from each site annually, by treatment arm (%; 95% CI; if >50 events reported)
- For skeletal related events present type, by treatment arm

8.2.6 PSA response

PSA response data within the first 24 weeks on trial are used to determine a patient's biochemical progression category. Of interest is whether this differs by allocated treatment arm, therefore the following data should be presented by treatment arm on an intention-to-treat basis:

- Boxplot of Log of PSA nadir (to 24 weeks)
 - Give data underpinning this including N, min, Q, Max, mean, SD
- PSA nadir category, N (%)

8.2.7 Toxicity/Safety

Toxicity data will be reported using the NCI Common Toxicity Criteria for Adverse Events (see **Section 7.14**) and are to be presented in the **ITT population for interim analysis reports and annual review reports** to the IDMC; however breakdown of safety/toxicity data should be presented in the **safety population for final analysis publication and reports**, with only headline figures shown for the ITT population to demonstrate comparability of the populations.

Data presented by treatment arm should be:

- KM plot of time to first G3-5 toxicity reported on the FU or SAE CRF; include maximum SAE grade
- Worst toxicity grade in any category (Overall & subgroup: metastatic status); in each category; proportion with grade 3-5, for the following time points/periods:
 - Within 6m from randomisation (up to and including week 30)
 - At 1 year (+/- 12 weeks)
 - At 2 years (+/- 12 weeks)
 - Ever on trial
- Time to first grade 3-5 SAE

- ☐ Time to any grade SAE
- ☐ Time to first grade 3-5 SAR
- ☐ Time to any grade SAR
- ☐ Time to first grade 3-5 SUSAR
- ☐ Time to any grade SUSAR
- ☐ Table of all reported SAEs reviewed as being related to treatment (classified as a SAR or SUSAR); table to include the following details ordered by trial arm and Patient ID:
 - Patient ID
 - Date of randomisation
 - Date of onset
 - Main diagnosis and associated symptom(s)
 - Grade
 - Why the event is serious
 - SAE status
 - Causal relationship to HT/trial drug; related? expected?
 - Reviewed relatedness; SAR or SUSAR

For Arm H patients the following additional RT acute toxicity data should be presented (data collected on the RT acute toxicity CRF):

- ☐ Table of worst bladder and bowel toxicity grade, split by intended dose
- ☐ Graphs of worst RT acute toxicity:
 - All patients, split by bladder/bowel
 - Bladder, split by intended RT dose
 - Bowel, split by intended RT dose
 - Bladder, split by week number and intended RT dose
 - Bowel, split by week number and intended RT dose

8.2.8 Data Returns

- ☐ Forms returned compared to expectations (routine forms only; randomisation, baseline (incl. cardiovascular & BD Risk), follow-up, radiotherapy detail/acute toxicity)
- ☐ Death, progression and end of trial participation forms received for the control arm
- ☐ FU forms expected vs number of FU forms received, for each patient
- ☐ Scatter plot of timing of most recent received FU for each patient vs time from randomisation, by treatment arm (including detail on number of patients randomised, number of deaths, number of withdrawals, number okay and number late for follow-up)
- ☐ Number of FU forms dated within each week since patient's randomisation
- ☐ Time since last FU form (KM graph; subgroup: randomised <24wks vs 24+wks)
- ☐ Duration of follow-up after randomisation, with median using reverse KM plot
- ☐ For surviving patients the time since last follow-up form received, by treatment arm
- ☐ Details of patients Lost-to-FU or for whom early stopping of follow-up and data collection has been reported

8.3 Prognostic Factors

Investigational analyses may be put together considering prognostic factors for FFS and OS where relevant.

8.4 Quality of Life analysis

Quality of life data, for those patients participating in the QoL study, is collected at baseline and at each follow-up visit, up to first progression (or longer if the patient wishes).

The questionnaires used to collect this data are:

- EORTC QLQ-30 Quality of life form
- EORTC PR-25 Quality of life form
- EQ-5D Health Economics form

Expert advice will be sought before planning and conducting any quality of life analyses in reference to "Clinical Trials in Cancer" – D. Girling, M. Parmar, S. Stenning, R. Stephens and L. Stewart.

A separate SAP is to be developed for these analyses.

8.5 Health Economics

Health economics (HE) analyses will be performed by Mark Sculpher et al at the University of York. That data used for a research comparison's final efficacy analysis will be released for such HE analysis, along with the relevant quality of life data.

HE data will encompass baseline characteristics, investigational and SOC treatment data, subsequent therapies, trial outcomes, Quality of life data (notably EQ-5D) and background costs data. Full details of the data requested can be found in the SMF and also in [S:\Stamperde Stats\Data\Data Requests\York HE Analysis](#).

A separate SAP is to be developed for these analyses.

8.6 Exploratory Analyses

Exploratory analyses will be clearly noted as such in all reports and presentations.

These will not be predetermined but will be driven by data emerging from within the trial or from external sources. This may include a look at consistency of effect by centre or centre size; the relationship of baseline characteristics to toxicity and toxicity to outcomes; the relationship of baseline characteristics to the “missingness” of later date points.

8.7 Comparing Effective Research Arms (Closed Test)

Direct comparison of research arms will be undertaken in a closed test approach with only those research arms shown to be better than the control arm being compared. Only contemporaneously randomised patients will contribute to this analysis.

The definitive primary outcome will remain the key outcome and comparisons only carried out in contemporaneously randomised patients (see [Section 7.3](#)). The reference arm of any such direct comparison will be that research arm with the less favourable observed hazard ratio for research arm vs control arm on overall survival.

Indirect comparison, of research treatments better than SOC, will be undertaken in the context of meta-analyses described in a separate document.

8.8 Reporting

Intermediate analyses will be collated and reported to the IDMC. Data will be released to the TSC if required and on to the TMG if deemed appropriate. Not all comparisons will be featured in every intermediate reporting, depending on the stage of the trial, the need for the analysis and time pressures.

The following table gives details of the intermediate analyses for all research comparisons.

Table 11: Intermediate Activity Stage Analyses – Targeted vs Observed events

Research Comparison	AS#	Control Arm Events		IDMC Report	IDMC Recommendation
		Target	Observed		
Original (Arms B-F)	1	113	129	30-Mar-2010	Continue as planned
(Arms B-F)	2	216	209	31-Mar-2011	Stop recruitment to arms D&F
(Arms B,C,E)	3	334	341	30-May-2012	Continue as planned
Abiraterone (Arm G)	1	75	88	30-Aug-2013	Continue as planned
	2	142	162	07-Feb-2014	Continue as planned
	3	221	241	06-Jun-2014	Continue as planned
M1 RT (Arm H)	1	75	69	06-Jun-2014	Continue as planned
	2	142	144	21-Nov-2014	Continue as planned
	3	221	255	13-May-2015	Continue as planned
Enzalutamide+abiraterone (Arm J)	1	66	85	05-Nov-2015	Continue as planned
	2	139	136	22-Mar-2016	Continue as planned

For each research comparison a final study report will be constructed. This will be the basis of each survival-based paper (see [Publication Plan](#)²⁰). These will be shared with the appropriate groups and as determined by contracts with external parties.

²⁰ See most recent version in S:\Stampede\Presentations_Publications\Publication Plan\ Current_version

9 ADDITIONAL INFORMATION IN THE CLINICAL PAPERS

Methodological papers will be developed and referenced. Clinical data will be presented across many papers.

Since the trial is ongoing, results of the interim analysis for each research comparison will be reviewed confidentially by the IDMC initially and they will make recommendations as to whether any data should be released at this time to the Trial Management Group and Trial Steering Committee (in the first instance) and whether external presentation/publication is appropriate. In this scenario, details of data to be included in any resulting publication would be based on advice from the IDMC in discussion with the Trial Management Group.

Details of any additional information to be included in the clinical paper following the primary analysis for a research comparison will be finalised prior to that analysis.

10 SIGNATURES OF APPROVAL

Date: 31-Jan-2017

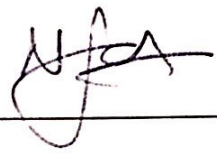
Version: 2.00

Signatures

Name	Trial Role	Signature	Date
Nick James	Chief Investigator* "abiraterone comparison" CI		
Chris Parker	"M1 RT comparison" CI		
Gert Attard	"enzalutamide + abiraterone comparison" CI	 Digitally signed by Gert Attard DN: cn=Gert Attard, o=Institute of Cancer Research, ou=Cancer Therapeutics, email=gert.attard@icr.ac.uk, c=GB Date: 2017.02.02 11:28:12 Z	
Matt Sydes	Trial Statistician	 Digitally signed by Matthew R Sydes DN: cn=Matthew R Sydes, o=University College London (UCL), London, UK, ou=MRC Clinical Trials Unit at UCL, Institute of Clinical Trials and Methodology, email=m.sydes@ucl.ac.uk, c=GB Date: 2017.02.09 09:03:26 Z	
Melissa Spears	Statistician	 Digitally signed by Melissa Spears DN: cn=Melissa Spears, o=MRC CTU @ UCL, ou=MRC CTU, email=m.spears@ucl.ac.uk, c=GB Date: 2017.02.09 09:05:06 Z	

*On behalf of the STAMPEDE Trial Management Group

10 SIGNATURES OF APPROVAL**Date:** 31-Jan-2017**Version:** 2.00**Signatures**

Name	Trial Role	Signature	Date
Nick James	Chief Investigator* "abiraterone comparison" CI		8 Feb 2017
Chris Parker	"M1 RT comparison" CI		
Gert Attard	"enzalutamide + abiraterone comparison" CI		
Matt Sydes	Trial Statistician		
Melissa Spears	Statistician		

*On behalf of the STAMPEDE Trial Management Group

10 SIGNATURES OF APPROVAL**Date:** 31-Jan-2017**Version:** 2.00**Signatures**

Name	Trial Role	Signature	Date
Nick James	Chief Investigator* "abiraterone comparison" CI		
Chris Parker	"M1 RT comparison" CI		1/2/17
Gert Attard	"enzalutamide + abiraterone comparison" CI		
Matt Sydes	Trial Statistician		
Melissa Spears	Statistician		

*On behalf of the STAMPEDE Trial Management Group